TRACK P (PLANETARY): BLUEPRINT

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Abstract. A proof plan, step by step, for the three frozen statements of track P: the properly degenerate KAM theorem of Chierchia–Pinzari (specialized as frozen), the analytic planar two-planet theorem with a moving Sun, and the fixed-Sun n-planet variant. Part 0 fixes conventions, restates every frozen definition and the three frozen theorems; the remaining parts are listed in `PLAN.md`. Results of track K's blueprint are cited as [K:`label`].

Contents

1. Conventions, definitions and the frozen statements 3
1.1. Conventions 3
1.2. Track-K definitions used by track P 4
1.3. Elementary planar and index definitions 4
1.4. Definitions for the properly degenerate theorem 5
1.5. Definitions for the physical theorems 6
1.6. The frozen theorems 9
2. Hamiltonian tools for tori: charts, period conversion, Diophantine conversion, Liouville 10
2.1. Conventions and external facts 10
2.2. Conformally symplectic charts 11
2.3. Transport of tori 12
2.4. Period conversion 14
2.5. Diophantine conversion 15
2.6. Liouville measure and unions of tori 15
2.7. The ODE layer (cited) 17
2.8. Sources, deviations, forward references 17
3. Averaging with two frequency scales 18
3.1. Conventions: norm, widths, dictionary 18
3.2. N's step lemma under a direct divisor hypothesis 18
3.3. The averaging proposition in N's conventions 20
3.4. The interface of Lemma 4.8 21
3.5. Averaging with an oscillator pair (CP10 §4, Step 1) 22
3.6. Suspected errata in CP10 and open points 25
4. Two-scale KAM theorem I: statement and the iterative lemma 26
4.1. Conventions, norms, constants 26
4.2. The theorem 27
4.3. External facts and imported tools 29
4.4. Two elementary lemmas 29
4.5. Averaging with two frequency scales (the case $m = 0$, $\Lambda = \{0\}$ of CP10 Proposition 1) 30
4.6. One step of the scheme 30
4.7. The iterative lemma 32
5. Two-scale KAM theorem II: the limit, the tori and the measure 33
5.1. The limit actions (CP10 Step 1) 34
5.2. The limit symplectic map (CP10 Step 2) 34

5.3. The tori (CP10 Steps 2–3 and the estimates (35)) 36
5.4. Measure estimate (CP10 Step 4) 37
5.5. Proof of Proposition 4.3 38
5.6. What Part 5 consumes 38
5.7. Source gaps and suspected misprints (CP10, Prop. 3, Appendix B) 39
5.8. References and imported facts 40
6. Measure estimates for P-G 41
6.1. External facts 41
6.2. The boundary collar 41
6.3. Uniform Lipschitz facts 42
6.4. The two-scale Diophantine measure (CP10 Lemma 3.1) 42
6.5. Collar of the domain $D = \bar{D}_t \times A$ 45
6.6. The cylinder cell and the singular shell 45
6.7. The budgets consumed by Parts 3 and 5 46
6.8. Suspected misprints and gaps in CP10 (Lemma 3.1, Appendix C, §3 end) 47
6.9. References and imported facts 47
7. P-G assembly: uniform remainder, parameters, application of Parts 2–4, tori, measure, strictness 47
7.1. Conventions and the order of the constants 48
7.2. Uniform complex extension and uniform remainder 48
7.3. The fast frequency map on a complex neighbourhood; the nondegenerate frequency map of the averaged system (domain reduction, CP10 (58)) 50
7.4. The parameters 52
7.5. Step 1 of CP10 §4: averaging (Part 2) 52
7.6. The KAM Hamiltonian in oscillator action–angle variables 53
7.7. The scale constants, the KAM condition, and r_K (replacing [P:bpP:p4:use5]) 54
7.8. Location, the tori, and the Kolmogorov set 55
7.9. The measure budget (replacing [P:bpP:p4:budget] and [P:bpP:p4:cell-budget] for $D = \check{D} \times A_{\mathrm{sh}}$) 56
7.10. Strictness and positivity (ours) 57
7.11. Proof of the frozen theorem 58
7.12. Errata, deviations, external facts, forward references 58
8. Newtonian models and physical reconstruction 59
8.1. Conventions and external facts 59
8.2. Planar length 60
8.3. The two reduced models 60
8.4. Derivation of H_{mov} from the barycentric Hamiltonian 62
8.5. Reconstruction of the physical motion 63
8.6. Tube geometry: collision-free, uniform pair bounds 65
8.7. Positive finite measure 66
8.8. Assembly of the physical clauses 67
8.9. Totality, sources, deviations, forward references 67
9. Kepler problem and Poincaré coordinates 68
9.1. Conventions, external facts, notation 68
9.2. The planar Kepler problem with mass m and strength κ 69
9.3. Kepler's equation in the eccentric longitude 70
9.4. The Poincaré chart 72
9.5. The product chart for the two physical models 75
9.6. The D25 secular definitions in terms of the chart 77
9.7. Sources, deviations, errata, forward references 78

10. Averaged perturbation and the secular quadratic form 79
10.1. Conventions, external facts, the secular polydisc 79
10.2. Analyticity of the interaction and of its average 80
10.3. Symmetries and evenness 81
10.4. Exact orthogonality of momenta (Fej04r Lemma 64) 82
10.5. Laplace coefficients 82
10.6. The quadratic part: C_0 and Q_h 83
10.7. The exact identification of the two models 85
10.8. Quartic data for the planar torsion (interface for Part 11b) 86
10.9. Sources, deviations, errata, forward references 87
11. The spectrum of the secular matrix 88
11.1. Conventions and external facts 88
11.2. Eigenvalue perturbation: Fej04r Lemmas 70, 71, Corollary 72 89
11.3. Negative definiteness of Q_h: Fej04r Proposition 73, Lemmas 74, 75 90
11.4. Local analytic diagonalisation; the symplectic rotation 92
11.5. Distinct and non-resonant spectrum: the interface for Parts 10–11 94
11.6. Sources, deviations, errata, forward references 96

1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the three frozen theorems of track P (owner approval D20, 27 October 2026) together with every frozen definition they use: the 33 definitions of `KAM/Frozen/Planetary/Defs/` and the three track-K definitions `KAM.IsDiophantine`, `KAM.hamiltonianVectorField`, `KAM.symplecticForm`. The Lean text is authoritative and the statements below transcribe it, with the same quantifier order and the same order in which constants are chosen. The D22 additions (P-EXP-SAFE, P-AMD-CONFINEMENT, P-SECULAR-BOX) are not frozen and are deliberately absent; a later part is reserved for them (`PLAN.md`, Part 14).

1.1. Conventions.

- *Ambient spaces.* All coordinate spaces are $\mathbb{R}^d = (\mathtt{Fin}\, d \to \mathbb{R})$ with the max norm $|v|_\infty = \max_i |v_i|$ and Lebesgue measure; $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ carries the max norm of the two components and the product Lebesgue measure `volume`. Fourier indices are $k \in \mathbb{Z}^d$. The Lean block conventions are: for $v \in \mathbb{R}^{a+b}$, the *fast* part is $v^{\mathrm{f}} = (v_i)_{i<a}$ and the *slow* part is $v^{\mathrm{s}} = (v_{a+j})_{j<b}$.
- *Phase order.* A phase point is $z = (z_1, z_2)$ with positions (angles) z_1 and momenta (actions) z_2. In the properly degenerate setting $z = (\varphi, q; I, p)$, that is $z_1 = (\varphi, q)$ and $z_2 = (I, p)$, with $\varphi \in \mathbb{R}^a$, $q \in \mathbb{R}^b$ and $I \in \mathbb{R}^a$, $p \in \mathbb{R}^b$ (`assemble`).
- *Physical distance.* On $\mathbb{R}^2$ the physical length is $|x|_2 = \sqrt{x_1^2 + x_2^2}$, never the max norm; $x \cdot y = x_1 y_1 + x_2 y_2$. Neighbourhoods in phase space and the norms $|\cdot|$ of tori data use the max norm; physical formulas use $|\cdot|_2$.
- *Bodies.* Positions of n planets are $q \in \mathbb{R}^{2n}$; planet i has position $q^{(i)} = (q_{(i,1)}, q_{(i,2)}) \in \mathbb{R}^2$, where the index of the pair $(i, k) \in \mathtt{Fin}\, n \times \mathtt{Fin}\, 2$ is $k + 2i$ (Lean `finProdFinEquiv`).
- *Torus parameters have period one* for every torus predicate below. For the cylinder tori of P-G (Definition 1.15) the *canonical fast angles keep period* 2π (CP10 footnote 1, p. 546), so the parameterization winds by 2π in the fast directions. Consequently the frequency vector w of a torus is the physical angular frequency divided by 2π (it changes frequency constants, not the dynamics or the measure ratio; `statement.md`, "Conventions"; owner question Q-P2). For the physical tori the parameter lattice is $\mathbb{Z}^d$ and the angles $2\pi\phi$ enter only through `circularPoint`.

- *Constants and quantifier order.* In each theorem, all data precede the small parameters; the proofs must keep the order of Theorems 1.28–1.30. In particular the thresholds r_*, C, c of P-G depend on the whole fixed family $(V, W, \dots, P, \dots)$, and μ_0, d, D (resp. ε_0, d, D) are chosen before μ (resp. ε).

1.2. **Track-K definitions used by track P.** The three frozen track-P theorems import exactly these frozen track-K definitions; no definition of track I or N is used.

Definition 1.1 (Diophantine vectors)**.** For $\gamma, \tau \in \mathbb{R}$ and $\omega \in \mathbb{R}^d$, ω is (γ, τ)-*Diophantine* if $\gamma/|k|_1^{\tau} \le |k \cdot \omega|$ for every $k \in \mathbb{Z}^d \setminus \{0\}$, where $|k|_1 = \sum_i |k_i|$ and $k \cdot \omega = \sum_i k_i \omega_i$.
Frozen Lean declaration: `KAM.IsDiophantine`
Source: HL19 (3.1), p. 14; Pos01 (1), p. 5; same as [K:`bp:p0:diophantine`].

Definition 1.2 (Hamiltonian vector field and symplectic form)**.** For $H : \mathrm{Ph}(d) \to \mathbb{R}$ and $z = (z_1, z_2)$,

$$X_H(z) = \big(\partial_{z_2} H(z),\, -\partial_{z_1} H(z)\big),$$

that is $X_H(z)_{1,i} = DH(z)(0, e_i)$ and $X_H(z)_{2,i} = -DH(z)(e_i, 0)$, with D the Fréchet derivative (`fderiv`). For $v = (v_1, v_2)$, $w = (w_1, w_2) \in \mathrm{Ph}(d)$,

$$\Omega(v, w) = \sum_i \big(v_{2,i} w_{1,i} - v_{1,i} w_{2,i}\big).$$

With z_1 the position and z_2 the momentum, $\dot z_1 = \partial_{z_2} H$, $\dot z_2 = -\partial_{z_1} H$ are the usual Hamilton equations, and $\Omega = \sum dz_{2,i} \wedge dz_{1,i}$. The condition $\Omega(Dv, Dw) = 0$ defining a Lagrangian torus is insensitive to the sign of Ω.
Frozen Lean declaration: `KAM.hamiltonianVectorField, KAM.symplecticForm`
Source: HL19 Rem. 2.1, p. 6; same as [K:`bp:p0:hamiltonian`].

1.3. **Elementary planar and index definitions.**

Definition 1.3 (Phase space)**.** $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ (canonical coordinates *(position, momentum)*).
Frozen Lean declaration: `KAM.Planetary.Phase`

Definition 1.4 (Planar length and scalar product)**.** For $v, w \in \mathbb{R}^2$: $\mathrm{sq}(v) = \sum_k v_k^2$, $|v|_2 = \sqrt{\mathrm{sq}(v)}$ and $v \cdot w = \sum_k v_k w_k$.
Frozen Lean declaration: `KAM.Planetary.planeSq, KAM.Planetary.planeLength, KAM.Planetary.planeDot`
Source: Fej04r (28), pp. 46–47 (physical Euclidean length; independent of the ambient max norm).

Definition 1.5 (Body vectors)**.** For $q \in \mathbb{R}^{2n}$ and $i < n$, $\mathrm{body}(q, i) = q^{(i)} = (q_{(i,1)}, q_{(i,2)}) \in \mathbb{R}^2$.
Frozen Lean declaration: `KAM.Planetary.bodyVector`

Definition 1.6 (Fast and slow blocks, assembly)**.** For $v \in \mathbb{R}^{a+b}$, $v^{\mathrm{f}} \in \mathbb{R}^a$ and $v^{\mathrm{s}} \in \mathbb{R}^b$ are the first a and last b coordinates. For $I, \varphi \in \mathbb{R}^a$ and $p, q \in \mathbb{R}^b$, $\mathrm{assemble}(I, \varphi, p, q) = \big((\varphi, q), (I, p)\big) \in \mathrm{Ph}(a+b)$, with the concatenated coordinate vectors.
Frozen Lean declaration: `KAM.Planetary.fast, KAM.Planetary.slow, KAM.Planetary.assemble`
Source: CP10 (1), p. 546; canonical order $(\varphi, q; I, p)$.

Definition 1.7 (Oscillator actions)**.** For $p, q \in \mathbb{R}^b$ and $j \le b$: $R_j(p, q) = (p_j^2 + q_j^2)/2$.
Frozen Lean declaration: `KAM.Planetary.oscillatorAction`
Source: CP10 (A2") and CP11 (A_2), pp. 547 and 51.

Definition 1.8 (Fast shift)**.** For $k \in \mathbb{Z}^a$, $\mathrm{shift}(k) = \big(((2\pi k_i)_{i<a}, 0_b),\, 0\big) \in \mathrm{Ph}(a+b)$: the fast angles are translated by $2\pi k$, the slow position coordinates and all momenta are unchanged.
Frozen Lean declaration: `KAM.Planetary.fastShift`

Definition 1.9 (Fast frequency map)**.** For $H_0 : \mathbb{R}^a \to \mathbb{R}$ and $I \in \mathbb{R}^a$, $\omega_{H_0}(I)_i = DH_0(I)(e_i)$ (the Fréchet derivative is 0 where H_0 is not differentiable).
Frozen Lean declaration: `KAM.Planetary.fastFrequency`
Source: CP10 (A1), p. 546.

1.4. **Definitions for the properly degenerate theorem.**

Definition 1.10 (Cylinder and its fundamental cell)**.** For $V \subset \mathbb{R}^a$ and $r \in \mathbb{R}$,

$$\mathcal{C}(V,r) = \Big\{ z \in \mathrm{Ph}(a+b) : \ z_2^{\mathrm{f}} \in V, \ \ (z_2^{\mathrm{s}})_j^2 + (z_1^{\mathrm{s}})_j^2 < r^2 \ \ \forall j \le b \Big\},$$

that is $I \in V$ and $p_j^2 + q_j^2 < r^2$ for every j: a *product of discs*, not a box. Its fundamental cell for the fast angles of period 2π is

$$\mathcal{F}(V,r) = \big\{ z \in \mathcal{C}(V,r) : \ 0 \le z_{1,i}^{\mathrm{f}} < 2\pi \ \ \forall i \le a \big\}.$$

Uses require $r > 0$.
Frozen Lean declaration: `KAM.Planetary.cylinder, KAM.Planetary.cylinderCell`
Source: CP10 p. 561, below (55) (disc region; audit P1); CP10 footnote 1 (period 2π).

Definition 1.11 (The family $H_0 + \mu P$)**.** For $H_0 : \mathbb{R}^a \to \mathbb{R}$ and $P : \mathbb{R} \times \mathrm{Ph}(a+b) \to \mathbb{R}$, $H_\mu(z) = H_0(z_2^{\mathrm{f}}) + \mu P(\mu, z)$.
Frozen Lean declaration: `KAM.Planetary.degenerateHamiltonian`
Source: CP10 (1), p. 546.

Definition 1.12 (Averaged perturbation)**.** For $\mu \in \mathbb{R}$, $I \in \mathbb{R}^a$, $p, q \in \mathbb{R}^b$,

$$P_{\mathrm{av}}(\mu, I; p, q) = (2\pi)^{-a} \int_{[0,2\pi]^a} P\big(\mu, \mathrm{assemble}(I, \varphi, p, q)\big)\, d\varphi$$

(Bochner integral; $= 0$ for a non-integrable integrand, so every use asserts integrability).
Frozen Lean declaration: `KAM.Planetary.secularAverage`
Source: CP10 (3), p. 547.

Definition 1.13 (Fast diffeomorphism)**.** For H_0 and $W \subset \mathbb{R}^a$, HasFastDiffeo(H_0, W) holds if (i) H_0 is real analytic at every point of W (`AnalyticOnNhd`); (ii) $\omega_{H_0}(W)$ is open; (iii) there is $g : \mathbb{R}^a \to \mathbb{R}^a$, of class C^1 on $\omega_{H_0}(W)$, with $g(\omega_{H_0}(I)) = I$ for $I \in W$, and $g(w) \in W$ and $\omega_{H_0}(g(w)) = w$ for $w \in \omega_{H_0}(W)$. Thus ω_{H_0} is a bijection of W onto its open image with C^1 inverse g (explicit, not a totalized inverse).
Frozen Lean declaration: `KAM.Planetary.HasFastDiffeomorphism`
Source: CP10 (A1), p. 547; neighbourhood version by audit P2.

Definition 1.14 (Secular Birkhoff form (A2"))**.** Let $V \subset \mathbb{R}^a$, $r_0, \mu_0 \in \mathbb{R}$, and functions $P_0 : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}$, $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$, $\beta : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^{b \times b}$. HasBirkhoff$(P, V, r_0, \mu_0, P_0, \Omega, \beta)$ holds if for every $\mu \in [0, \mu_0]$ and $I \in V$:

(i) for all $p, q \in \mathbb{R}^b$ with $p_j^2 + q_j^2 < r_0^2$ for all j, $\varphi \mapsto P(\mu, \mathrm{assemble}(I, \varphi, p, q))$ is integrable on $[0, 2\pi]^a$;

(ii) for every $\epsilon > 0$ there is δ with $0 < \delta \le r_0$ such that for all p, q with $p_j^2 + q_j^2 < \delta^2$ for all j,

$$\Big| P_{\mathrm{av}}(\mu, I; p, q) - \Big(P_0(\mu, I) + \sum_j \Omega_j(\mu, I) R_j + \tfrac{1}{2} \sum_{i,j} \beta_{ij}(\mu, I) R_i R_j \Big) \Big| \le \epsilon \, |(p,q)|_\infty^4,$$

where $R_j = R_j(p, q)$ and $|(p,q)|_\infty = \max(|p|_\infty, |q|_\infty)$. (The case $p = q = 0$ is included and fixes $P_{\mathrm{av}}(\mu, I; 0, 0) = P_0(\mu, I)$.)

Note that δ may depend on (μ, I, ϵ) in this definition; uniformity must be derived from the joint analyticity hypothesis of Theorem 1.28 (see `PLAN.md`, Part 5).
Frozen Lean declaration: `KAM.Planetary.HasSecularBirkhoffForm`
Source: CP10 (A2"), p. 547 (CP11 (A_2), p. 51), with the oscillator-disc guards of audit P1.

Definition 1.15 (Cylinder torus)**.** Let $H : \mathrm{Ph}(a+b) \to \mathbb{R}$, $U \subset \mathrm{Ph}(a+b)$, $w \in \mathbb{R}^{a+b}$, $K : \mathbb{R}^{a+b} \to \mathrm{Ph}(a+b)$. IsCylTorus$(H, U, w, K)$ holds if:

(i) K is real analytic on $\mathbb{R}^{a+b}$;
(ii) $K(\theta) \in U$ and H is differentiable at $K(\theta)$, for all θ;
(iii) $K(\theta + k) = K(\theta) + \mathrm{shift}\big((k_i)_{i<a}\big)$ for all $\theta \in \mathbb{R}^{a+b}$, $k \in \mathbb{Z}^{a+b}$;
(iv) if $K(\theta') = K(\theta) + \mathrm{shift}(k)$ for some $k \in \mathbb{Z}^a$, then $\theta' - \theta \in \mathbb{Z}^{a+b}$;
(v) $DK(\theta)$ is injective for all θ;
(vi) $\Omega(DK(\theta)u, DK(\theta)v) = 0$ for all θ, u, v;
(vii) for all $\theta, t \in \mathbb{R}$, $s \mapsto K(\theta + sw)$ has derivative $X_H(K(\theta + tw))$ at $s = t$.

No Diophantine condition is part of this predicate (see Definition 1.16).
Frozen Lean declaration: `KAM.Planetary.IsCylinderTorus`
Source: CP11 Thm. 11.1, pp. 50–51 (analytic conjugacy to a linear flow, Lagrangian); CP10 Thm. 1.3, p. 548.

Definition 1.16 (Cylinder Kolmogorov set)**.** For $H : \mathrm{Ph}(a+b) \to \mathbb{R}$, $V \subset \mathbb{R}^a$, $r, \mu, C \in \mathbb{R}$ and $S \subset \mathrm{Ph}(a+b)$, IsCylKol$(H, V, r, \mu, C, S)$ holds if S is Lebesgue measurable, $S \subset \mathcal{C}(V, r)$, and for every $z \in S$ there are $w \in \mathbb{R}^{a+b}$, $K : \mathbb{R}^{a+b} \to \mathrm{Ph}(a+b)$ and θ with: $K(\theta) = z$; $K(\mathbb{R}^{a+b}) \subset S$ (a whole torus); IsCylTorus$(H, \mathcal{C}(V, r), w, K)$; there are $\gamma > 0$ and $\tau > a + b$ with w (γ, τ)-Diophantine (the constant may depend on the torus); $|w^{\mathrm{f}}|_\infty \le C$ and $|w^{\mathrm{s}}|_\infty \le C\mu$.
Frozen Lean declaration: `KAM.Planetary.IsCylinderKolmogorovSet`
Source: CP11 Thm. 11.1, pp. 50–51: $\omega_1 = O(1)$, $\omega_2 = O(\mu)$, Diophantine.

1.5. **Definitions for the physical theorems.** Throughout, $G, m_0 > 0$ and the planet weights $m_i > 0$.

Definition 1.17 (Reference circular configurations and the tube)**.** For $m, a \in \mathbb{R}^n$ and $\phi \in \mathbb{R}^n$ (phases in $\mathbb{R}/\mathbb{Z}$),

$$\mathrm{circ}(\phi) = \Big(\big(a_i(\cos 2\pi\phi_i, \sin 2\pi\phi_i)\big)_i,\ \big(m_i\sqrt{Gm_0/a_i}\,(-\sin 2\pi\phi_i, \cos 2\pi\phi_i)\big)_i \Big) \in \mathrm{Ph}(2n)$$

(circular prograde Kepler configurations at coupling 0; $a_i > 0$, $G, m_0 > 0$ guard the square root and the division). For $\eta \in \mathbb{R}$ the *circular tube* is the open η-neighbourhood in the phase max norm, $U_\eta = \{z \in \mathrm{Ph}(2n) : \exists \phi \in \mathbb{R}^n,\ |z - \mathrm{circ}(\phi)|_\infty < \eta\}$.
Frozen Lean declaration: `KAM.Planetary.circularPoint, KAM.Planetary.circularTube`
Source: Fej04r Thm. 63, p. 49 (near circular coplanar Kepler motions); ours for the explicit reference point.

Definition 1.18 (No collision, pair bounds)**.** For $q \in \mathbb{R}^{2n}$: NoColl(q) means $|q^{(i)}|_2 > 0$ for all i and $|q^{(i)} - q^{(j)}|_2 > 0$ for all $i \ne j$ (heliocentric configurations, the Sun at the origin). For N points $Q_0, \dots, Q_{N-1} \in \mathbb{R}^2$, PairBounds$(d, D; Q)$ means $d \le |Q_j - Q_i|_2 \le D$ for all $i \ne j$.
Frozen Lean declaration: `KAM.Planetary.NoCollision, KAM.Planetary.PairBounds`
Source: Fej04r §6.1, pp. 45–47.

Definition 1.19 (Actual masses, barycentric reconstruction)**.** For $\mu \in \mathbb{R}$ let $\mathrm{mass}_\mu = (m_0, \mu m_1, \dots, \mu m_n) \in \mathbb{R}^{n+1}$ (index 0 is the Sun). For heliocentric positions $q \in \mathbb{R}^{2n}$ and scaled momenta $p \in \mathbb{R}^{2n}$,

$$x_0(q) = -\frac{\mu}{m_0 + \mu\sum_i m_i}\sum_i m_i q^{(i)}, \qquad x_i(q) = x_0(q) + q^{(i)},$$

$$v_0(p) = -\frac{\mu}{m_0}\sum_i p^{(i)}, \qquad v_i(p) = \frac{1}{m_i}p^{(i)} \qquad (1 \le i \le n),$$

with the Sun at index 0 and planet i at index i in the order of `Fin.cons`. The denominators are positive for $m_i > 0$, $\mu > 0$.
Frozen Lean declaration: `KAM.Planetary.actualMasses, KAM.Planetary.barycentricPositions, KAM.Planetary.barycentricVelocities`
Source: Fej04r §6.1, (28), pp. 46–47; zero total momentum and barycentre; Fej07e p. 3.

Definition 1.20 (Newton solutions)**.** Let $N \in \mathbb{N}$, $M \in \mathbb{R}^N$ and $Q, V : \mathbb{R} \to (\mathbb{R}^2)^N$. IsNewton$(G, M, Q, V)$ holds if for all $t \in \mathbb{R}$ and $i \neq j$ (all indices $< N$): (i) $|Q_j(t) - Q_i(t)|_2 > 0$; (ii) Q_i has derivative $V_i(t)$ at t; (iii) V_i has derivative

$$\sum_{j \neq i} \frac{GM_j}{|Q_j(t) - Q_i(t)|_2^3} \, (Q_j(t) - Q_i(t)) \quad \text{at } t$$

(the all-time Newton equations; `HasDerivAt`).
Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution`
Source: Fej04r (1), §6.1, with G restored.

Definition 1.21 (Fixed-Sun solutions)**.** For $Q, V : \mathbb{R} \to (\mathbb{R}^2)^n$ and $\varepsilon \in \mathbb{R}$, the predicate IsFixedSol$(G, m_0, m, \varepsilon, Q, V)$ holds if for all t and $i \neq j$: $|Q_i(t)|_2 > 0$, $|Q_j(t) - Q_i(t)|_2 > 0$, Q_i has derivative $V_i(t)$ and V_i has derivative

$$-\frac{Gm_0}{|Q_i(t)|_2^3} \, Q_i(t) + \varepsilon \sum_{j \neq i} \frac{Gm_j}{|Q_j(t) - Q_i(t)|_2^3} (Q_j(t) - Q_i(t)).$$

The Sun is the constant origin.
Frozen Lean declaration: `KAM.Planetary.IsFixedSolution`
Source: Ours (fixed-center law obtained by differentiating H_{fix}), after Fej04r (28).

Definition 1.22 (Moving-Sun Hamiltonian)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $\mu \in \mathbb{R}$,

$$H_{\mathrm{mov}}(z) = \sum_i \Big[\big(\tfrac{1}{m_i} + \tfrac{\mu}{m_0}\big) \tfrac{|p^{(i)}|_2^2}{2} - \frac{Gm_0 m_i}{|q^{(i)}|_2} \Big] + \mu \sum_{i<j} \Big[\frac{p^{(i)} \cdot p^{(j)}}{m_0} - \frac{Gm_i m_j}{|q^{(i)} - q^{(j)}|_2} \Big].$$

It is the reduced Hamiltonian divided by μ, with G restored, and every term is kept: the Sun moves, and neither the self kinetic term $\mu|p^{(i)}|_2^2/(2m_0)$ nor the cross term is dropped. Equivalently $H_{\mathrm{mov}} = H_{\mathrm{fix}}\big|_{\varepsilon=\mu} + \mu|\sum_i p^{(i)}|_2^2/(2m_0)$ (a calculation for a later part).
Frozen Lean declaration: `KAM.Planetary.movingHamiltonian`
Source: Fej04r (28), pp. 46–47, with Fej07e p. 3 (indices $m_j m_k$ corrected) and G restored.

Definition 1.23 (Fixed-Sun Hamiltonian)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $\varepsilon \in \mathbb{R}$,

$$H_{\mathrm{fix}}(z) = \sum_i \Big[\frac{|p^{(i)}|_2^2}{2m_i} - \frac{Gm_0 m_i}{|q^{(i)}|_2} \Big] - \varepsilon \sum_{i<j} \frac{Gm_i m_j}{|q^{(i)} - q^{(j)}|_2}.$$

The Sun is the origin; only the direct planet–planet interaction is kept, with coefficient ε.
Frozen Lean declaration: `KAM.Planetary.fixedHamiltonian`
Source: Ours (Newtonian fixed-center model; "our variant of Fej04r (28)"); `statement.md`, P-FIXED.

Definition 1.24 (Physical torus)**.** For $H : \mathrm{Ph}(d) \to \mathbb{R}$, $w \in \mathbb{R}^d$ and $K : \mathbb{R}^d \to \mathrm{Ph}(d)$, IsPhysTorus$(H, w, K)$ holds if:

(i) K is real analytic on $\mathbb{R}^d$;
(ii) H is differentiable at $K(\theta)$ for all θ;
(iii) $K(\theta + k) = K(\theta)$ for $k \in \mathbb{Z}^d$;
(iv) $K(\theta) = K(\theta')$ implies $\theta' - \theta \in \mathbb{Z}^d$;
(v) $DK(\theta)$ is injective;
(vi) $\Omega(DK(\theta)u, DK(\theta)v) = 0$;
(vii) there are $\gamma > 0$ and $\tau > d - 1$ with w (γ, τ)-Diophantine;
(viii) for all θ, t, $s \mapsto K(\theta + sw)$ has derivative $X_H(K(\theta + tw))$ at $s = t$.

(For such K the image is a compact embedded analytic Lagrangian torus. Track K's primary-torus predicate has affine winding and is not reused. Note $\tau > d-1$ here against $\tau > a+b$ in Definition 1.16.)

Frozen Lean declaration: `KAM.Planetary.IsPhysicalTorus`

Source: HL19 (2.1), (2.3), pp. 6–7 (modified to period-one embeddings); CP09 Thm. 4, p. 8; ours.

Definition 1.25 (Moving-Sun Kolmogorov set)**.** For $G, m_0 > 0$, $m, a \in \mathbb{R}^n$, $\mu, \eta, d, D \in \mathbb{R}$, $S \subset \mathrm{Ph}(2n)$: $\mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S)$ holds if S is measurable, $S \subset U_\eta$, and every $z \in S$ has w, K, θ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$, $\mathrm{IsPhysTorus}(H_{\mathrm{mov}}, w, K)$ and, for every $\phi \in \mathbb{R}^{2n}$:

(a) $\mathrm{NoColl}\big(K(\phi)_1\big)$;

(b) $\mathrm{IsNewton}\big(G, \mathrm{mass}_\mu,\ t \mapsto x(K(\phi+tw)_1),\ t \mapsto v(K(\phi+tw)_2)\big)$, where x, v are the barycentric reconstructions of Definition 1.19;

(c) $\mathrm{PairBounds}\big(d, D;\ x(K(\phi)_1)\big)$ (all pairs including the Sun; as ϕ is arbitrary this covers every point $K(\phi + tw)$ of every orbit, hence all times).

Frozen Lean declaration: `KAM.Planetary.MovingKolmogorovSet`

Source: Fej04r Prop. 77 and Fej07e §7, pp. 3–4, with CP09 Thm. 4; the physical conclusions are ours (`statement.md`).

Definition 1.26 (Fixed-Sun Kolmogorov set)**.** For ε, η, d, D and $S \subset \mathrm{Ph}(2n)$ the predicate $\mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S)$ holds if S is measurable, $S \subset U_\eta$, and every $z \in S$ has w, K, θ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$, $\mathrm{IsPhysTorus}(H_{\mathrm{fix}}, w, K)$ and for every ϕ: $\mathrm{NoColl}(K(\phi)_1)$; $\mathrm{IsFixedSol}\big(G, m_0, m, \varepsilon,\ t \mapsto (K(\phi + tw)_1^{(i)})_i,\ t \mapsto (\frac{1}{m_i} K(\phi + tw)_2^{(i)})_i\big)$; $\mathrm{PairBounds}\big(d, D;\ (0, K(\phi)_1^{(1)}, \dots, K(\phi)_1^{(n)})\big)$, with the Sun at the origin as point 0.

Frozen Lean declaration: `KAM.Planetary.FixedKolmogorovSet`

Source: Ours (new deduction; Fej04r Lemma 64 and Prop. 77, CP09 Thm. 4).

Remark 1.27 (Lean totality). The frozen definitions use Lean total functions, and the frozen theorems guard each of them as follows; proofs must establish these guarded facts before using the defining formulas.

- *Fréchet derivatives* (ω_{H_0}, X_H, DK) are 0 at points of non-differentiability. ω_{H_0} is used only through `HasFastDiffeomorphism`, which asserts analyticity of H_0 on $W \supset \overline{V}$; X_H appears only in the clauses (vii)/(viii) of the torus predicates, which also assert differentiability of H on the range of K (for the cylinder tori at points of $\mathcal{C}(V, r)$, where H_μ is analytic because $r < r_0$ and $\mu \in [0, \mu_0]$, by the hypotheses on P and H_0); DK is guarded by analyticity of K. Injectivity of DK and $\Omega(DK\cdot, DK\cdot) = 0$ are asserted for the genuine derivative.
- *Bochner integrals* (P_{av}) are 0 for non-integrable integrands; Definition 1.14 (i) asserts integrability on the disc $p_j^2 + q_j^2 < r_0^2$, and every point used in (ii) has $\delta \le r_0$.
- *Square roots and divisions*: $\sqrt{\cdot}$ in $|\cdot|_2$ has nonnegative radicand; $1/|\cdot|_2$ and $1/|\cdot|_2^3$ occur only after the collision-free clauses (a), IsNewton (i), IsFixedSol (the first two conjuncts) in the same conjunction; in circ, $\sqrt{Gm_0/a_i}$ and $1/a_i$ are guarded by $G, m_0, a_i > 0$; in x_0 and v_0 the denominators $m_0 + \mu \sum m_i$ and m_0 are positive since $\mu > 0$; the factor $1/m_i$ needs $m_i > 0$.
- *Real powers and logarithms*: $r^6/(\log(1/r))^c$ with $0 < r < r_* < 1$ has positive base $\log(1/r) > 0$; $\sqrt{r}$ has $r > 0$; $|k|_1^\tau$ in the Diophantine condition has $|k|_1 \ge 1$ for $k \ne 0$.
- *Truncated subtraction*: the lower bound is written with `ENNReal.ofReal`, so it is the real number $1 - C\sqrt{r}$ when that is positive and 0 otherwise; the strict inequality $\mathrm{ofReal}(1 - C\sqrt{r})\ \mathrm{meas}\,\mathcal{F} < \mathrm{meas}(S \cap \mathcal{F})$ is then nontrivial only when $1 - C\sqrt{r} > 0$, but must be proved for every $r < r_*$ as stated (it follows from $0 < \mathrm{meas}(S \cap \mathcal{F})$ when $1 - C\sqrt{r} \le 0$).
- *Measure*: meas is Lebesgue (outer) measure; the sets $\mathcal{F}(V, r)$ are Borel and S is measurable by the conclusion; $\mathrm{meas}\,\mathcal{F}(V, r) < \infty$ since V is bounded.

1.6. **The frozen theorems.**

Theorem 1.28 (P-G: corrected properly degenerate KAM)**.** ***Data (all fixed first).*** *Integers $a, b \geq 1$. A set $V \subset \mathbb{R}^a$ that is open, connected and bounded; an open set $W \subset \mathbb{R}^a$ with $\overline{V} \subset W$; a boundary collar condition: there are $A, d > 0$ such that for all $0 < t < d$*

$$\operatorname{meas}\{I \in \mathbb{R}^a : \ \exists J \in \partial V, \ |I - J|_\infty < t\} \leq A\, t.$$

Reals $r_0, \mu_0, B > 0$. Functions $H_0 : \mathbb{R}^a \to \mathbb{R}$, $P : \mathbb{R} \times \mathrm{Ph}(a+b) \to \mathbb{R}$, $P_0 : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}$, $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$, $\beta : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^{b \times b}$.

Hypotheses.

(H1) $(\mu, z) \mapsto P(\mu, z)$ *is real analytic at every point of* $[0, \mu_0] \times \overline{\mathcal{C}(V, r_0)}$ *(jointly, as a function on* $\mathbb{R} \times \mathrm{Ph}(a+b)$*);*

(H2) for $\mu \in [0, \mu_0]$, $z \in \mathcal{C}(V, r_0)$ *and* $k \in \mathbb{Z}^a$*:* $P(\mu, z + \mathrm{shift}(k)) = P(\mu, z)$*;*

(H3) $\mathrm{HasFastDiffeo}(H_0, W)$ *(Definition 1.13);*

(H4) $\mathrm{HasBirkhoff}(P, V, r_0, \mu_0, P_0, \Omega, \beta)$ *(Definition 1.14);*

(H5) $|\det \beta(\mu, I)| \geq B$ *for all* $\mu \in [0, \mu_0]$, $I \in V$*;*

(H6) $\beta(\mu, I)$ *is symmetric for all* $\mu \in [0, \mu_0]$, $I \in V$*.*

Conclusion. *There exist r_*, C, c with $0 < r_* < r_0$, $r_* < 1$, $C > 0$, $c > 0$ such that for all r, μ with*

$$0 < r < r_*, \qquad 0 < \mu < \mu_0, \qquad \mu < \frac{r^6}{(\log(1/r))^c},$$

there is $S \subset \mathrm{Ph}(a+b)$ with $\mathrm{IsCylKol}(H_\mu, V, r, \mu, C, S)$ *(with $H_\mu = H_0 + \mu P(\mu, \cdot)$ of Definition 1.11) and*

$$0 < \operatorname{meas}(S \cap \mathcal{F}(V, r)), \qquad \mathrm{ofReal}(1 - C\sqrt{r})\ \operatorname{meas} \mathcal{F}(V, r) < \operatorname{meas}(S \cap \mathcal{F}(V, r)) < \operatorname{meas} \mathcal{F}(V, r).$$

Here r_*, C, c depend on all the data and hypotheses above and not on r or μ; C is simultaneously the frequency bound of Definition 1.16 and the measure constant.

Frozen Lean declaration: `KAM.Planetary.properlyDegenerateKAM`

Source: CP11 Thm. 11.1, pp. 50–51, whose proof is CP10 Thm. 1.3, p. 548 (with $c_* = 2b$, $\rho = r$). It is not *the printed theorem: the neighbourhood W (CP10 p. 556, (58) p. 562), the collar (CP10 p. 561, below (56)), the product of discs (CP10 p. 561, below (55)), joint analyticity of the family (CP10 footnote 3, p. 546 asks only C^1 uniform dependence) and the 2π period are corrected-contract premises (audit P1–P4; D-PG-1 to D-PG-5).*

Source gap: CP11 Thm. 11.1 states $\operatorname{meas} P_\epsilon > \operatorname{meas} K$ *without proof; the strict inequality* $\operatorname{meas}(S \cap \mathcal{F}) < \operatorname{meas} \mathcal{F}$ *is obtained in the plan by deleting a small positive-measure subfamily of tori (Part 5), which is possible because the lower bound has slack. CP10 Thm. 1.3 also needs $P_{\mathrm{av}} - N = O(\epsilon^5)$ uniformly (CP10 (60), p. 562), which follows from (H1), (H2), (H4) and compactness but is not proved in the source.*

Theorem 1.29 (P-PLANAR: two planar planets with a moving Sun)**.** *Let $G, m_0 > 0$; let $m, a : \mathtt{Fin}\,2 \to \mathbb{R}$ (planets $0, 1$; m the planet weights, a the axes) with $m_i > 0$ and $a_i > 0$ for $i = 0, 1$ and $a_0 < a_1$ (strictly ordered axes, no numerical separation); let $\eta > 0$. Then there exist μ_0, d, D with $0 < \mu_0$ and $0 < d < D$ such that for every μ with $0 < \mu < \mu_0$ there is $S \subset \mathrm{Ph}(2 \cdot 2)$ with*

$$0 < \operatorname{meas} S < \infty \quad \text{and} \quad \mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S) \qquad (n = 2).$$

The order is: data (G, m_0, m, a, η), then (μ_0, d, D), then μ, then S. The tori are 4-dimensional in the 8-dimensional $\mathrm{Ph}(4)$*.*

Frozen Lean declaration: `KAM.Planetary.planarTwoPlanetKAM`

Source: CP09 Thm. 4, p. 8 with Fej04r Prop. 77, pp. 56–58, using the recorded source-proof repairs (`statement.md`, P5–P7); Fej07e p. 3–4 supplies the smooth counterpart. No threshold from CP11 §8.1 (audit P4).

Source gap: The route is not signed off as proved: the analytic weak-nondegeneracy gate (CP09 Thm. 4 resting on Rüssmann's theorem, CP09 Thm. 3, whose source is not pinned), the continuation of eigenvalue

branches (Fej04r Lemma 76 is false as printed) and the closeness of the tori to the circular torus (CP09 Thm. 4 says only "close to"); see `PLAN.md`.

Theorem 1.30 (P-FIXED: n planar planets around a fixed Sun)**.** *Let $n \geq 2$, $G, m_0 > 0$, $m, a \in \mathbb{R}^n$ with all $m_i > 0$ and $a_i > 0$, $a_i < a_j$ whenever $i < j$, and $\eta > 0$. Then there exist ε_0, d, D with $0 < \varepsilon_0$ and $0 < d < D$ such that for every ε with $0 < \varepsilon < \varepsilon_0$ there is $S \subset \mathrm{Ph}(n \cdot 2)$ with*

$$0 < \operatorname{meas} S < \infty \quad \textit{and} \quad \mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S).$$

The order is data (n, G, m_0, m, a, η), then (ε_0, d, D), then ε, then S.
Frozen Lean declaration: `KAM.Planetary.fixedSunPlanetaryKAM`
Source: Ours: "our proposed Newtonian fixed-center deduction, not a published fixed-Sun statement"; Fej04r Lemma 64 (p. 49–50) and Prop. 77, CP09 Thm. 4.
Source gap: The secular identification for the fixed Sun (equal secular quadratic forms and frequency map up to functions of the actions) and the general-n induction of Fej04r Prop. 77 ("par un raisonnement analogue", p. 58) are proof obligations; no secular nondegeneracy is a premise.

Remark 1.31 (Scope of this part). The three statements are the only frozen theorems of track P. P-SPATIAL is a sketch only and has no Lean statement. The D22 additions are under statement audit and are treated in the reserved last part.

2. Hamiltonian tools for tori: charts, period conversion, Diophantine conversion, Liouville

Track P produces tori in coordinates that are not the frozen Cartesian ones: CP's action–angle variables with angles of period 2π, oscillator action–angle discs, the Poincaré chart. This part supplies, once, the four tools that carry a torus found in such a chart to a frozen predicate ([P:`bpP:p0:cyltorus`], [P:`bpP:p0:phystorus`]):

- `torus-transport` (Theorem 2.9): an analytic embedding Ψ with $\Psi^*\Omega = c\,\Omega$ that is compatible with the angle lattice sends a lattice torus of $(H \circ \Psi)/c$ to a torus of H, clause by clause of the two frozen predicates;
- `period-convert` (Lemma 2.11 and Corollaries 2.12, 2.13): the passage between 2π-angles and the frozen period-one parameter is the chart $S_{2\pi}$, so the frozen frequency is $w = \tilde{\omega}/2\pi$ and the dictionary of [P:`bpP:p3:period-dictionary`] is a special case;
- `diophantine-convert` (Lemmas 2.15–2.16): CP10 (19) and CP09 (6) imply the frozen `KAM.IsDiophantine` with the exponents the two torus predicates require;
- `measure-symplectic` (Lemmas 2.18–2.20): conformally symplectic charts scale Lebesgue measure by c^d, and a union of whole tori that is a Lipschitz image is measurable with controlled measure.

The `ode` item is cited, not repeated (§2.7). Everything here is standard; the point of writing it is to fix signs, lattices and Lean totality exactly as frozen.

2.1. **Conventions and external facts.** Phase space is $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$, points $z = (z_1, z_2)$ (position, momentum), max norms, Lebesgue measure ([P:`bpP:p0:conventions`]). Let $M_0 = \left(\begin{smallmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$ and $J_0 = -M_0$. The frozen form ([P:`bpP:p0:hamsymp`]) is $\Omega(v, w) = \sum_i (v_{2,i} w_{1,i} - v_{1,i} w_{2,i}) = v^\top M_0 w$. For a linear form $\ell : \mathrm{Ph}(d) \to \mathbb{R}$ put $X_\ell := \big((\ell(0, e_i))_i, (-\ell(e_i, 0))_i\big)$; thus $X_H(z) = X_{DH(z)}$, and $X_H(z) = 0$ if H is not differentiable at z (Lean totality, [P:`bpP:p0:totality`]). Write $\iota_k := \big((k_i)_{i<d}, 0\big) \in \mathrm{Ph}(d)$ for $k \in \mathbb{Z}^d$ (integers coerced to reals: an integer translation of the positions; the nested form $((k_f, 0_b), 0)$ is reserved for the split $d = a + b$ and for `fastShift`) and, for $c > 0$, $S_c(z_1, z_2) := (c z_1, z_2)$.

(X1) Chain rule for `fderiv` and `HasDerivAt`; compositions of real analytic maps are real analytic (`AnalyticAt.comp` and `AnalyticOnNhd.comp`, `Analysis/Analytic/Composition.lean`;

the latter needs the range of the inner map inside the set where the outer map is analytic, here $K(\mathbb{R}^d) \subset O$); sums and products of analytic maps are analytic.

(X2) Inverse function theorem: a C^1 map Ψ with $D\Psi(z_0)$ invertible is a homeomorphism from a neighbourhood of z_0 onto an open set, with C^1 inverse (Mathlib: `HasStrictFDerivAt.toOpenPartialHomeomorph` and `HasStrictFDerivAt.to_localInverse` (`Analysis/Calculus/InverseFunctionTheorem/FDeriv.lean`); for C^1 regularity of the inverse `ContDiffAt.to_localInverse` (`.../ContDiff.lean`). A C^1 map has a strict derivative, and a bijective linear map of $\mathbb{R}^{2d}$ is a continuous linear equivalence).

(X3) Area formula (`MeasureTheory/Function/Jacobian.lean`; the derivative field is a family of continuous linear maps with `HasFDerivWithinAt` on the set, supplied by analyticity of Ψ on O): `MeasureTheory.addHaar_image_le_lintegral_abs_det_fderiv` (A *Borel*, no injectivity) gives $\operatorname{meas}\Psi(A) \le \int_A |\det D\Psi|$; `MeasureTheory.lintegral_abs_det_fderiv_eq_addHaar_image` ($A$ Borel, Ψ injective on A) and its subscript-0 version (name with the suffix `image`$_0$, i.e. `addHaar_image` followed by the subscript-zero character; A `NullMeasurableSet`) give equality. Also `addHaar_image_eq_zero_of_differentiableOn_of_addHaar_eq_zero`: differentiable images of null sets are null.

(X4) A Lebesgue measurable subset of $\mathbb{R}^N$ is the union of an F_σ set and a null set (inner regularity, `MeasureTheory/Measure/Regular.lean`); a closed set is σ-compact; continuous images of compact sets are compact. *Null sets need not be Borel*: "measurable" below means `NullMeasurableSet` (Lebesgue) unless the word Borel is used, and the frozen predicates need `MeasurableSet`, i.e. Borel. Injective continuous images of Borel sets are Borel: `MeasurableSet.image_of_continuousOn_injOn` (`MeasureTheory/Constructions/Polish/Basic.lean`).

(X5) $\det(AB) = \det A \det B$, $\det A^\top = \det A$, and $\det M_0 = 1$.

Track K: [K:`bp:p1:holomorphic-analytic`], [K:`bp:p1:real-restriction`], [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:lip-image`], [K:`bp:p1:mcshane`].

Lemma 2.1 (Algebra of Ω and X_H)**.** *(a) $M_0^\top = -M_0$, $M_0^2 = -\mathbf{1}$, Ω is nondegenerate, and $\Omega(w, X_\ell) = \ell(w)$ for all w and all linear ℓ. Hence X_ℓ is the unique vector with this property, and $\Omega(w, X_H(z)) = DH(z)w$ for every z (both sides vanish if H is not differentiable at z). (b) If A is a real $2d \times 2d$ matrix with $\Omega(Au, Av) = c\,\Omega(u, v)$ for all u, v ($c \neq 0$), then A is invertible, $|\det A| = |c|^d$, and $X_{\ell\circ A} = c\,A^{-1}X_\ell$ for every linear form ℓ. (c) The relation $\Omega(Au, Av) = c\Omega(u, v)$ is equivalent to $A^\top(-M_0)A = c(-M_0)$: it does not depend on the sign convention for the symplectic form (CP10's $dI \wedge d\varphi$ is Ω; CP09 Thm. 3 prints $dx \wedge dy$ with x the angles, which is $-\Omega$; both define the same class of maps, and the vector field is the frozen X_H, not a function of the printed sign).*

Proof. **Step S2.1: Matrices.** Direct from the block form; $X_\ell = J_0\nabla\ell$ so $M_0X_\ell = -M_0^2\nabla\ell = \nabla\ell$, i.e. $\Omega(w, X_\ell) = w^\top M_0 X_\ell = \ell(w)$. Uniqueness: M_0 invertible.

Step S2.2: (b). $A^\top M_0 A = cM_0$ and (X5) give $(\det A)^2 = c^{2d}$. For all w: $\Omega(w, cA^{-1}X_\ell) = \Omega(Aw, AA^{-1}X_\ell) = \Omega(Aw, X_\ell) = \ell(Aw) = (\ell \circ A)(w)$, so $cA^{-1}X_\ell = X_{\ell\circ A}$ by uniqueness. (c) is the displayed matrix identity. □

Remark 2.2 (Reordering of coordinates). Part 3 writes points as (J, θ) (action, angle). The relabelling $\pi(J, \theta) = (\theta, J)$ to the frozen order (position, momentum) = (angle, action) turns $\dot\theta = \partial_J H$, $\dot J = -\partial_\theta H$ into X_H of [P:`bpP:p0:hamsymp`] and the form $dJ \wedge d\theta$ into Ω; no sign changes. All statements of Part 3 are read in the frozen order after applying π.

2.2. **Conformally symplectic charts.**

Definition 2.3 (Chart)**.** Let $O \subset \mathrm{Ph}(d)$ be open, $c > 0$. A *c-chart on O* is a map $\Psi : \mathrm{Ph}(d) \to \mathrm{Ph}(d)$ (total) that is real analytic at every point of O and satisfies $\Omega(D\Psi(z)u, D\Psi(z)v) = c\,\Omega(u,v)$ for $z \in O$, $u, v \in \mathrm{Ph}(d)$. A 1-chart is a *symplectic* chart. For $H : \mathrm{Ph}(d) \to \mathbb{R}$ the *pulled-back Hamiltonian* is $H^{\Psi} := c^{-1}\,(H \circ \Psi)$.

By Lemma 2.1(b), $D\Psi(z)$ is invertible on O, so Ψ is a local analytic diffeomorphism there ((X2); analyticity of the local inverse is not needed below).

Lemma 2.4 (Charts and Hamiltonian fields)**.** *Let Ψ be a c-chart on O, $H : \mathrm{Ph}(d) \to \mathbb{R}$, $z \in O$. Then H^{Ψ} is differentiable at z iff H is differentiable at $\Psi(z)$, and then*

$$D\Psi(z)\, X_{H^{\Psi}}(z) = X_H(\Psi(z)).$$

If H is not differentiable at $\Psi(z)$ then both fields are 0 (totality) and the identity still holds.

Proof. **Step S2.3: Differentiability.** ($\Leftarrow$) chain rule (X1). ($\Rightarrow$) by (X2) there is a C^1 local inverse Ψ_{loc}^{-1} near $\Psi(z)$ and $H = c\,H^{\Psi} \circ \Psi_{\mathrm{loc}}^{-1}$ near $\Psi(z)$, a composition of differentiable maps.

Step S2.4: The field. With $\ell = DH(\Psi z)$ and $A = D\Psi(z)$, $D(H^{\Psi})(z) = c^{-1}\ell \circ A$, so $X_{H^{\Psi}}(z) = c^{-1}X_{\ell \circ A} = A^{-1}X_{\ell}$ by Lemma 2.1(b). Hence $AX_{H^{\Psi}}(z) = X_{\ell} = X_H(\Psi z)$. In the non-differentiable case both $D(H^{\Psi})(z)$ and $DH(\Psi z)$ are 0 by the Lean convention. □

Corollary 2.5 (Orbits transfer)**.** *Let $y : J \to O$ ($J \subset \mathbb{R}$ an interval) satisfy: H^{Ψ} is differentiable at $y(t)$ and y has derivative $X_{H^{\Psi}}(y(t))$ at t, for all $t \in J$. Then $x = \Psi \circ y$ has derivative $X_H(x(t))$ at every $t \in J$ and H is differentiable at $x(t)$.*

Proof. Chain rule (X1) and Lemma 2.4. □

Remark 2.6 (What is new compared with track N). [N:`bpN:p5:symplectic`] and [N:`bpN:p5:canonical`](b),(c) give $D\Phi X_{H\circ\Phi} = X_H \circ \Phi$ and the orbit transfer for the time-one map $\Phi = X_{\phi}^1$ of a *holomorphic* generating function on a complex domain, with symplecticity *derived* from the flow. Here the chart is an arbitrary real analytic map (Poincaré chart, polar chart of oscillators, scalings) with symplecticity *assumed*, in the frozen real setting, with the factor c, with the frozen sign of Ω (Lemma 2.1) and with the Lean-totality guard (differentiability of H is transported, not assumed). The algebra is that of [N:`bpN:p5:canonical`](b); it is repeated only because the hypotheses differ. Time-one maps of Part N are charts in the sense of Definition 2.3 with $c = 1$ (Lemma 2.7).

Lemma 2.7 (Holomorphic charts, real restriction)**.** *Let $c > 0$, $V \subset \mathbb{C}^{2d}$ be open, $\Psi : V \to \mathbb{C}^{2d}$ holomorphic, real at real points ($\Psi(V \cap \mathbb{R}^{2d}) \subset \mathbb{R}^{2d}$), and assume $D\Psi(z)^{\top} M_0 D\Psi(z) = cM_0$ (complex matrices) for $z \in V \cap \mathbb{R}^{2d}$. Identify $\mathbb{R}^{2d} \subset \mathbb{C}^{2d}$ through the real embedding; $O = V \cap \mathbb{R}^{2d}$ is open in $\mathrm{Ph}(d)$. Then any total map $\mathrm{Ph}(d) \to \mathrm{Ph}(d)$ that agrees with Ψ on O (choose any extension off O; no property off O is ever used) is a c-chart on O, with real derivative $D\Psi(z)|_{\mathbb{R}^{2d}}$.*

Proof. [K:`bp:p1:real-restriction`] (via [K:`bp:p1:holomorphic-analytic`]) gives analyticity and that the real derivative is the restriction of the complex one; the identity of matrices is the defining condition of Definition 2.3 (Lemma 2.1(c)). □

2.3. **Transport of tori.**

Definition 2.8 (Lattice torus in a chart)**.** Let $O \subset \mathrm{Ph}(d)$ be open, $G : \mathrm{Ph}(d) \to \mathbb{R}$, $w \in \mathbb{R}^d$, $K : \mathbb{R}^d \to \mathrm{Ph}(d)$. $\mathrm{LT}(G, O, w, K)$ holds if

(L1) K is real analytic on $\mathbb{R}^d$, $K(\theta) \in O$ and G is differentiable at $K(\theta)$ for all θ;
(L2) $K(\theta + k) = K(\theta) + \iota_k$ for $\theta \in \mathbb{R}^d$, $k \in \mathbb{Z}^d$;
(L3) $K(\theta) = K(\theta')$ implies $\theta' - \theta \in \mathbb{Z}^d$;
(L4) $DK(\theta)$ is injective;

(L5) $\Omega(DK(\theta)u, DK(\theta)v) = 0$ for all u, v;
(L6) $s \mapsto K(\theta + sw)$ has derivative $X_G(K(\theta + tw))$ at $s = t$, for all θ, t.

Note that (L2) is the period-one form of "angles are shifted by integers" in the chart; the frozen cylinder predicate differs from it only by the image shift in clause (iii), and the frozen physical predicate by periodicity.

Theorem 2.9 (Torus transport)**.** *Let $c > 0$, $O \subset \mathrm{Ph}(d)$ open, Ψ a c-chart on O, $H : \mathrm{Ph}(d) \to \mathbb{R}$, $G = H^\Psi = c^{-1} H \circ \Psi$, $w \in \mathbb{R}^d$, and K with $\mathrm{LT}(G, O, w, K)$. Assume the lattice compatibility*

(Q1) $O + \iota_k = O$ and $\Psi(z + \iota_k) = \Psi(z) + \sigma_k$ for $z \in O$, $k \in \mathbb{Z}^d$, for vectors $\sigma_k \in \mathrm{Ph}(d)$;
(Q2) (injectivity modulo the lattice) if $z, z' \in O$, $k \in \mathbb{Z}^d$ and $\Psi(z') = \Psi(z) + \sigma_k$, then $z' = z + \iota_m$ for some $m \in \mathbb{Z}^d$.

Put $\mathcal{K} := \Psi \circ K$. Then:

(a) (cylinder) *Let $d = a + b$ and $\sigma_k = \mathrm{shift}((k_i)_{i<a})$ (Definition 1.8). If $U \supset \Psi(K(\mathbb{R}^d))$, then* $\mathrm{IsCylTorus}(H, U, w, \mathcal{K})$ *(Definition 1.15).*
(b) (physical) *Let $\sigma_k = 0$ and assume w is (γ, τ)-Diophantine for some $\gamma > 0$, $\tau > d - 1$. Then* $\mathrm{IsPhysTorus}(H, w, \mathcal{K})$ *(Definition 1.24).*

In both cases $\mathcal{K}(\theta + tw)$ is the transport of the chart orbit, $\mathcal{K}(\mathbb{R}^d) = \Psi(K(\mathbb{R}^d))$, and $D\mathcal{K}(\theta) = D\Psi(K\theta)DK(\theta)$.

Frozen Lean declaration: `KAM.Planetary.IsCylinderTorus`, `KAM.Planetary.IsPhysicalTorus`

Source: Standard (change of canonical coordinates). Ours as a statement for the frozen predicates; no source page.

Proof. Write $\mathcal{K} = \Psi \circ K$.

Step S2.5: (i) Analyticity, range, differentiability. $K(\mathbb{R}^d) \subset O$ and Ψ is analytic on O, so $\mathcal{K}$ is analytic on $\mathbb{R}^d$ by (X1). $G = H^\Psi$ is differentiable at $K\theta$ by (L1), hence H is differentiable at $\mathcal{K}(\theta)$ by Lemma 2.4. Range: $\mathcal{K}(\theta) \in \Psi(K(\mathbb{R}^d)) \subset U$ in (a).

Step S2.6: (ii) Lattice behaviour. By (L2) and (Q1): $\mathcal{K}(\theta + k) = \Psi(K\theta + \iota_k) = \mathcal{K}(\theta) + \sigma_k$. In (a) this is clause (iii) of Def. 1.15 (with $k \in \mathbb{Z}^{a+b}$, shift by $k_f = (k_i)_{i<a}$); in (b) it is periodicity, clause (iii) of Def. 1.24.

Step S2.7: (iii) Injectivity modulo the lattice. (a): let $k \in \mathbb{Z}^a$ and $\mathcal{K}(\theta') = \mathcal{K}(\theta) + \mathrm{shift}(k)$. Then $\Psi(K\theta') = \Psi(K\theta) + \sigma_{(k,0)}$, so by (Q2) $K\theta' = K\theta + \iota_m = K(\theta + m)$ for some $m \in \mathbb{Z}^d$ (by (L2)), and (L3) gives $\theta' - (\theta + m) \in \mathbb{Z}^d$, i.e. $\theta' - \theta \in \mathbb{Z}^d$. (b): $\mathcal{K}(\theta') = \mathcal{K}(\theta)$ is the case $k = 0$ ($\sigma_0 = 0$) of the same argument.

Step S2.8: (iv) Derivative and Lagrangian property. $D\mathcal{K}(\theta) = D\Psi(K\theta)DK(\theta)$; $D\Psi(K\theta)$ is invertible (Lemma 2.1(b)), so $D\mathcal{K}(\theta)$ is injective by (L4). $\Omega(D\mathcal{K}u, D\mathcal{K}v) = c\,\Omega(DKu, DKv) = 0$ by Definition 2.3 and (L5).

Step S2.9: (v) The flow clause. Fix θ, t and let $y(s) = K(\theta + sw) \in O$. By (L6) y has derivative $X_G(y(t))$ at t and G is differentiable at $y(t)$; Corollary 2.5 (pointwise) gives that $s \mapsto \Psi(y(s)) = \mathcal{K}(\theta + sw)$ has derivative $X_H(\mathcal{K}(\theta + tw))$ at t.

Step S2.10: (vi) Diophantine clause. In (b) it is the hypothesis; in (a) none is part of IsCylTorus (see Def. 1.16 for where it is imposed; Lemmas 2.15–2.16). □

Remark 2.10 (Typical charts). (1) The polar chart of one oscillator $(\psi, R) \mapsto (q, p) = \sqrt{2R}(\sin\psi, \cos\psi)$ on $\{R > 0\}$ is a 1-chart ($dp \wedge dq = dR \wedge d\psi$, which is Ω in the frozen order), invariant under $\psi \mapsto \psi + 2\pi$; Part 5 composes it with $S_{2\pi}$ below. (2) S_c is a c-chart (Lemma 2.11). (3) Composition: if Ψ_1 is a c_1-chart on O_1, Ψ_2 a c_2-chart on $O_2 \supset \Psi_1(O_1)$, then $\Psi_2 \circ \Psi_1$ is a c_1c_2-chart on O_1 (chain rule), and $(H \circ \Psi_2)^{\Psi_1}$ with the factor c_1^{-1} followed by c_2^{-1} equals $H^{\Psi_2 \circ \Psi_1}$.

2.4. **Period conversion.** CP10 (footnote 1, p. 546: "$\mathbb{T}^n$ denotes the standard n-dimensional flat torus $\mathbb{R}^n/(2\pi\mathbb{Z}^n)$") and CP09 (Thm. 3, p. 6: "$\mathbb{T}^n$ the usual n-dimensional torus $\mathbb{R}^n/2\pi\mathbb{Z}^n$") use angles of period 2π. The frozen predicates use a parameter $\theta \in \mathbb{R}^d$ of period one. The relation is one explicit chart.

Lemma 2.11 (Period conversion)**.** *Let $c > 0$ and $S_c(z_1, z_2) = (cz_1, z_2)$.*

- *(a) S_c is a c-chart on all of* $\mathrm{Ph}(d)$ *(globally analytic, $\Omega(S_c u, S_c v) = c\,\Omega(u, v)$), $S_c(z + \iota_k) = S_c(z) + (ck, 0)$, and $|\det DS_c| = c^d$.*
- *(b)* (Hamiltonian dictionary) *For $\tilde{H} : \mathrm{Ph}(d) \to \mathbb{R}$ the pulled-back Hamiltonian is $\tilde{H}^{S_c}(z_1, z_2) = c^{-1}\tilde{H}(cz_1, z_2)$, and for every z: $S_c\, X_{\tilde{H}^{S_c}}(z) = X_{\tilde{H}}(S_c z)$. With $c = 2\pi$ and $(I, \phi) = (z_2, 2\pi z_1)$ this is $H(I, \theta) = \tilde{H}(I, 2\pi\theta)/2\pi$ of Lemma 4.5: orbits of H (angle θ) and of $\tilde{H}$ (angle $\phi = 2\pi\theta$) correspond with the* same *time; $\omega = \tilde{\omega}/2\pi$; Lagrangian for Ω iff after S_c (factor c).*
- *(c)* (Parameter conversion) *Let $\tilde{\kappa} : \mathbb{R}^d \to \mathrm{Ph}(d)$ (parameter of period 2π), $\tilde{\omega} \in \mathbb{R}^d$, $K(\theta) := \tilde{\kappa}(2\pi\theta)$, $w := \tilde{\omega}/2\pi$. Then K analytic iff $\tilde{\kappa}$ is; $DK(\theta) = 2\pi D\tilde{\kappa}(2\pi\theta)$, so injectivity of DK and the Lagrangian property (factor $4\pi^2$) hold for K iff for $\tilde{\kappa}$; $\tilde{\kappa}(\vartheta + 2\pi k) = \tilde{\kappa}(\vartheta) + \tau_k$ for all ϑ, k iff $K(\theta + k) = K(\theta) + \tau_k$; $\tilde{\kappa}(\vartheta) = \tilde{\kappa}(\vartheta') \Rightarrow \vartheta' - \vartheta \in 2\pi\mathbb{Z}^d$ iff $K(\theta) = K(\theta') \Rightarrow \theta' - \theta \in \mathbb{Z}^d$; and $s \mapsto K(\theta + sw)$ has derivative v at t iff $s \mapsto \tilde{\kappa}(2\pi\theta + s\tilde{\omega})$ has derivative v at t. In particular, for every Hamiltonian F, the frozen flow clause for (K, w) is CP's flow clause for $(\tilde{\kappa}, \tilde{\omega})$ and the frozen frequency is $w = \tilde{\omega}/2\pi$.*
- *(d)* (Frozen shift) *The frozen fast shift* $\mathrm{shift}(k_f) = ((2\pi k_f, 0_b), 0)$ *(Def. 1.8) equals $S_{2\pi}$ applied to $((k_f, 0_b), 0)$.*

Proof. **Step S2.11: (a).** $DS_c = \mathrm{diag}(c\mathbf{1}, \mathbf{1})$, so $\Omega(S_c u, S_c v) = \sum_i (u_{2,i}\, cv_{1,i} - cu_{1,i}v_{2,i}) = c\Omega(u, v)$; $\det = c^d$; the lattice formula is linearity.

Step S2.12: (b). Lemma 2.4 gives $DS_c\, X_{\tilde{H}^{S_c}} = X_{\tilde{H}} \circ S_c$. Explicitly $\partial_{z_2}\tilde{H}^{S_c} = c^{-1}\partial_2\tilde{H}$, $\partial_{z_1}\tilde{H}^{S_c} = \partial_1\tilde{H}$, which are Part 3's equations $\dot{\theta} = \partial_I H = \dot{\phi}/2\pi$, $\dot{I} = -\partial_\theta H = -\partial_\phi\tilde{H}$.

Step S2.13: (c). All claims are composition with the linear bijection $\theta \mapsto 2\pi\theta$ of $\mathbb{R}^d$ (the chain rule is not even needed for the flow clause: $K(\theta + sw) = \tilde{\kappa}(2\pi\theta + s\tilde{\omega})$ exactly). (d) is (a). □

Corollary 2.12 (Cylinder pipeline (consumed by Part 5))**.** *Let $d = a + b$, $H : \mathrm{Ph}(d) \to \mathbb{R}$, Φ a 1-chart on an open $O_0 \subset \mathrm{Ph}(d)$ with $O_0 + (2\pi k, 0) = O_0$ and $\Phi(z + (2\pi k, 0)) = \Phi(z) + \mathrm{shift}(k_f)$ for $z \in O_0$, $k \in \mathbb{Z}^d$, and ($\Phi(z') = \Phi(z) + \mathrm{shift}(k_f) \Rightarrow z' = z + (2\pi m, 0)$ for some $m \in \mathbb{Z}^d$); put $O = S_{2\pi}^{-1}(O_0)$, $\Psi = \Phi \circ S_{2\pi}$, $G_3 = H^\Psi = (2\pi)^{-1}(H \circ \Phi)(2\pi z_1, z_2)$. If $\mathrm{LT}(G_3, O, w, K_3)$ (for instance, K_3 the torus of Prop. 4.3(a)–(d) with π of Remark 2.2 applied, and the injectivity modulo translations of that proposition for (L3)) and $U \supset \Psi(K_3(\mathbb{R}^d))$, then* $\mathrm{IsCylTorus}(H, U, w, \Psi \circ K_3)$, with the same *$w$: the frozen frequency equals Part 3's frequency $\omega \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$, and CP's own angular frequency is $2\pi w$.*

Proof. Ψ is a 2π-chart on O (Remark 2.10(3)); (Q1) holds because $\Psi(z + \iota_k) = \Phi(S_{2\pi}z + (2\pi k, 0)) = \Psi(z) + \mathrm{shift}(k_f)$; (Q2) because $\Psi(z') = \Psi(z) + \mathrm{shift}(k_f)$ gives $S_{2\pi}z' = S_{2\pi}z + (2\pi m, 0)$, i.e. $z' = z + \iota_m$. Apply Theorem 2.9(a). □

Corollary 2.13 (Physical pipeline (consumed by Parts 12, 13))**.** *Let $H : \mathrm{Ph}(d) \to \mathbb{R}$, Φ a 1-chart on an open O_0 with $O_0 + (2\pi k, 0) = O_0$, $\Phi(z + (2\pi k, 0)) = \Phi(z)$ and ($\Phi(z') = \Phi(z) \Rightarrow z' = z + (2\pi m, 0)$, some $m \in \mathbb{Z}^d$). Let $\tilde{\kappa} : \mathbb{R}^d \to O_0$ be a torus in the 2π convention of CP09 Thm. 3 for $\tilde{G} = H \circ \Phi$: i.e. the clauses (L1)–(L5) hold for $\tilde{\kappa}$ with period 2π as in Lemma 2.11(c) ($\tau_k = (2\pi k, 0)$) and $s \mapsto \tilde{\kappa}(\vartheta + s\tilde{\omega})$ has derivative $X_{\tilde{G}}(\tilde{\kappa}(\vartheta + t\tilde{\omega}))$ at t, and $\tilde{\omega}$ is $(\tilde{\gamma}, \tau)$-Diophantine with $\tilde{\gamma} > 0$ and $\tau > d - 1$ (the positivity of $\tilde{\gamma}$ is a hypothesis: `KAM.IsDiophantine` itself has no positivity guard, and $\tilde{\omega} = 0$ is $(0, \tau)$-Diophantine). Then* $\mathrm{IsPhysTorus}(H, \tilde{\omega}/2\pi,\ \theta \mapsto \Phi(\tilde{\kappa}(2\pi\theta)))$, *and $K(\mathbb{R}^d) = \Phi(\tilde{\kappa}(\mathbb{R}^d))$.*

Proof. By Lemma 2.11(c), $K_0(\theta) = \tilde{\kappa}(2\pi\theta)$ satisfies (L1)–(L6) for $G = \tilde{G}$ and $w = \tilde{\omega}/2\pi$ with ι_k replaced by τ_k; the argument of Theorem 2.9(b) with $\Psi = \Phi$ (a 1-chart, $\sigma = 0$, lattice τ_k in place of ι_k) applies verbatim. The frequency w is $(\tilde{\gamma}/2\pi, \tau)$-Diophantine with $\tilde{\gamma}/2\pi > 0$ (Lemma 2.15(c)). □

2.5. **Diophantine conversion.**

Definition 2.14 (Two-scale Diophantine set)**.** For $n = n_1 + n_2$, $\gamma_1, \gamma_2, \tau > 0$: $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ is the set of $\omega = (\omega_1, \omega_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$ with $|k \cdot \omega| \geq \gamma_1 |k|_1^{-\tau}$ whenever $k_1 \neq 0$ and $|k \cdot \omega| \geq \gamma_2 |k|_1^{-\tau}$ whenever $k_1 = 0$, $k \neq 0$ ($k = (k_1, k_2) \in \mathbb{Z}^{n_1} \times \mathbb{Z}^{n_2}$). This is the set of Part 3 (§4.1); for $k_1 = 0$, $|k|_1 = |k_2|_1$.
Source: CP10 (19), p. 551 (with $\gamma_2 = \mu\bar{\gamma}_2$; norm $|k| = \sum |k_j|$, CP10 footnote 5, p. 547), and the definition of $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$, p. 551.

Lemma 2.15 (Elementary properties of `KAM.IsDiophantine`)**.** *Let $\omega \in \mathbb{R}^d$, $\gamma, \tau > 0$ (the frozen definition is $\gamma/|k|_1^\tau \leq |k \cdot \omega|$ for $k \neq 0$, Def. 1.1).*

- *(a) (monotone) (γ, τ)-Diophantine implies (γ', τ')-Diophantine for $0 < \gamma' \leq \gamma$, $\tau' \geq \tau$. In particular $\tau > d - 1$ and $\tau > a + b$ can be arranged by raising τ.*
- *(b) (norms) If $|k \cdot \omega| \geq \gamma\, \nu(k)^{-\tau}$ for all $k \neq 0$ with $0 < \nu(k) \leq |k|_1$ (e.g. $\nu = |\cdot|_2$ or $|\cdot|_\infty$, which are positive on $k \neq 0$), then ω is (γ, τ)-Diophantine.*
- *(c) (scaling) For $s > 0$: ω is (γ, τ)-Diophantine iff ω/s is $(\gamma/s, \tau)$-Diophantine.*
- *(d) (two scales) $\mathcal{D}_{\gamma_1,\gamma_2,\tau} \subset \{\omega :\ \omega$ is $(\min\{\gamma_1, \gamma_2\}, \tau)$-Diophantine$\}$; in particular, if $\gamma_1 \geq \gamma_2$, the set of (γ_2, τ)-Diophantine vectors.*

Proof. (a) For $k \neq 0$, $|k|_1 \geq 1$, so $|k|_1^\tau \leq |k|_1^{\tau'}$ (real powers of a base ≥ 1 are nondecreasing in the exponent; the Lean base is ≥ 1, so no totality issue). (b) On the positive bases $0 < \nu(k) \leq |k|_1$ one has $\nu(k)^{-\tau} \geq |k|_1^{-\tau}$ ($\tau > 0$). The positivity guard is necessary: Lean has $0^{-\tau} = 0$, and $\nu \equiv 0$ would satisfy the hypothesis for $\omega = 0$. (c) $|k \cdot \omega/s| = |k \cdot \omega|/s \geq (\gamma/s)|k|_1^{-\tau}$. (d) For each $k \neq 0$ the bound in the definition is one of the two lines, each $\geq \min\{\gamma_1, \gamma_2\}|k|_1^{-\tau}$. □

Lemma 2.16 (CP10 and CP09 give the frozen Diophantine clauses)**.**

- *(a)* (CP10, cylinder tori) *Let $\omega \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ with $\gamma_1, \gamma_2 > 0$ and $\tau > n = a + b$ (CP10: $\tau_* > n$, footnote 12, p. 550; Prop. 4.3 assumes $\tau > n$). Then ω is $(\min\{\gamma_1, \gamma_2\}, \tau)$-Diophantine, so the clause "$\exists \gamma > 0, \tau > a + b$, w (γ, τ)-Diophantine" of Def. 1.16 holds for $w = \omega$. If $w = \tilde{\omega}/2\pi$ with $\tilde{\omega} \in \mathcal{D}_{\tilde{\gamma}_1,\tilde{\gamma}_2,\tau}$ then $w \in \mathcal{D}_{\tilde{\gamma}_1/2\pi,\tilde{\gamma}_2/2\pi,\tau}$.*
- *(b)* (CP09, physical tori) *Let $\tilde{\omega} = \omega^\star(b)$ satisfy $|\langle k, \tilde{\omega}\rangle| \geq \gamma |k|_2^{-\tau}$ for all $k \in \mathbb{Z}^n \setminus \{0\}$ (CP09 Thm. 3(iv), (6), p. 7: $|k|_2$ the Euclidean norm), with $\gamma > 0$ and $\tau > n\mu \geq n$ ($\mu \in \mathbb{Z}_{\geq 1}$ by CP09 Def. 2, p. 6). Then $\tilde{\omega}$ is (γ, τ)-Diophantine, $w = \tilde{\omega}/2\pi$ is $(\gamma/2\pi, \tau)$-Diophantine, and $\tau > n - 1$: the clause (vii) of Def. 1.24 holds with $d = n$.*

Proof. (a) Lemma 2.15(d) and (c). (b) Lemma 2.15(b) with $\nu = |\cdot|_2 \leq |\cdot|_1$, then (c); $\tau > n\mu \geq n > n - 1$. □

Remark 2.17 (Dimension). For the planar theorems the torus has $d = 2n$ frequencies; CP09 Thm. 3 is applied with its n equal to this d, so its $\tau > n\mu \geq d$ exceeds $d - 1$ as the frozen clause requires.

2.6. **Liouville measure and unions of tori.**

Lemma 2.18 (Charts scale Lebesgue measure)**.** *Let Ψ be a c-chart on O and $A \subset O$ Lebesgue measurable. Then $\Psi(A)$ is Lebesgue measurable (null-measurable) and* meas $\Psi(A) \leq c^d$ meas A*, with equality if Ψ is injective on A. An* injective *symplectic chart ($c = 1$) preserves the measure of A; without injectivity only the upper bound is asserted (the polar chart of Remark 2.10(1) maps $[0, 4\pi) \times [1, 2]$, of area 4π, onto an annulus of area 2π). If A is Borel then $\Psi(A)$ is Borel.*

Proof. **Step S2.14: Determinant.** $|\det D\Psi| = c^d$ on O (Lemma 2.1(b)).

Step S2.15: Measurability. By (X2) cover O by countably many open O_j on which Ψ is a homeomorphism onto an open set with continuous inverse Ψ_j^{-1} (Lindelöf). Write $A\cap O_j = B\cup N$, B F_σ, N null ((X4)); $\Psi(B)$ is σ-compact, $\Psi(N)$ is null because Ψ is locally Lipschitz ([K:`bp:p1:lip-image`] on small cubes, countable union). So $\Psi(A) = \bigcup_j \Psi(A\cap O_j)$ is Lebesgue measurable.

Step S2.16: Borel sets. If A is Borel, $\Psi(A\cap O_j) = (\Psi_j^{-1})^{-1}(A\cap O_j)$ is the preimage of a Borel set under the continuous map Ψ_j^{-1} on the open set $\Psi(O_j)$, hence Borel; the countable union is Borel.

Step S2.17: Measure. For Borel A, (X3) gives $\operatorname{meas}\Psi(A) \le \int_A |\det D\Psi| = c^d \operatorname{meas} A$, with equality if Ψ is injective on A. For Lebesgue measurable A write $A = B\cup N$ with B Borel, N null: $\operatorname{meas}\Psi(A) \le \operatorname{meas}\Psi(B) + 0$, and for equality under injectivity use the subscript-0 version of (X3). □

Lemma 2.19 (Quotient version for whole tori)**.** *Assume the hypotheses of Theorem 2.9 with $\sigma_k = 0$. Let $A \subset O$ be measurable with $A + \iota_k = A$ for all $k \in \mathbb{Z}^d$, and $Q = \{z : z_{1,i} \in [0,1)\ \forall i\}$. Then $\Psi(A)$ is Lebesgue measurable, $\Psi(A) = \Psi(A\cap Q)$ and $\operatorname{meas}\Psi(A) = c^d \operatorname{meas}(A\cap Q)$. Consequently $0 < \operatorname{meas}\Psi(A)$ iff $0 < \operatorname{meas} A$, and $\operatorname{meas}\Psi(A) < \infty$ if $A\cap Q$ is bounded (e.g. the momenta of A are bounded).*

Proof. $A = \bigcup_m (A\cap Q) + \iota_m$, and $\Psi((A\cap Q)+\iota_m) = \Psi(A\cap Q)$ by (Q1) with $\sigma = 0$. If $z, z' \in A\cap Q$ and $\Psi(z) = \Psi(z')$, (Q2) gives $z' = z + \iota_m$ and $z, z' \in Q$ forces $m = 0$: Ψ is injective on $A\cap Q$; Lemma 2.18 gives $\operatorname{meas}\Psi(A\cap Q) = c^d \operatorname{meas}(A\cap Q)$. *Borel version:* if A is Borel, so is $A\cap Q$, and Ψ is continuous and injective on it, so $\Psi(A) = \Psi(A\cap Q)$ is Borel by (X4) (`MeasurableSet.image_of_continuousOn_injOn`); this is the form needed for the frozen `MeasurableSet` clauses. Finally $\operatorname{meas} A = \sum_m \operatorname{meas}((A\cap Q)+\iota_m)$ is 0 iff $\operatorname{meas}(A\cap Q) = 0$. □

Lemma 2.20 (Unions of whole tori as Lipschitz images)**.** *Let $E \subset \mathbb{R}^d$ be Lebesgue measurable, $A = E\times[0,1)^d \subset \mathbb{R}^{2d}$, $\mathcal{F} : A \to \mathbb{R}^{2d}$ with $|\mathcal{F}(x) - \mathcal{F}(y)| \le \Lambda|x-y|$ (max norms). Then $\mathcal{F}(A)$ is Lebesgue measurable (`NullMeasurableSet`; not asserted Borel) and $\operatorname{meas}\mathcal{F}(A) \le \Lambda^{2d}\operatorname{meas} E$. **Borel consumers (the frozen predicates need `MeasurableSet`):** (B1) if E is σ-compact (e.g. compact, as $\mathcal{K}_\star$ in CP09 Thm. 3), then $\mathcal{F}(A)$ is σ-compact, hence Borel, with no injectivity needed; (B2) if E is Borel and $\mathcal{F}$ is continuous and injective on A, then $\mathcal{F}(A)$ is Borel ((X4)). In both cases the saturated union $\bigcup_k(\mathcal{F}(A)+\iota_k)$ below is a countable union of Borel sets. If moreover $|\mathcal{F}(x)-\mathcal{F}(y)| \ge \lambda|x-y|$ with $\lambda > 0$, then $\operatorname{meas}\mathcal{F}(A) \ge \lambda^{2d}\operatorname{meas} E$; in particular $0 < \operatorname{meas}\mathcal{F}(A)$ iff $0 < \operatorname{meas} E$, and $\operatorname{meas}\mathcal{F}(A) < \infty$ iff $\operatorname{meas} E < \infty$. If $\mathcal{F}(\omega,\vartheta) = \varphi_\omega(\vartheta)$ with $\varphi_\omega(\vartheta+k) = \varphi_\omega(\vartheta)+\iota_k$, then $\bigcup_{\omega\in E}\varphi_\omega(\mathbb{R}^d) = \bigcup_{k\in\mathbb{Z}^d}(\mathcal{F}(A)+\iota_k)$ is measurable.*

Proof. **Step S2.18: Extension.** McShane ([K:`bp:p1:mcshane`], componentwise in the max norm) gives a Λ-Lipschitz $\hat{\mathcal{F}} : \mathbb{R}^{2d} \to \mathbb{R}^{2d}$ extending $\mathcal{F}$.

Step S2.19: Measurability. A is measurable; $A = B\cup N$ with B F_σ, N null ((X4)); $\hat{\mathcal{F}}(B)$ is σ-compact and $\operatorname{meas}\hat{\mathcal{F}}(N) = 0$ by [K:`bp:p1:lip-image`]; so $\mathcal{F}(A) = \hat{\mathcal{F}}(B)\cup\hat{\mathcal{F}}(N)$ is Lebesgue measurable (completeness of Lebesgue measure; this does not give a Borel set).

Step S2.20: Borel cases. (B1): $E\times[0,1)^d$ is a countable union of compact sets and continuous images of compact sets are compact. (B2): (X4).

Step S2.21: Bounds. [K:`bp:p1:lip-image`]: $\operatorname{meas}\mathcal{F}(A) \le \Lambda^{2d}\operatorname{meas} A = \Lambda^{2d}\operatorname{meas} E$. Under the lower bound $\mathcal{F}$ is injective with λ^{-1}-Lipschitz inverse on $\mathcal{F}(A)$, so $\operatorname{meas} E = \operatorname{meas}\mathcal{F}^{-1}(\mathcal{F}(A)) \le \lambda^{-2d}\operatorname{meas}\mathcal{F}(A)$ (same lemma). The last claim is the periodicity and countability of $\mathbb{Z}^d$. □

Remark 2.21 (Use). *P-PLANAR/P-FIXED.* The chart-level family is $X_0(b,\xi)$ of CP09 Thm. 3(ii), bi-Lipschitz on $\mathcal{K}_\star\times\mathbb{T}^d$ with $\mathcal{K}_\star$ compact, so the chart-level union is compact (B1), hence Borel, and of positive finite measure (Lemma 2.20, applied *at the chart level*). The Cartesian union is its

image under the symplectic chart Φ of Corollary 2.13; Lemma 2.19 gives meas = meas$(A \cap Q_{2\pi})$ with $c = 1$ and the cell $Q_{2\pi} = \{z_{1,i} \in [0, 2\pi)\}$ (the lattice of the chart is $2\pi\mathbb{Z}^d$ there). In the period-one formulation $\Psi = \Phi \circ S_{2\pi}$ one has $c = 2\pi$ and the cell Q_1, and the two formulas agree because $S_{2\pi}$ multiplies cell volume by $(2\pi)^d$. The lower Lipschitz bound of Lemma 2.20 must *not* be applied to a wrapped (periodic) Cartesian embedding, which cannot satisfy it on a whole cell. Borelness of the Cartesian union follows from Lemma 2.19 (Borel version). *P-G.* The family is that of Prop. 4.3; its parameter set $E = \omega_0(D) \cap \mathcal{D}$ must be Borel (or σ-compact) for (B2)/(B1), and injectivity of the family modulo translations is part of that proposition; Part 5 must retain whole tori and Borelness when deleting parameters; the measure bounds of the cell are Part 4.

2.7. **The ODE layer (cited).** Track N's Part 3 supplies, for the three tracks, existence, uniqueness, continuation, energy and the identification of the frozen solution predicates. Track P uses only the following, always by these labels.

[N:bpN:p3:gen-uniqueness]	uniqueness on every interval for C^1 fields (orbits of a torus are *the* solutions; Lemma 2.22)
[N:bpN:p3:ham-smooth], [N:bpN:p3:ham-energy], [N:bpN:p3:ham-periodic]	X_H is C^1 where H is C^2; energy conservation; angle periodicity
[N:bpN:p3:newton-reduction], [N:bpN:p3:newton-continuation]	`IsNewtonSolution`, `IsFixedSolution` as `IsSolIn` (Part 6)
[N:bpN:p3:gen-continuation], [N:bpN:p3:gen-confinement]	only if a later part needs continuation (not used in Part 1)
[N:bpN:p5:symplectic], [N:bpN:p5:canonical]	time-one maps are symplectic; $D\Phi X_{H\circ\Phi} = X_H \circ \Phi$ (Remark 2.6)

Lemma 2.22 (Orbits on a torus)**.** *Let $H : \mathrm{Ph}(d) \to \mathbb{R}$ be C^2 on an open $W \supset \mathcal{K}(\mathbb{R}^d)$ and let $\mathcal{K}, w$ satisfy the flow clause of Defs. 1.15 (vii) or 1.24 (viii). Then $t \mapsto \mathcal{K}(\theta + tw)$ is a solution of $\dot{z} = X_H(z)$ in W on $\mathbb{R}$, and every solution $y : \mathbb{R} \to W$ with $y(0) = \mathcal{K}(\theta)$ equals it. $H \circ \mathcal{K}(\cdot + tw)$ is constant in t.*

Proof. The flow clause is the solution property (with values in W by hypothesis). X_H is C^1 on W ([N:bpN:p3:ham-smooth]); uniqueness is [N:bpN:p3:gen-uniqueness] on $J = \mathbb{R}$; energy is [N:bpN:p3:ham-energy]. □

Remark 2.23 (Orbit transfer to Newton). That the frozen clause (b) of MovKol and the fixed-Sun clause hold along torus orbits is Part 6: the barycentric reconstruction is a linear map composed with the orbit, the chain rule is Corollary 2.5 or elementary, and the translation of the Newton predicates is [N:bpN:p3:newton-reduction]; nothing of the existence theory is used.

2.8. **Sources, deviations, forward references.**

Sources. CP10 footnote 1 (p. 546), (19) (p. 551), footnote 5 (p. 547) and footnote 12 (p. 550), all read; CP09 Def. 2 (p. 6), Thm. 3 and (6) (pp. 6–7), read. The rest is standard (the algebra of symplectic matrices, the inverse function theorem, the area formula).

Remarks on the sources. (1) CP09 Thm. 3 prints the symplectic form as $dx \wedge dy$ with $x \in \mathbb{T}^n$ the angles; with the frozen $\Omega = \sum dz_2 \wedge dz_1$ this is $-\Omega$; the class of symplectic maps and the Lagrangian property are unaffected (Lemma 2.1(c)), and the printed Hamiltonian field in CP09 Remark 3(i) ($\xi \mapsto \xi + \omega^\star t$) is the frozen one. (2) The exponent in CP09 (6) is typeset as $|k|_2^\tau$ with a Euclidean norm; we only use $|k|_2 \le |k|_1$. (3) CP09 Thm. 3 gives no statement that $X(b, \cdot, \cdot)$ is defined on all of $\mathbb{R}^{2n}$ or injective modulo 2π translations of ξ; (L3) for $X_0(b, \cdot)$ follows from (ii) (a homeomorphism of $\mathcal{K}_\star \times \mathbb{T}^n$ onto its image).

Forward references. [P:bpP:p5:kam-application] (cylinder pipeline), Parts 6, 12, 13 (physical pipeline, orbit transfer) cite Theorem 2.9 and Corollaries 2.12, 2.13. Part 3 cites [P:bpP:p1:period-convert] (Lemma 2.11; its Lemma 4.5 is the case (b) with $c = 2\pi$). Every other cited label exists in tracks K and N.

Label index. bpP:p1:omega-algebra, chart, chart-field, chart-orbit, hol-chart, lattice-torus, torus-transport, period-convert, cyl-pipeline, phys-pipeline, two-scale, dio-basic, dio-sources, liouville, liouville-lattice, torus-union, torus-orbit.

3. Averaging with two frequency scales

This part proves the averaging statement that Part 3 consumes, Lemma 4.8 (CP10 Proposition 1 with $m = 0$ and two non-resonance scales $\alpha_1 \geq \alpha_2$), and the averaging step of CP10 §4, Step 1 (p. 563) that Part 5 consumes. CP10 proves its Lemma A.1 only by reference to the unpinned [4, Lemma A.5]; here everything is derived from track N's Parts 4–6.

What track N gives, and what it does not. The one-scale Normal Form Lemma [N:bpN:p6:normal-form] cannot simply be applied with $\alpha = \alpha_2$, for four reasons, each of which is repaired below. (1) *Exponential averaging needs an iteration*: one homological solve and one flow leave an angle-dependent quadratic remainder; the exponential bound $e^{-Ks/6}E$ comes from the finite iteration of [N:bpN:p6:iterative] (as in [N:bpN:p6:normal-form], CP10 p. 565), not from one step. (2) *The strip.* [N:bpN:p6:normal-form] shrinks the strip from S to $S/6$; the interface needs $\bar{s}+s \to \bar{s}+s/6$, that is, a loss budget s that is unrelated to the initial width. The iteration is therefore re-run with N's step lemma as a black box (Proposition 3.4). (3) *Two scales.* The first-action displacement with α_1 is not a consequence of the scalar statement (since α_1/α_2 is unbounded); it comes from the splitting of the generating function (Lemma 3.3, CP10 (70)–(71)). (4) *Hypotheses.* [N:bpN:p6:iterative] asks for non-resonance on the base set plus a Lipschitz bound L with $KLr \leq \alpha/p$; Part 3 provides non-resonance on the whole complex neighbourhood, and no Lipschitz constant enters the statement. Lemma 3.2 records that N's proof only uses the divisor bound on the neighbourhood.

3.1. Conventions: norm, widths, dictionary.

Norm ruling. Track P uses track N's norm ([N:bpN:p4:norm], ruling of the coordinator after review P3-03): for u holomorphic on $\mathcal{D}_{r,s} = T_s \times V_r D$ and 1-periodic in θ, $|u|_{r,s} = \sup_{I \in V_r D} \sum_k |\hat{u}_k(I)| e^{2\pi |k|_1 s}$, where s is the *geometric* half-width of the strip $T_s = \{|\operatorname{Im}\theta|_\infty < s\}$ and $V_r D = \{I \in \mathbb{C}^n : \operatorname{dist}_\infty(I, D) < r\}$ ([N:bpN:p4:domains]). This is not the norm $\sum_k \sup_I$ used in Part 3 as printed; only $|\cdot|, \leq \sum_k \sup_I$ holds.

Dictionary to Part 3. (Lemma 4.5 and [N:bpN:p4:dictionary]). Part 3 writes widths as *coefficient radii* σ (weight $e^{\sigma|k|_1}$, strip $\{|\operatorname{Im}\theta| < \sigma/2\pi\}$, notation $\mathbb{T}_\sigma$). Hence $\mathbb{T}^{\mathrm{P3}}_\sigma = T_{\sigma/2\pi}$, and Part 3's $(\bar{s}, s)$ correspond to $(\bar{s}/2\pi, s/2\pi)$ here. Frequencies, divisors, the Hamilton equations $\dot{\theta} = \partial_I H$, $\dot{I} = -\partial_\theta H$ and the period-one angles are the same in both; N's $\bar{\alpha} = 2\pi\alpha$ and $\kappa = 2\pi K s$ are the quantities $2\pi\alpha$ and "Ks of Part 3". Throughout this part

$$s \text{ (geometric)}, \qquad \kappa := 2\pi K s \quad (\text{so } \kappa = K s_{\mathrm{P3}}), \qquad \bar{\alpha}_j := 2\pi\alpha_j, \qquad Z_K = \{k \in \mathbb{Z}^n : |k|_1 \leq K\}.$$

The statement for Part 3 in its own notation is Corollary 3.6.

Splitting of indices. $n = n_1 + n_2$, $k = (k_1, k_2)$, $\theta = (\theta_1, \theta_2)$, $I = (I_1, I_2)$. A function is *independent of* θ_1 if all its Fourier modes have $k_1 = 0$.

Cauchy estimates used. After the repair of N Part 4 ([N:bpN:p4:cauchy](a)) the angular estimate holds at each fixed parameter point for the *sum* over j and, globally, for each single j; the sum of global norms over j is not available. This part never uses it: Lemma 3.3 and the field bounds use one coordinate at a time ([N:bpN:p5:vf-bound]), and the brackets are taken at a fixed parameter point (Lemma 3.8).

External facts (all inherited): those of [N:bpN:p4:s-ext], [N:bpN:p6:s-ext]; $12 \ln 2 > 8$ and $\sum_{i \geq 1} 4^{1-i} = \frac{4}{3}$ ([N:bpN:p6:ext-numerics]); $|k \cdot v| \leq |k|_1 |v|_\infty$.

3.2. N's step lemma under a direct divisor hypothesis.

Definition 3.1 (Direct divisor setting)**.** Let $D \subset \mathbb{C}^n$, $r > 0$, $K \geq 0$, $\alpha > 0$, $q \geq 1$, $\Lambda \subset \mathbb{Z}^n$, and h holomorphic on $V_r D$, $\omega = \partial_I h$. The data satisfy (Div_q) if $|k \cdot \omega(I)| \geq \alpha/q$ for all $I \in V_r D$ and $k \in Z_K \setminus \Lambda$.

Lemma 3.2 (Iterative Lemma, direct form)**.** [N:bpN:p6:homological] *and* [N:bpN:p6:iterative] *(with its Remark* [N:bpN:p6:iterative-special]*) hold verbatim, with the same* $\phi = \sum_{k \in Z_K \setminus \Lambda} \frac{\hat{f}_k}{2\pi i k \cdot \omega} \chi_k$*, the same constants and the same estimates, if the hypotheses* [N:bpN:p6:cond-nr]*,* [N:bpN:p6:cond-L]*,* $KLr \leq \alpha/p$ *are replaced by* (Div_q)*,* $q \geq 1$ *arbitrary. In particular* $q = 1$ *is allowed.*

Source: ours; N Part 6 as written.

Source gap: This is a variant of the written N statement. N's own proof is unchanged (check below); N may adopt (Div_q) *as its primary hypothesis.*

Proof. **Step S3.1: Where the hypotheses are used.** In [N:bpN:p6:homological], [N:bpN:p6:cond-nr], [N:bpN:p6:cond-L] and $KLr \leq \alpha/p$ enter only through [N:bpN:p6:divisor], whose conclusion is exactly (Div_q); and [N:bpN:p6:divisor] is used only in Steps [N:stepN:p6:hom-def] (holomorphy, nonvanishing denominators) and [N:stepN:p6:hom-norm] (the bound $|\hat{\phi}_k| \leq q|\hat{f}_k|/\bar{\alpha}$). The proof of [N:bpN:p6:iterative] uses the hypotheses only through [N:bpN:p6:homological]; the identity $p^{-1} + q^{-1} = 1$ occurs only in [N:bpN:p6:divisor]. The rest (Lemmas [N:bpN:p5:flow], [N:bpN:p5:composition], [N:bpN:p5:canonical-map], [N:bpN:p5:transform], [N:bpN:p4:bracket]) never mentions ω. □

Lemma 3.3 (Two-scale step: splitting of ϕ and componentwise displacement)**.** *Let* $D \subset \mathbb{C}^n$*,* $r, s > 0$*,* $0 < \alpha_2 \leq \alpha_1$*,* $q \geq 1$*,* $\Lambda = \{0\}$*,* $K \geq 0$*, and assume*

$$(\mathrm{Div}_q^{(2)}) \qquad |k\cdot\omega(I)| \geq \alpha_1/q \ \ (k_1 \neq 0), \qquad |k\cdot\omega(I)| \geq \alpha_2/q \ \ (k_1 = 0,\ k \neq 0), \qquad I \in V_r D,\ k \in Z_K.$$

Let $f \in \mathcal{O}(r,s)$ *with* $\delta = |f|_{r,s}$*,* $0 < \rho < r/2$*,* $0 < \sigma < s/2$*,* $u = q\delta/(\bar{\alpha}_2 \rho\sigma) \leq \frac{1}{4}$ *(the hypothesis of Lemma 3.2 with* $\alpha = \alpha_2$*),* ϕ *and* $\Phi = X_\phi^1$ *as there. Put* $\phi^{(1)} = \sum_{k \in Z_K, k_1 \neq 0} \hat{\phi}_k \chi_k$*,* $\phi^{(2)} = \phi - \phi^{(1)}$*. Then*

(a) $\phi^{(2)}$ *is independent of* θ_1*;* $|\phi^{(1)}|_{r,s} \leq q\delta/\bar{\alpha}_1$ *and* $|\phi|_{r,s} \leq q\delta/\bar{\alpha}_2$*;*

(b) on $\mathcal{D}_{r-\frac{3}{2}\rho, s-\frac{3}{2}\sigma}$*:*

$$|\Phi_{I_1} - I_1| \leq \frac{q\delta}{e\,\bar{\alpha}_1 \sigma}, \qquad |\Phi_{I_2} - I_2| \leq \frac{q\delta}{e\,\bar{\alpha}_2 \sigma}, \qquad |\Phi_\theta - \theta| \leq \frac{q\delta}{\bar{\alpha}_2 \rho}.$$

Source: CP10 (70)–(71), p. 565 (splitting of H_ϕ*, bound for* $|I_1 - I_1'|$*); the displacement bookkeeping is ours.*

Proof. **Step S3.2: Coefficients.** By the formula for ϕ ([N:bpN:p6:homological]), $\hat{\phi}_k = \hat{f}_k/(2\pi i k \cdot \omega)$, a finite family of holomorphic functions. For $k_1 \neq 0$, $(\mathrm{Div}_q^{(2)})$ gives $|\hat{\phi}_k| \leq q|\hat{f}_k|/\bar{\alpha}_1$; for every $k \in Z_K \setminus \{0\}$, $\alpha_1 \geq \alpha_2$ gives $|\hat{\phi}_k| \leq q|\hat{f}_k|/\bar{\alpha}_2$. Multiply by $e^{2\pi|k|_1 s}$, sum, take the supremum: (a) except independence, which holds because $\phi^{(2)}$ has only modes with $k_1 = 0$. Both $\phi^{(i)}$ lie in $\mathcal{O}(r,s)$ with finite norm ([N:bpN:p4:coefficients](c)).

Step S3.3: Field. Put $\varepsilon = |\phi|_{r,s}/(\rho\sigma) \leq u \leq \frac{1}{4}$, $\Omega = \mathcal{D}_{r-\rho, s-\sigma}$, $\nu = \nu_{\rho,\sigma}$ ([N:bpN:p5:setting]). By [N:bpN:p5:flow](a) and Step [N:stepN:p5:cm-ab] of [N:bpN:p5:canonical-map], for $z \in \mathcal{D}_{r-\frac{3}{2}\rho, s-\frac{3}{2}\sigma}$ the trajectory $\gamma(t) = X_\phi^t(z)$, $0 \leq t \leq 1$, stays in $\bar{B}_\nu(z,\varepsilon) \subset \Omega$ and solves $\dot{\gamma} = X_\phi(\gamma) = (\partial_I \phi, -\partial_\theta \phi)$. The first n_1 components of $\dot{\gamma}_I$ are $-\partial_{\theta_1}\phi = -\partial_{\theta_1}\phi^{(1)}$ because $\partial_{\theta_1}\phi^{(2)} = 0$. By [N:bpN:p5:vf-bound] applied to $\phi^{(1)}$, $|\partial_{\theta_1}\phi^{(1)}| \leq |\phi^{(1)}|_{r,s}/(e\sigma) \leq q\delta/(e\bar{\alpha}_1\sigma)$ on Ω. Likewise [N:bpN:p5:vf-bound] for ϕ bounds $|\partial_{\theta_2}\phi| \leq q\delta/(e\bar{\alpha}_2\sigma)$ and $|\partial_I \phi| \leq q\delta/(\bar{\alpha}_2\rho)$ on Ω.

Step S3.4: Integrate. The mean value inequality along γ ([K:bp:p1:ext-mvt]) gives (b) with $t = 1$. □

3.3. **The averaging proposition in N's conventions.**

Proposition 3.4 (Averaging with a loss budget and two scales)**.** *Let $n = n_1 + n_2$, $D \subset \mathbb{R}^n$, $r > 0$, $\bar{s} \geq 0$, $s > 0$, $K > 0$ with $\kappa = 2\pi K s \geq 6$, $0 < \alpha_2 \leq \alpha_1$. Let h be holomorphic on $V_r D$ and real on $V_r D \cap \mathbb{R}^n$, $\omega = \partial_I h$, and assume*

(A1) $|k \cdot \omega(I)| \geq \alpha_1$ if $k_1 \neq 0$, $\geq \alpha_2$ if $k_1 = 0$, for all $I \in V_r D$ and $k \in Z_K \setminus \{0\}$;
(A2) $f \in \mathcal{O}(r, \bar{s}+s)$, real on real points, with $E := |f|_{r,\bar{s}+s} \leq \dfrac{\alpha_2 r}{128K}$.

Then there is a map $\Psi : \mathcal{D}_{r/2,\bar{s}+s/6} \to \mathcal{D}_{r,\bar{s}+s}$, $\Psi(\theta', I') = (\theta, I)$, and functions g, f_* on $\mathcal{D}_{r/2,\bar{s}+s/6}$ such that

(a) Ψ is holomorphic, injective, real on real points, $\Psi(\theta' + k, I') = \Psi(\theta', I') + (k, 0)$ ($k \in \mathbb{Z}^n$), and $(D\Psi)^\top J\, D\Psi = J$ (Ψ preserves $d\theta \wedge dI$, hence $dI \wedge d\theta$);
(b) $H \circ \Psi = h + g + f_*$, $H = h + f$, with $g = g(I')$ independent of θ', real on real points, and $\sup_{V_{r/2}D} |g - \hat{f}_0| \leq |g - \hat{f}_0|_{r/2,\bar{s}+s/6} \leq E/4$ (indeed $\leq E/\kappa$);
(c) f_* is holomorphic, periodic, real on real points and $|f_*|_{r/2,\bar{s}+s/6} \leq e^{-\kappa/6} E$;
(d) $\bar{\alpha}_1 s\, |I_1 - I_1'| \leq 3E$, $\bar{\alpha}_2 s\, |I_2 - I_2'| \leq 3E$, $\bar{\alpha}_2 r\, |\theta - \theta'| \leq 9E$ on $\mathcal{D}_{r/2,\bar{s}+s/6}$.

Constants depend on nothing: no n, τ, q or L enter.
Source: CP10 Prop. 1 (p. 552) with $m = 0$, Lemma A.1 (p. 564), proof pp. 564–565; Pöschel 1993 pp. 193–196 via [N:`bpN:p6:normal-form`]*. Constants recomputed (see Remark 3.5).*

Proof. Throughout $q = 1$: by (A1), ($\mathrm{Div}_1^{(2)}$) holds on $V_r D$ with $\Lambda = \{0\}$, and for every $r' \leq r$ and $D' \subset D$-neighbourhoods inside $V_r D$. Write $\bar{\alpha} = \bar{\alpha}_2$, $\varepsilon := E$, $S_i := \bar{s} + s_i$ for the strip widths, where s_i (the *budget* still to be spent) is chosen below. From (A2): $\varepsilon \leq \alpha_2 r/(128K) = \bar{\alpha} r s/(128\kappa)$. Every application of Lemma 3.2 below is on $\mathcal{D}_{r_i,S_i}$ with $r_i \leq r$, so its non-resonance hypothesis is (A1); its requirements $2\rho < r_i$, $2\sigma < S_i = \bar{s} + s_i$ are checked directly at each step (below; at the last step $s_N = s/6 + 2\sigma > 2\sigma$ and $r_N > 2\rho$). Because the weights $|g|_*$ of [N:`bpN:p6:setting`] contain the widths only through the *differences* $s_j - s_i$, the strip offset $\bar{s}$ plays no role in any estimate.

Step S3.5: Step 0. Take $\rho_0 = r/8$, $\sigma_0 = s/6$ and $g = 0$ in Lemma 3.2. Then $y_0 := \varepsilon/(\bar{\alpha}\rho_0\sigma_0) = 48\varepsilon/(\bar{\alpha} r s) \leq \frac{3}{8\kappa} \leq \frac{1}{16}$, so $u = y_0 \leq \frac{1}{4}$. Lemma 3.2(d) with $|g|_* = 0$ gives

$$|f_1|_{\frac{3}{4}r,\bar{s}+\frac{2}{3}s} \leq (1 - 2y_0)^{-1}\big(\tfrac{2}{e} y_0 + e^{-\kappa/6}\big)\varepsilon, \qquad (1 - 2y_0)^{-1} \leq \tfrac{8}{7}$$

($2\pi K \sigma_0 = \kappa/6 \geq 1$). *Case A: $e^{-\kappa/6} \leq y_0$.* Then

$$\varepsilon_1 := |f_1|, \leq \tfrac{8}{7}\big(1 + \tfrac{2}{e}\big) y_0 \varepsilon \leq 1.984 \cdot \tfrac{3}{8\kappa}\, \varepsilon \leq \frac{3\varepsilon}{4\kappa} \leq \frac{\varepsilon}{8}. \tag{2.1}$$

Case B: $e^{-\kappa/6} > y_0$. Repeat Step 0 with the doubled losses $\rho' = r/4$, $\sigma' = s/3$ ($u' = y_0/4$): $|f_1|_{r/2,\bar{s}+s/3} \leq (1 - y_0/2)^{-1}(\frac{2}{e}\frac{y_0}{4} + e^{-\kappa/3})\varepsilon \leq \frac{32}{31}(\frac{1}{2e} + \frac{1}{e})e^{-\kappa/6}\varepsilon \leq e^{-\kappa/6}\varepsilon$, using $e^{-\kappa/3} \leq e^{-1}e^{-\kappa/6}$. We stop with $\Psi = \Phi_0'$ on $\mathcal{D}_{r/2,\bar{s}+s/3} \supset \mathcal{D}_{r/2,\bar{s}+s/6}$, $g = \hat{f}_0$, $f_* = f_1$. Clauses (a)–(c) follow from Lemma 3.2; (d): Lemma 3.3(b) gives $|\Delta I_j| \leq \varepsilon/(e\bar{\alpha}_j\sigma') = 3\varepsilon/(e\bar{\alpha}_j s) < 3\varepsilon/(\bar{\alpha}_j s)$ and $|\Delta\theta| \leq \varepsilon/(\bar{\alpha}\rho') = 4\varepsilon/(\bar{\alpha} r)$. From now on Case A.

Step S3.6: Number of steps. $N := \lfloor \kappa/(12\ln 2) \rfloor$, so $N < \kappa/8$. If $N = 0$: $\kappa < 12\ln 2$, hence $e^{-\kappa/6} > \frac{1}{4}$, and $\varepsilon_1 \leq \varepsilon/8 \leq e^{-\kappa/6}\varepsilon$ by (2.1); set $\Psi = \Phi_0$, $g = \hat{f}_0$, $f_* = f_1$ on $\mathcal{D}_{3r/4,\bar{s}+2s/3}$. Displacements: $|\Delta I_j| \leq \varepsilon/(e\bar{\alpha}_j\sigma_0) = 6\varepsilon/(e\bar{\alpha}_j s) \leq 2.21\,\varepsilon/(\bar{\alpha}_j s)$, $|\Delta\theta| \leq 8\varepsilon/(\bar{\alpha} r)$. (a)–(d) follow. Let $N \geq 1$ and for $1 \leq i \leq N$

$$\rho_i = \rho = \frac{r}{8N}, \quad \sigma_i = \sigma = \frac{s}{4N}, \qquad r_i = \tfrac{3}{4}r - 2(i-1)\rho, \quad s_i = \tfrac{2}{3}s - 2(i-1)\sigma, \quad S_i = \bar{s} + s_i;$$

then $r_{N+1} = r/2$, $s_{N+1} = s/6$. $2\pi K\sigma = \kappa/(4N) \geq 3\ln 2$, so $e^{-2\pi K\sigma} \leq \frac{1}{8}$; $\rho\sigma = rs/(32N^2)$. From $\kappa > 8N$, (A2) and (2.1):

$$\varepsilon \leq \frac{\bar\alpha rs}{2^{10}N}, \qquad \varepsilon_1 \leq \frac{3}{4\kappa}\varepsilon < \frac{3\varepsilon}{32N} \leq \frac{3\,\bar\alpha rs}{2^{15}N^2} < \frac{\bar\alpha\rho\sigma}{2^8}. \tag{2.2}$$

Step S3.7: Induction. *Claim:* for $1 \leq i \leq N$ step i (Lemma 3.2 on $\mathcal{D}_{r_i,S_i}$, losses (ρ,σ)) is admissible, and $\varepsilon_{i+1} := |f_{i+1}|_{r_{i+1},S_{i+1}} \leq \varepsilon_i/4$; hence $\varepsilon_i \leq 4^{1-i}\varepsilon_1$ $(1 \leq i \leq N+1)$. Here f_{i+1}, $g_i = g_{i-1} + (\hat f_i)_0$ are produced by the step, $g_0 := \hat f_0$, $|g_i - g_{i-1}| \leq \varepsilon_i$ ([N:bpN:p4:projections]). *Admissibility:* $u_i = \varepsilon_i/(\bar\alpha\rho\sigma) \leq 2^{-8}$ by (2.2). *Weights:* decompose $g_{i-1} = g_0 + \sum_{j=1}^{i-1}(g_j - g_{j-1})$; the term $j \geq 1$ lives on $\mathcal{D}_{r_j,S_j}$ with $r_j - r_i \geq 2\rho$, $s_j - s_i \geq 2\sigma$, weight $\leq \frac12(\frac13+\frac13) \leq \frac12$; the term g_0 lives on $\mathcal{D}_{r,\bar s+s}$ with $r - r_i \geq r/4$, $s - s_i \geq s/3$, weight $\leq \frac12(\frac{1}{2N}+\frac{3}{4N}) \leq \frac1N$. Thus $|g_{i-1}|_* \leq \frac12\cdot\frac43\varepsilon_1 + \varepsilon/N \leq \varepsilon_1 + \varepsilon/N \leq \bar\alpha\rho\sigma\,(2^{-8}+2^{-5}) \leq \bar\alpha\rho\sigma/2^4$ (by (2.2) and $\varepsilon/N \leq \bar\alpha rs/(2^{10}N^2) = 2^{-5}\bar\alpha\rho\sigma$). *The step:* Lemma 3.2(d) with $\delta = \varepsilon_i$ gives $\varepsilon_{i+1} \leq \frac{128}{127}\big(\frac{2}{e}(\frac{1}{16}+\frac{1}{2^8})+\frac18\big)\varepsilon_i \leq \frac{128}{127}\cdot 0.1915\,\varepsilon_i < \frac14\varepsilon_i$. (This is the computation of [N:bpN:p6:normal-form], Step [N:stepN:p6:nf-induct], with $q = 1$.)

Step S3.8: Conclusion of the construction. Put $\Psi = \Phi_0 \circ \cdots \circ \Phi_N$ on $\mathcal{D}_{r/2,\bar s+s/6}$, $g = g_N$, $f_* = f_{N+1}$. Since $H_{i+1} = H_i\circ\Phi_i$, $H\circ\Psi = h+g+f_*$. g is a finite sum of θ-independent functions (each $(\hat f_i)_0$), so $g = g(I')$; $\hat f_0 = g_0$. [N:bpN:p6:telescope] with $U_i = \mathcal{D}_{r_i,S_i}$ gives (a) (holomorphic, canonical, injective; periodicity and reality are inherited from each Φ_i, [N:bpN:p6:iterative](a),(e)), and reality of g, f_* is [N:bpN:p6:iterative](e). (c): $\varepsilon_{N+1} \leq 4^{-N}\varepsilon_1 \leq 4^{-N-1}\varepsilon \leq e^{-\kappa/6}\varepsilon$, since $N+1 > \kappa/(12\ln 2)$ gives $4^{-(N+1)} < e^{-\kappa/6}$. (b): $|g - g_0| \leq \sum_{i=1}^N \varepsilon_i \leq \frac43\varepsilon_1 \leq \varepsilon/\kappa \leq \varepsilon/6$ by (2.1); this is the bound of CP10 (25) in which no K appears in the numerator (review P3-04). The sup bound follows from [N:bpN:p4:sup-weighted](a).

Step S3.9: (d) Displacement. By [N:bpN:p6:telescope] (seminorms $|\Pi_{I_1}\cdot|$, $|\Pi_{I_2}\cdot|$, $|\Pi_\theta\cdot|$) and Lemma 3.3(b) for each step (valid on $V_{i+1} \subset \mathcal{D}_{r_i-\frac32\rho_i,S_i-\frac32\sigma_i}$, [N:bpN:p6:iterative](a)):

$$|\Delta I_j| \leq \frac{\varepsilon}{e\bar\alpha_j\sigma_0} + \sum_{i=1}^N \frac{\varepsilon_i}{e\bar\alpha_j\sigma} \leq \frac{1}{e\bar\alpha_j s}\Big(6\varepsilon + 4N\cdot\tfrac43\varepsilon_1\Big), \qquad |\Delta\theta| \leq \frac{\varepsilon}{\bar\alpha\rho_0} + \sum_{i=1}^N \frac{\varepsilon_i}{\bar\alpha\rho} \leq \frac{1}{\bar\alpha r}\Big(8\varepsilon + 8N\cdot\tfrac43\varepsilon_1\Big).$$

By (2.1) and $N < \kappa/8$: $N\varepsilon_1 \leq \frac{3N}{4\kappa}\varepsilon < \frac{3}{32}\varepsilon$. Hence $\frac1e(6 + \frac{16}{3}\cdot\frac{3}{32}) = \frac{6.5}{e} < 2.4 \leq 3$ and $8 + \frac{32}{3}\cdot\frac{3}{32} = 9$ (strict inequality, since $N < \kappa/8$ strictly). Both bounds are $\leq$ the claimed ones. □

Remark 3.5 (Where the constants come from). Smallness: (A2) is the one-scale hypothesis (i) of [N:bpN:p6:normal-form] with $q = 1$ ($2^7 qK\varepsilon \leq \alpha r$). The weaker requirement of CP10, $64e\,KE < \alpha_2 r$ (CP10 (23) with $m = 0$, $c_0 = e/2$, $d = rs$), implies it since $64e > 128$. The number of steps N, the losses, 4^{1-i} and $e^{-\kappa/6}$ are those of [N:bpN:p6:normal-form]; the improvements (E/κ instead of $16c\varepsilon^2$, $3E$ instead of $9E$ for the actions) come from keeping the exact expressions $6/e$ and $y_0 \leq 3/(8\kappa)$ instead of the cruder $\kappa \geq 6 \Rightarrow 1/(\bar\alpha s) \leq K/(6\alpha)$ of [N:bpN:p6:normal-form](c), which is a bound in K and is useless for the interface (the interface displacement is independent of K). The deviation of [N:bpN:p6:deviations](1) is respected: all steps satisfy $|\phi| \leq \rho\sigma/4$.

3.4. The interface of Lemma 4.8.

Corollary 3.6 (Averaging with two scales, in Part 3's notation)**.** *Let $0 < \alpha_2 \leq \alpha_1$, $D \subset \mathbb{R}^n$, $r > 0$, $\bar s, s > 0$, $K > 0$ with $Ks \geq 6$. Let h be real analytic on D_r (holomorphic on $D_r = V_rD$, real on real points) with $\omega = \partial h$, and let f be holomorphic on $D_r \times \mathbb{T}_{\bar s+s}$, real analytic, 1-periodic in θ, with the* N-norm $E := \sup_{I\in D_r}\sum_k |\hat f_k(I)|e^{(\bar s+s)|k|_1} < \infty$. *Assume (A1) (that is, (N1) of Lemma 4.8) and*

$$64e\,KE < \alpha_2 r.$$

Then there is a holomorphic, injective, real, symplectic ($\Psi^*(dI\wedge d\theta) = dI'\wedge d\theta'$) map $\Psi : D_{r/2} \times \mathbb{T}_{\bar s+s/6} \to D_r \times \mathbb{T}_{\bar s+s}$ with $\Psi - \mathrm{id}$ $\mathbb{Z}^n$-periodic in θ', and $g = g(I')$, f_* with $H\circ\Psi = h + g + f_*$ on

$D_{r/2} \times \mathbb{T}_{\bar{s}+s/6}$ and, in the N-norm with weight $e^{(\bar{s}+s/6)|k|_1}$ on $D_{r/2}$,

$$\sup_{D_{r/2}} |g - f_0| \le E/4, \qquad \|f_*\|_{r/2,\bar{s}+s/6} \le e^{-Ks/6}E,$$
$$\alpha_1 s|I_1 - I_1'| \le 3E \ (\le 9E), \quad \alpha_2 s|I_2 - I_2'| \le 3E \ (\le 9E), \quad 2\pi\alpha_2 r|\theta - \theta'| \le 9E.$$

Source: CP10 Prop. 1 (p. 552) with $m = 0$, $\Lambda = \{0\}$; proof: Proposition 3.4.

Proof. **Step S3.10: Dictionary.** Put $\bar{s}_N = \bar{s}/2\pi$, $s_N = s/2\pi$ (geometric). Then $\mathbb{T}_{\bar{s}+s} = T_{\bar{s}_N+s_N}$, the weight $e^{(\bar{s}+s)|k|_1} = e^{2\pi|k|_1(\bar{s}_N+s_N)}$, so $E = |f|_{r,\bar{s}_N+s_N}$ in the notation of Proposition 3.4; $\kappa = 2\pi K s_N = Ks \ge 6$; $\bar{\alpha}_j s_N = \alpha_j s$ and $\bar{\alpha}_2 r = 2\pi\alpha_2 r$.

Step S3.11: Hypotheses. $64e > 128$, so $64eKE < \alpha_2 r$ gives $E \le \alpha_2 r/(128K)$. The remaining hypotheses are identical. Proposition 3.4(a)–(d) are the claims; $e^{-\kappa/6} = e^{-Ks/6}$. □

Remark 3.7 (What Part 3 must change (the interface report)). Lemma 4.8 is *provable as stated, with the single change of norm* (made in revised Part 3, which now uses N's norm; the list below records what the consumer relies on): the former Part 3 norm $\|\cdot\|_{r,\sigma} = \sum_k \sup_I |f_k| e^{\sigma|k|}$ by N's $\sup_I \sum_k |\hat{f}_k(I)| e^{\sigma|k|_1}$ (this is Corollary 3.6). Precisely:

(1) Every occurrence of the norm (done in the revision; hypothesis $E \ge \|f\|_{\rho,\bar{s}+s}$ of [P:`bpP:p3:two-scale-kam`], E_j in the iteration, outputs $\|f_*\|$) is read in N's norm. Used properties: monotone under restriction of r, σ and of the base set; $\sup|f| \le \|f\|$ ([N:`bpN:p4:sup-weighted`](a)); Cauchy estimates for g use only sup. Nothing in Part 3 uses the finer $\sum_k \sup_I$ structure, except the printed definition.

(2) The numerical constants: hypothesis (N2) $64eKE < \alpha_2 r$ and the conclusions $E/4$, $e^{-Ks/6}E$, $\alpha_2 r 2\pi|\Delta\theta| \le 9E$ are unchanged; $\alpha_i s|\Delta I_i| \le 9E$ may be replaced by $3E$ (so the constant $\frac{3}{4}$ of `st-avg` may become $\frac{1}{4}$); no other constant of Part 3 changes.

(3) Hypothesis (N1) is used on the full complex neighbourhood D_r, exactly as in Lemma 4.8; *no Lipschitz hypothesis is added* (Lemma 3.2). Reality of h and f on real points is used, as in Part 3's setting.

(4) The one-flow proof plan of Lemma 4.8 is replaced by Proposition 3.4 (a finite iteration; done in the revision); the Fourier-tail remark is not the source of the exponential.

3.5. **Averaging with an oscillator pair (CP10 §4, Step 1).** CP10 §4, Step 1 (p. 563) averages over the n_1 fast angles while the oscillator variables $(p, q) \in \mathbb{C}^{n_2} \times \mathbb{C}^{n_2}$ are dynamical: the transformation moves them, symplectically for $dI \wedge d\theta + dp \wedge dq$. Track N's Parts 4–6 are written for (θ, I) only. The extension is mechanical but must be recorded; we do it with its own constants.
Setting. $n, m \ge 1$, $D \subset \mathbb{R}^n$, $v = (r, r_p, r_q)$, $s > 0$, $\bar{s} \ge 0$. $\mathcal{D}_{v,S} := T_S \times V_r D \times B_{r_p} \times B_{r_q} \ni (\theta, I, p, q)$ with $B_a = \{x \in \mathbb{C}^m : |x|_\infty < a\}$ (the complex neighbourhood of $\{0\}$ in the max norm; the Euclidean ball of CP10 contains it up to $\sqrt{m}$ in radius), $\mathcal{O}(v, S)$ and $|\cdot|_{v,S}$ as in [N:`bpN:p4:norm`] with the sup over (I, p, q) (I, p, q together play the rôle of N's parameter I; only θ is Fourier expanded). The field of u is $X_u = (\partial_I u, -\partial_\theta u;\ \partial_p u, -\partial_q u)$ in the order $(\dot{\theta}, \dot{I}; \dot{q}, \dot{p})$ and $\{u, w\} = \sum_i (\partial_{\theta_i} u\, \partial_{I_i} w - \partial_{I_i} u\, \partial_{\theta_i} w) + \sum_j (\partial_{q_j} u\, \partial_{p_j} w - \partial_{p_j} u\, \partial_{q_j} w) = Du \cdot X_w$. Put $d := \min\{rs, r_p r_q\}$ and $b := 1 + m$. For losses $\nu = (\rho, \rho_p, \rho_q, \sigma)$ put $\delta_\nu := \min\{\rho\sigma, \rho_p\rho_q\}$ and the scaled norm $\nu(a, b_I, c_q, c_p) = \max(|a|/\sigma, |b_I|/\rho, |c_q|/\rho_q, |c_p|/\rho_p)$, whose balls are the polydiscs of the four domain losses.

Lemma 3.8 (Extended bracket and field estimates)**.** *Let $u \in \mathcal{O}(v_0, S_0)$, $w \in \mathcal{O}(v, S)$ with finite norms, $0 < \rho < r$, $0 < \rho_p < r_p$, $0 < \rho_q < r_q$, $0 < \sigma < S$, $v - \nu < v_0$ componentwise, $S - \sigma < S_0$. Then $\{u, w\} \in \mathcal{O}(v - \nu, S - \sigma)$ and*

$$|\{u, w\}|_{v-\nu,S-\sigma} \le \frac{W}{\delta_\nu} |u|_{v_0,S_0} |w|_{v,S}, \quad W = \tfrac{1}{e}\Big(\tfrac{\rho}{r_0 - r + \rho} + \tfrac{\sigma}{S_0 - S + \sigma}\Big) + m\Big(\tfrac{\rho_q}{r_{q0} - r_q + \rho_q} + \tfrac{\rho_p}{r_{p0} - r_p + \rho_p}\Big) \le 2b\,R,$$

where R is the largest of the four ratios. Moreover for $\phi \in \mathcal{O}(v, S)$: on $\mathcal{D}_{v-\nu,S-\sigma}$, $\nu(X_\phi) \le |\phi|_{v,S}/\delta_\nu$.

Proof. **Step S3.12: Derivative bounds.** Write $x = (I, p, q)$ for the parameter. [N:bpN:p4:cauchy](a) gives, at every fixed x, $\sum_i |\partial_{\theta_i} u|_{x,S} \le |u|_{x,S_0}/(e(S_0 - S))$, and individually $|\partial_{\theta_i} u|_{,S} \le |u|_{/,(}e(S_0 - S))$ globally; [N:bpN:p4:cauchy](b) is a one-variable Cauchy integral in one coordinate of x and holds for each coordinate with its own loss: $|\partial_{I_i} u|_{,} \le |u|_{/,(}r_0 - r + \rho)$, $|\partial_{p_j} u|_{,} \le |u|_{/,(}r_{p0} - r_p + \rho_p)$, $|\partial_{q_j} u|_{,} \le |u|_{/,(}r_{q0} - r_q + \rho_q)$ (global norms, each coordinate separately), and likewise for w with losses ρ, ρ_p, ρ_q.

Step S3.13: Products. Fix x. By [N:bpN:p4:algebra] at this x and the triangle inequality, $|\{u, w\}|_{x,S-\sigma} \le \big(\sum_i |\partial_{\theta_i} u|_{x,}\big) \max_i |\partial_{I_i} w|_{,} + \max_i |\partial_{I_i} u|_{,}\big(\sum_i |\partial_{\theta_i} w|_{x,}\big) + \sum_{j \le m} \big(|\partial_{q_j} u|_{,}|\partial_{p_j} w|_{,} + |\partial_{p_j} u|_{,}|\partial_{q_j} w|_{,}\big)$, and the last sum is at most m times the product of the maxima (this is the factor m). Insert the bounds of the previous step and take the supremum over x last. The (θ, I) part is $\frac{1}{e}\big(\frac{1}{(r_0 - r + \rho)\sigma} + \frac{1}{(S_0 - S + \sigma)\rho}\big)$, the (q, p) part $m\big(\frac{1}{(r_{q0} - r_q + \rho_q)\rho_p} + \frac{1}{(r_{p0} - r_p + \rho_p)\rho_q}\big)$. Since $\frac{1}{(r_0 - r + \rho)\sigma} = \frac{1}{\rho\sigma} \cdot \frac{\rho}{r_0 - r + \rho}$ etc. and $\rho\sigma, \rho_p\rho_q \ge \delta_\nu$, the bound follows; $\frac{1}{e} + \frac{1}{e} \le 2$ gives $W \le 2bR$.

Step S3.14: Field. As in [N:bpN:p5:vf-bound]: $|\partial_I \phi|/\sigma \le |\phi|_{,}/(\rho\sigma)$, $|\partial_\theta \phi|/\rho \le |\phi|_{,}/(e\sigma\rho)$, $|\partial_p \phi|/\rho_q \le |\phi|_{,}/(\rho_p\rho_q)$, $|\partial_q \phi|/\rho_p \le |\phi|_{,}/(\rho_q\rho_p)$. □

Lemma 3.9 (Iterative Lemma with an oscillator pair)**.** [N:bpN:p6:iterative] *(in direct form, Lemma 3.2, $\Lambda = \{0\}$, $q = 1$) holds on $\mathcal{D}_{v,S}$ with all losses ν, with $\rho\sigma$ replaced by δ_ν, the factor $\frac{2}{e}$ in the bracket estimate replaced by $2b$ (so $|g|_*$ is replaced by $|g|_R := \sum_j R_j |g_j|_{,}$), and $\Phi = X^1_\phi$ symplectic for $d\theta \wedge dI + dq \wedge dp$:*

(a) if $|f|_{v,S} = \delta \le \bar\alpha\delta_\nu/(16b)$ then $|\Phi_\theta - \theta| \le \delta/(\bar\alpha\rho)$, $|\Phi_I - I| \le \delta/(e\bar\alpha\sigma)$, $|\Phi_q - q| \le \delta/(\bar\alpha\rho_p)$, $|\Phi_p - p| \le \delta/(\bar\alpha\rho_q)$;

(b) $|f_+|_{v-2\nu,S-2\sigma} \le (1 - 2eb\,\delta/(\bar\alpha\delta_\nu))^{-1}\big(\frac{2b}{\bar\alpha\delta_\nu}(|g|_R + \delta) + e^{-2\pi K\sigma}\big)\delta$.

Proof. **Step S3.15: Transfer.** [N:bpN:p6:homological] involves θ and the parameters only through coefficients and is unchanged ($|\phi|_{,} \le \delta/\bar\alpha$). [N:bpN:p5:flow] is for an arbitrary complex norm and open set; [N:bpN:p5:symplectic] uses only $DF = JS$ with S symmetric and $J^\top = -J$, true for the enlarged J; Corollary [N:bpN:p5:canonical] follows. Lemma [N:bpN:p5:adjoint], Theorem [N:bpN:p5:composition], Propositions [N:bpN:p5:transform], [N:bpN:p5:canonical-map] use the geometry only through the bracket bound and the field bound: replace [N:bpN:p4:bracket] and [N:bpN:p5:vf-bound] by Lemma 3.8, and $\rho\sigma$ by δ_ν.

Step S3.16: The adjoint ratio. N's iterated-bracket estimate is $|\mathrm{ad}^h_\phi u|_{,}/h! \le (e\eta/4)^h |u|_{,}$, $\eta = 2|\phi|_{,}/(\rho\sigma)$, obtained by h bracket estimates with losses divided by h, each costing the bracket constant $2/(e\rho'\sigma')$. With the constant $2b/\delta'$ each bracket costs a factor eb more, so the ratio becomes $eb \cdot e\eta/4 = e^2 b\eta/4 \le eb\eta$ (as $e/4 < 1$), now with $\eta := 2|\phi|_{,}/\delta_\nu$ (we use this cruder bound; $e^2/4 > 1$, so the ratio $b\eta$ is *not* claimed). Put $\theta_\phi := eb\eta = 2eb|\phi|_{,}/\delta_\nu$. Then the Lie series converges with $|\mathrm{ad}^h_\phi u|_{,}/h! \le \theta^h_\phi |u|_{,}$ and $|u \circ X^1_\phi|_{,} \le (1 - \theta_\phi)^{-1}|u|_{,}$, and the same for the integral remainders ([N:bpN:p5:transform], [N:bpN:p5:composition-integral]).

Step S3.17: Smallness range. The equality of the Lie series with the composition needs the ratio, evaluated at the shrunken losses $(1 - \varepsilon_\phi)\nu$ with $\varepsilon_\phi = |\phi|_{,}/\delta_\nu \le \frac{1}{4}$ ([N:bpN:p5:composition], remainder step: factor $\le \frac{16}{9}$), to be < 1. We *assume* $\delta \le \bar\alpha\delta_\nu/(16b)$ (this replaces $u \le \frac{1}{4}$). Then $\theta_\phi \le 2eb\delta/(\bar\alpha\delta_\nu) \le e/8 < 0.34$ and $\frac{16}{9}\theta_\phi < 0.61 < 1$. The assumption holds at every step of Proposition 3.10 (there $\delta/(\bar\alpha\delta_\nu) \le y_0 \le 1/(64b\kappa)$ or is smaller by (2.3)). Then (a) and (b) follow as in [N:bpN:p6:iterative] with $\frac{2}{e} \mapsto 2b$ and $1 - \frac{2q\delta}{\bar\alpha\rho\sigma} \mapsto 1 - \theta_\phi$, $\theta_\phi \le 2eb\delta/(\bar\alpha\delta_\nu)$.

Step S3.18: Displacement. As in Step S3.3: components of X_ϕ and Lemma 3.8: $|\partial_I \phi| \le |\phi|_{,}/\rho$, $|\partial_\theta \phi| \le |\phi|_{,}/(e\sigma)$, $|\partial_p \phi| \le |\phi|_{,}/\rho_p$, $|\partial_q \phi| \le |\phi|_{,}/\rho_q$. □

Source gap: Lemma 3.9 extends N Parts 5–6 by an oscillator pair; the extension is stated by its (short) list of changes, not rewritten in N. It is the oscillator part of CP10 Lemma A.1 / [4, Lemma A.5]. If track N prefers, it can be absorbed into N Part 5 ($\{\cdot,\cdot\}$ on $\mathbb{C}^n\times\mathbb{C}^n\times\mathbb{C}^m\times\mathbb{C}^m$). Constants differ from CP10's $c_m = e(1+em)/2$.

Proposition 3.10 (Averaging with oscillators, one scale)**.** *In the setting above, let h be holomorphic on V_rD, real on real points, $\omega=\partial h$ (h independent of p,q), $\alpha>0$, $K>0$, $\kappa=2\pi Ks\ge 6$,*

(O1) $|k\cdot\omega(I)|\ge\alpha$ for $I\in V_rD$, $0<|k|_1\le K$;

(O2) $f\in\mathcal{O}(v,\bar s+s)$ real on real points, $E:=|f|_{v,\bar s+s}$ with $2^{12}\,b\,KE\,s\le\alpha d$.

Then there is $\Psi:\mathcal{D}_{v/2,\bar s+s/6}\to\mathcal{D}_{v,\bar s+s}$ holomorphic, injective, real, $\Psi-\mathrm{id}$ periodic in θ, symplectic for $d\theta\wedge dI+dq\wedge dp$, with $H\circ\Psi=h+g+f_*$ on $\mathcal{D}_{v/2,\bar s+s/6}$, $H=h+f$, $g=g(I,p,q)$ independent of θ, real, and with $\bar\alpha=2\pi\alpha$

$$|g-\hat f_0|_{v/2,\bar s+s/6}\le\frac{400\,b\,E^2}{\bar\alpha d},\qquad |f_*|_{v/2,\bar s+s/6}\le e^{-\kappa/6}E,$$
$$\bar\alpha s|I-I'|\le 3E,\quad \bar\alpha r|\theta-\theta'|\le 9E,\quad \bar\alpha r_q|p-p'|\le 9E,\quad \bar\alpha r_p|q-q'|\le 9E.$$

Source: CP10 Prop. 1 (p. 552), Lemma A.1 (p. 564) with $\ell_2=0$, $\Lambda=\{0\}$; (25), (26). Constants ours.

Proof. Use the notation of Proposition 3.4 ($q=1$, $\varepsilon=E$, $\bar\alpha=2\pi\alpha$); (O2) reads $\varepsilon\le\bar\alpha d/(2^{12}b\kappa)$. Step 0 losses $\rho_0=r/8$, $\rho_{p0}=r_p/8$, $\rho_{q0}=r_q/8$, $\sigma_0=s/6$; then $\delta_{\nu_0}=\min\{rs/48,r_pr_q/64\}\ge d/64$ and $y_0:=\varepsilon/(\bar\alpha\delta_{\nu_0})\le 64\varepsilon/(\bar\alpha d)\le\frac{1}{64b\kappa}\le\frac{1}{16}$.

Step S3.19: Step 0. Lemma 3.9(b) with $|g|_R=0$: $\varepsilon_1\le(1-2eby_0)^{-1}(2by_0+e^{-\kappa/6})\varepsilon$. Here $by_0\le\frac{1}{64\kappa}\le\frac{1}{384}$ (the factor $b=1+m\ge 2$ is kept; it is absorbed by the b^{-1} in the smallness), so $(1-2eby_0)^{-1}\le(1-\frac{e}{192})^{-1}<1.015\le\frac{8}{7}$. *Case A:* $e^{-\kappa/6}\le 2by_0$. $\varepsilon_1\le\frac{32}{7}by_0\varepsilon\le\frac{32}{7}\cdot\frac{1}{64\kappa}\varepsilon\le 0.08\,\varepsilon/\kappa$, and $\varepsilon_1\le\frac{32}{7}\cdot 64\,b\varepsilon^2/(\bar\alpha d)\le 294\,b\varepsilon^2/(\bar\alpha d)$. *Case B:* $e^{-\kappa/6}>2by_0$. Doubled losses ($\delta_{\nu'}=4\delta_{\nu_0}$): $|f_1|,\le(1-eby_0/2)^{-1}(2b\frac{y_0}{4}+e^{-\kappa/3})\varepsilon$, with $(1-eby_0/2)^{-1}\le(1-\frac{e}{768})^{-1}<\frac{32}{31}$ (this is the denominator $1-2eb\,y_0/4$ with the doubled losses), and $\frac{by_0}{2}<\frac14 e^{-\kappa/6}$, so the bound is $\le\frac{32}{31}(\frac14+\frac1e)e^{-\kappa/6}\varepsilon\le e^{-\kappa/6}\varepsilon$; stop, $g=\hat f_0$, displacements $\le\varepsilon/(e\bar\alpha\sigma')<3\varepsilon/(\bar\alpha s)$, $\le 4\varepsilon/(\bar\alpha r)$, $\le 4\varepsilon/(\bar\alpha r_q),4\varepsilon/(\bar\alpha r_p)$.

Step S3.20: Iteration (Case A). $N=\lfloor\kappa/(12\ln 2)\rfloor<\kappa/8$; $N=0$: stop, since then $e^{-\kappa/6}>\frac14>\varepsilon_1/\varepsilon$. $N\ge 1$: losses $\rho=r/8N$, $\rho_p=r_p/8N$, $\rho_q=r_q/8N$, $\sigma=s/4N$, all four radii and the budget following the schedule of Step S3.6 (so every ratio R_j is $\le\frac13$ for $j\ge1$ and $\le\frac{3}{4N}$ for g_0). Then $\delta_\nu\ge d/(64N^2)$ and by $\kappa>8N$

$$\frac{\varepsilon}{N}\le\frac{\bar\alpha d}{2^{15}bN^2}\le\frac{\bar\alpha\delta_\nu}{2^9 b}\quad(\text{since } d\le 64N^2\delta_\nu),\qquad \varepsilon_1\le\frac{0.08\,\varepsilon}{8N}.\tag{2.3}$$

Then $|g_{i-1}|_R+\varepsilon_i\le 2\varepsilon_1+\varepsilon/N\le 1.02\,\varepsilon/N$ and $x:=\frac{2b}{\bar\alpha\delta_\nu}(|g_{i-1}|_R+\varepsilon_i)\le 0.004$; Moreover $u_i:=\varepsilon_i/(\bar\alpha\delta_\nu)\le 0.01/(2^9b)$ by (2.3), so $2ebu_i<1.1\cdot10^{-4}$ and $(1-2ebu_i)^{-1}<\frac{128}{127}$. Lemma 3.9(b) gives $\varepsilon_{i+1}\le\frac{128}{127}(0.004+\frac18)\varepsilon_i<\frac14\varepsilon_i$ (using $e^{-2\pi K\sigma}\le\frac18$). Hence $\varepsilon_i\le 4^{1-i}\varepsilon_1$.

Step S3.21: Conclusion. As Step S3.8: $|f_*|,\le 4^{-(N+1)}\varepsilon\le e^{-\kappa/6}\varepsilon$; $|g-\hat f_0|,\le\frac43\varepsilon_1\le\frac43\cdot 294\,b\varepsilon^2/(\bar\alpha d)\le 400\,b\varepsilon^2/(\bar\alpha d)$. Displacements by Lemma 3.9(a) and [N:bpN:p6:telescope]: I: $\frac{1}{e\bar\alpha s}(6\varepsilon+4N\frac43\varepsilon_1)\le 2.3\,\varepsilon/(\bar\alpha s)$; θ: $\frac{1}{\bar\alpha r}(8\varepsilon+\frac{32}{3}N\varepsilon_1)\le 8.2\,\varepsilon/(\bar\alpha r)$ since $N\varepsilon_1\le 0.01\varepsilon$; the same for p,q with r_q, r_p. □

Proposition 3.11 (Averaged remainder for [P:bpP:p0:pg]: CP10 §4, Step 1)**.** *Let $n=n_1\ge1$, $m=n_2\ge1$, $\bar D\subset\mathbb{R}^n$, $\bar\rho,\epsilon_0>0$, $s_0>0$ (geometric), $0<\epsilon<e^{-1/5}$, $\mu>0$, $v_0=(\bar\rho,\epsilon_0,\epsilon_0)$. Let $H=H_0(I)+\mu P(\theta,I,p,q)$, H_0 holomorphic on $V_{\bar\rho}\bar D$ and real, $P\in\mathcal{O}(v_0,s_0)$ real with $|P|_{v_0,s_0}\le1$, and $\langle P\rangle(I,p,q):=\int_{\mathbb{T}^n}P\,d\theta=\hat P_0$. Choose K by $e^{-2\pi Ks_0/6}=\epsilon^5$, i.e. $K=\frac{15}{\pi s_0}\log\frac1\epsilon$ (so $\kappa=30\log\frac1\epsilon\ge6$; CP10 (36) with 2π-conversion $K_{\mathrm{CP}}=K$, $s_{\mathrm{CP}}=2\pi s_0$). Assume, with $\bar\alpha=2\pi\alpha$, $d=\min\{\bar\rho s_0,\epsilon_0^2\}$,*

(R1) $|k\cdot\partial H_0(I)|\ge\alpha$ *for* $I\in V_{\bar\rho}\bar D$, $0<|k|_1\le K$ *(non-resonance on the complex neighbourhood, CP10 (39));*

(R2) $2^{12}(1+m)\,Ks_0\,\mu\le\alpha d$ *(CP10 (65), implied by (61)).*

Then there is a map $\bar\varphi:\mathcal{D}_{v_0/2,s_0/6}\to\mathcal{D}_{v_0,s_0}$ *as in Proposition 3.10 (with* $\bar s=0$, $s=s_0$*) such that on* $\mathcal{D}_{v_0/2,s_0/6}$

$$H\circ\bar\varphi=H_0+\mu\langle P\rangle+\widetilde P,\qquad |\widetilde P|_{v_0/2,s_0/6}\le\frac{400(1+m)\,\mu^2}{\bar\alpha d}+\mu\,\epsilon^5,$$

where $\widetilde P=(g-\mu\langle P\rangle)+f_*$, $g=g(I,p,q)$, $|f_*|,\le\mu\epsilon^5$ *(so* $f_*=\mu\epsilon^5\bar P$ *with* $|\bar P|,\le 1$*, in CP10's (40) notation), and*

$$\bar\alpha s_0|I-\bar I|\le 3\mu,\quad \bar\alpha\bar\rho|\theta-\bar\theta|\le 9\mu,\quad \bar\alpha\epsilon_0|p-\bar p|,\ \bar\alpha\epsilon_0|q-\bar q|\le 9\mu.$$

Source: CP10 §4, Step 1, p. 563 (36)–(40), (62)–(63); the estimates (41)–(42) of Step 1 of §3 (p. 557). Perturbation size: $E=\mu$ *before averaging and* $\mu\epsilon^5$ *after (CP10 p. 560:* $f=\mu\epsilon^5\breve P$, $E=C\mu\epsilon^5$*; not* μ^5*).*

Proof. **Step S3.22: Apply.** $f=\mu P$, $E=\mu$, $\bar s=0$, $s=s_0$, $r=\bar\rho$, $r_p=r_q=\epsilon_0$, $h=H_0$; (R1), (R2) are (O1), (O2). Proposition 3.10 gives $H\circ\bar\varphi=H_0+g+f_*$, and $\hat f_0=\mu\langle P\rangle$.

Step S3.23: Collect. $\widetilde P:=(g-\mu\langle P\rangle)+f_*$; its norm is bounded by the sum of the two bounds of Proposition 3.10, and $e^{-\kappa/6}\mu=\epsilon^5\mu$ by the choice of K. □

Remark 3.12 (How Part 5 uses this). (1) The two terms of CP10 (63) correspond to $\mu^2/(\bar\alpha d)$ (the average-correction) and $\mu\epsilon^5$ (the tail). Part 5 must verify $\mu/(\bar\alpha d)\le C\epsilon^5$ from its choice of $\bar\gamma$ (CP10 (61), with $\mu<\epsilon^6/(\log\frac{1}{\epsilon})^c$ of [P:bpP:p0:pg]); this is a different inequality from CP10's (since the proof here has no $K^{2\tau+1}/\bar\gamma^2$), and it is implied by (61) together with $\mu\le\epsilon^5$. (2) Distinguish three radii: the physical oscillator radius ϵ (only through K and the later domain), the ambient radius ϵ_0 of P, and Part 3's enlargement radius $\hat\rho$. (3) *Period dictionary with Cartesian oscillators remaining.* The pure all-angle conversion of Lemma 4.5 ($\tilde H\mapsto\tilde H(I,2\pi\theta)/2\pi$ with I fixed) must not be applied to (ϕ,I,p,q): it pulls the form back to $2\pi\,dI\wedge d\theta+dp\wedge dq$, which is not a single conformal multiple of the standard form, and dividing the Hamiltonian by 2π would also rescale the oscillator speeds. Use instead the *canonical* chart (a c-chart of [P:bpP:p1:period-convert] with $c=1$, acting on the angle–action block only)

$$(\theta,I,p,q)=\big(\phi_{\rm CP}/2\pi,\ 2\pi I_{\rm CP},\ p_{\rm CP},\ q_{\rm CP}\big),\qquad dI\wedge d\theta=dI_{\rm CP}\wedge d\phi_{\rm CP}.$$

The Hamiltonian is the same function, μ is unchanged, the frequency is $\omega=\omega_{\rm CP}/2\pi$ and $\alpha=\alpha_{\rm CP}/2\pi$; $\bar\rho=2\pi\bar\rho_{\rm CP}$, $s_0=s_{\rm CP}/2\pi$, ϵ_0 unchanged. The Fourier coefficients and the norm are unchanged (weight $e^{|k|_1 s_{\rm CP}}$). Invariants: $\bar\alpha=2\pi\alpha=\alpha_{\rm CP}$, $\bar\alpha s_0=\alpha_{\rm CP}s_{\rm CP}/2\pi$, $d=\min\{\bar\rho s_0,\epsilon_0^2\}=\min\{\bar\rho_{\rm CP}s_{\rm CP},\epsilon_0^2\}$, $Ks_0/\alpha=Ks_{\rm CP}/\alpha_{\rm CP}$, so (R2) is the same inequality in both conventions; and the displacement bounds become exactly CP10's shape: $\alpha_{\rm CP}s_{\rm CP}|\Delta I_{\rm CP}|\le 6\mu$ (from $\bar\alpha s_0|\Delta I|\le 3\mu$, $\Delta I=2\pi\Delta I_{\rm CP}$), $\alpha_{\rm CP}\bar\rho_{\rm CP}|\Delta\phi|\le 9\mu$, $\alpha_{\rm CP}\epsilon_0|\Delta p|,\alpha_{\rm CP}\epsilon_0|\Delta q|\le 9\mu$. After the oscillators have been put in action–angle form (Part 5), all angles have period 2π again and Lemma 4.5 applies to the whole system. The max-norm polydisc B_{ϵ_0} replaces CP10's Euclidean ball.

3.6. Suspected errata in CP10 and open points.

(1) CP10 p. 564, Lemma A.1, (66): the target is printed as $W_{v,\sigma}$; it must be $W_{v,s}$ (the map lands in the original domain).

(2) CP10 p. 563, Step 1 (rendered page checked): the text prints $r_p=r_q=\epsilon_0$, as on p. 556; an earlier version of this part misquoted "$r_p=r_q=0$" from the text extraction (the subscript is lost) and that claim is *withdrawn*. The genuine inconsistency on that page is the action loss: it takes "$\rho=\bar\rho$" and "$r=\bar\rho$" while Lemma A.1 (p. 564) requires $\rho<r/2$ and the output action radius claimed is $\bar\rho/2$. The loss $\rho=\bar\rho/4$ would fit the claimed output radius; this

is our inferred repair, not a printed statement. The finite loss schedule of Proposition 3.4 (resp. 3.10) does not depend on it.

(3) CP10 p. 565 proves the exponential bound by "applying A.1 once and then proceeding as in (A21)–(A26) of [4]": the iteration is not in the pinned sources; it is Proposition 3.4, Steps S3.6–S3.8.

(4) The original one-flow proof plan of Lemma 4.8 did not give $e^{-Ks/6}E$ (review P3-01); the finite iteration above replaces it.

Forward and cross-track references used: [N:bpN:p4:norm], [N:bpN:p4:domains], [N:bpN:p4:cauchy], [N:bpN:p4:bracket], [N:bpN:p4:projections], [N:bpN:p4:sup-weighted], [N:bpN:p4:coefficients], [N:bpN:p4:algebra], [N:bpN:p4:dictionary], [N:bpN:p5:setting], [N:bpN:p5:vf-bound], [N:bpN:p5:flow], [N:bpN:p5:symplectic], [N:bpN:p5:canonical], [N:bpN:p5:adjoint], [N:bpN:p5:composition], [N:bpN:p5:transform], [N:bpN:p5:canonical-map], [N:bpN:p6:setting], [N:bpN:p6:homological], [N:bpN:p6:divisor], [N:bpN:p6:iterative], [N:bpN:p6:iterative-special], [N:bpN:p6:telescope], [N:bpN:p6:normal-form], [N:bpN:p6:deviations], [K:bp:p1:ext-mvt], [P:bpP:p0:pg].

4. Two-scale KAM theorem I: statement and the iterative lemma

This part (3a) states the two-scale KAM theorem of Chierchia–Pinzari (CP10 Proposition 3, pp. 554–555) in the frozen conventions and proves the iterative lemma (CP10 Lemma B.1 with Lemma B.2, pp. 565–570). Part 3b (limit, Lipschitz bounds, measure) completes the proof. *Every constant is recomputed here*: the numerical constants differ from those of CP10 (max norms, track N's weighted norm, no use of the unpinned reference [4]), the shape of $K, \hat{\rho}, L, \hat{E}$ and of the KAM condition is the same. Results of track K are cited as [K:label]; the averaging step is proved in Part 2 ([P:bpP:p2:avg], [P:bpP:p2:nfb]), which rests on track N, Parts 4–6.

4.1. Conventions, norms, constants.

- *Period one.* $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ with coordinates θ; Fourier characters $e^{2\pi i k\cdot\theta}$, $k \in \mathbb{Z}^n$, $|k|_1 = \sum_i |k_i|$. The symplectic form is $dJ \wedge d\theta$, the Hamilton equations are $\dot\theta = \partial_J H$, $\dot J = -\partial_\theta H$ (this is the order (action, angle) used by CP10; the frozen order $(z_1, z_2) =$ (position, momentum) is restored in Part 5). A frequency $\omega \in \mathbb{R}^n$ is a rotation number per unit time. Part 0 keeps period 2π for the canonical fast angles of P-G; the passage is the one-line dictionary of Lemma 4.5, under which *all constants of this part are invariant.*
- *Norms.* $|\cdot|$ is the max norm on $\mathbb{R}^n, \mathbb{C}^n$; $\|A\| = \max_i \sum_j |A_{ij}|$ the induced operator norm (also for rectangular matrices). Hence for a bilinear form entries and operator norm are comparable ($|A_{ij}| \le \|A\|$).
- *Complex neighbourhoods.* For $D \subset \mathbb{R}^n$ and $r > 0$: $D_r = \{I \in \mathbb{C}^n : \exists I_0 \in D,\ |I - I_0| < r\}$ and $\mathrm{Re}(D_r) = D_r \cap \mathbb{R}^n = \{x \in \mathbb{R}^n : \mathrm{dist}_\infty(x, D) < r\}$; $(D_r)_{r'} \subset D_{r+r'}$ and $D_r \subset D_{r'}$ for $r \le r'$. For $\sigma > 0$, $\mathbb{T}_\sigma = \{\theta \in \mathbb{C}^n/\mathbb{Z}^n : |\mathrm{Im}\,\theta| < \sigma/2\pi\}$ (weight σ, strip width $\sigma/2\pi$).
- *Weighted norm (track N's norm, ruling P3-03).* For $f = \sum_k \hat f_k(I) e^{2\pi i k\cdot\theta}$ analytic on $D_r \times \mathbb{T}_\sigma$: $\|f\|_{r,\sigma} := \sup_{I\in D_r} \sum_{k\in\mathbb{Z}^n} |\hat f_k(I)|\, e^{\sigma|k|_1}$ ([N:bpN:p4:norm]; `docs/fourier-conventions.md`). *Width dictionary:* σ here is the coefficient radius, $\sigma = 2\pi\sigma_N$ with σ_N the geometric half-width ($\mathbb{T}_\sigma = T_{\sigma/2\pi}$ in the notation of Part 2). The printed CP10 norm is $\sum_k \sup_I$, which dominates this one; the two are not equivalent. Properties used in Part 3: monotone ($\|f\|_{r',\sigma'} \le \|f\|_{r,\sigma}$ for smaller base set, $r' \le r$, $\sigma' \le \sigma$), $\sup_{D_r\times\mathbb{T}_\sigma}|f| \le \|f\|_{r,\sigma}$, $\sup_{D_r}|f_0| \le \|f\|_{r,\sigma}$, and Cauchy estimates, applied to sup only; the averaging output (Corollary 3.6) is stated in this norm. Every constant below was rechecked for this change (none used $\sum_k \sup_I$; Part 2, Remark 3.7).
- *Two-scale Diophantine set.* For $n = n_1 + n_2$, $k = (k_1, k_2) \in \mathbb{Z}^{n_1} \times \mathbb{Z}^{n_2}$, $\gamma_1 \ge \gamma_2 > 0$, $\tau > 0$: $\mathcal{D}_{\gamma_1,\gamma_2,\tau} = \{\omega \in \mathbb{R}^n :\ |k\cdot\omega| \ge \gamma_1 |k|_1^{-\tau}$ if $k_1 \ne 0$, $\ |k\cdot\omega| \ge \gamma_2 |k|_1^{-\tau}$ if $k_1 = 0, k \ne 0\}$. For

$\gamma_1=\gamma_2$ this is the set of Definition 1.1; $\mathcal{D}_{\gamma_1,\gamma_2,\tau}\subset\mathcal{D}_{\gamma_2,\gamma_2,\tau}$. (CP10 (19) with $\bar\gamma_2\mu=\gamma_2$; for $k_1=0$ CP10 writes $|k_2|^\tau$, the same as $|k|_1^\tau$.) For $D\subset\mathbb{R}^n$ and $\omega_0: D_\rho\to\mathbb{C}^n$ put $D^{\mathrm{dio}}:=\omega_0^{-1}(\mathcal{D}_{\gamma_1,\gamma_2,\tau})\cap D$ (this is $\mathcal{D}_{\gamma_1,\gamma_2,\tau_*}$ of CP10, "the ω_0-preimage in D").

- *Blocks.* $U=\partial^2 h$ is the $n\times n$ Hessian; $\hat U$ is the $n_2\times n$ matrix of its last n_2 rows (the derivatives of $\omega_2=\partial_{I_2}h$ with respect to all actions); T_1, T_2 are the first n_1 and last n_2 rows of U^{-1}.

Remark 4.1 (Misprint in the definition of $\hat U$). CP10 p. 554 defines $\hat U$ as "the $n\times n_1$ submatrix of U, i.e. the matrix with entries $\hat U_{ij}=U_{ij}$, for $n_1+1\le i\le n$, $1\le j\le n$". The size and the index ranges disagree. Only the reading above ($n_2\times n$, rows $n_1+1,\dots,n$, all columns) is used in the proof: $\hat M$ bounds the variation of ω_2 (CP10 (82), via the choice of $\hat\rho$). We adopt it.

Definition 4.2 (Constants of the two-scale theorem)**.** Let $n=n_1+n_2$, $\tau>0$, $\gamma_1\ge\gamma_2>0$, $0<s$, $\rho>0$, and $E,M,\hat M,\bar M>0$. Put

$$L:=\max\{\bar M,\ M^{-1},\ \hat M^{-1}\},\qquad x:=\frac{EM^2L}{\gamma_1^2},\qquad K:=\frac{6}{s}\log_+\!\big(x^{-1}\big),\quad \log_+ a:=\max\{1,\log a\},$$

$$\hat\rho:=\min\Big\{\frac{\gamma_1}{3MK^{\tau+1}},\ \frac{\gamma_2}{3\hat MK^{\tau+1}},\ \rho\Big\},\qquad \hat E:=\frac{EL}{\hat\rho^{\,2}},\qquad \hat c:=\max\Big\{2^{13}n,\ \frac{24^{\tau+1}}{3}\Big\}.$$

The *KAM condition* is $\hat c\hat E<1$. (CP10 (31): $\hat c=\max\{2^8n,24^{\tau+1}/6\}$; the larger constants come from the max-norm Cauchy estimates and the Lipschitz bookkeeping of Parts 3a–3b. Only $\hat c=\hat c(n,\tau)$, an absolute function of n and τ, matters downstream.) *Totality.* $E>0$ guards $\log(1/x)$, and $\log_+\ge1$ gives $Ks\ge6$; $K,\hat\rho>0$.

Source: CP10 Prop. 3, p. 555, equations (31), (32), with the changes stated.

4.2. The theorem.

Proposition 4.3 (Two-scale KAM theorem)**.** ***Data.*** *Integers $n_1,n_2\ge1$, $n=n_1+n_2$; $\tau>n$; $\gamma_1\ge\gamma_2>0$; $s,\bar s$ with $0<4s\le\bar s<1$; $\rho>0$; an arbitrary set $D\subset\mathbb{R}^n$; $A:=D_\rho$. A function $H(J,\theta)=h(J)+f(J,\theta)$,* real analytic *on $A\times\mathbb{T}_{\bar s+s}$, 1-periodic in θ (real on $\mathbb{R}^n\times\mathbb{R}^n$). Let $\omega_0:=\partial h$; assume ω_0 is an injective holomorphic map $A\to\mathbb{C}^n$ with $U=\partial^2h$ nonsingular on A (hence a diffeomorphism of A onto an open set). Constants $M,\hat M,\bar M,\bar M_1,\bar M_2,E>0$ with*

$$M\ge\sup_A\|U\|,\quad \hat M\ge\sup_A\|\hat U\|,\quad \bar M\ge\sup_A\|U^{-1}\|,\quad \bar M_i\ge\sup_A\|T_i\|\ (i=1,2),\quad E\ge\|f\|_{\rho,\bar s+s},$$

and (normalizations, always satisfiable by lowering the constants; see Remark 4.4) $\hat M\le M$, $\bar M_i\le\bar M$. Let $L,K,\hat\rho,\hat E,\hat c$ be as in Definition 4.2 and assume the ***KAM condition*** *$\hat c\hat E<1$.*

Conclusions. *Put $r:=32\,\hat E\hat\rho$, $\Omega_*:=\omega_0(D)\cap\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ and $D^{\mathrm{dio}}:=\omega_0^{-1}(\Omega_*)\cap D$. There are maps φ_ω, $\omega\in\Omega_*$,*

$$\varphi_\omega:\mathbb{R}^n\to\mathrm{Re}(D_r)\times\mathbb{R}^n,\quad \varphi_\omega(\vartheta)=\big(v(\vartheta;\omega),\ \vartheta+u(\vartheta;\omega)\big),$$

with v,u 1-periodic in ϑ, such that for every $\omega\in\Omega_*$:

(a) (analyticity) φ_ω is real analytic and extends holomorphically to $\{|\mathrm{Im}\,\vartheta|<\bar s/2\pi\}$, with $|u|\le\frac{3}{4\pi}\hat Es$ there;

(b) (conjugacy) $\frac{d}{dt}\varphi_\omega(\vartheta+t\omega)=X_H(\varphi_\omega(\vartheta+t\omega))$ for all $\vartheta\in\mathbb{R}^n,t\in\mathbb{R}$, where $X_H=(-\partial_\theta H,\partial_J H)$ (Hamilton equations for (J,θ));

(c) (embedded Lagrangian torus) $D_\vartheta\varphi_\omega$ is injective (indeed $\|D_\vartheta u\|\le\frac34 n\hat E$) and $\sum_i\big(\partial_{\vartheta_a}v_i\,\partial_{\vartheta_b}(\vartheta_i+u_i)-\partial_{\vartheta_b}v_i\,\partial_{\vartheta_a}(\vartheta_i+u_i)\big)=0$ for all a,b;

(d) (estimates) with $I^0(\omega):=\omega_0^{-1}(\omega)\in D^{\mathrm{dio}}$ and $v=(v_1,v_2)\in\mathbb{R}^{n_1}\times\mathbb{R}^{n_2}$,

$$|v_1-I_1^0(\omega)|\le20\Big(\frac{\bar M_1}{\bar M}+\frac{\hat M}{M}\Big)\hat E\hat\rho,\quad |v_2-I_2^0(\omega)|\le20\Big(\frac{\bar M_2}{\bar M}+1\Big)\hat E\hat\rho,\quad |u|\le\frac{3}{4\pi}\hat Es\quad (\text{on }\mathbb{R}^n).$$

Moreover the map $(\vartheta,\omega)\mapsto\varphi_\omega(\vartheta)$, $\mathbb{R}^n\times\Omega_*\to\mathbb{R}^{2n}$, *is injective modulo the translations* $\vartheta\mapsto\vartheta+k$, $k\in\mathbb{Z}^n$ *(precisely:* $\varphi_\omega(\vartheta)-\varphi_{\omega'}(\vartheta')\in\{0\}\times\mathbb{Z}^n$ *implies* $\omega=\omega'$ *and* $\vartheta-\vartheta'\in\mathbb{Z}^n$*), and is Lipschitz in the following sense: it factors as* $\Phi\circ(\check{\imath}\circ\omega_0^{-1},\mathrm{id})$ *with* $\check{\imath}$ *of Lipschitz constant* $\le 1+2560n\hat{E}$ *and* Φ *bi-Lipschitz in the rescaled norm of §5.1;* ω_0^{-1} *is* $\bar{M}$*-Lipschitz on* Ω_* *for pairs at distance* $<\rho/\bar{M}$ *(globally* $\bar{M}\max\{1,\operatorname{diam}D/\rho\}$*-Lipschitz if* D *is bounded). Finally, with* $K_\Omega:=\bigcup_{\omega\in\Omega_*}\varphi_\omega(\mathbb{R}^n)$ *(a union of tori),* $c_n:=1+(1+2^n)(1+(4/3)^n)$ *and* meas *the Lebesgue measure on* $\mathbb{R}^n\times\mathbb{T}^n$*,*

$$\operatorname{meas}\big((\mathrm{Re}(D_r)\times\mathbb{T}^n)\setminus K_\Omega\big)\le c_n\Big(\operatorname{meas}\big((D\setminus D^{\mathrm{dio}})\times\mathbb{T}^n\big)+\operatorname{meas}\big((\mathrm{Re}(D_r)\setminus D)\times\mathbb{T}^n\big)\Big). \tag{3.1}$$

Constant dependence. $\hat{c},c_n$ *depend only on* n,τ*;* $K,\hat{\rho},\hat{E},L$ *depend only on* $(n_1,n_2,\tau,\gamma_1,\gamma_2,s,\rho,E,M,\hat{M},\bar{M})$*, in the order of Definition 4.2. The sets* D,h,f *enter only through those constants and the hypotheses;* $\bar{s}$ *enters only through* $4s\le\bar{s}<1$ *and through the width of analyticity in (a).*
Not frozen itself: Part 5 consumes this statement (§5.6) to prove the frozen theorem [P:bpP:p0:pg].
Source: CP10 Prop. 3, pp. 554–555, proved in Appendix B, pp. 565–573. Constants recomputed; (35) of CP10 is corrected in (d).
Source gap: Prop. 3 has no other source ("not available in literature", CP10 p. 550); its printed proof relies on Prop. 1/Lemma A.1 through the unpinned [4, Lemma A.5]; that input is supplied by Part 2 (Lemma 4.8 is proved there). The scheme (Lemma B.1) is proved inductively on CP10 pp. 567–570; the pointer to Arnold [3, p. 144] in Remark 3 is historical, not a dependency. Misprints and repairs: §5.7.

Remark 4.4 (Normalizations $\hat{M}\le M$, $\bar{M}_i\le\bar{M}$). $\hat{U}$ and T_i are sub-blocks of U,U^{-1}, so $\|\hat{U}\|\le\|U\|$ and $\|T_i\|\le\|U^{-1}\|$ on A: the hypotheses may be stated with $\hat{M}\le M$ and $\bar{M}_i\le\bar{M}$ *if one replaces* $\hat{M}$, $\bar{M}_i$ *by* $\min\{\hat{M},M\}$, $\min\{\bar{M}_i,\bar{M}\}$; this only decreases $\hat{M}$ (which can only increase $\hat{\rho}$, hence is not harmless for $\hat{E}$: one should lower after computing $\hat{E}$). We state the proposition with the normalization as a hypothesis, as CP10 uses it implicitly in (74), (77) and (35) ("$\le 5n\hat{E}\hat{\rho}$" needs $\bar{M}_i\le\bar{M}$). In P-G, $\hat{M}=C\mu\le M=C$ and $\bar{M}_i\le\bar{M}=C\mu^{-1}$.

Lemma 4.5 (Period dictionary)**.** *This is* [P:bpP:p1:period-convert] *with* $c=2\pi$ *(chart* $S_{2\pi}(z_1,z_2)=(2\pi z_1,z_2)$*; in Part 3's order,* $Q(I,\theta)=(I,2\pi\theta)$*). Let* $\tilde{H}(I,\phi)=\tilde{h}(I)+\tilde{f}(I,\phi)$ *be analytic on* $A\times\{|\mathrm{Im}\,\phi|<\sigma\}$, $\phi\in(\mathbb{R}/2\pi\mathbb{Z})^n$ *(CP10 convention: form* $dI\wedge d\phi$*, characters* $e^{ik\cdot\phi}$*, norm* $\sup_I\sum_k|\tilde{f}_k|e^{\sigma|k|_1}$*,* $\tilde{\omega}=\partial\tilde{h}$*; constants* $\tilde{M},\hat{\tilde{M}},\bar{\tilde{M}},\tilde{E},\tilde{\gamma}_i$ *and* $\tilde{L},\tilde{K},\hat{\tilde{\rho}},\hat{\tilde{E}}$ *computed by Definition 4.2 with* $\tilde{\gamma}_i$ *the Diophantine constants of* $\tilde{\omega}$*). Put* $H(I,\theta):=\tilde{H}(I,2\pi\theta)/2\pi$*. Then* $h=\tilde{h}/2\pi$, $f=\tilde{f}(\cdot,2\pi\theta)/2\pi$, $\omega=\tilde{\omega}/2\pi$, $\|f\|_{\rho,\sigma}=\|\tilde{f}\|_{\rho,\sigma}/2\pi$ *and, with* $M=\tilde{M}/2\pi$, $\hat{M}=\hat{\tilde{M}}/2\pi$, $\bar{M}=2\pi\bar{\tilde{M}}$, $\bar{M}_i=2\pi\bar{\tilde{M}}_i$, $E=\tilde{E}/2\pi$, $\gamma_i=\tilde{\gamma}_i/2\pi$*:* $L=2\pi\tilde{L}$ *and* $K=\tilde{K}$, $\hat{\rho}=\hat{\tilde{\rho}}$, $\hat{E}=\hat{\tilde{E}}$, $\hat{c}=\hat{\tilde{c}}$ identically. Tori. *If* $\varphi_\omega=(v,\mathrm{id}+u)$ *is a torus of Proposition 4.3 for* H *(coordinates* (I,θ)*, frequency* ω*), then* $\tilde{\varphi}(\phi):=Q\big(\varphi_\omega(\phi/2\pi)\big)=(v(\phi/2\pi),\ \phi+2\pi u(\phi/2\pi))$ *is a torus for* $\tilde{H}$ *in CP's coordinates with flow* $\phi\mapsto\phi+t\tilde{\omega}$, $\tilde{\omega}=2\pi\omega$*, and* $\tilde{u}:=2\pi u$ *satisfies* $|\tilde{u}|\le\frac{3}{2}\hat{E}s$ *(the* output angle is scaled*). The frozen parameterization is* $K=\Psi\circ(\pi\circ\varphi_\omega)$ *with the pipeline chart* $\Psi=\Phi\circ S_{2\pi}$ *of* [P:bpP:p1:cyl-pipeline] *(the* unscaled *torus* $\pi\circ\varphi_\omega$ *is the input;* Q *is only the CP-coordinate description of the same torus, never composed again), with frozen frequency* $w=\omega=\tilde{\omega}/2\pi$*:* one *division by* 2π*, already contained in the passage* $\tilde{H}\mapsto H$*;* w *is not divided again (test:* $\tilde{h}(I)=2\pi I$ *has* $\tilde{\omega}=2\pi$ *and* $\omega=w=1$*).*

Proof. **Step S4.1: Substitution.** $\tilde{f}(I,\phi)=\sum_k\tilde{f}_k(I)e^{ik\cdot\phi}$ and $\phi=2\pi\theta$ give $f(I,\theta)=\frac{1}{2\pi}\sum_k\tilde{f}_k(I)e^{2\pi ik\cdot\theta}$: same weights, coefficients divided by 2π. $U=\tilde{U}/2\pi$, $U^{-1}=2\pi\tilde{U}^{-1}$, sub-blocks likewise.

Step S4.2: Invariants. $EM^2L/\gamma_1^2=\frac{\tilde{E}}{2\pi}\frac{\tilde{M}^2}{4\pi^2}2\pi\tilde{L}\frac{4\pi^2}{\tilde{\gamma}_1^2}=\tilde{E}\tilde{M}^2\tilde{L}/\tilde{\gamma}_1^2$, hence $K=\tilde{K}$; $\gamma_1/(3MK^{\tau+1})=\tilde{\gamma}_1/(3\tilde{M}K^{\tau+1})$ (same for $\gamma_2,\hat{M}$), so $\hat{\rho}=\hat{\tilde{\rho}}$; $EL/\hat{\rho}^2=\tilde{E}\tilde{L}/\hat{\rho}^2$.

Step S4.3: Dynamics and tori. $\dot\phi = \partial_I \tilde H = 2\pi\partial_I H = 2\pi\dot\theta$, $\dot I = -\partial_\phi \tilde H = -\partial_\theta H$ ([P:`bpP:p1:period-convert`](b)); composing the conjugacy with Q gives $\tilde\varphi$, and [P:`bpP:p1:period-convert`](c) gives $w = \tilde\omega/2\pi = \omega$. □

4.3. **External facts and imported tools.**

(X1) *Holomorphic inverse function theorem.* An injective holomorphic $F : \Omega \to \mathbb{C}^n$ (Ω open) with $\det DF \neq 0$ is open, and $F^{-1} : F(\Omega) \to \Omega$ is holomorphic with $D(F^{-1}) = (DF)^{-1} \circ F^{-1}$.

(X2) *Fourier coefficients of periodic continuous functions* ([K:`bp:p1:ext-fourier`]): $\hat F_k = \int_{[0,1]^n} F(\vartheta) e^{-2\pi i k\cdot\vartheta} d\vartheta$, used in Lemma 5.4 (which replaces a several-variable Vitali–Montel theorem; review P3-11).

(X3) *Weierstrass.* Locally uniform limits of holomorphic functions are holomorphic and derivatives converge locally uniformly ([K:`bp:p1:uniform-limits`]).

(X4) Banach fixed point ([K:`bp:p1:ext-banach`]); mean value inequality on convex sets ([K:`bp:p1:ext-mvt`]); Cauchy estimates on polydiscs ([K:`bp:p1:cauchy-open`]).

(X5) Neumann series: $\|B\| \le \frac12$ implies $1 + B$ invertible with $\|(1+B)^{-1}\| \le 2$.

(X6) Lipschitz measure lemmas [K:`bp:p1:lip-image`], McShane extension [K:`bp:p1:mcshane`] (max norm).

4.4. **Two elementary lemmas.**

Lemma 4.6 (Balls in the image of a diffeomorphism)**.** *Let $\Omega \subset \mathbb{C}^n$ be open, $F : \Omega \to \mathbb{C}^n$ injective holomorphic with $\|(DF)^{-1}\| \le m$ on Ω, and $B(x_0, \delta) \subset \Omega$ (max-norm ball). Then $F(B(x_0,\delta)) \supset B(F(x_0), \delta/m)$. Moreover, if $w_1, w_2 \in B(F(x_0), \delta/m)$ then $|F^{-1}(w_1) - F^{-1}(w_2)| \le m|w_1 - w_2|$, with F^{-1} evaluated in $B(x_0,\delta)$.*

Proof. **Step S4.4: Inverse.** By (X1), $G := F^{-1} : F(\Omega) \to \Omega$ is holomorphic with $\|DG\| \le m$ and $F(B(x_0,\delta))$ is open.

Step S4.5: Path lifting. Let $w \in B(F(x_0), \delta/m)$, $\gamma(t) = F(x_0) + t(w - F(x_0))$, $\delta' = m|w - F(x_0)| < \delta$ and $T = \{t \in [0,1] : \gamma([0,t]) \subset F(B(x_0,\delta))\}$. $0 \in T$ and T is relatively open. For $t \in T$ the path $G \circ \gamma$ on $[0,t]$ is C^1 with $|(G\circ\gamma)'| \le m|w - F(x_0)|$, so $|G(\gamma(t)) - x_0| \le t\delta'$ ((X4), mean value inequality along the path).

Step S4.6: Closedness. If $t_k \uparrow t^*$, $t_k \in T$, then $G(\gamma(t_k))$ is Cauchy (Lipschitz in t), with limit x^*, $|x^* - x_0| \le \delta' < \delta$, $F(x^*) = \gamma(t^*)$ by continuity of F, so $t^* \in T$. Hence $T = [0,1]$ and $w = F(G(w)) \in F(B(x_0,\delta))$.

Step S4.7: Lipschitz bound. The segment $[w_1, w_2]$ lies in the convex ball $B(F(x_0), \delta/m) \subset F(B(x_0,\delta))$, so the mean value inequality for G gives the second claim. □

Lemma 4.7 (Lipschitz constants from sup bounds, non-convex sets)**.** *Let $S \subset \mathbb{R}^N$ be arbitrary, $r > 0$, $S_r \subset \mathbb{C}^N$ its complex r-neighbourhood in the max norm, and $\Delta : S_r \to \mathbb{C}^m$ holomorphic with $\sup_{S_r}|\Delta| \le A$. Then $|\Delta(x) - \Delta(y)| \le \frac{4NA}{r}|x - y|$ for all $x, y \in S$.*

Proof. **Step S4.8: Close pairs.** If $|x - y| < r/2$, every point z of the real segment $[x,y]$ has $|z - x| < r/2$, so the closed polydisc $\bar P(z, r/2) \subset S_r$ (z is within $r/2$ of $x \in S$, hence $\bar P(z,r/2) \subset S_r$ up to the open/closed boundary; use radius $r/2 - \epsilon$ and let $\epsilon \to 0$). By [K:`bp:p1:cauchy-open`], $|\partial_\ell \Delta_i(z)| \le 2A/r$, so $\|D\Delta(z)\| \le 2NA/r$, and the mean value inequality on the segment gives $|\Delta(x) - \Delta(y)| \le (2NA/r)|x - y|$.

Step S4.9: Distant pairs. If $|x - y| \ge r/2$, then $|\Delta(x) - \Delta(y)| \le 2A \le (4A/r)|x - y|$. Since $N \ge 1$, both are $\le 4NA/r$. □

4.5. **Averaging with two frequency scales (the case $m = 0$, $\Lambda = \{0\}$ of CP10 Proposition 1).**

Lemma 4.8 (One averaging step with two scales)**.** *Let $0 < \alpha_2 \le \alpha_1$, $D \subset \mathbb{R}^n$, $r > 0$, $\bar{s}, s > 0$, $K > 0$ with $Ks \ge 6$. Let h be analytic on D_r, real on real points, $\omega = \partial h$, f analytic on $D_r \times \mathbb{T}_{\bar{s}+s}$, real analytic, 1-periodic in θ, and $E := \|f\|_{r,\bar{s}+s}$ (N-norm, §4.1). Assume*

- *(N1) (non-resonance on the complex neighbourhood) for all $I \in D_r$ and $k \in \mathbb{Z}^n$ with $0 < |k|_1 \le K$: $|k \cdot \omega(I)| \ge \alpha_1$ if $k_1 \ne 0$ and $\ge \alpha_2$ if $k_1 = 0$;*
- *(N2) (smallness) $64e\,KE < \alpha_2 r$; equivalently $E < \alpha_2 d/(2^7 c_0 Ks)$ with $d = rs$, $c_0 = e/2$ (CP10 (23), $m = 0$).*

Then there is $\Psi : D_{r/2} \times \mathbb{T}_{\bar{s}+s/6} \to D_r \times \mathbb{T}_{\bar{s}+s}$ ($\Psi(I', \theta') = (I, \theta)$), holomorphic, injective, real on real points, $\Psi - \mathrm{id}$ $\mathbb{Z}^n$-periodic in θ', symplectic ($\Psi^*(dI \wedge d\theta) = dI' \wedge d\theta'$), with $H \circ \Psi = h + g + f_*$ on $D_{r/2} \times \mathbb{T}_{\bar{s}+s/6}$, $g = g(I')$ independent of θ', and

$$\sup_{D_{r/2}} |g - f_0| \le E/4 \quad (\text{indeed} \le E/(Ks)), \qquad \|f_*\|_{r/2,\bar{s}+s/6} \le e^{-Ks/6} E,$$

$$\alpha_1 s |I_1 - I_1'| \le 9E, \quad \alpha_2 s |I_2 - I_2'| \le 9E, \quad 2\pi\alpha_2 r |\theta - \theta'| \le 9E \qquad \text{on } D_{r/2} \times \mathbb{T}_{\bar{s}+s/6}.$$

Here $f_0(I) = \int_{\mathbb{T}^n} f(I, \theta) d\theta$ and $I = (I_1, I_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$.
Source: CP10 Prop. 1 (p. 552), $m = 0$, $\Lambda = \{0\}$, (23)–(26); Lemma A.1 (p. 564).

Proof. This is Corollary 3.6 of Part 2 verbatim (there the first two displacement constants are even $3E$; $9E$ is kept because Lemma 4.9 was computed with it). Its proof is Proposition 3.4: a *finite iteration* of track N's Iterative Lemma (the exponential $e^{-Ks/6}$ comes from the iteration, not from one flow), with the loss schedule $\rho_i = r/8N$, $\sigma_i = s/4N$, $N = \lfloor Ks/12 \ln 2 \rfloor$, and the generating-function splitting of Lemma 3.3 that gives the α_1-bound for I_1 at every stage. The bound on $g - f_0$ has denominator $\alpha_2 rs$ and no K in the numerator ($|g - f_0| \le E/\kappa$, $\kappa = Ks$; CP10 (25); review P3-04). Only (N1), (N2) are assumed: no Lipschitz hypothesis (Part 2, Remark 3.7). □

4.6. **One step of the scheme.**

Lemma 4.9 (One KAM step)**.** *Let the data of Proposition 4.3 be given with $H = h + f$ on $D_\rho \times \mathbb{T}_{\bar{s}+s}$, constants $M, \hat{M}, \bar{M}, \bar{M}_i, L, E, \ldots, K, \hat{\rho}, \hat{E}$ as in Definition 4.2, and in addition*

- *(S1) $\hat{c}\hat{E} < 1$ (so $\hat{E} < 2^{-13}/n$);*
- *(S2) $L \ge \max\{\bar{M}, M^{-1}, \hat{M}^{-1}\}$ (this is the definition; it is stated to allow L larger, used in the iteration: replace L by any larger number, and then use the same formulas);*
- *(S3) $\omega_0(D) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$.*

Put $\rho_+ := \hat{\rho}/16$, $s_+ := s/12$, $M_+ := 2M$, $\hat{M}_+ := 2\hat{M}$, $\bar{M}_+ := 2\bar{M}$, $\bar{M}_{i+} := 2\bar{M}_i$, $L_+ := 2L$, $E_+ := E^2 L M^2/\gamma_1^2$ and let $K_+, \hat{\rho}_+, \hat{E}_+$ be given by Definition 4.2 with the + data. Then there are a map Ψ_+, a function h_+ and a set $D_+ \subset D_{\hat{\rho}/16}$, with

- *(i) $\Psi_+ : (D_+)_{\rho_+} \times \mathbb{T}_{\bar{s}+s_+} \to D_{\hat{\rho}} \times \mathbb{T}_{\bar{s}+s}$ holomorphic, real on reals, $\Psi_+ - \mathrm{id}$ periodic, symplectic, and $H_+ := H \circ \Psi_+ = h_+ + f_+$ with $h_+ = h + g$, $g = g(I)$ analytic on $D_{\hat{\rho}/2}$, $\sup_{D_{\hat{\rho}/2}} |g| \le \frac{5}{4}E$, and $\|f_+\|_{\rho_+,\bar{s}+s_+} \le E_+$;*
- *(ii) $\omega_+ = \partial h_+$ is injective on $D_{\hat{\rho}/8}$ with $U_+ = \partial^2 h_+$ nonsingular there; $(D_+)_{\rho_+} \subset D_{\hat{\rho}/8}$, $\omega_+(D_+) = \omega_0(D)$ (so (S3) holds again);*
- *(iii) $M_+ \ge \sup_{D_{\hat{\rho}/4}} \|U_+\|$, $\hat{M}_+ \ge \sup \|\hat{U}_+\|$, $\bar{M}_+ \ge \sup \|U_+^{-1}\|$, $\bar{M}_{i+} \ge \sup \|T_{i+}\|$ (sup over $D_{\hat{\rho}/4}$), and $L_+ \ge \max\{\bar{M}_+, M_+^{-1}, \hat{M}_+^{-1}\}$;*
- *(iv) the map $\hat{\imath}_+ := (\omega_+|_{D_{\hat{\rho}/8}})^{-1} \circ \omega_0$ is defined and holomorphic on $D_{\hat{\rho}/16}$, $D_+ = \hat{\imath}_+(D)$, and*

$$\sup_{D_{\hat{\rho}/16}} |\hat{\imath}_{+,i} - \mathrm{id}_i| \le 10 \frac{\bar{M}_i}{\bar{M}} \hat{E}\hat{\rho} \ (\le 10\hat{E}\hat{\rho}, \ i = 1, 2), \qquad L_D(\hat{\imath}_+ - \mathrm{id}) \le 640\, n \hat{E};$$

(where ω_0 here means the frequency map of the current h, and L_D is the Lipschitz constant on the set D);

(v) the displacement bounds, for $(I',\theta') \in D_{\hat\rho/2} \times \mathbb{T}_{\bar s+s/6}$ and $\Psi_+(I',\theta') = (I,\theta)$:

$$|I_1 - I_1'| \le \tfrac{3}{4}\tfrac{\hat M}{M}\hat E\hat\rho, \quad |I_2 - I_2'| \le \tfrac{3}{4}\hat E\hat\rho, \quad 2\pi|\theta-\theta'| \le \tfrac{3}{4}\hat E s;$$

(vi) $\hat\rho_+ \ge \hat\rho/(2\cdot 24^{\tau+1})$, $K_+ \le 24K$, $\hat E_+ \le \frac{8}{9}(4s)^{2(\tau+1)}\hat E^2 < \hat E^2$, and $\hat c\hat E_+ < 1$.

Proof. **Step S4.10: Non-resonance on $D_{\hat\rho}$.** Let $I \in D_{\hat\rho}$, pick $I_0 \in D$ with $|I - I_0| < \hat\rho \le \rho$; the segment $[I_0, I]$ lies in D_ρ, where $\|U\| \le M$, hence $|\omega_i(I) - \omega_i(I_0)| \le M\hat\rho$ for all i, and for $i > n_1$ (rows of $\hat U$) $\le \hat M\hat\rho$ ([K:bp:p1:ext-mvt]). For $0 < |k|_1 \le K$ and $k_1 \ne 0$: $|k\cdot\omega(I)| \ge \gamma_1|k|_1^{-\tau} - |k|_1 M\hat\rho \ge \gamma_1 K^{-\tau} - KM\hat\rho \ge \frac{2}{3}\gamma_1 K^{-\tau} =: \alpha_1$, since $\hat\rho \le \gamma_1/(3MK^{\tau+1})$ and $\omega(I_0) \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ by (S3). For $k_1 = 0$ only ω_2 enters: $|k\cdot\omega(I)| \ge \gamma_2 K^{-\tau} - K\hat M\hat\rho \ge \frac{2}{3}\gamma_2 K^{-\tau} =: \alpha_2$. Also $\alpha_2 \le \alpha_1$ and (from $\hat\rho \le \gamma_i/(3\cdot K^{\tau+1}\cdot(\mathrm{M}\text{ or }\hat M))$)

$$\alpha_1 \ge 2MK\hat\rho, \qquad \alpha_2 \ge 2\hat M K\hat\rho. \tag{3.2}$$

Step S4.11: Smallness and $Ks \ge 6$. $Ks = 6\log_+(x^{-1}) \ge 6$. By (3.2) and $L \ge \hat M^{-1}$: $64e\,KE/(\alpha_2\hat\rho) \le 64e\,E/(2\hat M\hat\rho^2) \le 32e\,\hat E < 87\hat E < \hat c\hat E < 1$. So Lemma 4.8 applies with $r = \hat\rho$.

Step S4.12: Averaging. Lemma 4.8 gives Ψ, g, f_* on $D_{\hat\rho/2}\times\mathbb{T}_{\bar s+s/6}$. Since $e^{-Ks/6} = \min\{e^{-1}, x\} \le x$ ($Ks/6 = \log_+(1/x)$), $\|f_*\| \le xE = E^2LM^2/\gamma_1^2 = E_+$. Define Ψ_+ as the restriction of Ψ to $(D_+)_{\rho_+}\times\mathbb{T}_{\bar s+s_+}$ once D_+ is defined below ($(D_+)_{\rho_+} \subset D_{\hat\rho/8} \subset D_{\hat\rho/2}$ and $s_+ \le s/6$; the norm is monotone). Also $\sup_{D_{\hat\rho/2}}|g| \le \sup|g - f_0| + \sup|f_0| \le E/4 + E$. This proves the first part of (i).

Step S4.13: Displacements. By the last line of Lemma 4.8, (3.2), $Ks \ge 6$ and $M\hat M^{-1}$-bounds: $|I_1 - I_1'| \le \frac{9E}{\alpha_1 s} \le \frac{9EK}{6\alpha_1} \le \frac{9E}{12M\hat\rho} = \frac{3}{4}\frac{E}{M\hat\rho} \le \frac{3}{4}\frac{\hat M}{M}\frac{EL}{\hat\rho}$, using $\hat ML \ge 1$; and $\frac{EL}{\hat\rho} = \hat E\hat\rho$. Likewise $|I_2 - I_2'| \le \frac{9E}{\alpha_2 s} \le \frac{3}{4}\frac{E}{\hat M\hat\rho} \le \frac{3}{4}\hat E\hat\rho$, $2\pi|\theta-\theta'| \le \frac{9E}{\alpha_2\hat\rho} \le \frac{9E}{2\hat MK\hat\rho^2} \le \frac{9}{2}\frac{\hat E}{K} \le \frac{3}{4}\hat E s$ (as $K \ge 6/s$). This is (v).

Step S4.14: Derivatives of g. For $I \in D_{\hat\rho/4}$ the polydisc of radius $\hat\rho/4$ around I lies in $D_{\hat\rho/2}$. [K:bp:p1:cauchy-open] with $A = \frac{5}{4}E$, $\delta = \hat\rho/4$ gives $|\partial_i g(I)| \le 5E/\hat\rho$ and $|\partial_i\partial_j g(I)| \le 4A/\delta^2 = 80E/\hat\rho^2$, so

$$\sup_{D_{\hat\rho/4}}\|\partial^2 g\| \le \frac{80nE}{\hat\rho^2} = \frac{80n\hat E}{L}, \qquad \sup_{D_{\hat\rho/4}}|\partial g| \le \frac{5E}{\hat\rho}. \tag{3.3}$$

Since $L^{-1} \le M$, $L^{-1} \le \hat M$ and $\bar M \le L$: $\|\partial^2 g\| \le 80n\hat E\min\{M,\hat M\} \le \frac{1}{100}\min\{M,\hat M\}$ and $\bar M\|\partial^2 g\| \le 80n\hat E \le \frac{1}{100}$.

Step S4.15: Bounds on U_+. $U_+ = U + \partial^2 g$ on $D_{\hat\rho/4}$ and $\hat U_+ = \hat U + (\partial^2 g)_{\mathrm{rows}>\mathrm{n_1}}$ with $\|\mathrm{rows}\| \le \|\partial^2 g\|$. Hence $\|U_+\| \le M + \frac{M}{100} \le 2M$ and $\|\hat U_+\| \le \hat M + \frac{\hat M}{100} \le 2\hat M$. Further $U_+ = (1 + \partial^2 g\,U^{-1})U$, so $U_+^{-1} = U^{-1}(1 + \partial^2 g\,U^{-1})^{-1}$ ($\|\partial^2 g\,U^{-1}\| \le 1/100$, (X5)), hence $\|U_+^{-1}\| \le 2\bar M$ and, the rows of a product T_iB being T_i times B, $\|T_{i+}\| \le 2\|T_i\| \le 2\bar M_i$ (the left factor of $U^{-1}\cdot(\dots)$ keeps row blocks). Nonsingular. And $L_+ = 2L \ge \max\{2\bar M, (2M)^{-1}, (2\hat M)^{-1}\}$. This is (iii) on $D_{\hat\rho/4}$.

Step S4.16: Injectivity of ω_+ on $D_{\hat\rho/8}$. Let $I, I' \in D_{\hat\rho/8}$ with $\omega_+(I) = \omega_+(I')$ and put $w = \omega_0(I)$, $w' = \omega_0(I')$. Then $w - w' = \partial g(I') - \partial g(I)$ and $|w - w'| \le 10E/\hat\rho$ by (3.3). The ball $B(I, \hat\rho/8) \subset D_{\hat\rho/4} \subset D_\rho$, and ω_0 is injective on D_ρ with $\|U^{-1}\| \le \bar M$; Lemma 4.6 gives $\omega_0(B(I,\hat\rho/8)) \supset B(w, \hat\rho/(8\bar M))$. Since $10E/\hat\rho < \hat\rho/(8\bar M)$ (equivalent to $80E\bar M < \hat\rho^2$, implied by $80\hat E < 1$ and $\bar M \le L$), $w' = \omega_0(I'')$ with $I'' \in B(I,\hat\rho/8)$ and, by injectivity of ω_0 on D_ρ, $I'' = I'$. Thus the segment $[I, I'] \subset D_{\hat\rho/4}$ and, by Lemma 4.6 (second part) and the mean value inequality for ∂g along $[I, I']$, $|I - I'| \le \bar M|w - w'| \le \bar M\sup\|\partial^2 g\||I - I'| \le 80n\hat E|I - I'|$, whence $I = I'$. With

(iii), ω_+ is a holomorphic injection with nonsingular derivative on $D_{\hat\rho/8}$; by (X1) an open map with holomorphic inverse, $\|D(\omega_+^{-1})\| \le 2\bar M$ (and $\|(D\omega_+^{-1})_i\| \le 2\bar M_i$ for the row blocks) on $\omega_+(D_{\hat\rho/8})$.

Step S4.17: The map $\hat\imath_+$ and D_+. For $I \in D_{\hat\rho/16}$, $B(I,\hat\rho/16) \subset D_{\hat\rho/8}$ and Lemma 4.6 for $F=\omega_+$, $m = 2\bar M$ gives $\omega_+(B(I,\hat\rho/16)) \supset B(\omega_+(I),\hat\rho/(32\bar M))$. But $\omega_0(I) = \omega_+(I) - \partial g(I)$ and $|\partial g(I)| \le 5E/\hat\rho < \hat\rho/(32\bar M)$ (as $160E\bar M < \hat\rho^2$, because $160\hat E < 1$). So $\hat\imath_+ := \omega_+^{-1}\circ\omega_0$ is defined and holomorphic on $D_{\hat\rho/16}$, with values in $B(I,\hat\rho/16)$, $\omega_+\circ\hat\imath_+ = \omega_0$. By Lemma 4.6 (second part, with $w_1=\omega_0(I)$, $w_2=\omega_+(I)$): $|\hat\imath_+(I) - I| \le 2\bar M|\partial g(I)| \le 10\bar M E/\hat\rho$, and the same estimate holds for the first n_1 and last n_2 components with $\bar M_i$ instead of $\bar M$ (use the row blocks of $D\omega_+^{-1}$). By $\bar M \le L$: $10\bar M_i E/\hat\rho \le 10\frac{\bar M_i}{\bar M}\frac{EL}{\hat\rho} = 10\frac{\bar M_i}{\bar M}\hat E\hat\rho \le 10\hat E\hat\rho$. This is the sup bound in (iv). Then $\sup|\hat\imath_+ - \mathrm{id}| \le 10\hat E\hat\rho < \hat\rho/16$ since $\hat E < 2^{-13}$. Lipschitz bound: Lemma 4.7 with $S=D$, $r=\hat\rho/16$, $N=n$, $A = 10\hat E\hat\rho$ gives $L_D(\hat\imath_+ - \mathrm{id}) \le 4n\cdot 10\hat E\hat\rho\cdot 16/\hat\rho = 640n\hat E$. Since h is real analytic, ω_0 and ω_+ are real on $\mathbb{R}^n$; if $\omega_+(I'') = \omega_0(I)$ with $I\in D\subset\mathbb{R}^n$, then $\overline{I''}$ is a second preimage in the same ball, so $I'' = \overline{I''} \in \mathbb{R}^n$ by injectivity. Hence $\hat\imath_+(D)\subset\mathbb{R}^n$. Put $D_+ := \hat\imath_+(D) \subset D_{\hat\rho/16}\cap\mathbb{R}^n$. Since $\omega_+\circ\hat\imath_+ = \omega_0$ on D and ω_0 is injective, $\hat\imath_+$ is injective on D and $\omega_+(D_+) = \omega_0(D)$. Then $(D_+)_{\rho_+} \subset D_{\hat\rho/16+\hat\rho/16} = D_{\hat\rho/8}$. This is (ii) and the rest of (iv); in particular $\omega_+(D_+) = \omega_0(D) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$.

Step S4.18: The new constants. Put $x = EM^2L/\gamma_1^2$; then $E_+ = Ex$ and $x_+ := E_+L_+M_+^2/\gamma_1^2 = Ex\cdot 2L\cdot 4M^2/\gamma_1^2 = 8x^2$. Hence $\log(1/x_+) = 2\log(1/x) - 3\log 2 \le 2\log_+(1/x)$ and $\log_+(1/x_+) \le 2\log_+(1/x)$ (as $1\le\log_+$), so

$$K_+ = \frac{6}{s_+}\log_+\frac{1}{x_+} \le \frac{72}{s}\cdot 2\log_+\frac{1}{x} = 24K.$$

Therefore $\gamma_1/(3M_+K_+^{\tau+1}) \ge \frac12 24^{-(\tau+1)}\gamma_1/(3MK^{\tau+1})$, likewise for $\gamma_2,\hat M$, and $\rho_+ = \hat\rho/16$; as $\hat\rho \le \min\{\gamma_1/(3MK^{\tau+1}),\gamma_2/(3\hat MK^{\tau+1})\}$, $\hat\rho_+ \ge \hat\rho\min\{\frac{1}{2\cdot 24^{\tau+1}},\frac{1}{16}\} = \hat\rho/(2\cdot 24^{\tau+1})$.

Step S4.19: The quadratic bound. $E_+L_+ = 2E^2L^2M^2/\gamma_1^2$ and $\hat\rho_+^{-2} \le 4\cdot 24^{2(\tau+1)}\hat\rho^{-2}$, so $\hat E_+ \le 8\cdot 24^{2(\tau+1)}\,\hat E^2\,\hat\rho^2M^2/\gamma_1^2$ (using $E^2L^2/\hat\rho^2 = \hat E^2\hat\rho^2$). From $\hat\rho\le\gamma_1/(3MK^{\tau+1})$ and $K\ge 6/s$: $\hat\rho^2M^2/\gamma_1^2 \le \frac19(s/6)^{2(\tau+1)}$. Hence $\hat E_+ \le \frac89(24s/6)^{2(\tau+1)}\hat E^2 = \frac89(4s)^{2(\tau+1)}\hat E^2 < \hat E^2$ since $4s\le\bar s<1$. Finally $\hat c\hat E_+ < \hat c\hat E\cdot\hat E < 1$. This is (vi). □

Remark 4.10 (Differences from CP10 in the step). CP10 (74)–(75), (86)–(87) have $5n$, 2^6n and 2^7 from a mixture of 1-, ∞- and "sup" norms; the above constants are those of max norms. CP10 injectivity argument (p. 569–570) uses "$\sup\|\partial(\omega^{-1})\|\sup|\omega(I)-\omega(I')|$", a mean value inequality that silently needs $[\omega(I),\omega(I')]\subset\omega(D_\rho)$; Lemma 4.6 supplies it. The Lipschitz bound of $\hat\imath_+ - \mathrm{id}$ on the (non-convex) set D is repaired by Lemma 4.7. CP10 (91) prints $(4s)^{\tau+1}$ where the computation gives $(4s)^{2(\tau+1)}$ (a sharpening: the printed bound is true, being weaker).

4.7. **The iterative lemma.**

Definition 4.11 (The sequences)**.** Given the data and constants of Proposition 4.3, put $D_0 = D^{\mathrm{dio}} = \omega_0^{-1}(\mathcal{D}_{\gamma_1,\gamma_2,\tau})\cap D$, $\rho_0=\rho$, $s_0=s$, $M_0=M$, $\hat M_0=\hat M$, $\bar M_0=\bar M$, $\bar M_{i,0}=\bar M_i$, $L_0=L$, $E_0=E$, $h_0=h$, $f_0=f$ and, recursively for $j\ge1$: $s_j = s_{j-1}/12$, $M_j = 2M_{j-1}$, $\hat M_j = 2\hat M_{j-1}$, $\bar M_j = 2\bar M_{j-1}$, $\bar M_{i,j} = 2\bar M_{i,j-1}$, $L_j = 2L_{j-1}$, $\rho_j = \hat\rho_{j-1}/16$, $E_j = E_{j-1}^2L_{j-1}M_{j-1}^2/\gamma_1^2$, and $K_j,\hat\rho_j,\hat E_j$ by Definition 4.2 with the data $(s_j,\rho_j,E_j,M_j,\hat M_j,\bar M_j)$ (so L_j there is $\max\{\bar M_j,M_j^{-1},\hat M_j^{-1}\}\le 2^jL$, and in the formulas the larger $L_j = 2^jL$ is used, as allowed by (S2) of Lemma 4.9).

Lemma 4.12 (Iterative lemma)**.** *Under the hypotheses of Proposition 4.3 (in particular $\hat c\hat E_0<1$), for every $j\ge0$ there are a set $D_j\subset\mathbb{R}^n$, an analytic symplectic map $\Phi_j = \Psi_1\circ\cdots\circ\Psi_j$ on $W_j := (D_j)_{\rho_j}\times\mathbb{T}_{\bar s+s_j}$ (with $\Phi_0=\mathrm{id}$, D_0 as above) and $H_j = H\circ\Phi_j = h_j + f_j$, such that*

(i) $D_j \subset (D_{j-1})_{\hat\rho_{j-1}/16}$, $\omega_j = \partial h_j$ *is a diffeomorphism of* $(D_j)_{\rho_j}$ *onto an open set,* $\omega_j(D_j) = \omega_0(D_0) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$*; the map* $\hat\imath_j := \omega_j^{-1} \circ \omega_{j-1} : D_{j-1} \to D_j$ *(defined and holomorphic on* $(D_{j-1})_{\hat\rho_{j-1}/16}$*) satisfies* $\sup|\hat\imath_{j,i} - \mathrm{id}_i| \le 10\frac{\bar M_i}{M}\hat E_{j-1}\hat\rho_{j-1}$ *and* $L_{D_{j-1}}(\hat\imath_j - \mathrm{id}) \le 640n\hat E_{j-1}$*;*
(ii) $\|f_j\|_{\rho_j,\bar s+s_j} \le E_j$*;*
(iii) $\Psi_j : W_j \to W'_{j-1}$ *with* $W'_{j-1} := (D_{j-1})_{\hat\rho_{j-1}} \times \mathbb{T}_{\bar s+s_{j-1}} \subset W_{j-1}$ *is holomorphic (so* $\Psi_j(W_j) \subset W_{j-1}$*), real on reals,* $\Psi_j - \mathrm{id}$ *periodic, and on* W_j

$$|\Delta I_1| \le \tfrac34\tfrac{\hat M}{M}\hat E_{j-1}\hat\rho_{j-1}, \quad |\Delta I_2| \le \tfrac34\hat E_{j-1}\hat\rho_{j-1}, \quad 2\pi|\Delta\theta| \le \tfrac34\hat E_{j-1}s_{j-1};$$

(iv) $\hat E_j < \hat E_{j-1}^2$ *for* $j \ge 1$*, whence* $\hat E_j \le \hat E_0^{2^j}$*, and* $\hat c\hat E_j < 1$*;*
(v) $\hat\rho_j \le \hat\rho_{j-1}/16$ *and* $\hat\rho_j \ge \hat\rho_{j-1}/(2\cdot 24^{\tau+1})$*,* $s_j = s/12^j$*,* $K_j \le 24^j K$*.*

Proof. **Step S4.20: Induction hypothesis.** We prove by induction on $j \ge 0$ the statement $\mathcal{P}_j$: the data $(D_j, h_j, f_j, \rho_j, s_j, M_j, \hat M_j, \bar M_j, \bar M_{i,j}, L_j, E_j)$ satisfy the hypotheses of Lemma 4.9 (on the domain $(D_j)_{\rho_j} \times \mathbb{T}_{\bar s+s_j}$): $\hat c\hat E_j < 1$, $L_j \ge \max\{\bar M_j, M_j^{-1}, \hat M_j^{-1}\}$, $\omega_j(D_j) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$, bounds $M_j, \hat M_j, \bar M_j, \bar M_{i,j}$ on $(D_j)_{\rho_j}$, ω_j injective on $(D_j)_{\rho_j}$ with nonsingular U_j, $\|f_j\| \le E_j$, $\hat M_j \le M_j$, $\bar M_{i,j} \le \bar M_j$, and $4s_j \le \bar s < 1$.

Step S4.21: Base. $\mathcal{P}_0$ is the hypothesis of Proposition 4.3 and $\omega_0(D_0) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ by definition of D_0 (note $D_0 \subset D$, so the sup bounds on D_ρ give those on $(D_0)_\rho$; injectivity is inherited).

Step S4.22: Induction. Apply Lemma 4.9 to $\mathcal{P}_{j-1}$: it produces $D_j := D_+$, $\Psi_j := \Psi_+$, $h_j := h_+$, $f_j := f_+$, and (i)–(vi) of that lemma are (i)–(iii) here and the recursion of Definition 4.11 (the quantities named $\hat\rho_+, K_+, \hat E_+$ there are $\hat\rho_j, K_j, \hat E_j$). Items (iii) and (v) of the step give $\mathcal{P}_j$ ($\hat M_j \le M_j$, $\bar M_{i,j} \le \bar M_j$ propagate by doubling).

Step S4.23: Composition. $\Phi_j = \Phi_{j-1} \circ \Psi_j$ is defined on W_j because $\Psi_j(W_j) \subset W'_{j-1} \subset W_{j-1}$: $W_j \subset D_{\hat\rho_{j-1}/8} \times \mathbb{T}_{\bar s+s_j}$ (more precisely $(D_j)_{\rho_j}$ with $D_j \subset (D_{j-1})_{\hat\rho_{j-1}/16}$), the displacement is $\le \frac34\hat E_{j-1}\hat\rho_{j-1} < \hat\rho_{j-1}/16$, so the action of $\Psi_j(y)$ lies in $(D_{j-1})_{\hat\rho_{j-1}/8+\hat\rho_{j-1}/16} \subset (D_{j-1})_{\hat\rho_{j-1}}$, and $\hat\rho_{j-1} \le \rho_{j-1}$ (definition of $\hat\rho$). The identity $H_j = H_{j-1}\circ\Psi_j = H\circ\Phi_j$ is then immediate, and Φ_j is symplectic, real, with $\Phi_j - \mathrm{id}$ periodic.

Step S4.24: (iv)–(v) of the Lemma. By Step (vi) of Lemma 4.9: $\hat E_j < \hat E_{j-1}^2$, $\hat\rho_j \ge \hat\rho_{j-1}/(2\cdot 24^{\tau+1})$, $\hat\rho_j \le \rho_j = \hat\rho_{j-1}/16$, $K_j \le 24K_{j-1}$. □

5. Two-scale KAM theorem II: the limit, the tori and the measure

We continue with the notation of Definition 4.11 and Lemma 4.12; $c := 24^{\tau+1}/6$ and $q := 2\cdot 24^{\tau+1} = 12c$. Since $\hat c \ge 2c$, $\hat c \ge 2^{13}n$, the KAM condition gives

$$\hat E_0 < \frac{1}{8192\,n}, \qquad c\hat E_0 < \frac12. \tag{3.4}$$

Lemma 5.1 (Sums)**.** *For* $m \ge 0$*:* $\hat E_k \le \hat E_m^{2^{k-m}}$ *for* $k \ge m$*,* $\sum_{k\ge m}\hat E_k \le 2\hat E_m$*, and* $\sum_{k\ge0} c^k\hat E_k \le 2\hat E_0$*. Also* $\hat\rho_k \le \hat\rho_0\,16^{-k}$*,* $s_k = s\,12^{-k}$*.*

Proof. **Step S5.1: Doubling.** By Lemma 4.12(iv), $\hat E_k < \hat E_{k-1}^2$, so $\hat E_k \le \hat E_m^{2^{k-m}} \le \hat E_m^{k-m+1}$ (as $2^i \ge i+1$ and $\hat E_m < 1$). Hence $\sum_{k\ge m}\hat E_k \le \hat E_m/(1-\hat E_m) \le 2\hat E_m$.

Step S5.2: Weighted sum. $c^k\hat E_k \le c^k\hat E_0^{k+1} = \hat E_0(c\hat E_0)^k \le \hat E_0 2^{-k}$ by (3.4); sum $\le 2\hat E_0$.

Step S5.3: Radii. $\hat\rho_k \le \rho_k = \hat\rho_{k-1}/16$ (Definition of $\hat\rho_k$), iterate. □

5.1. **The limit actions (CP10 Step 1).**

Lemma 5.2 (Limit actions). *Put $\check\imath_0 = \mathrm{id}$, $\check\imath_j := \hat\imath_j \circ \cdots \circ \hat\imath_1 : D_0 \to D_j$. Then $\check\imath_j$ converges uniformly on D_0 to $\check\imath = (\check\imath_1, \check\imath_2)$ and*

(i) $\sup_{D_0}|\check\imath_i - \mathrm{id}_i| \le 20\frac{\bar M_i}{\bar M}\hat E_0\hat\rho_0 \le 20\hat E_0\hat\rho_0$; $\sup_{D_0}|\check\imath - \check\imath_j| \le 20\hat\rho_j\hat E_j$;
(ii) $D_* := \check\imath(D_0) \subset (D_j)_{\rho_j/2}$ *for all j, in particular* $D_* \subset \mathrm{Re}(D_0)_{20\hat E_0\hat\rho_0}$;
(iii) $\lambda := L_{D_0}(\check\imath - \mathrm{id}) \le 2560\, n\hat E_0 < \frac13$; *so $\check\imath$ is injective,* $(1-\lambda)|x-y| \le |\check\imath x - \check\imath y| \le (1+\lambda)|x-y|$;
(iv) $\omega_j(\check\imath_j(I_0)) = \omega_0(I_0)$ *for all* $I_0 \in D_0$, $j \ge 0$.

Source: CP10 Step 1, pp. 570–571, (97)–(104); constants recomputed (CP10: $10n$, $2^8 n\hat E$).

Proof. **Step S5.4: Well-posedness.** $\hat\imath_j : D_{j-1} \to D_j$ (Lemma 4.12(i)); so $\check\imath_j(D_0) \subset D_j$ and the composition is defined. (iv) follows from $\omega_j \circ \hat\imath_j = \omega_{j-1}$ by induction.

Step S5.5: Telescoping. $|\check\imath_{j,i}(x) - \check\imath_{j-1,i}(x)| = |\hat\imath_{j,i}(y) - y_i|$ with $y = \check\imath_{j-1}(x) \in D_{j-1}$, so $\le 10\frac{\bar M_i}{\bar M}\hat E_{j-1}\hat\rho_{j-1}$ (the ratio $\bar M_{i,j}/\bar M_j = \bar M_i/\bar M$ is constant). Since $\hat\rho_{j-1} \le \hat\rho_0$, Lemma 5.1 gives the Cauchy property and (i): $\sum_{j\ge1} 10\hat E_{j-1}\hat\rho_{j-1} \le 20\hat E_0\hat\rho_0$; the tail is $\le 10\hat\rho_j\sum_{m\ge j}\hat E_m \le 20\hat\rho_j\hat E_j$.

Step S5.6: Location. $\check\imath_j(D_0) \subset D_j \subset \mathbb{R}^n$ and $20\hat\rho_j\hat E_j \le \rho_j/2$ ($\hat\rho_j \le \rho_j$, $\hat E_j < 1/40$), so $D_* \subset \mathrm{Re}(D_j)_{\rho_j/2}$; for $j = 0$ and $D_0 \subset D$ this gives the last claim of (ii).

Step S5.7: Lipschitz constant. Put $i_j = \check\imath_j - \mathrm{id}$. From $\check\imath_{j+1} = \hat\imath_{j+1}\circ\check\imath_j$: $i_{j+1} = (\hat\imath_{j+1} - \mathrm{id})\circ\check\imath_j + i_j$, so $1 + L(i_{j+1}) \le (1 + L(\hat\imath_{j+1} - \mathrm{id}))(1 + L(i_j))$ and, by Lemma 4.12(i), $1 + L(i_j) \le \prod_{k\le j}(1 + 640n\hat E_{k-1}) \le \exp(640n\sum_k \hat E_{k-1}) \le \exp(1280n\hat E_0)$. As $x := 1280n\hat E_0 < \frac16$, $e^x \le 1 + 2x$: $L(i_j) \le 2560n\hat E_0$ uniformly in j; the pointwise limit of Lipschitz maps keeps the constant. Injectivity and the two-sided bound follow from $|\check\imath x - \check\imath y| \ge |x-y| - \lambda|x-y|$. □

5.2. **The limit symplectic map (CP10 Step 2).** *Rescaling.* Let $S(I,\theta) := (I/\hat\rho_0,\ 2\pi\theta/s_0)$ and $|(I,\theta)|_* := |S(I,\theta)|$ (max norm); for a map G put $\check G = SGS^{-1}$ and L_* for Lipschitz constants in $|\cdot|_*$. Put $\mathcal{D}_j := \mathrm{Re}(D_j)_{\rho_j/2}$. All maps below are real on real points.

Lemma 5.3 (Domains and rescaled Lipschitz bounds for Ψ_k). *(i) $\Psi_k(\mathcal{D}_k \times \mathbb{R}^n) \subset \mathcal{D}_{k-1} \times \mathbb{R}^n$ ($k \ge 1$); hence $\Phi_j = \Psi_1 \circ \cdots \circ \Psi_j$ is defined on $\mathcal{D}_j \times \mathbb{R}^n$ with values in $\mathcal{D}_0 \times \mathbb{R}^n$.*
(ii) $L_*(\check\Psi_k - \mathrm{id}; \mathcal{D}_k \times \mathbb{R}^n) \le 192\, n\, c^{k-1}\hat E_{k-1} =: \varsigma_k$, *and* $\sum_{k\ge1}\varsigma_k \le 384n\hat E_0 < \frac{1}{20}$.
(iii) $|\Psi_k(y) - y|_* \le \frac34\hat E_{k-1}12^{-(k-1)} \le \frac34\hat E_{k-1}$ *for* $y \in \mathcal{D}_k \times \mathbb{R}^n$.

Proof. **Step S5.8: Domains.** Let $y = (I,\theta)$, $I \in \mathcal{D}_k$: $\mathrm{dist}(I, D_k) < \rho_k/2 = \hat\rho_{k-1}/32$ and $D_k \subset \mathrm{Re}(D_{k-1})_{\hat\rho_{k-1}/16}$. By Lemma 4.12(iii) the real displacement is $\le \frac34\hat E_{k-1}\hat\rho_{k-1}$. Hence $\Psi_k(y)_I$ is within $\hat\rho_{k-1}(\frac{1}{32} + \frac{1}{16} + \frac34\hat E_{k-1}) < \hat\rho_{k-1}/8$ of D_{k-1}; and $\hat\rho_{k-1}/8 < \rho_{k-1}/2$ (for $k = 1$ because $\hat\rho_0 \le \rho$; for $k \ge 2$ because $\hat\rho_{k-1} \le \hat\rho_{k-2}/16$).

Step S5.9: Bound on displacement. Scaled: $|\Delta I|/\hat\rho_0 \le \frac34\hat E_{k-1}\hat\rho_{k-1}/\hat\rho_0 \le \frac34\hat E_{k-1}16^{-(k-1)}$ (Lemma 5.1; $\hat M \le M$) and $2\pi|\Delta\theta|/s_0 \le \frac34\hat E_{k-1}s_{k-1}/s_0 = \frac34\hat E_{k-1}12^{-(k-1)}$. Thus $A_k := \sup_{W_k}|\check\Psi_k - \mathrm{id}| \le \frac34\hat E_{k-1}12^{-(k-1)}$; this is (iii).

Step S5.10: Lipschitz. Apply Lemma 4.7 in the scaled coordinates, $N = 2n$, $S = S(\mathcal{D}_k \times \mathbb{R}^n)$, radius $\varepsilon_k = \min\{\rho_k/(2\hat\rho_0),\ s_k/s_0\}$: the scaled complex ε_k-neighbourhood of a real point of S lies in $(D_k)_{\rho_k} \times \mathbb{T}_{s_k} \subset W_k$ ($\varepsilon_k\hat\rho_0 \le \rho_k/2$ and $\varepsilon_k s_0 \le s_k \le \bar s + s_k$), where the bound A_k of (iii) holds (Lemma 4.12(iii) holds on the complex W_k). Now $\rho_k/(2\hat\rho_0) = \hat\rho_{k-1}/(32\hat\rho_0) \ge q^{-(k-1)}/32$ (Lemma 4.12(v)) and $s_k/s_0 = 12^{-k} \ge q^{-(k-1)}/32$, so $\varepsilon_k \ge q^{-(k-1)}/32$ and

$$L_* \le \frac{4\cdot 2n\, A_k}{\varepsilon_k} \le 8n\cdot\tfrac34\hat E_{k-1}12^{-(k-1)}\cdot 32q^{k-1} = 192n\,(q/12)^{k-1}\hat E_{k-1} = 192n\, c^{k-1}\hat E_{k-1}.$$

Step S5.11: Sum. Lemma 5.1: $\sum_k \varsigma_k = 192n\sum_{m\ge 0} c^m \hat{E}_m \le 384n\hat{E}_0 < 384/8192 < \frac{1}{20}$. □

Lemma 5.4 (Compactness of periodic holomorphic functions on a strip)**.** *Let $F_j : \{\vartheta \in \mathbb{C}^n : |\operatorname{Im}\vartheta| < w\} \to \mathbb{C}$ be holomorphic, $\mathbb{Z}^n$-periodic, $|F_j| \le B$, and suppose F_j converges uniformly on $\mathbb{R}^n$ to F. Then for* every $0 < v < w$, *F_j converges uniformly, together with all derivatives, on $\{|\operatorname{Im}\vartheta| \le v\}$; the limit is a holomorphic periodic function on the whole open strip extending F, with $|F| \le B$ there.*

Proof. **Step S5.12: Coefficient decay.** For $0 < w' < w$ and $k \ne 0$ shift the integration contour to $\operatorname{Im}\vartheta = -w'\operatorname{sign}(k)$ (componentwise; Cauchy's theorem for the periodic integrand, [K:bp:p1:ext-rectangle]): $|\hat{F}_{j,k}| \le Be^{-2\pi w'|k|_1}$, uniformly in j.

Step S5.13: Convergence of coefficients. Uniform convergence on $\mathbb{R}^n$ gives $\hat{F}_{j,k} \to \hat{F}_k$ for every k ((X2)), and $|\hat{F}_k| \le Be^{-2\pi w'|k|_1}$ for every $w' < w$.

Step S5.14: Tail and finite part on $|\operatorname{Im}\vartheta| \le v$. Fix $0 < v < w$ and put $w' = (v+w)/2 > v$. There $|e^{2\pi ik\cdot\vartheta}| \le e^{2\pi v|k|_1}$, so for every $m \ge 0$ $\sum_{|k|_1>N}|(2\pi k)^m \hat{F}_{j,k}e^{2\pi ik\cdot\vartheta}| \le B\sum_{|k|_1>N}(2\pi|k|_1)^m e^{-2\pi(w'-v)|k|_1} =: \varepsilon_{N,m} \to 0$ $(N\to\infty)$, uniformly in j, and the same for F; the finitely many modes $|k|_1 \le N$ converge uniformly. Hence $F_j \to F^\sharp := \sum_k \hat{F}_k e^{2\pi ik\cdot\vartheta}$ with all derivatives, uniformly on $|\operatorname{Im}\vartheta| \le v$. The limits for different v coincide (same coefficients), so $F^\sharp$ is defined and holomorphic on the whole open strip ((X3)), equals F on $\mathbb{R}^n$ and, as a pointwise limit of the F_j, satisfies $|F^\sharp| \le B$. □

Lemma 5.5 (The limit map Φ)**.** *On $W_* := D_* \times \mathbb{R}^n$ the maps Φ_j converge uniformly to a map Φ with $\Phi - \mathrm{id}$ $\mathbb{Z}^n$-periodic in θ, real analytic in θ, and*

- *(i)* $\sup_{W_*}|\Pi_{I_1}\Phi - \mathrm{id}| \le \frac{3}{2}\frac{\hat{M}}{M}\hat{E}_0\hat{\rho}_0$, $\sup|\Pi_{I_2}\Phi - \mathrm{id}| \le \frac{3}{2}\hat{E}_0\hat{\rho}_0$, $\sup 2\pi|\Pi_\theta\Phi - \mathrm{id}| \le \frac{3}{2}\hat{E}_0 s_0$, *and the same bounds on complex θ with $|\operatorname{Im}\theta| < \bar{s}/2\pi$;*
- *(ii)* $L_*(\check{\Phi} - \mathrm{id}; W_*) \le 768\, n\hat{E}_0 < \frac{1}{10}$; *hence Φ is injective on W_* and $|x-y|_*\cdot\frac{9}{10} \le |\Phi x - \Phi y|_* \le \frac{11}{10}|x-y|_*$;*
- *(iii)* $\Phi(I,\theta+k) = \Phi(I,\theta) + (0,k)$ *for $k \in \mathbb{Z}^n$.*

Source: CP10 Step 2, pp. 571–572, (105)–(108).

Proof. **Step S5.15: Telescoping of the displacements.** Write $(I^{(k)},\theta^{(k)}) = \Psi_{k+1}\circ\cdots\circ\Psi_j(x)$ for $x \in W_* \subset W_j$ (Lemma 5.2(ii)); then $\Pi_I\Phi_j(x) - \Pi_I x = \sum_{k=1}^{j}(I^{(k-1)} - I^{(k)})$ and every summand is bounded by Lemma 4.12(iii). Summing with Lemma 5.1 ($\hat{\rho}_{k-1} \le \hat{\rho}_0$, $s_{k-1} \le s_0$, $\sum \hat{E}_{k-1} \le 2\hat{E}_0$) gives (i), uniformly in j and also for complex θ (Ψ_k is defined on the complex strip).

Step S5.16: Cauchy sequence. For $x \in W_*$: $\Phi_j(x) - \Phi_{j-1}(x) = \Phi_{j-1}(\Psi_j x) - \Phi_{j-1}(x)$, with $x \in \mathcal{D}_j \times \mathbb{R}^n$ and $\Psi_j x \in \mathcal{D}_{j-1}\times\mathbb{R}^n$ (Lemma 5.3(i)). By the composition rule $L_*(\check{\Phi}_{j-1} - \mathrm{id}) \le \prod_{k<j}(1+\varsigma_k) - 1 \le e^{\sum\varsigma_k} - 1 \le 2\sum\varsigma_k \le \frac{1}{10}$ (using $e^x \le 1+2x$, $x \le 1$; the composition formula is that of Lemma 5.2, proof step S5.7, valid because each Ψ_k maps its domain into the next one). Therefore $|\Phi_j(x) - \Phi_{j-1}(x)|_* \le \frac{11}{10}|\Psi_j x - x|_* \le \frac{11}{10}\cdot\frac{3}{4}\hat{E}_{j-1}$, which is summable: uniform convergence on W_*. The limit satisfies (ii) with $L_* \le 2\sum\varsigma_k \le 768n\hat{E}_0$ (the constant passes to the pointwise limit).

Step S5.17: Periodicity and injectivity. Each $\Psi_k - \mathrm{id}$ is $\mathbb{Z}^n$-periodic in θ (Lemma 4.12), so (iii) holds for Φ_j and in the limit. From (ii), $|\Phi x - \Phi y|_* \ge \frac{9}{10}|x-y|_*$.

Step S5.18: Analyticity in θ. Fix $I_* \in D_*$; $\vartheta \mapsto \Phi_j(I_*,\vartheta) - (0,\vartheta)$ is holomorphic on $\{|\operatorname{Im}\vartheta| < \bar{s}/2\pi\}$ (as $I_* \in (D_j)_{\rho_j}$), $\mathbb{Z}^n$-periodic, bounded uniformly in j by (i), and converges uniformly on $\mathbb{R}^n$. By Lemma 5.4 (componentwise, $w = \bar{s}/2\pi$) it converges, with all derivatives, uniformly on $\{|\operatorname{Im}\vartheta| \le v\}$ for every $v < \bar{s}/2\pi$, to a function holomorphic on the *whole* strip $\{|\operatorname{Im}\vartheta| < \bar{s}/2\pi\}$

whose real restriction is $\Phi(I_*,\cdot)-(0,\mathrm{id})$; the bounds (i), being pointwise, pass to the limit. This is claim (a) of Proposition 4.3. □

5.3. **The tori (CP10 Steps 2–3 and the estimates (35)).** Define, for $\omega\in\Omega_*=\omega_0(D)\cap\mathcal{D}_{\gamma_1,\gamma_2,\tau}$, $I_0(\omega):=\omega_0^{-1}(\omega)\in D_0$ (injectivity of ω_0; $\omega\in\omega_0(D)\cap\mathcal{D}$ means $I_0\in D^{\mathrm{dio}}=D_0$), $I_*(\omega):=\check\imath(I_0(\omega))\in D_*$ and

$$\varphi_\omega(\vartheta):=\Phi(I_*(\omega),\vartheta)=(v(\vartheta;\omega),\ \vartheta+u(\vartheta;\omega)).$$

Lemma 5.6 (Estimates and location). *For $\omega\in\Omega_*$: $|u|\le\frac{3}{4\pi}\hat E_0 s$ on $\{|\mathrm{Im}\,\vartheta|<\bar s/2\pi\}$; $|v_1-I_{0,1}|\le 20(\frac{\bar M_1}{\bar M}+\frac{\hat M}{M})\hat E_0\hat\rho_0$, $|v_2-I_{0,2}|\le 20(\frac{\bar M_2}{\bar M}+1)\hat E_0\hat\rho_0$ on $\mathbb{R}^n$, and $v\in\mathrm{Re}(D_r)$ with $r=32\hat E_0\hat\rho_0$, so $\varphi_\omega(\mathbb{R}^n)\subset\mathrm{Re}(D_r)\times\mathbb{R}^n$. This is Proposition 4.3(a), (d), and the location claim.*
Source: CP10 (35), (105)–(107), Step 2 end, p. 572. Misprint: *CP10 (35) bounds $|v_2-I_2^0|$ by $10n(\bar M_2/\bar M+\hat M/M)\hat E\hat\rho$ although (107) only gives $2\hat E\hat\rho$ for I_2 without the factor $\hat M/M$.*

Proof. **Step S5.19: Sup bounds.** $|v_i-I_{0,i}|\le|\Pi_{I_i}\Phi(I_*,\vartheta)-I_{*,i}|+|I_{*,i}-I_{0,i}|$, the two terms bounded by Lemma 5.5(i) and Lemma 5.2(i): $\frac32\frac{\hat M}{M}\hat E\hat\rho+20\frac{\bar M_1}{\bar M}\hat E\hat\rho$ for $i=1$ (and $\frac32+20\frac{\bar M_2}{\bar M}$ for $i=2$), both $\le$ the stated constants. $|u|\le\frac32\hat E s/2\pi$ is the angle bound.

Step S5.20: Location. $I_*\in\mathrm{Re}(D_0)_{20\hat E\hat\rho}$ and $|v-I_*|\le\frac32\hat E\hat\rho$, so $\mathrm{dist}(v,D)<22\hat E\hat\rho<r$. □

Lemma 5.7 (Immersion and Lagrangian property). *For $\omega\in\Omega_*$, $\|D_\vartheta u\|\le\frac34 n\hat E_0$ (so $D_\vartheta\varphi_\omega=\binom{D_\vartheta v}{1+D_\vartheta u}$ has injective lower block $1+D_\vartheta u$, since $\frac34 n\hat E_0<\frac12$), and $\varphi_\omega^*(\sum_i dJ_i\wedge d\theta_i)=0$.*

Proof. **Step S5.21: Cauchy in the strip.** u is holomorphic on $|\mathrm{Im}\,\vartheta|<\bar s/2\pi$ with $|u|\le 3\hat E s/4\pi$. At real ϑ the polydisc of radius $\bar s/4\pi$ lies in the strip, so $|\partial_a u_i|\le\frac{3\hat E s}{4\pi}\cdot\frac{4\pi}{\bar s}=3\hat E\frac{s}{\bar s}\le\frac34\hat E$ ($4s\le\bar s$), and the row sum is $\le\frac34 n\hat E$.

Step S5.22: Each Φ_j-torus is Lagrangian. Φ_j is symplectic and $\vartheta\mapsto(I_*,\vartheta)$ has pullback $\sum dI_i\wedge d\theta_i=0$ (the I-coordinates are constant), so $\varphi^{(j)}:=\Phi_j(I_*,\cdot)$ has $\Omega_j(a,b):=\sum_i(\partial_a v_i^{(j)}\partial_b\Theta_i^{(j)}-\partial_b v_i^{(j)}\partial_a\Theta_i^{(j)})=0$.

Step S5.23: Limit. By Lemma 5.5 (Step S5.18) the first derivatives of $\varphi^{(j)}$ converge to those of φ_ω (uniformly on $\mathbb{R}^n$, Lemma 5.4); Ω_j is bilinear in them, so $\Omega=\lim\Omega_j=0$. □

Lemma 5.8 (Invariance). *For $\omega\in\Omega_*$, $\vartheta\in\mathbb{R}^n$, $t\in\mathbb{R}$: $\frac{d}{dt}\varphi_\omega(\vartheta+t\omega)=X_H(\varphi_\omega(\vartheta+t\omega))$, $X_H=(-\partial_\theta H,\partial_J H)$.*
Source: CP10 Step 3, pp. 572–573.

Proof. Let $I_0=I_0(\omega)$, $I_*=\check\imath(I_0)$, $\theta_t=\vartheta+t\omega$, $\gamma(t)=\Phi(I_*,\theta_t)$, $\gamma_j(t)=\Phi_j(I_*,\theta_t)$.

Step S5.24: Symplectic conjugation. Since Φ_j is a symplectic diffeomorphism onto its image and $H_j=H\circ\Phi_j$: $X_H\circ\Phi_j=D\Phi_j\,X_{H_j}$ ([N:`bpN:p5:symplectic`]). With $X_{H_j}(I_*,\theta)=(-\partial_\theta f_j,\ \omega_j(I_*)+\partial_I f_j)$ we get $\gamma_j'(t)-X_H(\gamma_j(t))=D\Phi_j(I_*,\theta_t)\big(\partial_\theta f_j,\ \omega-\omega_j(I_*)-\partial_I f_j\big)$ (with $\gamma_j'=D\Phi_j(0,\omega)$).

Step S5.25: $D\Phi_j$ bounded. $\mathcal{D}_j\times\mathbb{R}^n$ is open and $L_*(\check\Phi_j)\le\frac{11}{10}$ (Lemma 5.5(ii) argument), so $|S\,D\Phi_j S^{-1}|\le\frac{11}{10}$: $\|D\Phi_j\|\le\kappa:=\frac{11}{10}\max\{1,\frac{2\pi\hat\rho_0}{s_0},\frac{s_0}{2\pi\hat\rho_0}\}$, independent of j.

Step S5.26: The three error terms. (1) $\omega=\omega_j(\check\imath_j(I_0))$ (Lemma 5.2(iv)); $|I_*-\check\imath_j(I_0)|\le 20\hat\rho_j\hat E_j$ and the segment lies in $(D_j)_{\rho_j}$ where $\|U_j\|\le M_j$, so $|\omega-\omega_j(I_*)|\le 20M_j\hat\rho_j\hat E_j\le 7\gamma_1\hat E_j$, using $M_j\hat\rho_j\le\gamma_1/(3K_j^{\tau+1})\le\gamma_1/3$. (2) $I_*\in\mathrm{Re}(D_j)_{\rho_j/2}$, so the polydisc of radius $\rho_j/2$ about I_* is in $(D_j)_{\rho_j}$ and [K:`bp:p1:cauchy-open`] gives $|\partial_I f_j|\le 2E_j/\rho_j\le 2E_j/\hat\rho_j=2\hat E_j\hat\rho_j/L_j\le 2\hat E_j M_j\hat\rho_j\le\gamma_1\hat E_j$ ($L_j\ge M_j^{-1}$). (3) For real θ the polydisc of radius $\bar s/4\pi$ in θ lies in $\mathbb{T}_{\bar s/2}$, where $|f_j|\le E_j$ (as $\bar s/2\le\bar s+s_j$),

so $|\partial_\theta f_j| \le 4\pi E_j/\bar{s}$ ([K:bp:p1:cauchy-open]); and $E_j = \hat{E}_j\hat{\rho}_j^2/L_j \le \hat{E}_j\hat{\rho}_j M_j\hat{\rho}_j \le \hat{E}_j\gamma_1\hat{\rho}_0$. All three tend to 0, uniformly in (ϑ, t), as $\hat{E}_j \le \hat{E}_0^{2^j}$. Hence $\sup_t|\gamma_j' - X_H(\gamma_j)| \to 0$.

Step S5.27: Passing to the limit. $\delta_j := \sup_t|\gamma_j(t) - \gamma(t)| \to 0$ (Lemma 5.5; unscaled distances are comparable to scaled ones with fixed constants). Only nearby pairs are needed (review P3-08). The action of $\gamma(t)$ lies in $\mathrm{Re}(D_r) \subset \mathrm{Re}(D_{\rho/4})$ ($r = 32\hat{E}\hat{\rho} \le \rho/4$ since $\hat{E} < 2^{-13}$, $\hat{\rho} \le \rho$) and its angle is real; for j so large that $\delta_j < \min\{\rho/4, \bar{s}/4\pi\}$, the real segment $[\gamma_j(t), \gamma(t)]$ lies in the convex polydisc of radius δ_j about $\gamma(t)$, hence in $D_{\rho/2} \times \mathbb{T}_{\bar{s}/2}$. There $\|DX_H\| \le B_0$, where DX_H consists of $U + \partial_J^2 f$, $\partial_\theta\partial_J f$, $\partial_\theta^2 f$ and $B_0 = M + C(E, \rho, \bar{s})$ by [K:bp:p1:cauchy-open] (applied to $\sup|f| \le E$ on $D_\rho \times \mathbb{T}_{\bar{s}}$). By the mean value inequality, $|X_H(\gamma_j(t)) - X_H(\gamma(t))| \le B_0\delta_j \to 0$ uniformly in t. A uniform limit of C^1 functions $\gamma_j \to \gamma$ with $\gamma_j' \to X_H(\gamma)$ uniformly is C^1 with $\gamma' = X_H(\gamma)$. Since $\gamma(t) = \varphi_\omega(\vartheta + t\omega)$, this is the claim. (Conjugation identity of Step S5.24: [N:bpN:p5:symplectic], [N:bpN:p5:canonical].) □

5.4. **Measure estimate (CP10 Step 4).** Let $K_\Omega = \bigcup_{\omega\in\Omega_*} \varphi_\omega(\mathbb{R}^n)$; by definition $K_\Omega = \Phi(D_* \times \mathbb{R}^n)$ ($\check{\imath}$ maps $D_0 = D^{\mathrm{dio}}$ onto D_* and $\omega \mapsto I_0(\omega)$ is a bijection $\Omega_* \to D_0$). Here and below $N_\rho(X)$ is the *closed* real max-norm ρ-neighbourhood $\{x : \mathrm{dist}_\infty(x, X) \le \rho\}$; every radius used is $< r$, so $N_\rho(X) \subset \mathrm{Re}(X_r)$. Put $a := \mathrm{meas}(D \setminus D_0)$, $b := \mathrm{meas}(\mathrm{Re}(D_r) \setminus D)$, $\lambda \le \frac{1}{3}$ as in Lemma 5.2, $\eta := \sup|\check{\imath} - \mathrm{id}| \le 20\hat{E}_0\hat{\rho}_0$.

Lemma 5.9 (Bi-Lipschitz extension and covering)**.** *Let $X \subset \mathbb{R}^N$ be arbitrary, $g : X \to \mathbb{R}^N$ with $L(g) \le \lambda < 1$ in the max norm and $\sup|g| \le \eta$. Then there is a bijection $F = \mathrm{id} + g_e : \mathbb{R}^N \to \mathbb{R}^N$, $F|_X = \mathrm{id}+g$, with $(1-\lambda)|x-y| \le |F(x)-F(y)| \le (1+\lambda)|x-y|$, and for $y \in X$: $\mathrm{dist}_\infty(F^{-1}(y), X) \le |F^{-1}(y) - y| \le \eta/(1-\lambda)$ (a distance inequality, valid also for $\eta = 0$).*

Proof. **Step S5.28: Extension.** g_e: McShane [K:bp:p1:mcshane], componentwise, with the same λ for the max norm.

Step S5.29: Bijectivity. $|F(x) - F(y)| \ge |x - y| - \lambda|x - y|$ gives injectivity and the lower bound; for $y \in \mathbb{R}^N$ the map $x \mapsto y - g_e(x)$ is a λ-contraction ((X4)), with fixed point $x = F^{-1}(y)$.

Step S5.30: Location of preimages. For $y \in X$ and $x = F^{-1}(y)$: $|x - y| = |g_e(x)| \le |g_e(y)| + \lambda|x - y| \le \eta + \lambda|x - y|$ so $|x - y| \le \eta/(1 - \lambda)$. □

Lemma 5.10 (Measure of the complement of D_*)**.** $\mathrm{meas}(D_0 \setminus D_*) \le (4/3)^n(a + b)$ *and hence* $\mathrm{meas}(D \setminus D_*) \le a + (4/3)^n(a + b)$.

Proof. **Step S5.31: Extension and preimage.** Apply Lemma 5.9 with $X = D_0$, $g = \check{\imath} - \mathrm{id}$, $N = n$: $F = \check{\imath}_e$. Let $\rho_1 := \eta/(1 - \lambda) \le 30\hat{E}_0\hat{\rho}_0 < r$. If $y \in D_0 \setminus D_*$, then $x = F^{-1}(y) \in N_{\rho_1}(D_0)$ and $x \notin D_0$ (else $y = F(x) = \check{\imath}(x) \in D_*$). So $D_0 \setminus D_* \subset F\big(N_{\rho_1}(D_0) \setminus D_0\big)$.

Step S5.32: Measure. By [K:bp:p1:lip-image] (F is $(1 + \lambda)$-Lipschitz), $\mathrm{meas}(D_0 \setminus D_*) \le (1 + \lambda)^n \,\mathrm{meas}(N_{\rho_1}(D_0) \setminus D_0)$, $(1 + \lambda) \le \frac{4}{3}$. As $D_0 \subset D$: $N_{\rho_1}(D_0) \setminus D_0 \subset (N_{\rho_1}(D) \setminus D) \cup (D \setminus D_0)$, of measure $\le b + a$ ($\rho_1 < r$). The second claim: $D \setminus D_* \subset (D \setminus D_0) \cup (D_0 \setminus D_*)$. □

Lemma 5.11 (Measure of the complement of K_Ω over D)**.** $\mathrm{meas}\big((D \times \mathbb{T}^n) \setminus K_\Omega\big) \le \big[1 + (4/3)^n + 2^n\big(1 + (4/3)^n\big)\big](a + b)$.

Proof. Identify $\mathbb{T}^n$-measure with the unit cell; K_Ω is $\mathbb{Z}^n$-invariant in the angle.

Step S5.33: Splitting. $(D \times \mathbb{T}^n) \setminus K_\Omega \subset \big((D \setminus D_*) \times \mathbb{T}^n\big) \cup \big(((D_* \cap D) \times \mathbb{T}^n) \setminus K_\Omega\big)$. The first set has measure $\le a + (4/3)^n(a + b)$ by Lemma 5.10.

Step S5.34: Rescaled extension. Work in scaled coordinates $x \mapsto Sx$; S is linear, multiplies the Lebesgue measure of every subset of $\mathbb{R}^n_I \times \mathbb{R}^n_\theta$ by the constant $\hat{\rho}_0^{-n}(2\pi/s_0)^n$ and maps products

to products, so all comparisons below between measures of subsets of $\mathbb{R}^n \times \mathbb{R}^n$ are unaffected. By Lemma 5.5, $g = \check{\Phi} - \mathrm{id}$ on $X = S(D_* \times \mathbb{R}^n) \subset \mathbb{R}^{2n}$ has $L_*(g) \le \lambda' := 768 n \hat{E}_0 < \frac{1}{10}$ and $\sup|g| \le \eta' := \frac{3}{2}\hat{E}_0$ (Lemma 5.5(i), scaled). Lemma 5.9 ($N = 2n$) gives the bijection $\check{\Phi}_e = \mathrm{id} + g_e$ of $\mathbb{R}^{2n}$ extending $\check{\Phi}$ on X, $(1+\lambda')$-Lipschitz, with $|\check{\Phi}_e^{-1}(y) - y|_* \le \eta'/(1-\lambda') \le 2\hat{E}_0$ for $y \in X$.

Step S5.35: Preimage structure. Let $y = (I_y, \vartheta_y)$ with $I_y \in D_* \cap D$ and $\vartheta_y \in [0,1)^n$ be a point of the second set, i.e. $y \notin K_\Omega$ (mod $\mathbb{Z}^n$ in the angle), and let $x = (I_x, \vartheta_x) := \Phi_e^{-1}(y)$ (unscaled notation; $y \in D_* \times \mathbb{R}^n = S^{-1}X$). Then $|I_x - I_y| \le \rho_2 := 2\hat{E}_0\hat{\rho}_0$ and $|\vartheta_x - \vartheta_y| \le \hat{E}_0 s_0/\pi \le \hat{E}_0/(4\pi) =: \epsilon$ (as $s_0 \le \frac{1}{4}$). If $I_x \in D_*$, then $x \in D_* \times \mathbb{R}^n$, $\Phi_e(x) = \Phi(x) = y$ and $y \in K_\Omega$, a contradiction. Hence $I_x \in N_{\rho_2}(D_*) \setminus D_*$ and $\vartheta_x \in [-\epsilon, 1+\epsilon)^n$.

Step S5.36: Measure. Thus the second set is contained in $\Phi_e\big((N_{\rho_2}(D_*) \setminus D_*) \times [-\epsilon, 1+\epsilon)^n\big)$. Φ_e is $(1+\lambda')$-Lipschitz for $|\cdot|_*$ on $\mathbb{R}^{2n}$, so by [K:bp:p1:lip-image] in scaled coordinates (the scaling constants cancel on both sides) its measure is $\le (1+\lambda')^{2n}(1+2\epsilon)^n \,\mathrm{meas}(N_{\rho_2}(D_*) \setminus D_*) \le 2^n \,\mathrm{meas}(N_{\rho_2}(D_*) \setminus D_*)$, because $(1.1)^{2n}(1 + \hat{E}_0/2\pi)^n \le (1.21 \cdot 1.01)^n < 2^n$.

Step S5.37: The shell. $D_* \subset N_\eta(D_0)$, so $N_{\rho_2}(D_*) \subset N_{\rho_2+\eta}(D_0) \subset N_r(D)$ ($\rho_2 + \eta \le 22\hat{E}_0\hat{\rho}_0 < r$), and $N_{\rho_2}(D_*) \setminus D_* \subset (N_r(D) \setminus D) \cup (D_0 \setminus D_*) \cup (D \setminus D_0)$: if $x \in N_{\rho_2}(D_*) \setminus D_*$ then either $x \notin D$ (first set), or $x \in D \setminus D_0$, or $x \in D_0 \setminus D_*$. Its measure is $\le b + a + (4/3)^n(a+b) = (1 + (4/3)^n)(a+b)$ by Lemma 5.10.

Step S5.38: Sum. Total $\le [a + (4/3)^n(a+b)] + 2^n(1 + (4/3)^n)(a+b) \le [1 + (4/3)^n + 2^n(1 + (4/3)^n)](a+b)$. □

5.5. Proof of Proposition 4.3.

Proof of Proposition 4.3. **Step S5.39: Construction.** The sequence of Lemma 4.12 exists (hypotheses (S1)–(S3) at $j = 0$ from the proposition's hypotheses and the definition of D_0); Lemma 5.2 defines $\check{\imath}$, Lemma 5.5 defines Φ, and φ_ω is defined in §5.3.

Step S5.40: (a), (d). Lemmas 5.5, 5.6.

Step S5.41: (b), (c). Lemmas 5.8, 5.7.

Step S5.42: Injectivity and Lipschitz structure. If $\varphi_\omega(\vartheta) - \varphi_{\omega'}(\vartheta') = (0, k)$, then $\Phi(I_*, \vartheta) = \Phi(I'_*, \vartheta' + k)$ by periodicity (Lemma 5.5(iii)), so $(I_*, \vartheta) = (I'_*, \vartheta' + k)$ by (ii); $I_* = I'_*$ gives $I_0 = I'_0$ ($\check{\imath}$ injective), so $\omega = \omega_0(I_0) = \omega'$. The factorization and the constants: Lemma 5.2(iii) and Lemma 5.5(ii). For ω_0^{-1}: if $\omega, \omega' \in \omega_0(D)$ with $|\omega - \omega'| < \rho/\bar{M}$, then Lemma 4.6 (with $F = \omega_0$ on D_ρ, $\delta = \rho$, $x_0 = \omega_0^{-1}(\omega)$) gives $|\omega_0^{-1}\omega - \omega_0^{-1}\omega'| \le \bar{M}|\omega - \omega'|$; for far pairs $|\omega_0^{-1}\omega - \omega_0^{-1}\omega'| \le \mathrm{diam}\, D \le \mathrm{diam}\, D \cdot \frac{\bar{M}}{\rho}|\omega - \omega'|$.

Step S5.43: Measure. $K_\Omega \subset \mathrm{Re}(D_r) \times \mathbb{T}^n$ (Lemma 5.6), so $(\mathrm{Re}(D_r) \times \mathbb{T}^n) \setminus K_\Omega \subset ((\mathrm{Re}(D_r) \setminus D) \times \mathbb{T}^n) \cup ((D \times \mathbb{T}^n) \setminus K_\Omega)$ (an inclusion; the shell may meet K_Ω, which only helps), and Lemma 5.11 gives (3.1) with $c_n = 1 + [1 + (4/3)^n + 2^n(1 + (4/3)^n)] = 1 + (1 + 2^n)(1 + (4/3)^n)$ ($a = \mathrm{meas}(D \setminus D^{\mathrm{dio}})$, $b = \mathrm{meas}(\mathrm{Re}(D_r) \setminus D)$, $\mathrm{meas}\, \mathbb{T}^n = 1$). □

5.6. What Part 5 consumes.

Part 5 ([P:bpP:p5:kam-application]) applies Proposition 4.3 to the Hamiltonian $\check{H}$ of CP10 Step 6 (p. 560) after Part 2's averaging ([P:bpP:p2:averaged-remainder]). *Three radii must not be confused:* ϵ is the oscillator radius of CP10 (the quantity renamed r in the frozen cylinder $\mathcal{C}(V, r)$); $\hat{\rho}$ is the action radius of Definition 4.2; $r = 32\hat{E}\hat{\rho}$ is the enlargement radius of this proposition. The identification (CP10 p. 560, review P3-05): $n_1 = a$, $n_2 = b$, $h = H_0(J_1) + \mu\check{N}(J)$, $f = \mu\,\epsilon^5\check{P}$ (*not* $\mu^5\check{P}$), $\rho = \check{\rho}$, $s = \check{s}/5$, $\bar{s} = 4\check{s}/5$ (in coefficient-radius widths of §4.1), $D = D_{\mathrm{CP}}$ of CP10 (48).

(1) *Before* ϵ, μ (depend only on the fixed family): n, τ (hence $\hat{c}, c_n$), and a constant $C \geq 1$ such that the true bounds satisfy $\sup\|U\| \leq C$, $\sup\|\hat{U}\| \leq C\mu$, $\sup\|U^{-1}\| \leq C\mu^{-1}$, $\sup\|T_1\| \leq C$, $\sup\|T_2\| \leq C\mu^{-1}$, $\|f\| \leq C\mu\epsilon^5$ for $0 < \mu \leq 1$, $\epsilon \leq 1$ (CP10 p. 560; the N-norm of $\mu\epsilon^5\breve{P}$ is $\leq C\mu\epsilon^5$).

(2) *Choice of the numerical bounds (review R2-01).* Since $M, \hat{M}, \bar{M}, \bar{M}_i, E$ need only be *upper* bounds, take them *equal* to these values: $M = C$, $\hat{M} = C\mu$, $\bar{M} = \bar{M}_2 = C\mu^{-1}$, $\bar{M}_1 = C$, $E = C\mu\epsilon^5$ (the normalizations $\hat{M} \leq M$, $\bar{M}_i \leq \bar{M}$ hold for $\mu \leq 1$, $C \geq 1$). Then, exactly, $L = \max\{C\mu^{-1}, C^{-1}, (C\mu)^{-1}\} = C\mu^{-1}$ (because $C \geq C^{-1}$ gives $C\mu^{-1} \geq (C\mu)^{-1}$, and $C\mu^{-1} \geq C^{-1}$ as $\mu \leq 1$), $EL = C^2\epsilon^5$ and $x = EM^2L/\gamma_1^2 = C^4\epsilon^5/\gamma_1^2$, so $K = \frac{6}{s}\log_+\big(\gamma_1^2/(C^4\epsilon^5)\big)$ *exactly* (no inequality is inferred from an upper bound on x: $\log_+(1/x)$ is nonincreasing in x, so an upper bound on x would only give a lower bound on K). Whenever $\gamma_1^2/(C^4\epsilon^5) \geq e$ this is $K = \frac{6}{s}\log(\gamma_1^2/(C^4\epsilon^5)) \leq C'\log(\epsilon^5/\gamma_1^2)^{-1}$ (CP10 p. 560), with C' depending on s, C only. The smaller true E can only be used if one re-chooses the numerical E *before* computing K; Part 5 should keep the choice above. Then $\hat{\rho} = \min\{\gamma_1/(3CK^{\tau+1}), \gamma_2/(3C\mu K^{\tau+1}), \rho\}$, $\hat{E} = C^2\epsilon^5/\hat{\rho}^2$, and the KAM condition $\hat{c}\hat{E} < 1$ is CP10 (53)–(54), checked by Part 5.

(3) *Outputs used:* the tori φ_ω, $\omega \in \Omega_*$, with (a)–(d); the measure inequality (3.1) with the *absolute* constant c_n; $r = 32\hat{E}\hat{\rho} \leq C\hat{\rho}$ so that $r \leq C\bar{\gamma}_2$ as in CP10 p. 561; $|v - I^0|$, $|u|$ for the reconstruction (CP10 (51), (35)).

(4) *Frozen frequencies and Part 4.* $H = \tilde{H}/2\pi$ (Lemma 4.5); $\omega = \omega_0(I_0) = \omega_\mu/2\pi$ with ω_μ the frequency map of [P:`bpP:p4:freq`]. Hence $\omega \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ iff $\omega_\mu \in \mathcal{D}_{\alpha_1,\alpha_2,\tau}$ with $\alpha_i = 2\pi\gamma_i$ (that is the dictionary of [P:`bpP:p4:freq`]), and $D \setminus D^{\mathrm{dio}}$ of (3.1) is the set Part 4 estimates; (3.1) with the *same* $c_n = 1 + (1+2^n)(1+(4/3)^n)$ and $\operatorname{meas}\mathbb{T}^n = 1$ is exactly the right side consumed by [P:`bpP:p4:budget`], whose radius r_K is $r = 32\hat{E}\hat{\rho}$ here. Part 4 needs $\tau \geq n+1$; this is within the hypothesis $\tau > n$ of the proposition (take $\tau \geq n+1$ in Part 5; nothing else of Part 3 depends on τ beyond $\hat{c}$).

(5) *Conversions, with the maps named* ([P:`bpP:p1:order`], [P:`bpP:p1:cyl-pipeline`]; Lemma 4.5). The torus handed to the pipeline is the *unscaled* period-one torus $K_3 := \pi \circ \varphi_\omega : \mathbb{R}^n \to \mathrm{Ph}(n)$, $\pi(J, \theta) = (\theta, J)$ (frozen order); no factor 2π is applied to it. The 2π enters exactly once, in the pipeline's chart $\Psi = \Phi \circ S_{2\pi}$ (Φ the original reconstruction map of Part 5/Part 2): $K = \Psi \circ K_3 = \Phi \circ S_{2\pi} \circ \pi \circ \varphi_\omega$, with frozen frequency $w = \omega$ (CP's angular frequency is $2\pi w$), and the same w is used in both the pipeline and $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$. The map Q of Lemma 4.5 describes the same torus in CP coordinates, $\tilde{\varphi} = Q \circ \varphi_\omega(\cdot/2\pi) = S_{2\pi} \circ (\cdot)$ in Part 1's language; it must not be fed to the pipeline as K_3 (this would scale the angles twice). The output angle $\tilde{u} = 2\pi u$ is used only when estimates are quoted in CP's coordinates.

Not conclusions of Part 3 (obligations of Part 5): the complex neighbourhood and uniform bounds from the literal premises (H1)–(H6), the injectivity/nondegeneracy of ω_μ on a complex neighbourhood, the volume losses, the reconstruction of the oscillators and the seven clauses of [P:`bpP:p0:cyltorus`], measurability and whole-torus saturation of S, positivity and the strict inequalities.

Source gap: CP10 Prop. 3 claims *"a unique real-analytic embedding"*; translations $\vartheta \mapsto \vartheta + c$ preserve all stated properties, so uniqueness is neither proved nor claimed here. The Lipschitz continuity of $(\vartheta, \omega) \mapsto \varphi_\omega(\vartheta)$ passes through ω_0^{-1}, which is globally Lipschitz only for bounded D (locally, at scale $\rho/\bar{M}$, in general: separated action islands at j^2 with local frequency maps centred at j and Hessian 1 have inverse derivative 1 but no global Lipschitz inverse; review E9).

5.7. **Source gaps and suspected misprints (CP10, Prop. 3, Appendix B).** Dispositions follow the independent review (`reviews/p3-review-1.md`): V = verified misprint, R = proof repair or convention change (not an erratum), T = tentative.

(E1) **V.** p. 554: "$\hat{U}$ denote the $n \times n_1$ submatrix of U, i.e. the matrix with entries $\hat{U}_{ij} = U_{ij}$, for $n_1 + 1 \le i \le n$, $1 \le j \le n$": size and index range disagree; the proof needs $n_2 \times n$ (Remark 4.1).

(E2) **V.** p. 565 (Lemma B.1): "$M_0 := M$, $\hat{M}_0 := M$, $\bar{M}_0 := \bar{M}$"; $\hat{M}_0$ should be $\hat{M}$.

(E3) **V.** p. 566, B.1(i): "$\hat{\imath}_j : I \in D_{j-1} \mapsto \omega_{j-1} \circ \omega_{j-1}^{-1}(I) \in D_j$" is the identity; it should read $\omega_j^{-1} \circ \omega_{j-1}$ (cf. B.2, p. 568, $\hat{\imath}_+ = \omega_+^{-1} \circ \omega|_D$).

(E4) **V, with scope.** (74) uses $\bar{M}_i/\bar{M} \le 1$ without specifying compatible bounds (compatible bounds can be chosen: normalization, Remark 4.4); and (35), second line, carries a factor $\hat{M}/M$ for $|v_2 - I_2^0|$ that p. 572 (107) does not give (replacing it by 1 gives what is proved). No counterexample to a sharper estimate is claimed.

(E5) **T (not verified).** p. 551 assigns the 1-norm to the action, the angular sup metric, sup norms to p, q and the entrywise sup norm to matrices; the proof uses operator norms and mixed conversions. The blueprint's constants are a legitimate *convention change* (max norms, N-norm). No source constant is exhibited as false; the allegation that the printed constants "cannot all hold" is withdrawn.

(E6) **V (sharpening).** p. 569 (91) prints $\frac{8}{9}(4s)^{\tau+1}\hat{E}^2$; the computation gives exponent $2(\tau+1)$. The printed inequality is true (weaker).

(E7) **R.** pp. 569–570: the mean value step for ω^{-1} needs the frequency segment inside the domain of the inverse; and Lipschitz bounds on non-convex D, D_j need near/far pairs. Repaired by Lemmas 4.6, 4.7; composition domains stated (Lemma 5.3(i)).

(E8) **R (expository supplement).** Step 3 (pp. 572–573) does use (108) ("Using (108)"); the derivative bound on the open domain and analyticity on a fixed strip are not spelled out; supplied by Steps S5.25, S5.18 and Lemma 5.4. Not a missing estimate.

(E9) **V.** pp. 555, 573: "a unique real-analytic embedding" is not true as stated (translations); global Lipschitz dependence on ω fails for unbounded disconnected D (example above); bounded D suffices.

(E10) **R.** p. 573 footnote 32: Kirszbraun (Euclidean) is replaced by max-norm McShane and the cube-covering bound $\operatorname{meas} F(A) \le \Lambda^d \operatorname{meas} A$ ([K:`bp:p1:mcshane`], [K:`bp:p1:lip-image`]); c_n is recomputed, not reproduced; not an erratum.

(E11) **V (dependency).** Prop. 1 is proved in CP10 only through the unpinned [4, Lemma A.5] and a finite induction "as in [4]"; supplied by Part 2. The pointer to Arnold [3, p. 144] in Remark 3 is historical (Appendix B has its own induction) and is not a dependency.

Two further verified suspected defects found by the review (for the orchestrator's ledger, not used here): CP10 p. 565 (70) and the following split print "$k \in \Lambda$" where $k \notin \Lambda$ is meant (with $\Lambda = \{0\}$ the zero mode would be divided by zero); CP10 p. 564 (66) prints the target $W_{v,\sigma}$ where the angle width is s, i.e. $W_{v,s}$.

5.8. **References and imported facts.** Written parts cited: [P:`bpP:p1:period-convert`], [P:`bpP:p1:cyl-pipeline`], [P:`bpP:p2:avg`], [P:`bpP:p2:nfb`], [P:`bpP:p2:split`], [P:`bpP:p2:averaged-remainder`]; track N: [N:`bpN:p4:norm`], [N:`bpN:p5:symplectic`], [N:`bpN:p5:canonical`]. Forward reference to an unwritten part: [P:`bpP:p5:kam-application`]. Track K: [K:`bp:p1:cauchy-open`], [K:`bp:p1:uniform-limits`], [K:`bp:p1:mcshane`], [K:`bp:p1:lip-image`], [K:`bp:p1:ext-mvt`], [K:`bp:p1:ext-banach`], [K:`bp:p1:ext-fourier`], [K:`bp:p1:ext-rectangle`]. The only ingredient proved elsewhere is the averaging step (Lemma 4.8, proved in Part 2 from track N); the rest is proved here from the external facts (X1)–(X6). Formalization notes: the ball lemma needs a finite-dimensional path-continuation package; Lemma 5.4 is a bounded stand-alone task (no several-variable Vitali–Montel theorem is assumed).

6. Measure estimates for P-G

This part supplies the three measure inputs of the proof of [P:bpP:p0:pg]: the loss from the boundary collar of V, the loss from the non-Diophantine frequencies (CP10 Lemma 3.1, p. 562, proved in Appendix C, pp. 574–578), and the loss from the singular shell near the oscillator centres, together with the exact volume of the cylinder cell. Part 3 produces a set of whole tori K_Ω with the measure bound (3.1), whose right side contains $\mathrm{meas}(D \setminus D^{\mathrm{dio}})$ and $\mathrm{meas}(\mathrm{Re}(D_r) \setminus D)$ (Proposition 4.3); the lemmas below bound both quantities for the action domain $D = \bar{D}_t \times A$ of CP10 (48), and convert the bound to the relative loss in the frozen cell $\mathcal{F}(V, r)$ (Proposition 6.15). Part 5 chooses the free parameters.

Route. CP10 estimates the non-Diophantine set by covering the frequency image with p sets $U(I_1^{(i)})$ and slicing each one. For the frequency map of P-G the number p grows as the oscillator radius shrinks, so the constants c_1, c_2 of (58) are *not* independent of ε (Remark 6.10, errata (E1), (E2)). We replace the covering by a direct slicing argument. In the variables $y = \omega_{H_0}(I)$ (bi-Lipschitz coordinates, by the frozen hypothesis HasFastDiffeo) and ξ (oscillator actions), a resonance with $k_1 \neq 0$ is a perturbation of the linear form $k_1 \cdot y$ whose sections along y_j are short, as long as $\mu|k|$ is small; for $\mu|k|$ large the resonance is cut by the nondegenerate slow block β, and the sum over such k carries the factor $\mu^{\tau-n}$. This is the use of the full strength $\tau > n$.

Conventions. $n = a + b$, $n_1 = a$, $n_2 = b$; $|\cdot|$ is the max norm on $\mathbb{R}^m$, $|k|_1$ the ℓ^1 norm of $k \in \mathbb{Z}^m$; meas is Lebesgue outer measure (as in [K:bp:p1:ext-lebesgue]); sets to which Tonelli is applied are Borel. $\mathcal{D}_{\alpha_1,\alpha_2,\tau}$ is the two-scale set of Definition 2.14. The frozen radius of the cylinder is r (Definition 1.10); *Part 3's enlargement radius is written* r_K ($= 32\hat{E}\hat{\rho}$ there). Oscillator actions are $\xi_j = \frac{1}{2}(p_j^2 + q_j^2)$, so the frozen disc condition is $\xi_j < r^2/2$.

6.1. External facts.

(X1) *Tonelli.* If $R \subset \mathbb{R}^N$ is Borel, $j \leq N$, every section $\{s \in \mathbb{R} : (x', s) \in R\}$ ($x' \in \mathbb{R}^{N-1}$, other coordinates fixed) has outer measure $\leq \delta$, and the projection of R forgetting the j-th coordinate lies in a set of measure $\leq P$, then $\mathrm{meas}\, R \leq \delta P$. A set of diameter $\leq \delta$ in $\mathbb{R}$ has outer measure $\leq \delta$.

(X2) Mean value inequality on convex sets ([K:bp:p1:ext-mvt]); Lipschitz images ([K:bp:p1:lip-image]); Borel images of Borel sets under injective continuous maps ((X4)).

(X3) The derivative of a real analytic function on an open set is real analytic (hence C^1); a C^1 function is continuously differentiable and locally Lipschitz.

(X4) A connected set meeting both V and its complement meets ∂V.

6.2. The boundary collar.

Let V, A, d be as in [P:bpP:p0:pg] ($V \subset \mathbb{R}^a$ open, connected, bounded; $\mathrm{meas}\{I : \exists J \in \partial V,\ |I - J| < t\} \leq At$ for $0 < t < d$). Then $0 < \mathrm{meas}\, V < \infty$ and $\partial V \neq \emptyset$ ($V \neq \emptyset$ is open and not $\mathbb{R}^a$, which is connected). For $t > 0$ put

$$\bar{D}_t := \{I \in V :\ \mathrm{dist}(I, \partial V) \geq t\}, \qquad N_s(X) := \{x \in \mathbb{R}^a : \exists J \in X,\ |x - J| < s\}.$$

Lemma 6.1 (Collar). *(a) $\bar{D}_t$ is compact, $\bar{D}_t \subset V$, and $V \setminus \bar{D}_t = \{I \in V : \mathrm{dist}(I, \partial V) < t\}$.*
(b) If $0 < t < d$ then $\mathrm{meas}(V \setminus \bar{D}_t) \leq At$; hence $\mathrm{meas}\, \bar{D}_t \geq \mathrm{meas}\, V - At$.
(c) If $0 < s < t$ then $N_s(\bar{D}_t) \subset V$, and $N_s(\bar{D}_t) \setminus \bar{D}_t \subset V \setminus \bar{D}_t$; so for $0 < s < t < d$, $\mathrm{meas}(N_s(\bar{D}_t) \setminus \bar{D}_t) \leq At$.

Source: Ours. The frozen hypothesis `hboundary` *replaces CP10 p. 561's unproved collar estimate "$\mathrm{meas}(\mathrm{Re}(V_r) \setminus V) \leq C\bar{\gamma}_2\, \mathrm{meas}\, V$" (audit P3, D-PG-5, errata (E3)); only the* inner *side is needed because every neighbourhood below stays inside V.*

Proof. **Step S6.1: (a) Compact.** $\bar{D}_t$ is bounded. If $J_m \in \bar{D}_t$, $J_m \to J$, then $J \in \overline{V}$ and $\mathrm{dist}(J, \partial V) \geq t > 0$ by continuity of the distance, so $J \notin \partial V$, i.e. $J \in V$ (V open): closed. The description of $V \setminus \bar{D}_t$ is the definition.

Step S6.2: (b) Measure. If $I \in V \setminus \bar{D}_t$ then $\operatorname{dist}(I, \partial V) < t$, so there is $J \in \partial V$ with $|I - J| < t$: $V \setminus \bar{D}_t$ lies in the set of the hypothesis. Then $\operatorname{meas} \bar{D}_t = \operatorname{meas} V - \operatorname{meas}(V \setminus \bar{D}_t)$.

Step S6.3: (c) Neighbourhoods. Let $J \in \bar{D}_t$, $B = \{x : |x - J| < s\}$, $s < t$. B is convex, hence connected, contains $J \in V$ and misses ∂V; by (X4) it misses V^c, so $B \subset V$. If $x \in N_s(\bar{D}_t) \setminus \bar{D}_t$ then $x \in V$ and $x \notin \bar{D}_t$, which is $x \in V \setminus \bar{D}_t$. □

6.3. **Uniform Lipschitz facts.**

Lemma 6.2 (Uniform Lipschitz bounds on compact sets)**.** *Let $\Theta \subset \mathbb{R}^p$ and $K \subset \mathbb{R}^m$ be compact, $O \subset \mathbb{R}^p \times \mathbb{R}^m$ open with $\Theta \times K \subset O$, and $f : O \to \mathbb{R}^q$ continuous with $x \mapsto f(\theta, x)$ of class C^1 and $D_x f$ jointly continuous on O ($p = 0$ allowed). Then there is Λ with $|f(\theta, x) - f(\theta, x')| \le \Lambda |x - x'|$ for $\theta \in \Theta$, $x, x' \in K$.*

Proof. **Step S6.4: Contradiction.** If not, there are $\theta_m, x_m \ne x'_m$ with $|f(\theta_m, x_m) - f(\theta_m, x'_m)| > m|x_m - x'_m|$. Since f is bounded on $\Theta \times K$, $|x_m - x'_m| \to 0$; pass to a subsequence with $\theta_m \to \theta^*$, $x_m, x'_m \to x^*$.

Step S6.5: Local bound. Choose $\delta > 0$ with $\overline{B_\delta(\theta^*)} \times \overline{B_\delta(x^*)} \subset O$ and $M_* := \sup \|D_x f\|$ over it (finite, continuity). For large m the segment $[x_m, x'_m]$ lies in $B_\delta(x^*)$ and $\theta_m \in B_\delta(\theta^*)$, so (X2) gives $|f(\theta_m, x_m) - f(\theta_m, x'_m)| \le M_* |x_m - x'_m|$, contradicting the choice for $m > M_*$. □

Lemma 6.3 (Lipschitz inverse of the fast frequency)**.** *Assume* HasFastDiffeo(H_0, W) *(Def. 1.13) and $\overline{V} \subset W$ with V bounded. Write $\omega_{H_0} =$ `fastFrequency` H_0, g the frozen inverse. Then ω_{H_0} is continuous on W; $Y_0 := \omega_{H_0}(\overline{V})$ is compact and contained in the open set $\omega_{H_0}(W)$; $g(\omega_{H_0}(I)) = I$ on W; and there is $L_g < \infty$ with $|g(y) - g(y')| \le L_g |y - y'|$ for $y, y' \in Y_0$. Put $R_y := \max_{Y_0} |y|$ and $\Lambda_g := \max\{1, L_g\}$. These depend only on (H_0, V, W).*

Source: Ours; this is the "Lipschitz inverse of ω_0" used in CP10 Lemma 3.1 ("extended to a diffeomorphism"), made explicit from the frozen C^1 inverse.

Proof. **Step S6.6: Continuity and compactness.** H_0 is analytic on W, so ω_{H_0} is analytic on W ((X3)); $\overline{V}$ is compact (bounded, closed), so Y_0 is compact, and it lies in $\omega_{H_0}(W)$, which is open by the frozen clause (ii).

Step S6.7: Lipschitz. g is C^1 on the open set $\omega_{H_0}(W) \supset Y_0$ (frozen clause (iii)): apply Lemma 6.2 with $p = 0$. □

Lemma 6.4 (Uniform inverse of the torsion block)**.** *Let $\beta(\mu, I)$ be symmetric $b \times b$ with $|\det \beta(\mu, I)| \ge B > 0$ and entries bounded by M_β for $\mu \in (0, \mu_0]$, $I \in V$. Then $|\nu| \le L_\beta |\beta(\mu, I)^\top \nu|$ for all $\nu \in \mathbb{R}^b$, with $L_\beta := b!\, M_\beta^{b-1}/B$.*

Proof. Cramer ([K:`bp:p1:ext-cramer`]): $(\beta^\top)^{-1} = \operatorname{adj}(\beta^\top)/\det \beta$; entries of the adjugate are $(b-1) \times (b-1)$ minors, bounded by $(b-1)! M_\beta^{b-1}$ (Leibniz); so the row-sum operator norm is $\le b! M_\beta^{b-1}/B$. □

The boundedness $M_\beta < \infty$ comes from joint analyticity of the family (Part 5, [P:`bpP:p5:uniform-remainder`]), which also gives the regularity (N) below by Lemma 6.2.

6.4. **The two-scale Diophantine measure (CP10 Lemma 3.1).**

Definition 6.5 (Frequency map of the averaged system)**.** Let $N_\mu(I, \xi) = P_0(\mu, I) + \Omega(\mu, I) \cdot \xi + \frac{1}{2} \xi \cdot \beta(\mu, I) \xi$ ($I \in \mathbb{R}^a$, $\xi \in \mathbb{R}^b$), assume $P_0(\mu, \cdot)$, $\Omega(\mu, \cdot)$, $\beta(\mu, \cdot)$ are C^1 near $\bar{D}$, and put $G_\mu := \nabla_I N_\mu \in \mathbb{R}^a$, $\omega_\mu^{\mathrm{s}}(I, \xi) := \Omega(\mu, I) + \beta(\mu, I)\xi = \nabla_\xi N_\mu \in \mathbb{R}^b$. For $\mu > 0$ the *frequency map* (CP angular convention) is

$$\omega_\mu(I, \xi) := \big(\omega_{H_0}(I) + \mu\, G_\mu(I, \xi),\ \mu\, \omega_\mu^{\mathrm{s}}(I, \xi)\big) = \nabla_{(I,\xi)} \big(H_0 + \mu N_\mu\big).$$

Dictionary with Part 3. Part 3's $h = (H_0 + \mu N_\mu)/2\pi$ and $\omega_0 = \partial h = \omega_\mu/2\pi$ (Lemma 4.5); since $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ is a cone condition, $\omega_\mu/2\pi \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ iff $\omega_\mu \in \mathcal{D}_{2\pi\gamma_1,2\pi\gamma_2,\tau}$ (Lemma 2.15(c) for each line of the definition). So with $\alpha_i = 2\pi\gamma_i$: $D \setminus D^{\mathrm{dio}} = \{J \in D : \omega_\mu(J) \notin \mathcal{D}_{\alpha_1,\alpha_2,\tau}\}$.

Definition 6.6 (Standing data (S4))**.** $\tau > n$. $\bar{D} \subset V$ compact with $\operatorname{meas}\bar{D} > 0$. $\lambda_0, \mu_1 > 0$, and constants $\Lambda_N \geq 1$, $L_\beta \geq 1$ such that for $\mu \in (0, \mu_1]$, $I, I' \in \bar{D}$, $|\xi|, |\xi'| \leq \lambda_0$:

(N1) $|G_\mu(I,\xi) - G_\mu(I',\xi)| \leq \Lambda_N |I - I'|$ and $|\omega^{\mathrm{s}}_\mu(I,\xi) - \omega^{\mathrm{s}}_\mu(I',\xi)| \leq \Lambda_N |I - I'|$;
(N2) $|G_\mu(I,\xi) - G_\mu(I,\xi')| \leq \Lambda_N |\xi - \xi'|$;
(N3) $|\nu| \leq L_\beta |\beta(\mu, I)^\top \nu|$ for $\nu \in \mathbb{R}^b$ (Lemma 6.4).

Constants of Lemma 6.3: R_y, L_g, Λ_g. Put $M := 2b\Lambda_N L_\beta$ and require $\mu \leq \mu_1 \leq \big(2(1+M)a\Lambda_N\Lambda_g\big)^{-1}$.

Lemma 6.7 (Slab with offset)**.** *For $\ell \in \mathbb{R}^N \setminus \{0\}$, $c \in \mathbb{R}$, $\lambda, a' > 0$: $\operatorname{meas}\{x \in [-\lambda,\lambda]^N : |\ell \cdot x + c| < a'\} \leq 2a'(2\lambda)^{N-1}/|\ell|$.*

Proof. Take i with $|\ell_i| = |\ell|$. Each section in x_i lies in an interval of length $2a'/|\ell_i|$; apply (X1) (cf. [K:bp:p4:slab]). □

Lemma 6.8 (Lattice sums)**.** *(a) $\sum_{k\in\mathbb{Z}^n\setminus 0} |k|_1^{-\tau} \leq S_1 := 2\cdot 3^{n-1}\frac{\tau+1-n}{\tau-n}$ ($\tau > n$); $\sum_{k_2\in\mathbb{Z}^b\setminus 0}|k_2|_1^{-\tau-1} \leq S_2 := 2\cdot 3^{b-1}\frac{\tau+2-b}{\tau+1-b}$. (b) For real $m \geq 1$: $\sum_{|k|_1>m}|k|_1^{-\tau-1} \leq C_T(2/m)^{\tau+1-n}$, $C_T := 2\cdot 3^{n-1}/(\tau+1-n)$.*

Proof. **Step S6.8: (a).** [K:bp:p4:lattice-sum] with $(n, \tau-1)$ (valid as $\tau - 1 > n - 1$) and with (b, τ).

Step S6.9: (b). By [K:bp:p4:lattice-count] the sum is $\leq 2\cdot 3^{n-1}\sum_{\ell>m}\ell^{-p}$, $p = \tau + 2 - n > 1$. By convexity $\ell^{-p} \leq \int_{\ell-1/2}^{\ell+1/2} x^{-p}dx$, and $\ell > m \geq 1$ gives $\ell - \frac{1}{2} \geq m/2$: the sum is $\leq \int_{m/2}^{\infty} x^{-p}dx = (m/2)^{1-p}/(p-1)$. □

Proposition 6.9 (Two-scale Diophantine measure)**.** *Assume (S4), $0 < \lambda \leq \lambda_0$, $Q \subset (-\lambda,\lambda)^b$ Borel, $\mu \in (0, \mu_1]$, $\alpha_1, \alpha_2 > 0$, and set*

$$\mathcal{R} := \{(I,\xi) \in \bar{D} \times Q :\ \omega_\mu(I,\xi) \notin \mathcal{D}_{\alpha_1,\alpha_2,\tau}\}, \qquad \bar{\gamma} := \alpha_2/\mu.$$

Then

$$\operatorname{meas}\mathcal{R} \leq \Big[c_1\alpha_1\Big(1 + \frac{\mu^{\tau-n}}{\lambda}\Big) + c_2\frac{\bar{\gamma}}{\lambda}\Big]\operatorname{meas}(\bar{D})\,(2\lambda)^b,$$

$$c_1 := \frac{\Lambda_g^n}{\operatorname{meas}\bar{D}}\max\Big\{4(2R_y)^{a-1}S_1,\ 4b^2L_\beta(2R_y)^a C_T(4\Lambda_N\Lambda_g)^{\tau+1-n}\Big\}, \qquad c_2 := bL_\beta S_2.$$

The constants depend on $(a, b, \tau, H_0, V, W, \Lambda_N, L_\beta, \bar{D})$ only, not on $\mu, \lambda, Q, \alpha_i$.
Source: CP10 Lemma 3.1, p. 562, proof pp. 574–577, in the form needed for (59). The statement and proof are not *CP10's: see Remark 6.10 and errata (E1), (E2). The constant $c(n,\tau) = \sum|k|^{-\tau}$ of (58) is S_1-type; the factor $\bar{\gamma}/\lambda = \gamma_2/(\mu\lambda)$ is CP10's $\gamma_2/\hat{r}$.*

Proof. For $k = (k_1, k_2) \neq 0$ put $F_k := k \cdot \omega_\mu = k_1 \cdot \omega_{H_0}(I) + \mu\big[k_1 \cdot G_\mu + k_2 \cdot \omega^{\mathrm{s}}_\mu\big]$ and $\alpha_k := \alpha_1|k|_1^{-\tau}$ if $k_1 \neq 0$, $\alpha_k := \alpha_2|k|_1^{-\tau}$ if $k_1 = 0$. Then $\mathcal{R} = \bigcup_k \mathcal{R}_k$, $\mathcal{R}_k = \{(I,\xi) \in \bar{D} \times Q : |F_k| < \alpha_k\}$ (Borel: F_k is continuous by (N1)–(N2) and Lemma 6.3). Split $\mathcal{R} = \mathcal{R}^{(0)} \cup \mathcal{R}^{(A)} \cup \mathcal{R}^{(B)}$: $k_1 = 0$; $k_1 \neq 0$ in $\mathcal{K}_A$; $k_1 \neq 0$ in $\mathcal{K}_B$, where

$$\mathcal{K}_A := \{k_1 \neq 0 :\ \mu\Lambda_N\Lambda_g|k|_1 \leq |k_1|_\infty/2\}, \qquad \mathcal{K}_B := \{k_1 \neq 0\} \setminus \mathcal{K}_A.$$

(Here $|k_1|_\infty$ is the max norm of the block $k_1 \in \mathbb{Z}^a$, $|k_1|_1$ its ℓ^1 norm, and $|k|_1$ the ℓ^1 norm of k.)

Step S6.10: Case $k_1 = 0$ (slow resonances). Here $F_k = \mu k_2 \cdot (\Omega + \beta\xi)$, $|k|_1 = |k_2|_1$. For fixed I, $\{\xi \in [-\lambda,\lambda]^b : |(\beta^\top k_2)\cdot\xi + k_2\cdot\Omega| < \bar{\gamma}|k_2|_1^{-\tau}\}$ has measure $\leq 2\bar{\gamma}|k_2|_1^{-\tau}(2\lambda)^{b-1}/|\beta^\top k_2|$ by Lemma 6.7 ($\beta^\top k_2 \neq 0$ by (N3)). By (N3) and $|k_2| \geq |k_2|_1/b$, $|\beta^\top k_2| \geq |k_2|_1/(bL_\beta)$. Tonelli in I gives $\operatorname{meas}\mathcal{R}_k \leq$

$\operatorname{meas}\bar D\, 2\bar\gamma b L_\beta (2\lambda)^{b-1}|k_2|_1^{-\tau-1}$, and by Lemma 6.8(a) $\operatorname{meas}\mathcal{R}^{(0)} \le \operatorname{meas}\bar D (2\lambda)^b\, bL_\beta S_2\,\bar\gamma/\lambda$. (No injectivity of ω_μ is used; μ cancels because the threshold is $\mu\bar\gamma$.)

Step S6.11: Change to frequency coordinates. Let $\Phi(I,\xi) = (\omega_{H_0}(I),\xi)$, injective and continuous on $\bar D \times [-\lambda,\lambda]^b$ with inverse $(y,\xi)\mapsto(g(y),\xi)$, which is Λ_g-Lipschitz on $Y_0\times\mathbb{R}^b$ (Lemma 6.3, max norm). Put $Y = \omega_{H_0}(\bar D)\subset[-R_y,R_y]^a$, $\mathcal{R}'_k = \Phi(\mathcal{R}_k)$ (Borel, (X2)). By [K:bp:p1:lip-image] in $\mathbb{R}^n$: $\operatorname{meas}\mathcal{R}_k \le \Lambda_g^n \operatorname{meas}\mathcal{R}'_k$. On $\mathcal{R}'_k$ the function is $\tilde F_k(y,\xi) = k_1\cdot y + \mu\tilde G_k(y,\xi)$, $\tilde G_k := k_1\cdot G_\mu(g(y),\xi) + k_2\cdot\omega_\mu^{\mathrm{s}}(g(y),\xi)$, and by (N1) and Lemma 6.3

$$|\tilde G_k(y,\xi)-\tilde G_k(y',\xi)| \le \Lambda_N\Lambda_g|k|_1|y-y'| \quad (y,y'\in Y).$$

Step S6.12: Case $\mathcal{K}_A$ (slicing in y_j). Let $k\in\mathcal{K}_A$ and j with $|k_{1,j}| = |k_1|_\infty \ge 1$. For two points $(y+se_j,\xi)$, (y,ξ) of $\mathcal{R}'_k$: $|\tilde F_k(y+se_j,\xi)-\tilde F_k(y,\xi)| \ge |k_{1,j}||s| - \mu\Lambda_N\Lambda_g|k|_1|s| \ge |k_{1,j}||s|/2$. Both values have modulus $<\alpha_k$, hence $|s|<4\alpha_k/|k_{1,j}|\le 4\alpha_k$: every y_j-section of $\mathcal{R}'_k$ has diameter $\le 4\alpha_k$ (no convexity of $\bar D$ is needed). The projection forgetting y_j lies in $[-R_y,R_y]^{a-1}\times[-\lambda,\lambda]^b$. By (X1), $\operatorname{meas}\mathcal{R}'_k \le 4\alpha_1|k|_1^{-\tau}(2R_y)^{a-1}(2\lambda)^b$, and summing (Lemma 6.8(a)), $\operatorname{meas}\mathcal{R}^{(A)} \le \Lambda_g^n\, 4S_1(2R_y)^{a-1}\alpha_1(2\lambda)^b$.

Step S6.13: Structure of $\mathcal{K}_B$. Let $k\in\mathcal{K}_B$. (i) $|k|_1 > K_\mu := (2\mu\Lambda_N\Lambda_g)^{-1}$, because $\mu\Lambda_N\Lambda_g|k|_1 > |k_1|_\infty/2 \ge \frac12$; and $K_\mu\ge 1$ since $\mu\le\mu_1$. (ii) $|k_2|_1 > M|k_1|_1$: otherwise $|k|_1\le(1+M)|k_1|_1\le(1+M)a|k_1|_\infty$ and $\mu\Lambda_N\Lambda_g|k|_1\le|k_1|_\infty/2$ by $\mu\le\mu_1$, i.e. $k\in\mathcal{K}_A$. Consequently $|k|_1<(1+M^{-1})|k_2|_1\le 2b|k_2|_\infty$ and $\Lambda_N|k_1|_1\le\Lambda_N b|k_2|_\infty/M = |k_2|_\infty/(2L_\beta)$.

Step S6.14: Case $\mathcal{K}_B$ (slicing in ξ_j). Fix $k\in\mathcal{K}_B$ and $j\le b$; let $\mathcal{R}_{k,j}$ be the part of $\mathcal{R}'_k$ with $|(\beta(\mu,g(y))^\top k_2)_j| \ge |k_2|_\infty/L_\beta$; by (N3) $\mathcal{R}'_k = \bigcup_{j\le b}\mathcal{R}_{k,j}$ (take j maximizing $|(\beta^\top k_2)_j|$). Along $\xi\mapsto\xi+se_j$ (y and the other ξ's fixed, so $I=g(y)$ fixed)

$$\tilde F_k(y,\xi+se_j)-\tilde F_k(y,\xi) = \mu s\,(\beta^\top k_2)_j + \mu\, k_1\cdot\big(G_\mu(I,\xi+se_j)-G_\mu(I,\xi)\big),$$

because ω_μ^{s} is affine in ξ. By (N2) and S6.13 the modulus is $\ge \mu|s|\big(|k_2|_\infty/L_\beta - \Lambda_N|k_1|_1\big) \ge \mu|s||k_2|_\infty/(2L_\beta)$. Two points of $\mathcal{R}_{k,j}$ on one line thus satisfy $|s| < 4L_\beta\alpha_k/(\mu|k_2|_\infty) \le 8bL_\beta\alpha_1|k|_1^{-\tau-1}/\mu$. The projection forgetting ξ_j lies in $[-R_y,R_y]^a\times[-\lambda,\lambda]^{b-1}$. Summing over $j\le b$, then over $|k|_1>K_\mu$, with Lemma 6.8(b):

$$\operatorname{meas}\mathcal{R}^{(B)} \le \Lambda_g^n\, b\,(2R_y)^a(2\lambda)^{b-1}\frac{8bL_\beta\alpha_1}{\mu}C_T\big(2/K_\mu\big)^{\tau+1-n} = \Lambda_g^n(2\lambda)^b\,4b^2L_\beta(2R_y)^aC_T(4\Lambda_N\Lambda_g)^{\tau+1-n}\alpha_1\frac{\mu^{\tau-n}}{\lambda}.$$

(here $2/K_\mu = 4\mu\Lambda_N\Lambda_g$.)

Step S6.15: Sum. Add the three bounds; each of the Λ_g^n-terms is $\le c_1\operatorname{meas}\bar D\,\alpha_1(2\lambda)^b$ times 1, resp. $\mu^{\tau-n}/\lambda$. □

Remark 6.10 (What is and is not claimed; status of CP10 Lemma 3.1). CP10 Lemma 3.1 (p. 562) states $\operatorname{meas}\mathcal{R} \le (c_1\gamma_1 + c_2\gamma_2/\hat r)\operatorname{meas}(\bar D\times B_{\hat r})$, with c_1 containing an integer p, "for a suitable integer p depending on $\bar D$ and ω_1", the number of sets $U(I_1^{(i)})$ of (111) needed to cover $\omega(\bar D\times B_{\hat r})$. The lemma is true for each $\hat r$, but $p=p(\hat r)$, and in general $p\gtrsim\hat r^{-1}$: for $n_1=n_2=1$, $\bar D=[0,1]$, $\omega(I_1,I_2)=(I_1,I_1+I_2)$ (a diffeomorphism, $\beta=1$), the resonance $k=(1,-1)$ gives $|k\cdot\omega| = |I_2| < \hat r$, so the whole domain is resonant as soon as $\gamma_1\ge 2^\tau\hat r$ (CP10's $|k|=|k|_1$) while γ_2 is arbitrary small: the lemma forces $c_1\ge 2^{-\tau}\hat r^{-1}$. CP10 then applies it with $\hat r = \check c_2\varepsilon^2$ and asserts "c_i independent of ε and μ" (p. 562, after (59)). That is false for the abstract lemma; Proposition 6.9 proves the needed estimate directly, using the *structure* of the frequency map (the fast Hessian is invertible because ω_{H_0} is bi-Lipschitz, and the slow block is $\mu(\Omega+\beta\xi)$ with β nondegenerate), and so the covering lemma is not used. The counting lemmas of track K are reused as follows: [K:bp:p4:lattice-count], [K:bp:p4:lattice-sum] (Lemma 6.8); [K:bp:p4:slab] (model of Lemma 6.7, which needs a real

normal and an offset); [K:bp:p4:preimage-measure] is *not* reusable (its hypotheses are the pull-back by the inverse frequency map g of a full-dimensional single-scale setting with one threshold; it is replaced by Step S6.11, which is the same Lipschitz-image argument with the product map $(y,\xi)\mapsto(g(y),\xi)$); [K:bp:p4:cover-mono] concerns the specific cover `kamCover` of track K and [K:bp:p4:dyadic-approx] serves HEAD-Q (no boundary hypothesis): neither is needed since `hboundary` is a hypothesis of P-G.

Remark 6.11 (The hypothesis $\tau>n$ and the factor $\mu^{\tau-n}$). The sums $\sum|k|_1^{-\tau}$ and $\sum|k|_1^{-\tau-1}$ converge iff $\tau>n$ and $\tau>n-1$ resp. The extra term $c_1\alpha_1\mu^{\tau-n}/\lambda$ is harmless when $\mu^{\tau-n}\le\lambda$; since $\mu<r^6$ and $\lambda=r^2/2$, taking $\tau\ge n+1$ gives $\mu^{\tau-n}/\lambda\le 2r^4$. Part 5 chooses τ (the frozen theorem only needs some $\tau>a+b$, Lemma 2.16).

6.5. **Collar of the domain $D=\bar D_t\times A$.** Fix $0<\lambda_1<\lambda_2$ and $A_{\mathrm{sh}}:=(\lambda_1,\lambda_2)^b$ (the oscillator shell $\check c_1\varepsilon^{5/2}<J_{2,i}<\check c_2\varepsilon^2$ of (47); $\lambda_2=r^2/2$ for the frozen radius r), $\ell:=\lambda_2-\lambda_1$, and

$$D:=\bar D_t\times A_{\mathrm{sh}}\subset\mathbb{R}^a\times\mathbb{R}^b.$$

Lemma 6.12 (Enlargement of D). *Let $0<t<d$ and $0<r_K<\min\{t,\lambda_1\}$. Then $\mathrm{Re}(D_{r_K})=N_{r_K}(\bar D_t)\times(\lambda_1-r_K,\lambda_2+r_K)^b\subset V\times(0,\infty)^b$, and*

$$\mathrm{meas}\big(\mathrm{Re}(D_{r_K})\setminus D\big)\le\mathrm{meas}(D)\Big[\frac{At}{\mathrm{meas}\,\bar D_t}\Big(1+\frac{2r_K}{\ell}\Big)^b+\frac{2br_K}{\ell}\Big(1+\frac{2r_K}{\ell}\Big)^{b-1}\Big]=:\mathrm{meas}(D)\,\eta_2.$$

Proof. **Step S6.16: Product structure.** For the max norm, $\mathrm{dist}(x,X\times Y)=\max(\mathrm{dist}(x_1,X),\mathrm{dist}(x_2,Y))$, so the real neighbourhood of a product is the product of the real neighbourhoods; $N_{r_K}(\bar D_t)\subset V$ by Lemma 6.1(c) and $\lambda_1-r_K>0$.

Step S6.17: Difference. $(X'\times Y')\setminus(X\times Y)\subset((X'\setminus X)\times Y')\cup(X\times(Y'\setminus Y))$. With $\mathrm{meas}(N_{r_K}(\bar D_t)\setminus\bar D_t)\le At$ (Lemma 6.1(c)), $\mathrm{meas}\,Y'=(\ell+2r_K)^b$ and $(\ell+2r_K)^b-\ell^b\le 2br_K(\ell+2r_K)^{b-1}$ (mean value), divide by $\mathrm{meas}\,D=\mathrm{meas}\,\bar D_t\,\ell^b$. □

6.6. **The cylinder cell and the singular shell.** Define the action–angle chart of the oscillators in the frozen coordinates $z=((\varphi,q),(I,p))$, $\varphi\in\mathbb{R}^a$, $I\in\mathbb{R}^a$, $(p_j,q_j)\in\mathbb{R}^2$:

$$\Pi(I,\xi;\varphi,\psi):=\big(\varphi,\,q=\sqrt{2\xi}\sin\psi\big),\ \big(I,\ p=\sqrt{2\xi}\cos\psi\big),\qquad (I,\xi)\in\mathbb{R}^a\times(0,\infty)^b,\ \varphi\in\mathbb{R}^a,\ \psi\in\mathbb{R}^b,$$

and, for a period-one angle $\vartheta=(\vartheta_f,\vartheta_s)\in[0,1)^n$, $\mathsf{E}(I,\xi,\vartheta):=\Pi(I,\xi;2\pi\vartheta_f,2\pi\vartheta_s)$ (the dictionary of Lemma 4.5 with Q; Part 3's $(J,\vartheta)=((I,\xi),\vartheta)$).

Lemma 6.13 (Cell and polar coordinates). *(a) Π is real analytic and injective on $\mathbb{R}^a\times(0,\infty)^b\times\mathbb{R}^a\times[0,2\pi)^b$ modulo $\varphi\mapsto\varphi+2\pi k$; $|\det D\Pi|=1$. Hence E is injective on $\mathbb{R}^a\times(0,\infty)^b\times[0,1)^n$ and $\mathrm{meas}\,\mathsf{E}(X)=(2\pi)^n\,\mathrm{meas}\,X$ for measurable X there. (b) $\mathcal{F}(V,r)\setminus\mathsf{E}(V\times(0,r^2/2)^b\times[0,1)^n)$ is null, and*

$$\mathrm{meas}\,\mathcal{F}(V,r)=(2\pi)^a\,\mathrm{meas}\,V\,(\pi r^2)^b=(2\pi)^n\,\mathrm{meas}\,V\,(r^2/2)^b.$$

(c) (Singular shell.) For $0<\lambda_1<\lambda_2=r^2/2$, the set of $z\in\mathcal{F}(V,r)$ with $\xi_j\le\lambda_1$ for some j has measure $\le b\frac{\lambda_1}{\lambda_2}\,\mathrm{meas}\,\mathcal{F}(V,r)$; with $\lambda_1=\kappa r^{5/2}$ this is $2b\kappa\sqrt{r}\,\mathrm{meas}\,\mathcal{F}$.

Proof. **Step S6.18: Jacobian.** For one oscillator, $(\xi,\psi)\mapsto(q,p)=\sqrt{2\xi}(\sin\psi,\cos\psi)$ has Jacobian determinant of modulus 1 ($dp\wedge dq=d\xi\wedge d\psi$, Remark 2.10(1); directly, $\partial_\xi=(\sin\psi,\cos\psi)/\sqrt{2\xi}$, $\partial_\psi=\sqrt{2\xi}(\cos\psi,-\sin\psi)$). It is a bijection of $(0,\infty)\times[0,2\pi)$ onto $\mathbb{R}^2\setminus 0$. Π is the identity on (I,φ), so $|\det D\Pi|=1$ after the permutation of coordinates (measure preserving), and the area formula ([P:bpP:p1:liouville], (X3)) gives equality of measures on injectivity sets. $\vartheta\mapsto 2\pi\vartheta$ has determinant $(2\pi)^n$.

Step S6.19: (b). Let $z\in\mathcal{F}(V,r)$ with $(p_j,q_j)\neq 0$ for all j (the excluded set lies in finitely many subspaces of codimension 2, hence is null). Then $\xi_j\in(0,r^2/2)$, $\psi_j\in[0,2\pi)$ is unique,

$\varphi \in [0,2\pi)^a$, $I \in V$: $z = \mathsf{E}(I,\xi,\vartheta)$ with $\vartheta \in [0,1)^n$; the converse inclusion is the same computation. So $\operatorname{meas}\mathcal{F} = (2\pi)^n \operatorname{meas} V (r^2/2)^b$.

Step S6.20: (c). In the chart, the set is $\mathsf{E}(V \times \{\xi : \exists j,\ \xi_j \le \lambda_1\} \times [0,1)^n)$ with $\{\xi \in (0,\lambda_2)^b : \exists j, \xi_j \le \lambda_1\}$ of measure $\le b\lambda_1\lambda_2^{b-1}$. Divide by (b). □

6.7. **The budgets consumed by Parts 3 and 5.**

Proposition 6.14 (Budget on the action side (the right side of (3.1))). *Assume (S4); $0 < t < d$; $0 < \lambda_1 \le \lambda_2/2$, $\lambda_2 \le \lambda_0$; $0 < r_K < \min\{t, \lambda_1\}$; $\mu \in (0, \mu_1]$; and $D = \bar{D}_t \times A_{\mathrm{sh}}$ with $\bar{D}_t \subset \bar{D}$-admissible, i.e. (S4) holds with $\bar{D} := \bar{D}_t$. Let $\gamma_1 \ge \gamma_2 > 0$ be the Diophantine constants of Proposition 4.3 (for $h = (H_0 + \mu N_\mu)/2\pi$), D^{dio} as there, and K_Ω any set with (3.1) for $r = r_K$. Then*

$$\operatorname{meas}\big((D \times \mathbb{T}^n) \setminus K_\Omega\big) \le c_n(\eta_1 + \eta_2) \operatorname{meas} D, \qquad \eta_1 := 4^b\Big[c_1\, 2\pi\gamma_1\Big(1 + \frac{\mu^{\tau-n}}{\lambda_2}\Big) + c_2\frac{2\pi\gamma_2}{\mu\lambda_2}\Big],$$

η_2 of Lemma 6.12, $c_n = 1 + (1+2^n)(1+(4/3)^n)$ of Proposition 4.3, c_1, c_2 with $\bar{D} = \bar{D}_t$.

Proof. **Step S6.21: Reading (3.1).** $D \subset \mathrm{Re}(D_{r_K})$, so $(D \times \mathbb{T}^n) \setminus K_\Omega \subset (\mathrm{Re}(D_{r_K}) \times \mathbb{T}^n) \setminus K_\Omega$, of measure $\le c_n(\operatorname{meas}(D \setminus D^{\mathrm{dio}}) + \operatorname{meas}(\mathrm{Re}(D_{r_K}) \setminus D))$ ($\operatorname{meas}\mathbb{T}^n = 1$).

Step S6.22: Non-Diophantine part. By the dictionary of Definition 6.5, $D \setminus D^{\mathrm{dio}} = \mathcal{R}$ for $Q = A_{\mathrm{sh}} \subset (-\lambda_2, \lambda_2)^b$, $\lambda = \lambda_2$, $\alpha_i = 2\pi\gamma_i$. Proposition 6.9 gives $\operatorname{meas}\mathcal{R} \le [\dots] \operatorname{meas}(\bar{D}_t)(2\lambda_2)^b$, and $(2\lambda_2)^b = (2\lambda_2/\ell)^b \ell^b \le 4^b \ell^b$ since $\ell \ge \lambda_2/2$; $\operatorname{meas}\bar{D}_t\, \ell^b = \operatorname{meas} D$.

Step S6.23: Collar part. Lemma 6.12. □

Proposition 6.15 (Budget in the frozen cell). *In the situation of Proposition 6.14 with $\lambda_2 = r^2/2$ (r the frozen radius), let $K \subset \mathbb{R}^a \times (0,\infty)^b \times \mathbb{T}^n$ be any measurable set with $\operatorname{meas}((D \times \mathbb{T}^n) \setminus K) \le c_n(\eta_1 + \eta_2) \operatorname{meas} D$ and let $\mathsf{S} := \mathsf{E}(K \cap (\mathbb{R}^a \times (0,\infty)^b \times [0,1)^n))$. Then*

$$\operatorname{meas}\big(\mathcal{F}(V,r) \setminus \mathsf{S}\big) \le \Big[\frac{At}{\operatorname{meas} V} + b\frac{\lambda_1}{\lambda_2} + c_n(\eta_1 + \eta_2)\Big] \operatorname{meas}\mathcal{F}(V,r).$$

Proof. **Step S6.24: Decomposition.** Up to a null set, $\mathcal{F} = \mathsf{E}(V \times (0,\lambda_2)^b \times [0,1)^n)$ (Lemma 6.13(b)), and $\mathcal{F}_{\mathrm{good}} := \mathsf{E}(D \times [0,1)^n) \subset \mathcal{F}$ (since $\bar{D}_t \subset V$, $A_{\mathrm{sh}} \subset (0,\lambda_2)^b$). So $\mathcal{F} \setminus \mathsf{S} \subset (\mathcal{F} \setminus \mathcal{F}_{\mathrm{good}}) \cup \mathsf{E}((D \times \mathbb{T}^n) \setminus K)$ up to a null set.

Step S6.25: Bad action set. By Lemma 6.13, $\operatorname{meas}\mathcal{F}_{\mathrm{good}} = (2\pi)^n \operatorname{meas}\bar{D}_t\, \ell^b$ and $\operatorname{meas}\mathcal{F} = (2\pi)^n \operatorname{meas} V \lambda_2^b$, so $\operatorname{meas}\mathcal{F}_{\mathrm{good}}/\operatorname{meas}\mathcal{F} = (\operatorname{meas}\bar{D}_t/\operatorname{meas} V)(1 - \lambda_1/\lambda_2)^b \ge 1 - \frac{At}{\operatorname{meas} V} - b\frac{\lambda_1}{\lambda_2}$ (Lemma 6.1(b); $(1-x)^b \ge 1 - bx$; $(1-u)(1-v) \ge 1 - u - v$).

Step S6.26: Lost tori. $\operatorname{meas}\mathsf{E}((D \times \mathbb{T}^n) \setminus K) = (2\pi)^n \operatorname{meas}((D \times \mathbb{T}^n) \setminus K) \le c_n(\eta_1 + \eta_2) \operatorname{meas}\mathcal{F}_{\mathrm{good}} \le c_n(\eta_1 + \eta_2) \operatorname{meas}\mathcal{F}$. □

Remark 6.16 (What Part 5 must supply and check). Part 5 chooses, as functions of r and in the order of Theorem 1.28: $\tau \ge n+1$; $t = \sqrt{r}$ (so $At/\operatorname{meas} V \le A\sqrt{r}/\operatorname{meas} V$, $t < d$ for $r < r_*$); $\lambda_2 = r^2/2$, $\lambda_1 = \kappa r^{5/2}$ (shell loss $2b\kappa\sqrt{r}$ by Lemma 6.13(c)); and γ_1, γ_2 so that $\eta_1 \le C\sqrt{r}$: $\gamma_1 \le Cr^{5/2}$ and $\gamma_2/\mu \le Cr^{5/2}$ (CP10 (20): $\gamma_1 = \bar{\gamma}_2 = \gamma_*^2\varepsilon^{5/2}$), and $r_K \le \min\{t, \lambda_1\}$ with $r_K/\ell \le C\sqrt{r}$ (from $r_K = 32\hat{E}\hat{\rho} \le C\hat{\rho} \le C\gamma_2$) (so $\eta_2 \le C\sqrt{r}$). It also checks (S4): (N1)–(N2) with Λ_N from joint analyticity of P_0, Ω, β near $[0,\mu_0] \times \overline{V}$ (Lemma 6.2), (N3) from `htorsion` (Lemma 6.4), $\mu \le \mu_1$ from $\mu < r^6$, injectivity and nondegeneracy of ω_μ on the complex neighbourhood required by Proposition 4.3, and Borelness of K_Ω (Lemma 2.20) for the frozen `MeasurableSet`. Proposition 6.15 then gives $\operatorname{meas}(\mathcal{F} \setminus \mathsf{S}) \le C\sqrt{r} \operatorname{meas}\mathcal{F}$. The strict inequalities and positivity are Part 5 ([P:`bpP:p5:strictness`]).

6.8. **Suspected misprints and gaps in CP10 (Lemma 3.1, Appendix C, §3 end).**

(E1) **V (gap).** p. 562, after (59): "with c_i independent of ε and μ". The constant c_1 of (58) contains the covering number p, which depends on $\hat{r} = \check{c}_2\varepsilon^2$ (the lemma, p. 562, says "for a suitable integer p depending on $\bar{D}$ and ω_1"); Remark 6.10 gives a counterexample to $\hat{r}$-independence for the abstract lemma. Repaired by Proposition 6.9.

(E2) **V.** p. 562, (59) and the line before: Lemma 3.1 is applied to $\omega_{\mathrm{resc}} = (\partial_{J_1}(H_0 + \mu\breve{N}), \partial_{J_2}\breve{N})$, whose $k_1 \neq 0$ resonances $|k_1 \cdot \omega_1 + k_2 \cdot \omega_2|$ differ from those of $\omega_\mu = (\omega_1, \mu\omega_2)$ (where the term is $\mu k_2 \cdot \omega_2$), which define $\mathcal{D}_{\gamma_1,\mu\bar{\gamma}_2,\tau}$. The two resonant sets are different; our proof treats ω_μ itself.

(E3) **T.** p. 561, below (55), footnote 25 and (57): "$\mathrm{meas}(\mathrm{Re}(V_r)\setminus V) \le C\bar{\gamma}_2 \,\mathrm{meas}\, V$" (as extracted from the pinned PDF; the exponent/subscript may be a typesetting artifact) is asserted without proof and cannot hold for an arbitrary open V; it is replaced by `hboundary` (audit P3, D-PG-5), and only the inner collar is needed.

(E4) **T (not verified).** p. 561 (56): the loss $C\varepsilon^{n_2/2}$ for the shell $\check{c}_1\varepsilon^{5/2} < J_{2,i}$. Our computation (Lemma 6.13(c)) gives $n_2\check{c}_1\varepsilon^{1/2}/\check{c}_2$, which is larger than $\varepsilon^{n_2/2}$ for $n_2 \ge 2$; CP10's displayed (18) and (20) are consistent with the $\sqrt{\varepsilon}$ rate only.

(E5) **R.** p. 575 (113) and p. 576: after the change of variables the Jacobian $1/|k_j|$ and the cover $\mathcal{C}^i_k$ are fine, and the integrals over the slice use the norm $|\cdot|$; the sum $c(n,\tau) = \sum|k|^{-\tau}$ needs $\tau > n$, which is why CP10 assumes $\tau_* > n$ (p. 550, footnote 12); Lemma 6.8 makes this explicit.

6.9. **References and imported facts.** Part 1: [P:`bpP:p1:two-scale`], [P:`bpP:p1:dio-basic`], [P:`bpP:p1:dio-sources`], [P:`bpP:p1:liouville`], [P:`bpP:p1:torus-union`], [P:`bpP:p1:typical`]. Part 3: [P:`bpP:p3:two-scale-kam`] (inequality (3.1), c_n), [P:`bpP:p3:period-dictionary`], §5.6. Track K: [K:`bp:p4:lattice-count`], [K:`bp:p4:lattice-sum`], [K:`bp:p4:slab`] (model), [K:`bp:p1:lip-image`], [K:`bp:p1:ext-cramer`], [K:`bp:p1:ext-lebesgue`], [K:`bp:p1:ext-mvt`]. Forward references to Part 5: [P:`bpP:p5:uniform-remainder`], [P:`bpP:p5:strictness`]. Formalization notes: Lemma 6.2 is a stand-alone compactness argument; Proposition 6.9 is three slicing estimates plus Tonelli on sets of short sections; the polar chart is a bounded stand-alone task.

7. P-G assembly: uniform remainder, parameters, application of Parts 2–4, tori, measure, strictness

This part proves the frozen theorem [P:`bpP:p0:pg`] (Theorem 7.29) from Parts 1–4. Write $n = a+b$. The route is CP10 §4, pp. 562–564, but the *hypotheses are the frozen ones* and everything the source asserts without proof is supplied: the uniform remainder (CP10 (60)), the nondegeneracy of the frequency map on a complex neighbourhood, the explicit parameter choice with the numerical bounds of [P:`bpP:p3:interface`] tracked, the conversion of tori to `IsCylinderTorus`, the cell measure, and the strict inequalities.

What is different from CP10. (1) Hypothesis (A''_2) is the Birkhoff form *already*, so Steps 2–4 of CP10 §3 (equilibrium, diagonalization, Birkhoff) are not needed: the averaged secular function N of [P:`bpP:p0:birkhoff`] is used as it stands, and the remainder after averaging is $\mu Q + \widetilde{P}$, with $Q = P_{\mathrm{av}} - N = O(|(p,q)|^5)$ and $\widetilde{P}$ of Part 2. (2) The KAM step is applied on the shell $\lambda_1 < \xi_j < \lambda'_2$ with the constant $K = K_0$ *bounded* (the two Diophantine scales $\gamma_1 = \bar{\gamma}_2 = \kappa_0 r^{5/2}$ make $K = \frac{6}{s}\log_+(\kappa_0^2/C^4)$ independent of r); this is why no power of $\log(1/r)$ enters the $\sqrt{r}$ loss of the measure. (3) The set of $\bar{\omega}_0$-Diophantine actions is a Cantor-type set $\check{D}$; its complex neighbourhood is not controlled by the collar, so Lemma 7.26 and Proposition 7.27 re-run the budget of [P:`bpP:p4:budget`], [P:`bpP:p4:cell-budget`] for $D = \check{D} \times A_{\mathrm{sh}}$ (replacing $\bar{D}_t$ by $\check{D} \subset \bar{D}_t$ in the *domain*, but not in the Diophantine count).

7.1. **Conventions and the order of the constants.** All coordinates are CP's: angles φ of period 2π, actions I, oscillators (p,q); the frozen fast frequency map is $\bar\omega_0 = \omega_{H_0} = \partial H_0$ (CP angular convention). Part 3 works with period-one angles and $H_3 = \tilde H(J, 2\pi\theta)/2\pi$ ([P:bpP:p3:period-dictionary]); all quantities of Part 3 with 2π factors are absorbed in the constants C below, and the dictionary $\gamma_i = \alpha_i/2\pi$ of [P:bpP:p4:freq] is used. Norms are max norms; $|\cdot|_*$ is the weighted norm of [N:bpN:p4:norm] with coefficient radius (CP's s). λ_0 denotes the sup over real points of the complex bounds that appear.

Remark 7.1 (Order in which every constant is chosen). In the proof of Theorem 7.29 the constants are chosen in this order; each depends only on the earlier ones.

(1) *Data and hypotheses* (fixed first): $a, b, V, W, A, d, r_0, \mu_0, B, H_0, P, P_0, \Omega, \beta$.
(2) *Analytic constants* (Lemmas 7.2, 7.4, 7.9): δ_c, s_*, ε_0, r_1, Λ_P, C_Q, ρ_c, M_β, L_β, Λ_N, c_0, $\bar M_0$, R_y, L_g, μ_* (data only; μ and r do not enter).
(3) $\tau := n+1$; the Part 3 numerical constant $C_1 \ge 1$ of [P:bpP:p3:interface] (Lemma 7.18); the widths $s, \bar s$ (functions of s_*).
(4) The scale constants κ_0, then κ_1, then G (Lemma 7.19, Definition 7.12); then the measure constant C_* (Prop. 7.27).
(5) r_* (finite list of smallness conditions, Definition 7.13); $C := \max\{C_*+1, C_w\}$ with C_w the frequency bound; $c := 2\tau + 2 = 2n+4$.
(6) Only *then* $r \in (0, r_*)$ and $\mu \in (0, \min\{\mu_0, r^6/(\log\frac{1}{r})^c\})$ are given; $K, \bar K, \bar\gamma, \bar\rho, \rho, \hat\rho, \hat E, r_K, t, \lambda_1, \lambda_2, \lambda_2'$ are explicit functions of (r,μ) (Definition 7.12); finally the set S (with the deleted tori of §7.10).

7.2. **Uniform complex extension and uniform remainder.**

Lemma 7.2 (Uniform holomorphic extension of the family)**.** *Assume (H1), (H2) of* [P:bpP:p0:pg] *and* HasFastDiffeo(H_0, W). *Put* $\varepsilon_0 := r_0/2$ *and, for* $\delta > 0$, *let* $\mathcal{X}_\delta \subset \mathbb{C}^a \times \mathbb{C}^a \times \mathbb{C}^b \times \mathbb{C}^b$ *be the set of* (I, φ, p, q) *with* $\operatorname{dist}_\infty\big((I, \operatorname{Re}\varphi, \operatorname{Re} p, \operatorname{Re} q),\ \overline{V} \times \mathbb{R}^a \times \{|(p,q)|_\infty \le \varepsilon_0\}\big) < \delta$ *and* $|\operatorname{Im}(I,\varphi,p,q)|_\infty < \delta$. *Then there are* $\delta_c \in (0,1)$ *and* $M_P < \infty$ *and a map* $\tilde P : [0,\mu_0] \times \mathcal{X}_{\delta_c} \to \mathbb{C}$ *with:* $\tilde P(\mu,\cdot)$ *holomorphic,* $\tilde P = P$ *at real points,* $\tilde P(\mu, \cdot + 2\pi k e_{\mathrm{f}}) = \tilde P(\mu,\cdot)$ *(e_{f} the fast angles),* $|\tilde P| \le M_P$; $\tilde H_0$ *holomorphic on* $\{I : \operatorname{dist}_\infty(I, \overline{V}) < \delta_c\}$ *with* $|\tilde H_0|, \|\partial \tilde H_0\|, \|\partial^2 \tilde H_0\| \le M_0$, *and* $\tilde P$ *jointly continuous in* μ.

Proof. **Step S7.1: Local extensions.** Let $X_0 = [0,\mu_0] \times \overline{V} \times [0,2\pi]^a \times \{|(p,q)|_\infty \le \varepsilon_0\}$, compact. Since $2\varepsilon_0^2 < r_0^2$ for the sup-norm ball, $X_0 \subset [0,\mu_0] \times \overline{\mathcal{C}(V, r_0)}$. By (H1) P is analytic at each point of X_0: a convergent power series in a polydisc, which defines a holomorphic extension there. By compactness a Lebesgue number gives δ_1 such that for every $x \in X_0$ the polydisc $B(x,\delta_1)$ carries a holomorphic extension P_x, with $|P_x| \le M$ uniformly (shrink δ_1 so that closed polydiscs lie in the domains of finitely many series).

Step S7.2: Gluing. If $x, y \in X_0$ and $B(x,\delta_1) \cap B(y,\delta_1) \ne \emptyset$ then $|x-y| < 2\delta_1$ and the real midpoint lies in both; the intersection is convex hence connected, $P_x = P_y$ on its real points (both equal P), so $P_x = P_y$ on the intersection (identity theorem). Thus the P_x define one holomorphic $\tilde P$ on $\bigcup_x B(x,\delta_1)$ (for each real μ; jointly holomorphic in $(\mu,\cdot)$ near real μ).

Step S7.3: Periodicity. By (H2) and continuity of P on $\overline{\mathcal{C}(V,r_0)}$ (density of $\mathcal{C}(V,r_0)$ in its closure; P analytic hence continuous) periodicity holds on the closure. For φ outside $[0,2\pi]^a$ define $\tilde P$ by translation by $2\pi k_{\mathrm{f}}$ from the fundamental box; two such definitions agree on overlaps by the argument of Step S7.2 (agreement at real points, convex intersection). This gives $\mathcal{X}_{\delta_c}$ for $\delta_c = \delta_1$.

Step S7.4: H_0**.** H_0 is analytic on $W \supset \overline{V}$ (compact): Steps S7.1–S7.2 (without periodicity) give $\tilde H_0$ on a complex δ_c-neighbourhood of $\overline{V}$ (decrease δ_c); derivatives bounded by shrinking and Cauchy ([K:bp:p1:cauchy-open]). □

Definition 7.3 (Strip, averaged function, constants)**.** $s_* := \min\{\delta_c, 1\}/2$ (a coefficient radius $\le 1/2$), $\hat{D}_c := \{I \in \mathbb{C}^a : \mathrm{dist}_\infty(I, \overline{V}) < \delta_c/2\}$. For (μ, I, p, q) with $I \in \hat{D}_c$, $|(p,q)|_\infty < \varepsilon_0$ complex, put

$$\tilde{P}_{\mathrm{av}}(\mu, I; p, q) = (2\pi)^{-a} \int_{[0,2\pi]^a} \tilde{P}(\mu, I, \varphi, p, q)\, d\varphi$$

(real φ; $= P_{\mathrm{av}}$ of [P:bpP:p0:secavg] at real points of the cylinder). The N-norm of $\tilde{P}(\mu, \cdot)$ (Fourier in φ, coefficient radius s_*, sup over (I, p, q) in the complex domain $\hat{D}_c \times B_{\varepsilon_0} \times B_{\varepsilon_0}$) satisfies

$$\Lambda_P := \sup_{\mu \in [0,\mu_0]} |\tilde{P}(\mu, \cdot)|_{,s_*} \le M_P \coth(\delta_c/4)^a < \infty \quad (\ge 1 \text{ after enlarging}).$$

Justification of finiteness. $|\hat{f}_k| \le \sup|f|\, e^{-\sigma_1|k|_1}$ on a strip of width σ_1 (Cauchy in the angle, $\hat{f}_k = (2\pi)^{-a} \int f(\varphi) e^{-ik\cdot\varphi} d\varphi$ shifted to $\mathrm{Im}\,\varphi = \mp\sigma_1 \operatorname{sgn} k$), so $\sum_k |\hat{f}_k| e^{s_*|k|_1} \le \sup|f| \sum_k e^{-(\delta_c - s_*)|k|_1} \le M_P \coth(\delta_c/4)^a$ since $\delta_c - s_* \ge \delta_c/2$. This is the elementary *sup-to-weighted conversion*, used again in Lemma 7.18 (cf. [N:bpN:p4:sup-weighted], [K:bp:p1:decay]). □

Lemma 7.4 (Birkhoff coefficients as derivatives; analytic extension to $\overline{V}$)**.** *Assume (H1), (H2), (H4), (H6) of* [P:bpP:p0:pg]. *For $(\mu, I) \in [0, \mu_0] \times \hat{D}_c$ define, with the holomorphic extension $\tilde{P}_{\mathrm{av}}$ (and $R_j = (p_j^2 + q_j^2)/2$)*

$$\tilde{P}_0 = \tilde{P}_{\mathrm{av}}\big|_{x=0}, \quad \tilde{\Omega}_j = \partial^2_{p_j} \tilde{P}_{\mathrm{av}}\big|_{x=0}, \quad \tilde{\beta}_{ij} = \partial^2_{p_i} \partial^2_{p_j} \tilde{P}_{\mathrm{av}}\big|_{x=0} \ (i \ne j), \quad \tilde{\beta}_{jj} = \tfrac{1}{3} \partial^4_{p_j} \tilde{P}_{\mathrm{av}}\big|_{x=0},$$

$x = (p, q)$, and $\tilde{N} = \tilde{P}_0 + \tilde{\Omega} \cdot R + \frac{1}{2} R \cdot \tilde{\beta} R$, $Q := \tilde{P}_{\mathrm{av}} - \tilde{N}$. Then: (a) $\tilde{P}_0, \tilde{\Omega}, \tilde{\beta}$ are holomorphic in $I \in \hat{D}_c$, continuous in $\mu \in [0, \mu_0]$, $\tilde{\beta}$ is symmetric, and they equal *P_0, Ω, β on $[0, \mu_0] \times V$; (b) for each $\mu \in [0, \mu_0]$ and $I \in \hat{D}_c$, the jet of $Q(\mu, I; \cdot)$ at $x = 0$ vanishes to order 4 (all derivatives of order ≤ 4); (c) $|\det \tilde{\beta}(\mu, I)| \ge B$ for all $(\mu, I) \in [0, \mu_0] \times \overline{V}$.*

Proof. **Step S7.5: Coefficients on V.** Fix $\mu \in [0, \mu_0]$, $I \in V$. By [P:bpP:p0:birkhoff](ii), $P_{\mathrm{av}} - N = o(|x|^4)$, and $P_{\mathrm{av}} - N$ is real analytic near $x = 0$ (analytic by Lemma 7.2 and differentiation under the integral; N polynomial). Hence the Taylor polynomial of $P_{\mathrm{av}}(\mu, I; \cdot)$ of degree ≤ 4 equals that of N, which is N. Compute from $R_j = (p_j^2 + q_j^2)/2$: $\partial^2_{p_j} N(0) = \Omega_j$; $R_i R_j = \frac{1}{4}(p_i^2 + q_i^2)(p_j^2 + q_j^2)$ gives $\partial^2_{p_i} \partial^2_{p_j} N(0) = \frac{1}{2}(\beta_{ij} + \beta_{ji}) = \beta_{ij}$ for $i \ne j$ *using* (H6); $\frac{1}{2}\beta_{jj} R_j^2 = \frac{1}{8}\beta_{jj}(p_j^2 + q_j^2)^2$ gives $\partial^4_{p_j} N(0) = 3\beta_{jj}$; $\tilde{P}_0 = P_{\mathrm{av}}|_{x=0}$. So (a) on V.

Step S7.6: Analyticity. The listed derivatives of $\tilde{P}_{\mathrm{av}}$ at $x = 0$ are holomorphic in I (integral of holomorphic functions, uniform bounds by Lemma 7.2), continuous in μ; symmetry of $\tilde{\beta}$ is symmetry of mixed derivatives.

Step S7.7: (b) on V, then everywhere. On V, $\tilde{N} = N$ and Step S7.5 give the vanishing jet. Each derivative $\partial^\alpha_x Q(\mu, \cdot; 0)$, $|\alpha| \le 4$, is holomorphic on $\hat{D}_c$, which is connected ($\overline{V}$ is connected as the closure of the connected V (hVc), and $\hat{D}_c$ is a union of balls centered on it) and vanishes on the real open set V; by the identity theorem it vanishes on $\hat{D}_c$.

Step S7.8: (c). On V by (H5) and (a); $\overline{V}$ by continuity of $\det \tilde{\beta}$ in I and V dense in $\overline{V}$. □

Proposition 7.5 (Uniform remainder: CP10 (60) from the frozen hypotheses)**.** *Under the hypotheses of Lemma 7.4 there are $r_1 \in (0, \varepsilon_0]$ and $C_Q < \infty$ (depending on the data only) such that for all $\mu \in [0, \mu_0]$, $I \in \hat{D}'_c$ (:= $\{\mathrm{dist}_\infty(I, \overline{V}) < \delta_c/4\}$) and complex $x = (p, q)$ with $|x|_\infty \le r_1$:*

$$|Q(\mu, I; x)| \le C_Q |x|^5_\infty.$$

In particular $|P_{\mathrm{av}} - N| \le C_Q |(p,q)|^5_\infty$ on $[0, \mu_0] \times \overline{V} \times \{|x|_\infty \le r_1\}$ (real), with β, Ω, P_0 continuous (indeed analytic) on $\overline{V}$ and $|\det \beta| \ge B$ there. Hypothesis (A_2'') with a δ depending on (μ, I, ϵ) is thus upgraded to a uniform $O(|x|^5)$ bound.

Source: CP10 (60), p. 563, asserted "for a suitable $C > 0$" from (A_2''); proof not given.

Source gap: CP10 (60) is asserted without the compactness and analyticity argument; supplied here.

Proof. **Step S7.9: Bound.** Take $r_1 := \varepsilon_0/2$. Q is holomorphic in x on $\{|x|_\infty < \varepsilon_0\}$ and $|Q| \le M_P + |\tilde N| \le M_1$ on $\hat D_c' \times \{|x|_\infty \le r_1\}$, $r_1 = \varepsilon_0/2$, with M_1 from the bounds of Lemma 7.2 and Cauchy estimates for $\tilde\Omega, \tilde\beta$ (derivatives at 0 of a function bounded by M_P on a polydisc of radius ε_0).

Step S7.10: Schwarz. Fix (μ, I) and $\hat x$ with $|\hat x|_\infty = 1$; $g(t) = Q(\mu, I; t\hat x)$ is holomorphic on $|t| < r_1$, $|g| \le M_1$, and by Lemma 7.4(b) g vanishes to order ≥ 5 at 0 (derivatives of g at 0 of order ≤ 4 are combinations of derivatives of Q of order ≤ 4). So $g(t)/t^5$ is holomorphic and bounded by M_1/r_1^5 on the boundary circles, hence on the disc (maximum principle): $|g(t)| \le M_1|t|^5/r_1^5$. Take $C_Q = M_1 r_1^{-5}$. □

7.3. **The fast frequency map on a complex neighbourhood; the nondegenerate frequency map of the averaged system (domain reduction, CP10 (58)).** The frozen hypothesis HasFastDiffeo(H_0, W) replaces CP10's assumption that $\bar\omega_0 = \partial H_0$ is a diffeomorphism of V_{ρ_0} with bounded inverse Jacobian (58). Write $Z_c := \hat D_c$-type neighbourhoods $X_\rho := \{z \in \mathbb{C}^N : \operatorname{dist}_\infty(z, X) < \rho\}$ for $X \subset \mathbb{R}^N$.

Lemma 7.6 (Lipschitz bound on a complex neighbourhood)**.** *Let $X \subset \mathbb{R}^N$, $\rho > 0$, $\Delta : X_{2\rho} \to \mathbb{C}^m$ holomorphic with $\sup|\Delta| \le A$. Then $|\Delta(z) - \Delta(z')| \le \frac{2NA}{\rho}|z - z'|$ for all $z, z' \in X_\rho$.*

Proof. Close pairs $|z - z'| < \rho$: every point w of the (complex) segment lies within ρ of $z \in X_\rho$, so the polydisc $B(w, \rho - \epsilon) \subset X_{2\rho}$ for all small ϵ; Cauchy ([K:bp:p1:cauchy-open]) gives $\|D\Delta(w)\| \le NA/\rho$; mean value inequality ([K:bp:p1:ext-mvt]). Distant pairs: $|\Delta(z) - \Delta(z')| \le 2A \le (2A/\rho)|z - z'|$.
Source: Ours; the complex-point version of [P:bpP:p3:lip-sup].

□

Lemma 7.7 (Complex bi-Lipschitz bound for $\bar\omega_0$)**.** *Under* HasFastDiffeo(H_0, W) *there are $\rho_c \in (0, \delta_c/16)$ and $c_0 > 0$ with $|\tilde\omega_0(z) - \tilde\omega_0(z')| \ge c_0|z - z'|$ for all $z, z' \in (\overline{V})_{\rho_c}$, where $\tilde\omega_0 = \partial\tilde H_0$; hence $\|(\partial\tilde\omega_0(z))^{-1}\| \le c_0^{-1}$ there. Moreover $\bar M_0 := \sup_{(\overline{V})_{2\rho_c}}\|\partial\tilde\omega_0\| < \infty$.*

Proof. **Step S7.11: Real facts.** By [P:bpP:p0:fastdiffeo], $g(\bar\omega_0(I)) = I$ on W, so $\bar\omega_0$ is injective on $W \supset \overline{V}$; differentiating (chain rule, $g \in C^1$, $\bar\omega_0$ analytic) $Dg\, D\bar\omega_0 = 1$, so $\partial^2 H_0 = D\bar\omega_0$ is invertible on W.

Step S7.12: Compactness. Suppose not: $z_m \ne z_m' \in (\overline{V})_{1/m}$ with $|\tilde\omega_0(z_m) - \tilde\omega_0(z_m')| < |z_m - z_m'|/m$. Pass to subsequences with $z_m \to x$, $z_m' \to x' \in \overline{V}$ (compact). If $x \ne x'$, continuity gives $\bar\omega_0(x) = \bar\omega_0(x')$, contradicting injectivity. If $x = x'$: let $m_0 = \|(\partial\tilde\omega_0(x))^{-1}\|$; choose δ with $\|\partial\tilde\omega_0(w) - \partial\tilde\omega_0(x)\| \le 1/(2m_0)$ on $B(x, \delta)$ (continuity); for $z, z' \in B(x, \delta)$, $|\tilde\omega_0(z) - \tilde\omega_0(z')| \ge |\partial\tilde\omega_0(x)(z - z')| - \frac{1}{2m_0}|z - z'| \ge \frac{1}{2m_0}|z - z'|$, contradicting the choice for $m > 2m_0$. The operator bound: differentiate along $z' \to z$. $\bar M_0 < \infty$ by Lemma 7.2. □

Lemma 7.8 (One-scale Diophantine measure on V (CP10 (38)))**.** *Let L_g, R_y, Λ_g be as in* [P:bpP:p4:lip-inverse] *and S_1 as in* [P:bpP:p4:lattice-tail] *(with $n = a + b$, $\tau > n$). For $\bar\gamma > 0$ put $\mathcal{D}^1_{\bar\gamma} := \{y \in \mathbb{R}^a : |k \cdot y| \ge \bar\gamma|k|_1^{-\tau}\ \forall k \in \mathbb{Z}^a \setminus 0\}$ (closed). Then $\operatorname{meas}\big(V \setminus \bar\omega_0^{-1}(\mathcal{D}^1_{\bar\gamma})\big) \le c_{\mathrm{dio}}\bar\gamma$ with $c_{\mathrm{dio}} := 2\Lambda_g^a S_1 (2R_y)^{a-1}$.*

Proof. $V \setminus \bar\omega_0^{-1}(\mathcal{D}^1_{\bar\gamma}) = \bigcup_k g(Y_k)$, $Y_k = \{y \in Y_0 : |k \cdot y| < \bar\gamma|k|_1^{-\tau}\}$ (since $g \circ \bar\omega_0 = \mathrm{id}$ on $\overline{V}$, [P:bpP:p4:lip-inverse]). By [P:bpP:p4:affine-slab] (with $c = 0$, $N = a$, $\lambda = R_y$, $\ell = k$, $|\ell| \ge 1$), $\operatorname{meas} Y_k \le 2\bar\gamma|k|_1^{-\tau}(2R_y)^{a-1}$; g is Λ_g-Lipschitz on Y_0, so [K:bp:p1:lip-image] (after McShane

extension [K:bp:p1:mcshane]) gives meas $g(Y_k) \le \Lambda_g^a \operatorname{meas} Y_k$. Sum over k using $\sum_{k\in\mathbb{Z}^a\setminus 0}|k|_1^{-\tau} \le S_1$ (the lattice $\mathbb{Z}^a \subset \mathbb{Z}^n$).
Source: CP10 (38), p. 556, "standard measure estimate"; proof ours.

□

Lemma 7.9 (Constants of the averaged frequency map)**.** *Let $\tilde N$ be as in Lemma 7.4. There are constants $A_N, \Lambda_N, M_\beta, L_\beta \ge 1$ (data only) such that on $\mathcal{E} := (\overline{V})_{\rho_c} \times \{|J_2|_\infty < 2\}$, $\tilde N$, $\partial_J \tilde N$, $\partial_J^2 \tilde N$ are bounded by A_N; with $G := \partial_{J_1}\tilde N$ and $\Theta(J_1,\xi) := \tilde\Omega(J_1) + \tilde\beta(J_1)\xi$, for $J_1, J_1' \in (\overline{V})_{\rho_c}$, $|\xi|, |\xi'| \le 1$:*

$$|G(J_1,\xi) - G(J_1',\xi)|,\ |\Theta(J_1,\xi) - \Theta(J_1',\xi)| \le \Lambda_N |J_1 - J_1'|, \quad |G(J_1,\xi) - G(J_1,\xi')| \le \Lambda_N |\xi - \xi'|,$$

and $|\nu| \le L_\beta |\tilde\beta(J_1)\nu|$ for $J_1 \in (\overline{V})_{\rho_c}$, $\nu \in \mathbb{C}^b$ (after decreasing ρ_c). The real-variable constants of [P:bpP:p4:data] *((N1)–(N3) with $\bar D = \bar D_t$, $\lambda_0 = 1$) hold with the same Λ_N, L_β and for all $\mu \in (0, \mu_0]$.*

Proof. **Step S7.13: Bounds.** $\tilde N$ is a polynomial in J_2 with coefficients $\tilde P_0, \tilde\Omega, \tilde\beta$ holomorphic and bounded on $\hat D_c$ (Lemma 7.4, Cauchy from M_P); derivatives in J_1 up to order 3 are bounded on $(\overline{V})_{2\rho_c}$ for $\rho_c \le \delta_c/16$.

Step S7.14: Lipschitz. In J_1: Lemma 7.6 applied to $\partial_{J_1}\tilde N(\cdot,\xi)$ and $\Theta(\cdot,\xi)$ ($|\xi| \le 1$ makes the sup bound uniform). In J_2: polynomial of degree ≤ 2 with bounded coefficients on the convex set $|\xi| \le 1$.

Step S7.15: Inverse of $\tilde\beta$. $M_\beta := \sup|\tilde\beta_{ij}|$ on $(\overline{V})_{\rho_c}$; $|\det\tilde\beta| \ge B$ on $\overline{V}$ (Lemma 7.4(c)), and det is Lipschitz, so $|\det\tilde\beta| \ge B/2$ on $(\overline{V})_{\rho_c}$ after decreasing ρ_c. Cramer as in [P:bpP:p4:beta-inverse] (the adjugate bound holds for complex matrices): $L_\beta := 2b! M_\beta^{b-1}/B$ (and $\tilde\beta^\top = \tilde\beta$). □

Lemma 7.10 (Nondegeneracy and injectivity of the frequency map on the complex neighbourhood)**.** *For $\mu \in (0, \mu_*]$, where $\mu_* \le 1$ is defined by*

$$\mu_* A_N \le \tfrac{c_0}{2}, \qquad 4\mu_* A_N^2 L_\beta \le c_0, \qquad 2\mu_* \Lambda_N^2 L_\beta < c_0,$$

let $\tilde h_\mu(J) = \tilde H_0(J_1) + \mu\tilde N(J)$ on $\mathcal{E}_1 := (\overline{V})_{\rho_c} \times \{|J_2| < 1\}$ and $F_\mu = \partial\tilde h_\mu = \big(\tilde\omega_0(J_1) + \mu G(J),\ \mu\Theta(J_1,J_2)\big)$, $U = \partial^2\tilde h_\mu$. Then

(a) F_μ is injective on $\mathcal{E}_1$;

(b) U is invertible on $\mathcal{E}_1$, and with $U = \begin{pmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{pmatrix}$ ($a+b$ blocks), $\hat U = (U_{21}, U_{22})$, $U^{-1} = \binom{T_1}{T_2}$:

$$\|U\| \le C_U, \quad \|\hat U\| \le C_U\mu, \quad \|U^{-1}\| \le C_U\mu^{-1}, \quad \|T_1\| \le C_U, \quad \|T_2\| \le C_U\mu^{-1},$$

with $C_U := \max\{\bar M_0 + A_N + 1,\ 4(A_N+1)^2 L_\beta c_0^{-1} + 4,\ 8L_\beta + 4\}$ (depends on data only).

Proof. **Step S7.16: (a).** Let $F_\mu(J) = F_\mu(J')$. First components: $c_0|J_1 - J_1'| \le |\tilde\omega_0(J_1) - \tilde\omega_0(J_1')| = \mu|G(J) - G(J')| \le \mu\Lambda_N(|J_1 - J_1'| + |J_2 - J_2'|)$ (Lemma 7.7, Lemma 7.9, splitting $G(J_1,J_2) - G(J_1',J_2')$ through $G(J_1',J_2)$). Since $\mu\Lambda_N \le c_0/2$ ($\Lambda_N \le A_N$ w.l.o.g.), $|J_1 - J_1'| \le (2\mu\Lambda_N/c_0)|J_2 - J_2'|$. Second components: $\tilde\beta(J_1)(J_2 - J_2') = -[\Theta(J_1,J_2') - \Theta(J_1',J_2')]$, so $|J_2 - J_2'| \le L_\beta\Lambda_N|J_1 - J_1'| \le (2\mu\Lambda_N^2 L_\beta/c_0)|J_2 - J_2'|$, hence $J_2 = J_2'$, then $J_1 = J_1'$.

Step S7.17: (b) Blocks. $U_{11} = \partial\tilde\omega_0 + \mu\tilde N_{11} =: A$, $U_{12} = \mu\tilde N_{12}$, $U_{21} = \mu\tilde N_{21} = U_{12}^\top$, $U_{22} = \mu\tilde\beta$. $\|A^{-1}\| \le 2/c_0$ (Lemma 7.7; Neumann series since $\mu A_N \le c_0/2$). Schur complement $S = \mu(\tilde\beta - \mu\tilde N_{21}A^{-1}\tilde N_{12})$; as $\mu A_N^2 \cdot \frac{2}{c_0}L_\beta \le \frac12$, $\|\tilde\beta^{-1}\| \le L_\beta$ gives $\|S^{-1}\| \le 2L_\beta/\mu$.

Step S7.18: Inverse. Block inversion: $T_{11} = A^{-1} + \mu^2 A^{-1}\tilde N_{12}S^{-1}\tilde N_{21}A^{-1}$, $T_{12} = -\mu A^{-1}\tilde N_{12}S^{-1}$, $T_{21} = -\mu S^{-1}\tilde N_{21}A^{-1}$, $T_{22} = S^{-1}$. Their norms are $\le 2/c_0 + 8\mu A_N^2 L_\beta c_0^{-2}$, $4A_N L_\beta/c_0$, $4A_N L_\beta/c_0$, $2L_\beta/\mu$. Row sums of $T_1 = (T_{11}, T_{12})$ are $O(1)$, of T_2 are $O(\mu^{-1})$, and the entries of $\hat U$ are $\mu\times$ bounded. The constant C_U dominates all of these. □

Remark 7.11. (a) gives injectivity on the *complex* neighbourhood that [P:`bpP:p3:interface`] lists as an obligation of Part 5 ("injectivity/nondegeneracy of ω_μ on a complex neighbourhood"); it uses only HasFastDiffeo (through Lemma 7.7), joint analyticity and (H5). The same computation, with $H_3 = \tilde{H}(\cdot, 2\pi\cdot)/2\pi$, gives the bounds for Part 3 with every number divided/multiplied by 2π as in [P:`bpP:p3:period-dictionary`]; those factors are absorbed in C_1 of Lemma 7.18.

7.4. **The parameters.**

Definition 7.12 (Parameters as functions of (r,μ))**.** Let $\tau := n+1$ and $c := 2\tau+2$. Fix the data constants of Remark 7.1 (2), $\sigma_{\mathrm{av}} := s_*/6$, $\sigma_{\mathrm{tot}} := s_*/12$, $s := \sigma_{\mathrm{tot}}/5$, $\bar{s} := 4\sigma_{\mathrm{tot}}/5$ (so $4s = \bar{s} < 1$ and $\bar{s} + s = \sigma_{\mathrm{tot}}$), and C_1, κ_0, κ_1, G as in Lemmas 7.18, 7.19. For $r \in (0,1)$ and $\mu \in (0, \mu_0)$ put ($\mu_2 := \Lambda_P \mu$)

$$\bar{K} := \tfrac{30}{s_*}\log\tfrac{1}{r}, \quad t := \sqrt{r}, \quad \lambda_2 := \tfrac{r^2}{2}, \quad \lambda_1 := \kappa_1 r^{5/2}, \quad \lambda_2' := \lambda_2 - 2\lambda_1, \quad \ell := \lambda_2' - \lambda_1 = \lambda_2 - 3\lambda_1, \quad \rho := \tfrac{\lambda_1}{2},$$

$$\bar{\gamma} := G\max\Big\{\sqrt{\mu_2}\,\bar{K}^{\tau+\frac12} r^{-5/2},\ r^{5/2}\bar{K}^{\tau+1}\Big\}, \quad \alpha_{\mathrm{av}} := \frac{\bar{\gamma}}{2\bar{K}^\tau}, \quad \bar{\rho} := \frac{\bar{\gamma}}{2\bar{M}_0\bar{K}^{\tau+1}},$$

$$\gamma_1 := \kappa_0 r^{5/2}, \quad \gamma_2 := \mu\kappa_0 r^{5/2}, \quad K := K_0 := \tfrac{6}{s}\log_+(\kappa_0^2/C_1^4),$$

$\hat{\rho}, \hat{E}, r_K$ as in [P:`bpP:p3:constants`] with the numerical bounds of Lemma 7.18. The sets: $\bar{D}_t$ of [P:`bpP:p4:s-collar`], $\check{D} := \bar{D}_t \cap \bar{\omega}_0^{-1}(\mathcal{D}^1_{\bar{\gamma}})$ (compact: $\mathcal{D}^1_{\bar{\gamma}}$ is closed, $\bar{\omega}_0$ continuous on $W \supset \bar{D}_t$), $A_{\mathrm{sh}} := (\lambda_1, \lambda_2')^b$, $D := \check{D} \times A_{\mathrm{sh}}$. The 2π-dictionary: $\alpha_i := 2\pi\gamma_i$.

Definition 7.13 (Smallness thresholds)**.** $r_* \in (0, \min\{r_0, e^{-1/5}, 1\})$ is so small that for $0 < r < r_*$ (all conditions involve only the constants of Remark 7.1, items 2–4, and C_* of Proposition 7.27):

- (r1) $\log\frac{1}{r_*} \geq \Lambda_P$ and $r_*^2(\log\frac{1}{r_*})^{\tau+1} \leq 1$, so that $\mu < r^6/(\log\frac{1}{r})^c$ implies $\mu_2 \leq r^6(\log\frac{1}{r})^{-(2\tau+1)}$; and $r_*^6 \leq \mu_1 := \min\{\mu_*,\ (2(1+M)a\Lambda_N\Lambda_g)^{-1}\}$, $M = 2b\Lambda_N L_\beta$ ([P:`bpP:p4:data`]);
- (r2) $\sqrt{r_*} < d$, $A\sqrt{r_*} \leq \frac12 \operatorname{meas} V$ (so $\operatorname{meas}\bar{D}_t \geq \frac12 \operatorname{meas} V$: positive inner core), $r_* \leq r_1/2$, $r_* \leq \varepsilon_0/4$;
- (r3) with $C_\gamma := G(30/s_*)^{\tau+1}$: $C_\gamma\sqrt{r_*} \leq 2\bar{M}_0\rho_c$ and $c_{\mathrm{dio}}C_\gamma\sqrt{r_*} \leq \frac14 \operatorname{meas} V$;
- (r4) $6\kappa_1\sqrt{r_*} \leq \frac12$;
- (r5) the displacement margin $\sqrt{2}\,\Delta_{\max} \leq \kappa_1 r^{3/2}$ of Lemma 7.22, and $(C_*+1)\sqrt{r_*} \leq \frac12$.

Lemma 7.14 (Consequences of the parameter choice)**.** *For $0 < r < r_*$, $0 < \mu < \min\{\mu_0, r^6/(\log\frac{1}{r})^c\}$: (a) $\mu_2 \leq r^6(\log\frac{1}{r})^{-(2\tau+1)}$; $\mu < \mu_1$; $\bar{K}s_* = 30\log\frac{1}{r} \geq 6$; $\bar{\gamma} \leq C_\gamma\sqrt{r}$. (b) $\bar{\rho} \leq \rho_c$, $\bar{\rho} \geq \lambda_1$ if $G \geq 2\bar{M}_0\kappa_1$ (so $\rho = \lambda_1/2 \leq \bar{\rho}/2$), and $\ell \geq \lambda_2/2 \geq r^2/4$. (c) $\operatorname{meas}\bar{D}_t \geq \frac12 \operatorname{meas} V$, and $\operatorname{meas}\check{D} \geq \frac14 \operatorname{meas} V$, $\operatorname{meas}\bar{D}_t \leq 4 \operatorname{meas}\check{D}$.*

Proof. (a) $\mu < r^6(\log\frac{1}{r})^{-c}$ and $\Lambda_P \leq \log\frac{1}{r}$ give the first; $\mu_2\bar{K}^{2\tau+1} \leq r^6(30/s_*)^{2\tau+1}$, so $\sqrt{\mu_2}\bar{K}^{\tau+1/2}r^{-5/2} \leq (30/s_*)^{\tau+1/2}\sqrt{r}$, and $r^{5/2}\bar{K}^{\tau+1} \leq (30/s_*)^{\tau+1}r^{1/2}(r^2\log^{\tau+1}\frac{1}{r})$, both $\leq (30/s_*)^{\tau+1}\sqrt{r}$ by (r1). (b) $\bar{\rho} \leq \bar{\gamma}/(2\bar{M}_0) \leq C_\gamma\sqrt{r}/(2\bar{M}_0) \leq \rho_c$ by (r3); $\bar{\rho} \geq Gr^{5/2}/(2\bar{M}_0) \geq \lambda_1$ from the second branch of $\bar{\gamma}$; $\ell = \lambda_2(1 - 6\kappa_1\sqrt{r}) \geq \lambda_2/2$ by (r4). (c) [P:`bpP:p4:collar`](b): $\operatorname{meas}\bar{D}_t \geq \operatorname{meas} V - At \geq \frac12 \operatorname{meas} V$ (r2); $\operatorname{meas}(\bar{D}_t \setminus \check{D}) \leq \operatorname{meas}(V \setminus \bar{\omega}_0^{-1}\mathcal{D}^1_{\bar{\gamma}}) \leq c_{\mathrm{dio}}\bar{\gamma} \leq \frac14 \operatorname{meas} V$ (Lemma 7.8, (r3)); here $\frac12 - \frac14 = \frac14$. □

7.5. **Step 1 of CP10 §4: averaging (Part 2).**

Proposition 7.15 (Averaged system in CP variables)**.** *For r, μ as above there is a map $\bar{\Psi}$ on $\mathcal{W} := \{(I,\varphi,p,q) \in \mathbb{C}^{2n} :\ I \in \check{D}_{\bar{\rho}/2},\ |\operatorname{Im}\varphi| < \sigma_{\mathrm{av}},\ |(p,q)|_\infty < \varepsilon_0/2\}$ (angles of period 2π), holomorphic, injective, real at real points, with $\bar{\Psi} - \mathrm{id}$ 2π-periodic in φ, mapping $\mathcal{W}$ into the domain of $\tilde{P}$ of Lemma 7.2, with $D\bar{\Psi}^\top M_0 D\bar{\Psi} = \pm M_0$ (it preserves $\pm\Omega$), and*

$$H_\mu \circ \bar{\Psi} = \tilde{H}_0(I) + \mu\tilde{N}(I,R) + \mu Q(\mu, I; p, q) + \widetilde{P}_{\mathrm{rem}}, \qquad |\widetilde{P}_{\mathrm{rem}}|_{,\sigma_{\mathrm{av}}} \leq 2\Lambda_P\mu r^5 \quad (\text{on } \mathcal{W}),$$

and, writing $(\bar I,\bar\varphi,\bar p,\bar q)\mapsto(I,\varphi,p,q)$*:* $|I-\bar I|\le\Delta_I:=\dfrac{12\mu_2\bar K^\tau}{s_*\bar\gamma}$, $|p-\bar p|,|q-\bar q|\le\Delta_{\max}:=\dfrac{18\mu_2\bar K^\tau}{\varepsilon_0\bar\gamma}$, *where, by the choice of* $\bar\gamma$, $\Delta_{\max}\le\dfrac{18}{G\varepsilon_0}r^{11/2}$, $\Delta_I\le\dfrac{12}{Gs_*}r^{11/2}$.

Proof. **Step S7.19: Dictionary.** Apply [P:`bpP:p2:averaged-remainder`] with $n=a$, $m=b$, $\bar D=\check D$, $\bar\rho=\bar\rho_{\rm CP}$, $\epsilon=r$, $\mu\to\mu_2$, $P\to\tilde P/\Lambda_P$ (so $|\cdot|,\le1$ by Definition 7.3, and $H_\mu=H_0+\mu_2(\tilde P/\Lambda_P)$), in CP variables through the canonical chart $(\theta,I,p,q)=(\varphi/2\pi,2\pi I_{\rm CP},p,q)$ of [P:`bpP:p2:use5`] (3): $s_0=s_*/2\pi$, $\bar\alpha=2\pi\alpha=\alpha_{\rm av}$, $d=\min\{\bar\rho s_*,\varepsilon_0^2\}=\bar\rho s_*$ ($\bar\rho\le\rho_c<\varepsilon_0^2/s_*$ after decreasing ρ_c), $\kappa=\bar K s_*=30\log\frac1r$ and $e^{-\kappa/6}=r^5$.

Step S7.20: (R1). Let $I'\in\check D_{\bar\rho}$, $I\in\check D$ with $|I'-I|<\bar\rho$ and $0<|k|_1\le\bar K$. As $\bar\omega_0(I)\in\mathcal D^1_{\bar\gamma}$, $|k\cdot\bar\omega_0(I)|\ge\bar\gamma\bar K^{-\tau}$; the mean value inequality on the segment $[I,I']\subset(\overline V)_{\rho_c}$ gives $|k\cdot(\tilde\omega_0(I')-\bar\omega_0(I))|\le|k|_1\bar M_0\bar\rho\le\bar K\bar M_0\bar\rho=\bar\gamma\bar K^{-\tau}/2$. So $|k\cdot\tilde\omega_0(I')|\ge\bar\gamma/(2\bar K^\tau)=\alpha_{\rm av}$ (CP10 (39)).

Step S7.21: (R2) and the remainder. $\alpha_{\rm av}d=\bar\gamma^2s_*/(4\bar M_0\bar K^{2\tau+1})\ge G^2s_*\mu_2r^{-5}/(4\bar M_0)$ (first branch of $\bar\gamma$). (R2): $2^{12}(1+b)\bar K s_*\mu_2\le\alpha_{\rm av}d$ holds if $G^2\ge2^{14}(1+b)\bar M_0$ (as $\bar Kr^5\le1$); and $400(1+b)\mu_2/(\alpha_{\rm av}d)\le r^5$ holds if $G^2\ge1600(1+b)\bar M_0/s_*$. Both are conditions on G (Lemma 7.19). Then $|\widetilde P_{\rm rem}|,\le\mu_2r^5+\mu_2r^5=2\Lambda_P\mu r^5$ ([P:`bpP:p2:averaged-remainder`]).

Step S7.22: Identification. In [P:`bpP:p2:averaged-remainder`], $\mu_2\langle P_2\rangle=\mu\tilde P_{\rm av}=\mu\tilde N+\mu Q$ (Lemma 7.4), which gives the displayed identity. **Displacements:** [P:`bpP:p2:use5`] (3) converts [P:`bpP:p2:averaged-remainder`]'s bounds to $\alpha_{\rm av}s_*|\Delta I|\le6\mu_2$, $\alpha_{\rm av}\varepsilon_0|\Delta p|,|\Delta q|\le9\mu_2$; insert $\alpha_{\rm av}=\bar\gamma/2\bar K^\tau$ and $\bar\gamma\ge G\sqrt{\mu_2}\bar K^{\tau+1/2}r^{-5/2}$: $\mu_2\bar K^\tau/\bar\gamma\le\sqrt{\mu_2}\,r^{5/2}\bar K^{-1/2}/G\le r^{11/2}/G$. □

7.6. The KAM Hamiltonian in oscillator action–angle variables.

Definition 7.16 (Polar chart, composite chart Φ)**.** $\mathrm{Pol}(\psi,J)=\big((\varphi,q),(I,p)\big)$ with $\varphi=\psi_{\rm f}$, $I=J_1$, $(q_j,p_j)=\sqrt{2J_{2,j}}(\sin\psi_{{\rm s},j},\cos\psi_{{\rm s},j})$ ($J_2\in(0,\infty)^b$): the map Π of [P:`bpP:p4:s-disc`], in the frozen order (position, momentum); it extends holomorphically to $\{\mathrm{Re}\,J_2>0\}\times\mathbb C^n$ (principal square root). Let $O_0:=\{(\psi,J)\in\mathrm{Ph}(n):\ J_1\in\mathrm{Re}(\check D_{\bar\rho/2}),\ J_2\in(\lambda_1/2,2\lambda_2)^b\}$ (open, invariant under $\psi\mapsto\psi+2\pi k$) and $\Phi:=\bar\Psi\circ\mathrm{Pol}$ (real restriction on O_0, any extension elsewhere), $\Phi:O_0\to\mathrm{Ph}(n)$.

Lemma 7.17 (Φ is a 1-chart with the lattice properties of [P:`bpP:p1:cyl-pipeline`])**.** *Φ is a symplectic chart on O_0 (Definition 2.3, $c=1$), $O_0+(2\pi k,0)=O_0$, $\Phi(z+(2\pi k,0))=\Phi(z)+\mathrm{shift}(k_{\rm f})$ for $k\in\mathbb Z^n$, $z\in O_0$, and $\Phi(z')=\Phi(z)+\mathrm{shift}(k_{\rm f})\Rightarrow z'=z+(2\pi m,0)$ for some $m\in\mathbb Z^n$. Moreover $H_\mu\circ\Phi=h_\mu+f_\mu$ on O_0, where, with $N(J)=\tilde N(J_1,J_2)$,*

$$h_\mu(J)=\tilde H_0(J_1)+\mu\tilde N(J),\qquad f_\mu=\big(\mu Q+\widetilde P_{\rm rem}\big)\circ\mathrm{Pol}.$$

Proof. **Step S7.23: Image.** For $z\in O_0$: $|\bar p|,|\bar q|\le\sqrt{2J_2}<2\sqrt{\lambda_2}\le2r<\varepsilon_0/2$ and $I\in\check D_{\bar\rho/2}$: $\mathrm{Pol}(O_0)\subset\mathcal W\cap\mathbb R^{2n}$, so $\bar\Psi$ is defined and holomorphic there. $\bar\Psi$ real analytic and symplectic: [P:`bpP:p1:hol-chart`] (with Proposition 7.15); Pol is a 1-chart on $\{J_2>0\}$: [P:`bpP:p1:typical`](1) for each oscillator and the identity on (φ,I); composition: [P:`bpP:p1:typical`](3).

Step S7.24: Lattice. $\mathrm{Pol}(\psi+2\pi k,J)=\mathrm{Pol}(\psi,J)+\mathrm{shift}(k_{\rm f})$ ($\psi_{\rm s}$ enters through 2π-periodic functions; shift moves only the fast angles) and $\bar\Psi(w+\mathrm{shift}(k_{\rm f}))=\bar\Psi(w)+\mathrm{shift}(k_{\rm f})$ (Proposition 7.15: $\bar\Psi-\mathrm{id}$ periodic). Injectivity modulo the lattice: if $\Phi(z')=\Phi(z)+\mathrm{shift}(k_{\rm f})=\Phi(z+(2\pi(k_{\rm f},0),0))$, injectivity of $\bar\Psi$ gives $\mathrm{Pol}(z')=\mathrm{Pol}(z+(2\pi(k_{\rm f},0),0))$, and Pol is injective modulo $\psi\mapsto\psi+2\pi k$ on $\{J_2>0\}$ ([P:`bpP:p4:disc-measure`](a)).

Step S7.25: Hamiltonian. $H_\mu\circ\bar\Psi$ is given by Proposition 7.15; compose with Pol, $R_j\circ\mathrm{Pol}=J_{2,j}$ ([P:`bpP:p0:oscaction`]). □

Lemma 7.18 (Hypotheses of the two-scale KAM theorem)**.** *Let $H_3(J,\theta) := \tilde{H}(J, 2\pi\theta)/2\pi$ where $\tilde{H} = h_\mu + f_\mu$ in (J,ψ), i.e. $h_3 = h_\mu/2\pi$, $f_3 = f_\mu(\cdot, 2\pi\theta)/2\pi$. There is $C_1 \geq 1$, depending on the data only (Remark 7.1, items 2–3), such that for $0 < r < r_*$, $0 < \mu < \min\{\mu_0, r^6/(\log\frac{1}{r})^c\}$, the hypotheses of* [P:bpP:p3:two-scale-kam] *hold with: $n_1 = a$, $n_2 = b$, $\tau = n+1$, $D = \check{D} \times A_{\mathrm{sh}}$, $\rho = \lambda_1/2$, $A = D_\rho$, $s, \bar{s}$ of Definition 7.12, and*

$$M = C_1, \quad \hat{M} = C_1\mu, \quad \bar{M} = \bar{M}_2 = C_1\mu^{-1}, \quad \bar{M}_1 = C_1, \quad E = C_1\mu r^5 \qquad (\textit{the values of } \text{[P:bpP:p3:interface]} \textit{ (2}$$

Explicitly $C_1 := \max\{1,\ 2\pi C_U,\ C_R \coth(\sigma_{\mathrm{av}}/4)^n/2\pi\}$ with $C_R := 2^5 C_Q + 2\Lambda_P$ and C_U of Lemma 7.10.

Proof. **Step S7.26: Domain.** $D_\rho = \check{D}_\rho \times (A_{\mathrm{sh}})_\rho$ (max norm) $\subset \mathcal{E}_1$, since $\rho \leq \bar{\rho}/2 \leq \rho_c$ and $|J_2| < \lambda_2' + \rho < 1$; also $\operatorname{Re} J_2 > \lambda_1/2$. The polar image of $D_\rho \times \{|\operatorname{Im}\psi| < \sigma_{\mathrm{av}}\}$: $|(\bar{p},\bar{q})|_\infty \leq \sqrt{2|J_2|}\, e^{\sigma_{\mathrm{av}}} < 2r \leq \varepsilon_0/2$ (as $|J_2| < \lambda_2$, $\sigma_{\mathrm{av}} \leq \frac{1}{12}$), $|\operatorname{Im}\varphi| < \sigma_{\mathrm{av}}$, $J_1 \in \check{D}_{\bar{\rho}/2}$: inside $\mathcal{W}$. So f_μ is holomorphic on $D_\rho \times \{|\operatorname{Im}\psi| < \sigma_{\mathrm{av}}\}$, 2π-periodic, real on real points; h_μ is holomorphic on D_ρ.

Step S7.27: h. Lemma 7.10: $\partial h_3 = F_\mu/2\pi$ injective, $U_3 = U/2\pi$ nonsingular on A, $\|U_3\| \leq C_U$, $\|\hat{U}_3\| \leq C_U\mu$, $\|U_3^{-1}\| \leq 2\pi C_U\mu^{-1}$, $\|T_1\| \leq 2\pi C_U$, $\|T_2\| \leq 2\pi C_U/\mu$ (the factors 2π from $U_3^{-1} = 2\pi U^{-1}$). $\mu < \mu_*$ by Lemma 7.14(a).

Step S7.28: f. On the polar image $|x|_\infty < 2r \leq r_1$: $|\mu Q| \leq \mu C_Q (2r)^5$ (Proposition 7.5; $I \in \hat{D}_c'$), $|\widetilde{P}_{\mathrm{rem}}| \leq |\widetilde{P}_{\mathrm{rem}}|_* \leq 2\Lambda_P \mu r^5$ ([N:bpN:p4:sup-weighted]); so $\sup|f_\mu| \leq C_R\mu r^5$ on the strip σ_{av}. Sup-to-weighted conversion (proof of Definition 7.3, all n angles, loss $\sigma_{\mathrm{av}} - \sigma_{\mathrm{tot}} = \sigma_{\mathrm{av}}/2$): $\|f_3\|_{\rho,\sigma_{\mathrm{tot}}} \leq C_R \coth(\sigma_{\mathrm{av}}/4)^n \mu r^5/2\pi \leq C_1\mu r^5 = E$.

Step S7.29: Normalizations. $\hat{M} = C_1\mu \leq M = C_1$, $\bar{M}_i \leq \bar{M}$ since $\mu \leq 1$. All numerical constants are upper bounds as required ([P:bpP:p3:rem-norm]). □

7.7. The scale constants, the KAM condition, and r_K (replacing [P:bpP:p4:use5]).

Lemma 7.19 (Scale constants and the KAM condition)**.** *Let $\hat{c} = \hat{c}(n,\tau) = \max\{2^{13}n, 24^{\tau+1}/3\}$ (*[P:bpP:p3:constants]*) and C_1, s as above. There are constants, chosen* in this order*:*

(1) $\kappa_0 \geq C_1^2$ with $\kappa_0^2 > 9\hat{c}C_1^4 K_0^{2\tau+2}$, $K_0 = \frac{6}{s}\log_+(\kappa_0^2/C_1^4)$ (exists: $\kappa_0^2/(\log\kappa_0)^{2\tau+2} \to \infty$);
(2) κ_1 with $\kappa_1^2 > 4\hat{c}C_1^2$;
(3) $G \geq \max\{2\bar{M}_0\kappa_1,\ (2^{14}(1+b)\bar{M}_0)^{1/2},\ (1600(1+b)\bar{M}_0/s_*)^{1/2}\}$.

With them, for all $r \in (0,1)$ small enough that $\rho = \lambda_1/2$ makes sense and all $\mu \in (0,1]$ (Part 3's formulas, with $E = C_1\mu r^5$ etc. of Lemma 7.18): $L = C_1/\mu$, $x = C_1^4 r^5/\gamma_1^2 = C_1^4/\kappa_0^2$, $K = K_0$ independent of r, μ,

$$\hat{\rho} = \min\Big\{\frac{\kappa_0 r^{5/2}}{3C_1 K_0^{\tau+1}},\ \frac{\kappa_1 r^{5/2}}{2}\Big\}, \qquad \hat{E} = \frac{EL}{\hat{\rho}^2} = \frac{C_1^2 r^5}{\hat{\rho}^2} = \max\Big\{\frac{9C_1^4 K_0^{2\tau+2}}{\kappa_0^2}, \frac{4C_1^2}{\kappa_1^2}\Big\} < \frac{1}{\hat{c}},$$

*that is, the **KAM condition** $\hat{c}\hat{E} < 1$ holds, $\gamma_1 \geq \gamma_2$, and $\tau > n$.*

Proof. **Step S7.30: Evaluation.** $L = \max\{\bar{M}, M^{-1}, \hat{M}^{-1}\} = \max\{C_1/\mu, 1/C_1, 1/(C_1\mu)\} = C_1/\mu$ ($C_1 \geq 1$, $\mu \leq 1$). $EM^2L = C_1^4 r^5$, so $x = C_1^4 r^5/\gamma_1^2 = C_1^4/\kappa_0^2 \leq 1$ and $K = \frac{6}{s}\log_+(x^{-1}) = K_0$ exactly ([P:bpP:p3:interface] (2)). Since $\gamma_2/(3\hat{M}K^{\tau+1}) = \mu\kappa_0 r^{5/2}/(3C_1\mu K_0^{\tau+1}) = \gamma_1/(3MK^{\tau+1})$, $\hat{\rho}$ is as displayed; $\hat{E} = EL/\hat{\rho}^2 = C_1^2 r^5/\hat{\rho}^2$; $\hat{\rho}^{-2} r^5 = \max\{(3C_1 K_0^{\tau+1}/\kappa_0)^2, 4/\kappa_1^2\}$.

Step S7.31: Smallness. Items 1, 2 give $\hat{c}\hat{E} < 1$. $\gamma_2 = \mu\gamma_1 \leq \gamma_1$. □

Lemma 7.20 (The radius r_K (what Part 4 should state instead of $r_K \le C\hat\rho \le C\gamma_2$))**.** *In the situation of Lemma 7.19, $r_K := 32\hat E\hat\rho$ (the enlargement radius of* [P:bpP:p3:two-scale-kam]*) satisfies, for all admissible (r,μ),*

$$r_K < \tfrac{1}{2^8}\hat\rho \le \tfrac{1}{2^9}\lambda_1, \qquad r_K < \min\{t,\lambda_1\}, \qquad r_K < \tfrac{\bar\rho}{2}, \qquad \frac{2r_K}{\ell} \le \frac{\kappa_1}{64}\sqrt{r}.$$

Proof. **Step S7.32: Direct.** $\hat c \ge 2^{13}$ and $\hat c\hat E < 1$ give $\hat E < 2^{-13}$, so $r_K = 32\hat E\hat\rho < 2^{-8}\hat\rho$, and $\hat\rho \le \rho = \lambda_1/2$. Further $\lambda_1 = \kappa_1 r^{5/2} < \sqrt{r} = t$ for $r < 1$ ($\kappa_1 r^2 < 1$ by (r4)), and $\lambda_1 \le \bar\rho$ (Lemma 7.14(b)). Finally $\ell \ge r^2/4$ gives $2r_K/\ell \le 2\kappa_1 r^{5/2}/(512)\cdot 4/r^2$. □

Remark 7.21 (Defect in [P:bpP:p4:use5] (review P4-02)). Part 4's remark claims $r_K \le C\hat\rho \le C\gamma_2$. This is false with $\hat M = C\mu$, since $\hat\rho \le \gamma_2/(3C\mu K^{\tau+1})$. The correct chain, used here, is $r_K \le 2^{-8}\hat\rho$, $\hat\rho \le \min\{\gamma_1,\gamma_2/\mu\}/(3C_1K^{\tau+1})$; with $\gamma_1 = \gamma_2/\mu = \kappa_0 r^{5/2}$ this gives $r_K/\ell = O(\sqrt{r})$ with a constant depending on κ_1 only. Part 4 should state: "$r_K \le C\hat\rho \le C\min\{\gamma_1,\gamma_2/\mu\}$". Also [P:bpP:p4:domain-collar] needs $\operatorname{meas}\bar D_t > 0$ (review P4-01): ensured here by $A\sqrt{r} \le \frac12 \operatorname{meas} V$.

7.8. Location, the tori, and the Kolmogorov set.

Lemma 7.22 (Location of the tori in the cylinder)**.** *Let $\varphi_\omega = (v, \vartheta + u)$ be a torus of* [P:bpP:p3:two-scale-kam] *(so $v \in \mathrm{Re}(D_{r_K})$), and let $z = \Phi(2\pi(\vartheta+u), v) \in \mathrm{Ph}(n)$. Then $v \in O_0$-range ($v_1 \in \mathrm{Re}(\check D_{\bar\rho/2})$, $v_2 \in (\lambda_1/2, 2\lambda_2)^b$), $I(z) \in V$ and $p_j^2 + q_j^2 < r^2$ for all j; thus $z \in \mathcal{C}(V,r)$. Here $\Delta_{\max}$ is that of Proposition 7.15, and (r5) is the condition $\sqrt{2}\Delta_{\max} \le \kappa_1 r^{3/2}$ (satisfied for small r since $\Delta_{\max} \le \frac{18}{G\varepsilon_0} r^{11/2}$).*

Proof. **Step S7.33: Range.** $v \in \mathrm{Re}(D_{r_K})$ means $\operatorname{dist}(v_1, \check D) < r_K < \bar\rho/2$ (Lemma 7.20) and $v_2 \in (\lambda_1 - r_K, \lambda_2' + r_K)^b \subset (\lambda_1/2, 2\lambda_2)^b$.

Step S7.34: Actions. $I = \bar I + (I - \bar I)$ with $\bar I = v_1$, $\operatorname{dist}(\bar I, \bar D_t) < r_K$ and $|I - \bar I| \le \Delta_I$ (Proposition 7.15); $r_K + \Delta_I < t$, hence $I \in N_{r_K+\Delta_I}(\bar D_t) \subset V$ by [P:bpP:p4:collar](c).

Step S7.35: Discs. For the polar point $(\bar p, \bar q)$: $\bar p_j^2 + \bar q_j^2 = 2v_{2,j} < 2(\lambda_2' + r_K) = r^2 - 4\lambda_1 + 2r_K \le r^2 - 3\lambda_1$ (Lemma 7.20), so $\sqrt{\bar p_j^2 + \bar q_j^2} < r - \frac{3\lambda_1}{2r} = r - \frac32\kappa_1 r^{3/2}$. With $|p - \bar p|, |q - \bar q| \le \Delta_{\max}$: $\sqrt{p_j^2 + q_j^2} \le \sqrt{\bar p_j^2 + \bar q_j^2} + \sqrt{2}\Delta_{\max} < r - \frac32\kappa_1 r^{3/2} + \kappa_1 r^{3/2} < r$. □

Definition 7.23 (Torus family and Kolmogorov set)**.** Let $\Omega_* := \omega_0(D) \cap \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ ($\omega_0 = \partial h_3 = F_\mu/2\pi$), $D_0 := D^{\mathrm{dio}} = \omega_0^{-1}(\Omega_*) \cap D$, and for each Borel $B \subset D_0$ with $\Omega_B := \omega_0(B)$ and φ_ω as in [P:bpP:p3:two-scale-kam],

$$K_\omega(\theta) := \Phi\big(S_{2\pi}\,\pi\,\varphi_\omega(\theta)\big) \quad (\omega \in \Omega_*), \qquad S_B := \bigcup_{\omega\in\Omega_B} K_\omega(\mathbb{R}^n), \qquad \mathsf{K}_B := \bigcup_{\omega\in\Omega_B} \varphi_\omega(\mathbb{R}^n) \subset \mathbb{R}^n\times\mathbb{R}^n \text{ (coordinates } (J,$$

$(\pi(J,\vartheta) = (\vartheta, J)$, $S_{2\pi}$ of [P:bpP:p1:period-convert].) D_0 is relatively closed in D ($\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ closed, F_μ continuous), $D = \check D \times A_{\mathrm{sh}}$ is σ-compact, so D_0 is σ-compact.

Proposition 7.24 (Each torus is a frozen cylinder torus with frozen frequencies)**.** *For $0 < r < r_*$ and admissible μ, $\omega \in \Omega_*$, $w := \omega$:* $\mathrm{IsCylTorus}(H_\mu, \mathcal{C}(V,r), w, K_\omega)$*; w is $(\min\{\gamma_1,\gamma_2\},\tau)$-Diophantine with $\tau = n+1 > a+b$; $|w^{\mathrm f}|_\infty \le C_w$, $|w^{\mathrm s}|_\infty \le C_w\mu$ with $C_w := \frac{M_\omega + A_N}{2\pi} + 1$, $M_\omega := \sup_{\overline V}|\bar\omega_0|$; $K_\omega(\mathbb{R}^n) \subset \mathcal{C}(V,r)$.*

Proof. **Step S7.36: Chart data.** Lemma 7.17 supplies O_0, Φ with the hypotheses of [P:bpP:p1:cyl-pipeline] ($d = n$); $G_3 = (2\pi)^{-1}(H_\mu \circ \Phi)(2\pi z_1, z_2)$, which by Lemma 7.17 equals H_3 in the frozen order π on $O = S_{2\pi}^{-1}(O_0)$.

Step S7.37: $\mathrm{LT}(G_3, O, w, K_3)$**,** $K_3 = \pi \circ \varphi_\omega$**.** (L1): φ_ω is real analytic ([P:bpP:p3:two-scale-kam](a)); $K_3(\mathbb{R}^n) \subset O$ by Lemma 7.22, Step S7.33; G_3 is differentiable at $K_3(\theta)$ (chart-field Lemma 2.4: H_μ is analytic near $\Phi(\dots) \in \mathcal{C}(V,r)$ since $r < r_0$, (H1), H_0 analytic on W). (L2): v, u are 1-periodic. (L3): injectivity modulo translations (Proposition 4.3, with equal ω). (L4), (L5): Proposition 4.3(c); the displayed vanishing sum is $\Omega(DK_3\,\cdot, DK_3\,\cdot)$ up to sign. (L6): Proposition 4.3(b), with π ([P:bpP:p1:order]), where $H_3 = G_3$ on the open set $\mathrm{Re}(D_\rho) \times \mathbb{R}^n \ni K_3(\theta)$ ($r_K < \rho$) so the fields agree.

Step S7.38: Pipeline. [P:bpP:p1:cyl-pipeline] with $U = \mathcal{C}(V,r) \supset \Psi(K_3(\mathbb{R}^n)) = K_\omega(\mathbb{R}^n)$ (Lemma 7.22) gives $\mathrm{IsCylTorus}(H_\mu, \mathcal{C}(V,r), w, K_\omega)$ with the same $w = \omega$.

Step S7.39: Frequencies. $\omega = \omega_0(I_0)$, $I_0 = (J_1, J_2) \in D$ real, $|J_2| \le \lambda_2 \le 1$: $w^{\mathrm{f}} = (\bar\omega_0(J_1) + \mu G)/2\pi$, $w^{\mathrm{s}} = \mu\Theta/2\pi$, bounded using $|G|, |\Theta| \le A_N$. Diophantine: [P:bpP:p1:dio-sources](a) ($\gamma_1, \gamma_2 > 0$, $\tau > n$). □

Lemma 7.25 (Borelness, whole tori, cell measure)**.** *For Borel $B \subset D_0$ with σ-compact B: (a) S_B is Borel, $S_B \subset \mathcal{C}(V,r)$, every point of S_B lies on some $K_\omega(\mathbb{R}^n) \subset S_B$. (b)* $\mathrm{meas}(S_B \cap \mathcal{F}(V,r)) = (2\pi)^n \,\mathrm{meas}\big(\mathsf{K}_B \cap (\mathbb{R}^n \times [0,1)^n)\big)$.

Proof. **Step S7.40: Cells.** Let $\mathsf{K}_B^{\mathrm{c}} := \Phi_3(\check\imath(B) \times [0,1)^n)$ in the notation of Part 3 ($\mathsf{K}_B = \bigcup_{k\in\mathbb{Z}^n}(\mathsf{K}_B^{\mathrm{c}} + (0,k))$, since $\varphi_\omega(\vartheta + k) = \varphi_\omega(\vartheta) + (0,k)$); $\check\imath$ and Φ_3 are continuous ([P:bpP:p3:limit-actions], [P:bpP:p3:limit-phi]), so $\mathsf{K}_B^{\mathrm{c}}$ is σ-compact (B is), as is $T := \Phi(S_{2\pi}\pi\mathsf{K}_B^{\mathrm{c}})$ (Φ is continuous on O_0). Hence $S_B = \bigcup_{k\in\mathbb{Z}^a}(T + \mathrm{shift}(k))$ is Borel (Lemma 7.17: $K_\omega(\theta + k) = K_\omega(\theta) + \mathrm{shift}(k_{\mathrm{f}})$ and slow shifts act trivially). (a) follows with Lemma 7.22.

Step S7.41: Disjointness. The sets $T + \mathrm{shift}(k)$, $k \in \mathbb{Z}^a$, are pairwise disjoint: if $\Phi(z) + \mathrm{shift}(k) = \Phi(z') + \mathrm{shift}(k')$ with $z, z' \in Q'' := \{\psi \in [0,2\pi)^n\}$, Lemma 7.17 gives $z' = z + (2\pi m, 0)$, hence $m = 0$, $k = k'$.

Step S7.42: Counting. $S_B \subset \mathcal{C}(V,r)$, so $S_B \cap \mathcal{F} = S_B \cap \{\varphi \in [0,2\pi)^a\}$ and, by translation invariance of meas, $\mathrm{meas}(S_B \cap \{\varphi \in [0,2\pi)^a\}) = \sum_k \mathrm{meas}\big(T \cap \{\varphi \in [-2\pi k, 2\pi(1-k))\}\big) = \mathrm{meas}\, T$ (the slabs partition $\mathbb{R}^a$).

Step S7.43: Liouville. Φ is a 1-chart, injective on $Q'' \cap O_0$ (Step S7.41); [P:bpP:p1:liouville]: $\mathrm{meas}\, T = \mathrm{meas}(S_{2\pi}\pi\mathsf{K}_B^{\mathrm{c}}) = (2\pi)^n \,\mathrm{meas}\, \mathsf{K}_B^{\mathrm{c}}$ ($S_{2\pi}$ multiplies the volume by $(2\pi)^n$, [P:bpP:p1:period-convert](a)). Finally $\mathsf{K}_B^{\mathrm{c}} = \mathsf{K}_B \cap (\mathbb{R}^n \times [0,1)^n)$. □

7.9. **The measure budget (replacing [P:bpP:p4:budget] and [P:bpP:p4:cell-budget] for $D = \check D \times A_{\mathrm{sh}}$).**

Lemma 7.26 (Budget on the action side)**.** *There is $C_{\mathrm{b}} < \infty$ (data and the constants κ_0, κ_1, G only) such that for admissible (r,μ), with $\mathsf{K} = \mathsf{K}_{D_0}$ and $c_n = 1 + (1+2^n)(1+(4/3)^n)$,*

$$\mathrm{meas}\big((D \times \mathbb{T}^n) \setminus \mathsf{K}\big) \le c_n C_{\mathrm{b}} \sqrt{r}\ \mathrm{meas}\, D, \qquad \mathrm{meas}\, D = \mathrm{meas}\, \check D\, \ell^b > 0.$$

Proof. **Step S7.44: Reading (3.1).** $\mathsf{K} = K_\Omega$ of [P:bpP:p3:two-scale-kam] for (Ω_*, r_K); $D \subset \mathrm{Re}(D_{r_K})$, so $(D \times \mathbb{T}^n) \setminus \mathsf{K} \subset (\mathrm{Re}(D_{r_K}) \times \mathbb{T}^n) \setminus \mathsf{K}$, of measure $\le c_n\big(\mathrm{meas}(D \setminus D_0) + \mathrm{meas}(\mathrm{Re}(D_{r_K}) \setminus D)\big)$.

Step S7.45: Diophantine part. $D \setminus D_0 \subset \mathcal{R} := \{(I,\xi) \in \bar D_t \times A_{\mathrm{sh}} : F_\mu(I,\xi) \notin \mathcal{D}_{\alpha_1,\alpha_2,\tau}\}$, $\alpha_i = 2\pi\gamma_i$ (dictionary [P:bpP:p4:freq]). The standing data (S4) of [P:bpP:p4:data] hold with $\bar D = \bar D_t$ (compact, meas > 0 by Lemma 7.14(c)), $\lambda_0 = 1$, μ_1 of (r1), and the constants of Lemma 7.9. [P:bpP:p4:two-scale-diophantine-measure] with $\lambda = \lambda_2' \in [\lambda_2/2, 1]$, $Q = A_{\mathrm{sh}}$, $\bar\gamma_{\mathrm{P4}} = \alpha_2/\mu = 2\pi\kappa_0 r^{5/2}$: $\mathrm{meas}\,\mathcal{R} \le [c_1\alpha_1(1 + \mu^{\tau-n}/\lambda) + c_2\bar\gamma_{\mathrm{P4}}/\lambda]\,\mathrm{meas}\,\bar D_t (2\lambda)^b$. Here c_1, c_2 (which depend on $\bar D$ only through $1/\mathrm{meas}\,\bar D$) are $\le \bar c_1, \bar c_2$, data-dependent, as $\mathrm{meas}\,\bar D_t \ge \frac12\,\mathrm{meas}\, V$; $\tau - n = 1$, $\mu \le r^6$, $\lambda \ge r^2/4$ give $\mu/\lambda \le 4$ and $\bar\gamma_{\mathrm{P4}}/\lambda \le 8\pi\kappa_0 r^{1/2}$. Also $(2\lambda)^b \le (4\ell)^b$ ($\lambda_1 \le \lambda_2'/2$), $\mathrm{meas}\,\bar D_t \le 4\,\mathrm{meas}\,\check D$. So $\mathrm{meas}(D \setminus D_0) \le 4^{b+1} 2\pi\kappa_0(5\bar c_1 + 4\bar c_2)\sqrt{r}\ \mathrm{meas}\, D$.

Step S7.46: Collar part. $\mathrm{Re}(D_{r_K}) = N_{r_K}(\check D) \times (\lambda_1 - r_K, \lambda_2' + r_K)^b$ (max norm, [P:bpP:p4:domain-collar], Step S6.16). $N_{r_K}(\check D) \setminus \check D \subset (N_{r_K}(\bar D_t) \setminus \bar D_t) \cup (\bar D_t \setminus \check D)$, of measure $\le At + c_{\mathrm{dio}}\bar\gamma \le (A + c_{\mathrm{dio}}C_\gamma)\sqrt{r}$ ([P:bpP:p4:collar](c) with $r_K < t < d$; Lemma 7.8). As in [P:bpP:p4:domain-collar], Step S6.17, and with $\operatorname{meas}\check D \ge \frac14 \operatorname{meas} V$ and $(1 + 2r_K/\ell)^b \le 2$ (Lemma 7.20, (r6)): $\operatorname{meas}(\mathrm{Re}(D_{r_K}) \setminus D) \le \operatorname{meas} D\,[8(A + c_{\mathrm{dio}}C_\gamma)/\operatorname{meas} V + b\kappa_1/16]\sqrt{r}$. (The denominator $\operatorname{meas}\check D > 0$ is part of the statement: review P4-01.) □

Proposition 7.27 (Lower bound for $\operatorname{meas}(S \cap \mathcal F)$)**.** *Let* $C_* := \dfrac{A + c_{\mathrm{dio}}C_\gamma}{\operatorname{meas} V} + 6b\kappa_1 + c_nC_{\mathrm{b}}$. *For* $0 < r < r_*$ *and admissible* μ *the set* $S := S_{D_0}$ *satisfies* $\operatorname{meas}(\mathcal F(V,r) \setminus S) \le C_*\sqrt{r}\operatorname{meas}\mathcal F(V,r)$, *i.e.* $\operatorname{meas}(S \cap \mathcal F) \ge (1 - C_*\sqrt{r})\operatorname{meas}\mathcal F$.
Source: CP10 §3, (55)–(59) and §4, p. 564: $\operatorname{meas}(P \setminus K_*) \le C(\bar\gamma + \gamma_1 + \bar\gamma_2/\epsilon^2 + \epsilon^{n_2/2})\operatorname{meas} P$, *here with the shell loss* $6b\kappa_1\sqrt{r}$ *(Part 4 erratum E4) and the collar term from* `hboundary`.

Proof. **Step S7.47: Set up.** By Lemma 7.25(b), $\operatorname{meas}(S\cap\mathcal F) = (2\pi)^n \operatorname{meas}\mathsf L$, $\mathsf L := \mathsf K \cap (\mathbb R^n \times [0,1)^n)$; by Lemma 7.22 (the (v_1, v_2) part) $\mathsf L \subset V \times (0,\lambda_2)^b \times [0,1)^n =: \mathsf F'$, and $\operatorname{meas}\mathcal F = (2\pi)^n \operatorname{meas}\mathsf F' = (2\pi)^n \operatorname{meas} V \lambda_2^b$ ([P:bpP:p4:disc-measure](b)). So $\operatorname{meas}(\mathcal F \setminus S) = (2\pi)^n \operatorname{meas}(\mathsf F' \setminus \mathsf L)$ (no passage through $\mathsf E$ is needed: the two measures are compared in the (J,ϑ) coordinates).

Step S7.48: Bad actions. $D \times [0,1)^n \subset \mathsf F'$ and $\operatorname{meas} V\lambda_2^b - \operatorname{meas}\check D\ell^b \le \lambda_2^b(\operatorname{meas} V - \operatorname{meas}\check D) + \operatorname{meas} V\, b\lambda_2^{b-1}\cdot 3\lambda_1$; $\operatorname{meas} V - \operatorname{meas}\check D \le At + c_{\mathrm{dio}}\bar\gamma$ ([P:bpP:p4:collar](b), Lemma 7.8); $3\lambda_1/\lambda_2 = 6\kappa_1\sqrt{r}$. Divide by $\operatorname{meas} V\lambda_2^b$: $\le [(A + c_{\mathrm{dio}}C_\gamma)/\operatorname{meas} V + 6b\kappa_1]\sqrt{r}$.

Step S7.49: Lost tori. $\operatorname{meas}((D \times [0,1)^n) \setminus \mathsf L) = \operatorname{meas}((D \times \mathbb T^n) \setminus \mathsf K) \le c_nC_{\mathrm{b}}\sqrt{r}\operatorname{meas} D \le c_nC_{\mathrm{b}}\sqrt{r}\operatorname{meas}\mathsf F'$ ($\mathsf K$ is $\mathbb Z^n$-invariant in ϑ, Lemma 7.26). Add. □

7.10. Strictness and positivity (ours).

Proposition 7.28 (Deleting a small positive-measure family of tori)**.** *For* $0 < r < r_*$ *and admissible* μ *there is a* σ*-compact* $B \subset D_0$ *such that* $S' := S_{D_0\setminus B}$ *satisfies*

$$0 < \operatorname{meas}(S' \cap \mathcal F), \qquad (1 - (C_* + \tfrac12)\sqrt{r})\operatorname{meas}\mathcal F \le \operatorname{meas}(S' \cap \mathcal F) < \operatorname{meas}\mathcal F.$$

Source: Ours; not in CP10 (12) or CP11 Thm. 11.1, where $\operatorname{meas} P > \operatorname{meas} K$ *is asserted without proof.*
Source gap: The strict inequality is obtained here from the slack $\frac12\sqrt{r}\operatorname{meas}\mathcal F > 0$ *in the lower bound and the bi-Lipschitz parametrisation of Part 3.*

Proof. **Step S7.50: Additivity.** For σ-compact $B \subset D_0$ put $m(B) := \operatorname{meas}(S_B \cap \mathcal F) = (2\pi)^n \operatorname{meas}(\mathsf K_B \cap (\mathbb R^n \times [0,1)^n))$ (Lemma 7.25). The tori $\varphi_\omega(\mathbb R^n)$, $\omega \in \Omega_*$, are pairwise disjoint (Proposition 4.3, injectivity modulo translations), so m is countably additive on σ-compact subsets of D_0.

Step S7.51: Slices are null. Let $x_1(I) = I_1$ (first coordinate) and $H_s = D_0 \cap \{x_1 = s\}$ (a null set). $\mathsf K^{\mathrm c}_{H_s} = \Phi_3(\check\imath(H_s) \times [0,1)^n)$ is the image of $\check\imath(H_s) \times [0,1)^n$ under a Lipschitz map ($\check\imath$ is Lipschitz, Φ_3 Lipschitz in the rescaled norm: [P:bpP:p3:limit-actions], [P:bpP:p3:limit-phi]); [P:bpP:p1:torus-union] gives $\operatorname{meas}\mathsf K^{\mathrm c}_{H_s} \le \Lambda^{2n}\operatorname{meas}\check\imath(H_s) = 0$ (Lipschitz image of a null set).

Step S7.52: Intermediate value. $f(s) := m(D_0 \cap \{x_1 \le s\})$ is nondecreasing, $f(-\infty) = 0$, $f(+\infty) = m(D_0) = \operatorname{meas}(S \cap \mathcal F) \ge (1 - C_*\sqrt{r})\operatorname{meas}\mathcal F > 0$ (Proposition 7.27, (r5)). It is right-continuous (decreasing intersections, finite measure, additivity) and left-continuous by Step S7.51. Put $\theta_0 := \min\{\frac12 m(D_0), \frac12\sqrt{r}\operatorname{meas}\mathcal F\} > 0$ and choose s with $f(s) = \theta_0$; $B := D_0 \cap \{x_1 \le s\}$ is closed in D_0, hence σ-compact, and $D_0 \setminus B = D_0 \cap \{x_1 > s\}$ is σ-compact.

Step S7.53: Conclusion. $\operatorname{meas}(S'\cap\mathcal F) = m(D_0) - \theta_0$. Upper: $\le \operatorname{meas}(S\cap\mathcal F) - \theta_0 \le \operatorname{meas}\mathcal F - \theta_0 < \operatorname{meas}\mathcal F$ ($S \cap \mathcal F \subset \mathcal F$). Lower: $\ge (1 - C_*\sqrt{r})\operatorname{meas}\mathcal F - \frac12\sqrt{r}\operatorname{meas}\mathcal F$; positive since $(C_* + \frac12)\sqrt{r} \le \frac12$ (r5) and $\operatorname{meas}\mathcal F > 0$. □

7.11. **Proof of the frozen theorem.**

Theorem 7.29 (P-G; [P:bpP:p0:pg]). *Frozen Lean declaration:* `KAM.Planetary.properlyDegenerateKAM`

Under the hypotheses and data of [P:bpP:p0:pg]*, with* $c := 2n+4$, $C := \max\{C_*+1, C_w\}$, *and* r_* *of Definition 7.13, the conclusion holds with exactly the printed quantifier order.*

Proof. **Step S7.54: Constants first.** Following Remark 7.1: the data constants of Lemmas 7.2–7.10 (Proposition 7.5); C_U, C_1; κ_0, κ_1, G (Lemma 7.19); $C_\gamma, C_{\mathrm{b}}, C_*$ (Lemma 7.26, Proposition 7.27); C_w; r_* (Definition 7.13, which is a finite list of conditions on these constants; $r_* < r_0$, $r_* < 1$ included); $c = 2n+4$ and C. None depends on r or μ, and $0 < r_* < r_0$, $r_* < 1$, $C > 0$, $c > 0$.

Step S7.55: Given (r, μ)**.** Let $0 < r < r_*$, $0 < \mu < \mu_0$, $\mu < r^6/(\log \frac{1}{r})^c$ (positive base: $r < 1$). All objects of Definition 7.12 are defined, Lemma 7.14 holds; Propositions 7.15, Lemmas 7.18, 7.19 verify the hypotheses of [P:bpP:p3:two-scale-kam] (including the complex injectivity, Lemma 7.10).

Step S7.56: The set. $S := S'$ of Proposition 7.28. By Lemma 7.25(a) it is Borel (so Lebesgue measurable), $S \subset \mathcal{C}(V, r)$, and every $z \in S$ is $K_\omega(\theta)$ for some $\omega \in \Omega_*$, $\theta \in \mathbb{R}^n$, with $K_\omega(\mathbb{R}^n) \subset S$; Proposition 7.24 supplies IsCylTorus$(H_\mu, \mathcal{C}(V,r), w, K_\omega)$, $\gamma > 0$, $\tau > a + b$, $|w^{\mathrm{f}}|_\infty \le C_w \le C$, $|w^{\mathrm{s}}|_\infty \le C_w \mu \le C\mu$. Hence IsCylKol$(H_\mu, V, r, \mu, C, S)$ ([P:bpP:p0:cylkol]).

Step S7.57: Measures. Proposition 7.28: $0 < \mathrm{meas}(S \cap \mathcal{F}) < \mathrm{meas}\,\mathcal{F}$ and $\mathrm{meas}(S \cap \mathcal{F}) \ge (1 - (C_* + \frac{1}{2})\sqrt{r})\,\mathrm{meas}\,\mathcal{F} > (1 - C\sqrt{r})\,\mathrm{meas}\,\mathcal{F}$ since $C \ge C_* + 1$ and $\mathrm{meas}\,\mathcal{F} > 0$. If $1 - C\sqrt{r} \le 0$, $\mathrm{ofReal}(1 - C\sqrt{r}) = 0 < \mathrm{meas}(S \cap \mathcal{F})$; otherwise $\mathrm{ofReal}(1 - C\sqrt{r})\,\mathrm{meas}\,\mathcal{F} = (1 - C\sqrt{r})\,\mathrm{meas}\,\mathcal{F} < \mathrm{meas}(S \cap \mathcal{F})$ ([P:bpP:p0:totality]). $\mathrm{meas}\,\mathcal{F} < \infty$ because V is bounded. □

Remark 7.30 (Quantifier audit). Frozen order: $\exists\, r_*, C, c\ \forall r, \mu\ \exists S$. Constants chosen before r: r_*, C, c and every constant of Remark 7.1, items 1–5 (all functions of the data). Chosen after (r, μ): $\bar{K}, t, \lambda_i, \bar{\gamma}, \bar{\rho}, \rho, \gamma_i, \hat{\rho}, \hat{E}, r_K$, the set D, the family φ_ω, the deleted parameters B, hence S. $K = K_0$ does not depend on (r, μ) at all. The Diophantine exponent $\tau = n + 1$ is data-independent. The frequency bound C and the measure constant coincide, as in the frozen statement.

7.12. **Errata, deviations, external facts, forward references.**

Suspected defects in the sources. (1) CP10 p. 562, (60): "$\sup|\tilde{P}_{\mathrm{av}}| \le C\epsilon^5$" is asserted from (A_2'') (a little-o statement with δ depending on (μ, I, ϵ)); supplied as Proposition 7.5 via joint analyticity, periodicity, the jet identification and Schwarz. (2) CP10 p. 563, Step 2: "we proceed as in Steps 5 and 6" and the measure estimate (56)–(57) uses $\mathrm{meas}(\mathrm{Re}(V_r) \setminus V) \le C\bar{\gamma}_2\,\mathrm{meas}\,V$, which cannot hold for arbitrary V; replaced by `hboundary`. (3) The neighbourhood of the Cantor set $\check{D}$ of $\bar{\omega}_0$-Diophantine actions is not controlled by the collar of V; CP10's (57) writes $\mathrm{Re}(D_r) \setminus D \subset (\mathrm{Re}(V_r) \setminus V) \cup (V \setminus D)$ without proof; Lemma 7.26, Step S7.46 proves it. (4) CP10 (20) $\bar{\gamma}_2 = \gamma_*^2\epsilon^{5/2}$ vs (17) lower bound with $(\log(\epsilon^5/\gamma_1^2)^{-1})^{\tau_*+1}$: consistent only because $\gamma_1 = \gamma_*^2\epsilon^{5/2}$ makes this logarithm a constant (our K_0); not an erratum, but the dependence must be tracked, as done in Lemma 7.19. (5) CP10 p. 564 "replacing ϵ by $\sqrt{2\check{c}_2}\epsilon$": in our frozen normalisation the disc is $\xi_j < r^2/2$ and the shell loss constant is $6b\kappa_1$.

Deviations. Steps 2–4 of CP10 §3 are skipped (the hypothesis is already Birkhoff form); the polynomial $\tilde{N}$ is defined from $\tilde{P}_{\mathrm{av}}$ through its fourth jet; $\check{D}$ replaces $\bar{D} \times A$ with an explicit Cantor-set measure estimate; the top of the shell is cut at $\lambda_2' = \lambda_2 - 2\lambda_1$ so that the tori stay in the open cylinder; $\mu_2 = \Lambda_P\mu$ absorbs the sup-norm of P (CP10 normalises $|P|_{\cdot} = 1$).

External facts used (beyond Parts 1–4). Identity theorem and Cauchy estimates for holomorphic functions of several variables; compactness of closed bounded sets; Cramer's rule; block inversion (Schur complement) and Neumann series; Schwarz lemma in one variable; continuity of measure along monotone sequences; Lebesgue number lemma. Tracks K, N as cited in Parts 1–4.

Forward and cross-track references. Cited from Part 1: `cyl-pipeline`, `hol-chart`, `chart-field`, `typical`, `liouville`, `torus-union`, `dio-sources`, `period-convert`; Part 2: `averaged-remainder`, `use5`; Part 3: `two-scale-kam`, `interface`, `period-dictionary`, `constants`, `rem-norm`, `limit-actions`, `limit-phi`; Part 4: `collar`, `lip-inverse`, `affine-slab`, `lattice-tail`, `two-scale-diophantine-measure`, `data`, `freq`, `domain-collar`, `disc-measure`. Tracks N: `bpN:p4:norm`, `bpN:p4:sup-weighted`. Parts 12–13 are not used by P-G. Part 4 must repair `use5` and `domain-collar` as stated in Remark 7.21.
Label index. `bpP:p5:ext`, `consts`, `birkhoff-ext`, `uniform-remainder`, `lip-complex`, `bilip`, `dio-one`, `freq-data`, `freq-inj`, `params`, `rstar`, `param-facts`, `avg`, `phi`, `phi-chart`, `kam-hyp`, `kam-cond`, `rK`, `location`, `kset`, `torus-conv`, `cell-transfer`, `budget`, `lower`, `strictness`, `main`.

8. Newtonian models and physical reconstruction

This part connects the Hamiltonians H_{mov}, H_{fix} of Part 0 (Definitions 1.22, 1.23) to the physical $(n+1)$-body problem, and supplies the three physical clauses of the frozen Kolmogorov sets (Definitions 1.25, 1.26) once the torus predicate is known. The results are:

- `reduction` (Prop. 8.8) and `split` (Lemma 8.3): H_{mov} is the barycentric N-body Hamiltonian divided by μ, and $H_{\mathrm{mov}} - H_{\mathrm{fix}}|_{\varepsilon=\mu} = \mu|\sum_i p^{(i)}|_2^2/(2m_0)$;
- `hamilton-eqs` (Prop. 8.6): explicit Hamilton equations of both models on the collision-free set;
- `reconstruction`, `reconstruction-fixed` (Props. 8.11, 8.13): an H_{mov}-orbit gives, through the frozen barycentric maps, a solution of Newton's equations with the actual masses; an H_{fix}-orbit gives a fixed-centre solution. *Both are stated pointwise in time, for an arbitrary set $J \subset \mathbb{R}$ of times*, so that the later additions (D22/D25: stopped solutions on $[-T,T]$, `KAM.Planetary.IsNewtonSolutionOn`, `KAM.Planetary.IsMovingSolutionOn`) reuse them verbatim; the all-time frozen predicates are the case $J=\mathbb{R}$;
- `tube-geometry` (Lemma 8.16): explicit constants d, D (chosen from the axes alone, before any coupling) and a tube radius ζ_* such that every point of U_{ζ_*} is collision free with all Sun–planet and planet–planet distances in $[d,D]$, *exactly independently of μ*;
- `orbit-transfer`, `movkol-assembly`, `fixkol-assembly`: the physical clauses of MovKol and FixKol follow from $S \subset U_{\zeta_*}$ and the torus predicate alone;
- `measure-positive` (Lemma 8.19): $0 < \operatorname{meas} S < \infty$.

Nothing here is proved by a hypothesis shaped like its conclusion: the inputs are the Hamilton equations along the orbit, the collision-free clause, and (for the tube) the position of the orbit. *Units.* The frozen definitions keep the gravitational constant G and the Sun's mass m_0 as parameters; Fej04r uses $G=1$. The statements below are for general $G, m_0 > 0$ (the frozen `IsMovingSolutionOn` and `IsNewtonSolutionOn` use $G = m_0 = 1$; see Remark 8.12).

8.1. **Conventions and external facts.** Throughout $n \geq 1$, $G, m_0 > 0$, $m_i > 0$ $(i < n)$, $\mu, \varepsilon \geq 0$ unless said otherwise; $\ell = |\cdot|_2$ is the planar length (Definition 1.4). For $q, p \in \mathbb{R}^{2n}$ write $q_i = q^{(i)}$, $p_i = p^{(i)} \in \mathbb{R}^2$ and $P = \sum_i p_i$. The set of *collision-free phase points* is

$$W = \{z = (q,p) \in \mathrm{Ph}(2n) : \mathrm{NoColl}(q)\} \qquad \text{(Definition 1.18)}.$$

For q with $\mathrm{NoColl}(q)$ put

$$g_i(q) = \frac{q_i}{\ell(q_i)^3}, \qquad f_i(q) = \sum_{j \neq i} m_j \frac{q_j - q_i}{\ell(q_j - q_i)^3} \qquad (i < n),$$

the central and the planet–planet force kernels. Planet i of the $(n+1)$-body system sits at index $\mathrm{succ}(i) \in \mathtt{Fin}\,(n+1)$ and the Sun at index 0 (the order of `Fin.cons`); I write Q_0, Q_i for the entries of a configuration $Q : \mathtt{Fin}\,(n+1) \to \mathbb{R}^2$.

External facts (Mathlib level, used by label).

(E1) Chain, product, quotient and finite-sum rules for `HasDerivAt` and `HasFDerivAt`; a map into a finite product $\mathbb{R}^N$ has a derivative iff every component does (`hasDerivAt_pi`); `fderiv` of a differentiable function is the derivative; a polynomial is C^∞.
(E2) $\sqrt{\cdot}$ is C^∞ on $(0,\infty)$ with derivative $x \mapsto 1/(2\sqrt{x})$; $x \mapsto 1/x$ is C^∞ on $\mathbb{R}\setminus\{0\}$.
(E3) Preimages of open sets under continuous maps are open; finite intersections of open sets are open; $\mathbb{R}$ is totally ordered and finite minima/maxima exist.
(E4) Finite-sum algebra: `Fin.sum_univ_succ`, `Finset.sum_comm`, splitting $\sum_j = \sum_{j<i} + \sum_{j>i}$, and $\sum_{j\neq i} = \sum_j -(\text{term } j=i)$.
(E5) The reverse triangle inequality $|\,|u| - |w|\,| \le |u-w|$ in a normed space.
(E6) Bounded sets of $\mathbb{R}^N$ have finite Lebesgue measure; the product measure on $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ (`volume`) is Lebesgue measure on $\mathbb{R}^{2d}$ and a box $\prod[-r_k, r_k]$ has measure $\prod 2r_k$; measure is monotone.

8.2. **Planar length.**

Lemma 8.1 (Planar length)**.** *For $v, w \in \mathbb{R}^2$, $c \in \mathbb{R}$:*

(a) $\ell(v) \ge 0$, $\ell(v)^2 = \mathrm{sq}(v) = v \cdot v$, $\ell(v) = 0 \iff v = 0$, $\ell(-v) = \ell(v)$, $\ell(cv) = |c|\ell(v)$, $\ell(v+w) \le \ell(v) + \ell(w)$.
(b) $|v|_\infty \le \ell(v) \le \sqrt{2}\,|v|_\infty$.
(c) On $\mathbb{R}^2 \setminus \{0\}$ the functions ℓ, $1/\ell$ are C^∞ with $D\ell(v)h = \dfrac{v\cdot h}{\ell(v)}$ and $D(1/\ell)(v)h = -\dfrac{v\cdot h}{\ell(v)^3}$.

Proof. **Step S8.1: (a).** Algebra of $\sqrt{v_1^2 + v_2^2}$; the triangle inequality is Cauchy–Schwarz $v \cdot w \le \ell(v)\ell(w)$ in $\ell(v+w)^2 = \ell(v)^2 + 2v\cdot w + \ell(w)^2$.

Step S8.2: (b). $v_k^2 \le v_1^2 + v_2^2 \le 2\max_k v_k^2$; take square roots.

Step S8.3: (c). sq is a polynomial with $D\,\mathrm{sq}(v)h = 2v\cdot h$, positive on $v \neq 0$; apply (E2) and (E1): $D\ell(v)h = \frac{2v\cdot h}{2\ell(v)}$, and $D(1/\ell) = -D\ell/\ell^2$. □

8.3. **The two reduced models.**

Lemma 8.2 (Collision-free domain; regularity)**.** *W is open, and $H_{\mathrm{mov}} =$ `movingHamiltonian` $G\,m_0\,m\,\mu$ and $H_{\mathrm{fix}} =$ `fixedHamiltonian` $G\,m_0\,m\,\varepsilon$ are C^∞ on W (hence differentiable at every point of W, with continuous Fréchet derivative). The Lean value of H off W is irrelevant to everything below, because every use is guarded by* NoColl *(Remark 1.27).*

Proof. **Step S8.4: Openness.** NoColl(q) is the finite conjunction of $\mathrm{sq}(q_i) > 0$ and $\mathrm{sq}(q_i - q_j) > 0$ $(i \neq j)$ (Lemma 8.1(a)); each sq $\circ$(linear map of q) is continuous, so W is open by (E3).

Step S8.5: Smoothness. On W each term of H_{mov}, H_{fix} is a polynomial in p or q-dependent $1/\ell(q_i)$, $1/\ell(q_i - q_j)$, which are C^∞ by Lemma 8.1(c) and (E1); finite sums and products of C^∞ maps are C^∞. □

Lemma 8.3 (Splitting; the identity of Part 0)**.** *Let $H_{\mathrm{Kep}}(q,p) = \sum_i \big[\frac{|p_i|_2^2}{2m_i} - \frac{Gm_0m_i}{\ell(q_i)}\big]$ (independent of μ, ε), and for $z \in W$*

$$f_{\mathrm{mov}}(z) = \frac{|P|_2^2}{2m_0} - G\sum_{i<j}\frac{m_i m_j}{\ell(q_i - q_j)}, \qquad f_{\mathrm{fix}}(z) = -G\sum_{i<j}\frac{m_i m_j}{\ell(q_i - q_j)}.$$

Then for all $z \in \mathrm{Ph}(2n)$ (with the Lean total functions) $H_{\mathrm{mov}} = H_{\mathrm{Kep}} + \mu f_{\mathrm{mov}}$, $H_{\mathrm{fix}} = H_{\mathrm{Kep}} + \varepsilon f_{\mathrm{fix}}$, and

$$H_{\mathrm{mov}}(z) - H_{\mathrm{fix}}(z)\big|_{\varepsilon=\mu} = \frac{\mu\,|P|_2^2}{2m_0}\ \Big(= \mu\,\frac{|\sum_i p^{(i)}|_2^2}{2m_0}\Big).$$

Proof. **Step S8.6: Algebra.** $\left(\frac{1}{m_i}+\frac{\mu}{m_0}\right)\frac{|p_i|_2^2}{2} = \frac{|p_i|_2^2}{2m_i} + \frac{\mu|p_i|_2^2}{2m_0}$, and $|P|_2^2 = \sum_i |p_i|_2^2 + 2\sum_{i<j} p_i \cdot p_j$ (Lemma 8.1(a), bilinearity of $\cdot$). Hence the kinetic part of H_{mov} is $\sum_i \frac{|p_i|_2^2}{2m_i} + \frac{\mu}{m_0}\left(\frac{1}{2}|P|_2^2\right)$; the potential parts agree term by term. At $\varepsilon = \mu$ the pair-potential terms of H_{mov} and H_{fix} are the same, which gives the difference. □

Remark 8.4 (Kepler part; mass dictionary). Lemma 8.3 is why both models share the unperturbed part H_{Kep} and have perturbations μf_{mov}, $\varepsilon f_{\mathrm{fix}}$ with $f_{\mathrm{mov}}, f_{\mathrm{fix}}$ independent of the coupling. Each summand of H_{Kep} is $m_i E_k(q_i, p_i/m_i)$ with $k = Gm_0$ in the notation of [I:bpI:p7:scaling], whose "mass dictionary" gives the Kepler orbit, action $m_i\sqrt{ka}$ and the mean values used in Part 7. (In the moving-Sun reduction the *reduced* mass $m_0 m_i/(m_0+\mu m_i)$ appears if one chooses to keep the term $\mu|p_i|_2^2/(2m_0)$ in the Kepler part, as Fej04r (28) does with μ_j, M_j; the frozen H_{mov} and Part 7 do not: that term belongs to μf_{mov}.)

Lemma 8.5 (Action–reaction)**.** *For q with* NoColl(q)*:* $\sum_i m_i f_i(q) = 0$.

Proof. $\sum_i m_i f_i = \sum_i \sum_{j\neq i} m_i m_j \frac{q_j - q_i}{\ell(q_j-q_i)^3} =: \sum_{i\neq j} a_{ij}$ with $a_{ji} = -a_{ij}$ (Lemma 8.1(a)). Swapping the names of i, j ((E4)) gives $\sum a_{ij} = \sum a_{ji} = -\sum a_{ij}$. □

Proposition 8.6 (Hamilton equations of H_{mov} and H_{fix})**.** *Let $z = (q,p) \in W$. Using the Lean convention $X_H = (\partial_p H, -\partial_q H)$ of Definition 1.2 and the block form $v = (v_0, \dots, v_{n-1})$, $v_i \in \mathbb{R}^2$, of vectors of $\mathbb{R}^{2n}$ (index $2i + c$):*

$$X_{H_{\mathrm{mov}}}(z) = \Big(\big(\tfrac{p_i}{m_i} + \tfrac{\mu}{m_0} P\big)_i,\ \big(-Gm_0 m_i\, g_i(q) + \mu G m_i\, f_i(q)\big)_i \Big),$$

$$X_{H_{\mathrm{fix}}}(z) = \Big(\big(\tfrac{p_i}{m_i}\big)_i,\ \big(-Gm_0 m_i\, g_i(q) + \varepsilon G m_i\, f_i(q)\big)_i \Big).$$

In particular a curve $z = (q,p) : J \to W$ has derivative $X_H(z(t))$ at t (with H differentiable there, automatic by Lemma 8.2) iff $q_i' = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$ (resp. $\frac{p_i}{m_i}$) and $p_i' = -Gm_0 m_i g_i(q) + \mu G m_i f_i(q)$ (resp. with ε), componentwise.

Proof. **Step S8.7: Directional derivative.** By Lemma 8.2 H_{mov} is differentiable at z. Split $H_{\mathrm{mov}} = T(p) + V(q)$, $T = \sum_i c_i |p_i|_2^2/2 + \frac{\mu}{m_0}\sum_{i<j} p_i \cdot p_j$ with $c_i = \frac{1}{m_i} + \frac{\mu}{m_0}$, a quadratic polynomial, and $V(q) = -Gm_0 \sum_i m_i/\ell(q_i) - \mu G \sum_{i<j} m_i m_j/\ell(q_i - q_j)$. For $(\delta q, \delta p) \in \mathrm{Ph}(2n)$, by (E1) and Lemma 8.1(c),

$$DT(p)\delta p = \sum_i c_i p_i \cdot \delta p_i + \tfrac{\mu}{m_0}\sum_{i<j}(p_i \cdot \delta p_j + \delta p_i \cdot p_j) = \sum_i \delta p_i \cdot \big(\tfrac{p_i}{m_i} + \tfrac{\mu}{m_0} P\big),$$

using $c_i p_i + \frac{\mu}{m_0}\sum_{j\neq i} p_j = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$, and

$$DV(q)\delta q = Gm_0 \sum_i m_i \frac{q_i \cdot \delta q_i}{\ell(q_i)^3} + \mu G \sum_{i<j} m_i m_j \frac{(q_i - q_j)\cdot(\delta q_i - \delta q_j)}{\ell(q_i - q_j)^3}.$$

Step S8.8: Collect coefficients. The coefficient of δq_i in the double sum is $\mu G m_i \sum_{j\neq i} m_j \frac{q_i - q_j}{\ell(q_i-q_j)^3} = -\mu G m_i f_i(q)$ (a pair $\{i,j\}$ with $j < i$ contributes with δq_i the same way, because $\ell(q_i - q_j) = \ell(q_j - q_i)$). So $\partial_{q_i} H_{\mathrm{mov}} = Gm_0 m_i g_i - \mu G m_i f_i$, $\partial_{p_i} H_{\mathrm{mov}} = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$. The first component of X_H at (i,c) is $DH(z)(0, e_{2i+c})$ and the second is $-DH(z)(e_{2i+c}, 0)$ (Definition 1.2); this is the claim.

Step S8.9: H_{fix}. The same computation: the kinetic part is $\sum_i |p_i|_2^2/(2m_i)$ with $\partial_{p_i} = p_i/m_i$, and the potential is V with μ replaced by ε.

Step S8.10: The curve statement. A curve into $\mathrm{Ph}(2n) = \mathbb{R}^{4n}$ has derivative (q', p') at t iff each coordinate does ((E1)); compare with the displayed vector field. □

8.4. **Derivation of H_{mov} from the barycentric Hamiltonian.**

Definition 8.7 (The $(n+1)$-body Hamiltonian and Newton equations)**.** Let $\mu > 0$, $M = \mathrm{mass}_\mu = (m_0, \mu m_1, \dots, \mu m_n)$ the actual masses (Definition 1.19), $X = (X_0, \dots, X_n)$, $Y = (Y_0, \dots, Y_n)$ with $X_k, Y_k \in \mathbb{R}^2$ (positions and actual momenta), and

$$F_\mu(X,Y) = \sum_{k=0}^{n} \frac{|Y_k|_2^2}{2M_k} - G \sum_{0 \le k < l \le n} \frac{M_k M_l}{\ell(X_k - X_l)}.$$

Newton's equations are $X_k' = Y_k/M_k$, $Y_k' = \sum_{l \neq k} \frac{G M_k M_l}{\ell(X_l - X_k)^3}(X_l - X_k)$; with $V_k = X_k'$ they are $V_k' = \sum_{l \neq k} \frac{G M_l}{\ell(X_l - X_k)^3}(X_l - X_k)$, the equations of `IsNewtonSolution` (Definition 1.20). They are Hamilton's equations of F_μ for $\Omega = \sum_k dY_k \wedge dX_k$ on $\{\ell(X_k - X_l) > 0\}$ (Lemma 8.1(c) as in Prop. 8.6).

Source: Fej04r §6.1, pp. 45–46 (Newton equations (1), Hamiltonian with $\|y_0\|^2/2m_0$, $\epsilon \sum \|y_j\|^2/2m_j$, potential $m_0 m_j$ and $\epsilon^2 m_j m_k$ terms), G restored; Fej04r writes momenta $y_0, \epsilon y_j$ and the symplectic form $\sum dx_0 \wedge dy_0 + \epsilon \sum dx_j \wedge dy_j$, so $Y_j = \epsilon y_j = \mu p_j$ below.

Proposition 8.8 (Heliocentric symplectic coordinates and reduction)**.** *Let $\mu > 0$. Define the linear map $\Theta(X,Y) = (Q_0, q, P_0, P)$ with $Q_0 = X_0$, $q_i = X_i - X_0$, $P_0 = \sum_{k=0}^n Y_k$, $P_i = Y_i$ $(1 \le i \le n$; planet i has index i here).*

(a) Θ is a linear isomorphism with $\sum_k X_k \cdot Y_k = Q_0 \cdot P_0 + \sum_i q_i \cdot P_i$ (the canonical pairing is preserved), hence it is symplectic: $\sum_k dY_k \wedge dX_k = dP_0 \wedge dQ_0 + \sum_i dP_i \wedge dq_i$.

(b) On $\{\mathrm{NoColl}(q)\}$ one has

$$F_\mu \circ \Theta^{-1}(Q_0, q, P_0, P) = \frac{|P_0 - \sum_i P_i|_2^2}{2m_0} + \sum_i \frac{|P_i|_2^2}{2\mu m_i} - G\mu \sum_i \frac{m_0 m_i}{\ell(q_i)} - G\mu^2 \sum_{i<j} \frac{m_i m_j}{\ell(q_i - q_j)},$$

independent of Q_0; so P_0 (total momentum) is conserved. On the invariant set $P_0 = 0$, writing $P_i = \mu p_i$:

$$F_\mu \circ \Theta^{-1}(Q_0, q, 0, \mu p) = \mu\, H_{\mathrm{mov}}(q,p).$$

*(c) (*Solutions correspond.*) If (X,Y) solves Newton's equations on a set $J \subset \mathbb{R}$ of times, with total momentum $P_0(t) = 0$ and $\mathrm{NoColl}(q(t))$ for $t \in J$, then $(q,p) = (X_i - X_0,\, Y_i/\mu)$ satisfies $(q,p)' = X_{H_{\mathrm{mov}}}(q,p)$ on J (the converse direction of Proposition 8.11, where X_0 is then recovered as $x_0(q)$ if the barycentre is 0).*

Source: Fej04r (28), pp. 46–47 ("coordonnées héliocentriques symplectiques", reduction to $Y_0 = 0$, $F = \sum(\|Y_j\|^2/2\mu_j - \mu_j M_j/\|X_j\|) + \epsilon \sum(-m_j m_k/\|X_j - X_k\| + Y_j \cdot Y_k/m_0)$), with the index correction of Fej07e p. 3 ($m_i m_j \to m_j m_k$); the revised printing already has $m_j m_k$ (read on p. 46); G restored. Ours: the statement as a lemma about H_{mov} and with μ-scalings written out.

Proof. **Step S8.11: (a).** The inverse is $X_0 = Q_0$, $X_i = q_i + Q_0$, $Y_i = P_i$, $Y_0 = P_0 - \sum_i P_i$. Then $X_0 \cdot Y_0 + \sum_i X_i \cdot Y_i = Q_0 \cdot (P_0 - \sum P_i) + \sum_i (q_i + Q_0) \cdot P_i = Q_0 \cdot P_0 + \sum_i q_i \cdot P_i$. Moreover the Liouville forms agree identically: $\sum_k Y_k\, dX_k = (P_0 - \sum_i P_i)\, dQ_0 + \sum_i P_i\, d(q_i + Q_0) = P_0\, dQ_0 + \sum_i P_i\, dq_i$; taking d gives the symplectic statement.

Step S8.12: (b). Substitute $Y_0 = P_0 - \sum P_i$, $X_i - X_0 = q_i$, $X_i - X_j = q_i - q_j$, $M_0 = m_0$, $M_i = \mu m_i$: the kinetic energy is $\frac{|P_0 - \sum P_i|_2^2}{2m_0} + \sum_i \frac{|P_i|_2^2}{2\mu m_i}$ and the potential is $-G\mu \sum_i \frac{m_0 m_i}{\ell(q_i)} - G\mu^2 \sum_{i<j} \frac{m_i m_j}{\ell(q_i - q_j)}$. At $P_0 = 0$, $P_i = \mu p_i$ the total is

$$\frac{\mu^2 |\sum_i p_i|_2^2}{2m_0} + \mu \sum_i \frac{|p_i|_2^2}{2m_i} - G\mu \sum_i \frac{m_0 m_i}{\ell(q_i)} - G\mu^2 \sum_{i<j} \frac{m_i m_j}{\ell(q_i - q_j)} = \mu\big(H_{\mathrm{Kep}} + \mu f_{\mathrm{mov}}\big) = \mu H_{\mathrm{mov}}$$

by Lemma 8.3. $F_\mu \circ \Theta^{-1}$ does not involve Q_0, so $P_0' = -\partial_{Q_0}(\cdot) = 0$.

Step S8.13: (c). With $P_0 = \sum_k Y_k = 0$ we have $Y_0 = -\sum_i Y_i = -\mu P$, $P = \sum_i p_i$, hence $q_i' = X_i' - X_0' = \frac{Y_i}{\mu m_i} - \frac{Y_0}{m_0} = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$ and, from $Y_i' = \sum_{l \neq i} \frac{G M_i M_l}{\ell(X_l - X_i)^3}(X_l - X_i)$ with $X_0 - X_i = -q_i$, $X_j - X_i = q_j - q_i$, $p_i' = \mu^{-1} Y_i' = G m_i \big[-m_0 \frac{q_i}{\ell(q_i)^3} + \mu \sum_{j \neq i} m_j \frac{q_j - q_i}{\ell(q_j - q_i)^3} \big]$. This is Proposition 8.6. □

Remark 8.9 (Role of this proposition). Part (b) is the derivation of Definition 1.22: H_{mov} is F_μ on the reduced space divided by μ (Fej04r's F with $\epsilon = \mu$). Nothing below uses (a)–(b): the forward reconstruction (Prop. 8.11) is verified directly from the Hamilton equations, so the dependence of the physical clauses on the Hamiltonian reduction is only through Prop. 8.6. Part (c) is needed only by consumers who start from a Newton solution (the uniqueness predicate `MovingUniqueOn` of D25 is the reverse direction).

8.5. **Reconstruction of the physical motion.**

Lemma 8.10 (Algebra of the barycentric maps)**.** *Let $\mu \geq 0$, $q, p \in \mathbb{R}^{2n}$, $M_* = m_0 + \mu \sum_i m_i > 0$, $M = \mathrm{mass}_\mu$ (the frozen `actualMasses`), $Q = $ `barycentricPositions` $m_0\, m\, \mu\, q$, $V = $ `barycentricVelocities` $m_0\, m\, \mu\, p$. Then, with $P = \sum_i p_i$:*

(a) $Q_0 = x_0 = -\frac{\mu}{M_*} \sum_i m_i q_i$, $Q_i - Q_0 = q_i$ and $Q_j - Q_i = q_j - q_i$ for planets i, j; hence $\ell(Q_i - Q_0) = \ell(q_i)$ and $\ell(Q_j - Q_i) = \ell(q_j - q_i)$ for every μ.
(b) $V_0 = -\frac{\mu}{m_0} P$, $V_i = p_i / m_i$, and $V_i - V_0 = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$ (the first component of $X_{H_{\mathrm{mov}}}$, Prop. 8.6).
(c) (barycentre and momentum) $\sum_{k=0}^{n} M_k Q_k = 0$ and $\sum_{k=0}^{n} M_k V_k = 0$.
(d) $\ell(x_0) \leq \frac{\mu}{m_0} \sum_i m_i \ell(q_i)$.

Proof. **Step S8.14: (a),(b).** Unfold the definitions (`Fin.cons`, `Fin.cons_zero`, `Fin.cons_succ`). $Q_j - Q_i = (x_0 + q_j) - (x_0 + q_i)$, and ℓ is even (Lemma 8.1(a)).

Step S8.15: (c). $\sum_k M_k Q_k = m_0 x_0 + \sum_i \mu m_i (x_0 + q_i) = M_* x_0 + \mu \sum m_i q_i = 0$; $\sum_k M_k V_k = m_0 V_0 + \mu \sum_i p_i = -\mu P + \mu P = 0$.

Step S8.16: (d). $M_* \geq m_0$ and Lemma 8.1(a). □

Proposition 8.11 (Reconstruction: moving Sun ⇒ Newton, on any set of times)**.** *Let $G, m_0 > 0$, $m_i > 0$, $\mu \geq 0$, $J \subset \mathbb{R}$ arbitrary, and $z = (q, p) : \mathbb{R} \to \mathrm{Ph}(2n)$ such that for all $t \in J$:*

(R1) $\mathrm{NoColl}(q(t))$*;*
(R2) z has derivative $X_{H_{\mathrm{mov}}}(z(t))$ at t, $H_{\mathrm{mov}} =$ `movingHamiltonian` $G\, m_0\, m\, \mu$.

(Differentiability of H_{mov} at $z(t)$ is automatic, Lemma 8.2; for $G = m_0 = 1$ conditions (R1)–(R2) say exactly `IsMovingSolutionOn` $m\, \mu\, J\, z$ of D25.) Put $M = \mathrm{mass}_\mu$, $Q(t) =$ `barycentricPositions` $m_0\, m\, \mu\, q(t)$, $V(t) =$ `barycentricVelocities` $m_0\, m\, \mu\, p(t)$. Then for every $t \in J$ and every pair $k \neq l$ of indices of `Fin`$(n+1)$:

(a) $\ell(Q_l(t) - Q_k(t)) > 0$;
(b) Q_k has derivative $V_k(t)$ at t;
(c) V_k has derivative $\displaystyle\sum_{l \neq k} \frac{G M_l}{\ell(Q_l(t) - Q_k(t))^3} \big(Q_l(t) - Q_k(t)\big)$ at t.

Consequently: `IsNewtonSolutionOn` $(G \cdot M)\, J\, Q\, V$ (the frozen interval predicate, whose mass argument absorbs G; for $G = 1$ it is `IsNewtonSolutionOn` $(\texttt{actualMasses}\, 1\, m\, \mu)\, J\, Q\, V$ as in D25), and for $J = \mathbb{R}$ the frozen all-time predicate `IsNewtonSolution` $G\, (\texttt{actualMasses}\, m_0\, m\, \mu)\, Q\, V$ (Definition 1.20). The conclusions are literally the three clauses of those predicates at each $t \in J$, with no continuity or uniformity in t used.

Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution`, `KAM.Planetary.IsNewtonSolutionOn`, `KAM.Planetary.IsMovingSolutionOn`

Source: Fej04r §6.1, p. 46 (the Newton curves are projections of the integral curves of the Hamiltonian; "Si l'on exclut les collisions"); G restored; ours as stated (pointwise for arbitrary J).

Proof. **Step S8.17: Coordinates of the equations.** Fix $t \in J$. By (R1)–(R2) and Prop. 8.6 (curve statement), at t: $q_i' = \frac{p_i}{m_i} + \frac{\mu}{m_0}P =: a_i$ and $p_i' = -Gm_0m_ig_i(q) + \mu Gm_if_i(q) =: b_i$, for all i, where $q = q(t)$, $p = p(t)$. Since `bodyVector` is a coordinate projection, $\texttt{bodyVector}(q(\cdot), i)$ has derivative a_i at t ((E1)), and so does every linear combination.

Step S8.18: (a). By Lemma 8.10(a), $\ell(Q_i - Q_0) = \ell(q_i) > 0$, $\ell(Q_j - Q_i) = \ell(q_j - q_i) > 0$ by (R1) (the Lean `NoCollision` gives positivity of $\ell(q_i - q_j)$ for $i \neq j$, and ℓ is even); and $\ell(Q_0 - Q_i) = \ell(Q_i - Q_0)$.

Step S8.19: (b), Sun. $Q_0 = x_0 = -\frac{\mu}{M_*}\sum_i m_iq_i$ is a constant multiple of a sum of the q_i, with M_* independent of t; so $Q_0' = -\frac{\mu}{M_*}\sum_i m_ia_i = -\frac{\mu}{M_*}\big(P + \frac{\mu}{m_0}P\sum_i m_i\big) = -\frac{\mu}{M_*}\cdot\frac{M_*}{m_0}P = -\frac{\mu}{m_0}P = V_0$, because $\sum_i m_ia_i = P + \frac{\mu}{m_0}P\sum_i m_i = P\,M_*/m_0$.

Step S8.20: (b), planets. $Q_i = Q_0 + q_i$ has derivative $V_0 + a_i = V_0 + (V_i - V_0) = V_i$ by Lemma 8.10(b).

Step S8.21: (c), planets. $V_i = p_i/m_i$ has derivative $b_i/m_i = -Gm_0g_i(q) + \mu Gf_i(q) = \frac{Gm_0\,(Q_0 - Q_i)}{\ell(Q_0 - Q_i)^3} + \sum_{j\neq i}\frac{G\mu m_j(Q_j - Q_i)}{\ell(Q_j - Q_i)^3}$, using $Q_0 - Q_i = -q_i$ and (a); the right-hand side is the sum over $l \neq i$ of $GM_l(Q_l - Q_i)/\ell(Q_l - Q_i)^3$ (with $M_0 = m_0$, $M_j = \mu m_j$; the sum over $l \in \texttt{Fin}(n+1)$ is split as $l = 0$ and $l = \mathrm{succ}(j)$, `Fin.sum_univ_succ`, (E4)).

Step S8.22: (c), Sun. $V_0 = -\frac{\mu}{m_0}\sum_i p_i$ has derivative $-\frac{\mu}{m_0}\sum_i b_i = \mu G\sum_i m_ig_i(q) - \frac{\mu^2 G}{m_0}\sum_i m_if_i(q) = \mu G\sum_i m_ig_i(q)$ by Lemma 8.5. By Lemma 8.10(a) ($Q_i - Q_0 = q_i$) this is $\sum_{l\geq 1} GM_l(Q_l - Q_0)/\ell(Q_l - Q_0)^3$ with $M_l = \mu m_l$.

Step S8.23: Packaging. The three bullet clauses are conjunctions over $t \in J$ of (a), (b), (c). For $J = \mathbb{R}$ and the frozen `IsNewtonSolution` (Definition 1.20) they are literally the three conjuncts; `IsNewtonSolutionOn` differs by the restriction to $t \in J$ and by having G absorbed in the mass, replace M by $G \cdot M$. The case $\mu = 0$ is included: then $M_l = 0$ for planets, $Q_0 = 0$ and the equations are the test-particle equations. □

Remark 8.12 (Units and the D25 predicates). The frozen `IsMovingSolutionOn` and `IsNewtonSolutionOn` carry no G, m_0: $G = m_0 = 1$ (the D25 normalisation, `OrbitalUnit`). Proposition 8.11 then states: if z satisfies `IsMovingSolutionOn` $m\,\mu\,J$ (Hamiltonian solution on J in B = univ, plus `NoCollision`), then `IsNewtonSolutionOn` $(\texttt{actualMasses}\,1\,m\,\mu)\,J\,Q\,V$ for $Q = \texttt{barycentricPositions}\,1\,m\,\mu\,q$, $V = \texttt{barycentricVelocities}\,1\,m\,\mu\,p$. The interval need not be open or connected; each time is treated separately. Track N's Prop. [N:`bpN:p3:newton-reduction`] covers only the all-time predicates (a `IsSolIn` reformulation) and is not needed here; the equivalence it gives for `IsNewtonSolutionOn` is the same two-line argument with J in place of $\mathbb{R}$.

Proposition 8.13 (Reconstruction: fixed Sun)**.** *Let $G, m_0 > 0$, $m_i > 0$, $\varepsilon \in \mathbb{R}$, $J \subset \mathbb{R}$ arbitrary, and $z = (q, p) : \mathbb{R} \to \mathrm{Ph}(2n)$ with, for $t \in J$:* $\mathrm{NoColl}(q(t))$ *and z has derivative $X_{H_{\mathrm{fix}}}(z(t))$ at t ($H_{\mathrm{fix}} = \texttt{fixedHamiltonian}\,G\,m_0\,m\,\varepsilon$). Put $Q_i(t) = q_i(t)$, $V_i(t) = \frac{1}{m_i}p_i(t)$. Then for all $t \in J$ and $i \neq j$: $\ell(Q_i(t)) > 0$, $\ell(Q_j(t) - Q_i(t)) > 0$, Q_i has derivative $V_i(t)$ at t, and V_i has derivative*

$$-\frac{Gm_0}{\ell(Q_i)^3}Q_i + \varepsilon\sum_{j\neq i}\frac{Gm_j}{\ell(Q_j - Q_i)^3}(Q_j - Q_i)$$

at t. For $J = \mathbb{R}$ this is $\texttt{IsFixedSolution}\,G\,m_0\,m\,\varepsilon\,Q\,V$ (Definition 1.21).
Frozen Lean declaration: `KAM.Planetary.IsFixedSolution`
Source: Ours (differentiating H_{fix}); after Fej04r (28).

Proof. **Step S8.24: Equations.** Prop. 8.6 gives $q_i' = p_i/m_i$ and $p_i' = -Gm_0m_ig_i + \varepsilon Gm_if_i$. Positivity: `NoCollision` and evenness of ℓ. $V_i = p_i/m_i$ has derivative $p_i'/m_i = -Gm_0\,q_i/\ell(q_i)^3 + \varepsilon Gf_i(q)$, which is the displayed expression since $f_i = \sum_{j\neq i} m_j(q_j - q_i)/\ell(q_j - q_i)^3$. □

Corollary 8.14 (Orbit transfer for tori)**.** *Let $d = 2n$, $K : \mathbb{R}^d \to \mathrm{Ph}(2n)$ and $w \in \mathbb{R}^d$.*

(a) If IsPhysTorus(H_{mov}, w, K) *and* NoColl$(K(\phi)_1)$ *for all ϕ, then for every ϕ*

$$\texttt{IsNewtonSolution}\, G\, (\texttt{actualMasses}\, m_0\, m\, \mu)\, (t \mapsto Q(K(\phi + tw)_1))\, (t \mapsto V(K(\phi + tw)_2)),$$

clause (b) of MovKol *(Definition 1.25); here $Q = \texttt{barycentricPositions}\, m_0\, m\, \mu$, $V = \texttt{barycentricVelocities}\, m_0\, m\, \mu$.*

(b) If IsPhysTorus(H_{fix}, w, K) *and* NoColl$(K(\phi)_1)$ *for all ϕ, then $\texttt{IsFixedSolution}\, G\, m_0\, m\, \varepsilon\, (t \mapsto \texttt{body}(K(\phi + tw)_1))\, (t \mapsto \frac{1}{m_i}\texttt{body}(K(\phi + tw)_2))$ for all ϕ.*

Proof. **Step S8.25: The curve.** Fix ϕ and put $z(t) = K(\phi + tw)$. Clause (viii) of Definition 1.24 (applied at $\theta = \phi$ and the time t) says precisely that $s \mapsto K(\phi + sw)$ has derivative $X_H(K(\phi + tw))$ at $s = t$, i.e. (R2) of Prop. 8.11 for every $t \in \mathbb{R}$; (R1) is the hypothesis at the angle $\phi + tw$.

Step S8.26: Apply. Proposition 8.11 (resp. 8.13) with $J = \mathbb{R}$. □

8.6. **Tube geometry: collision-free, uniform pair bounds.**

Definition 8.15 (Axis constants; tube radius; d, D)**.** Let $a \in \mathbb{R}^n$ with $a_i > 0$ pairwise distinct (in P-PLANAR $n = 2$, $0 < a_0 < a_1$; in P-FIXED $0 < a_0 < \cdots < a_{n-1}$), and $\eta > 0$. Put

$$a_{\max} = \max_i a_i, \qquad \gamma_a = \min\Big\{\min_i a_i,\ \min_{i \neq j} |a_i - a_j|\Big\} > 0$$

(for $n = 1$ the inner minimum over pairs is omitted), and

$$\zeta_* = \min\{\eta, \gamma_a/6\}, \qquad d = \gamma_a/2, \qquad D = 2a_{\max} + \gamma_a.$$

Then $0 < d < D$; d, D depend only on a (not on $\eta, G, m_0, m, \mu, \varepsilon$), and ζ_* only on (a, η).

Lemma 8.16 (Tube geometry)**.** *Let $G, m_0 > 0$, $m_i > 0$, a, η, ζ_*, d, D as in Definition 8.15 and $0 < \zeta \leq \zeta_*$. For every $z = (q, p) \in U_\zeta$ (Definition 1.17, tube radius ζ):*

(a) $U_\zeta \subset U_{\zeta_*} \subset U_\eta$.

*(b) (*Euclidean positions*) there is $\phi \in \mathbb{R}^n$ with $\ell\big(q_i - a_i e(\phi_i)\big) < \sqrt{2}\,\zeta$ for all i, $e(\phi) = (\cos 2\pi\phi, \sin 2\pi\phi)$.*

*(c) (*distances*) for all i and all $i \neq j$:*

$$a_i - \sqrt{2}\zeta < \ell(q_i) < a_i + \sqrt{2}\zeta, \qquad |a_i - a_j| - 2\sqrt{2}\zeta < \ell(q_j - q_i) < a_i + a_j + 2\sqrt{2}\zeta.$$

(d) $d \leq \ell(q_i) \leq D$ and $d \leq \ell(q_j - q_i) \leq D$; in particular NoColl$(q)$.

(e) For every $\mu \geq 0$*:* PairBounds$\big(d, D;\ \texttt{barycentricPositions}\, m_0\, m\, \mu\, q\big)$ *(all pairs including the Sun), with the same d, D* for all μ*; and* PairBounds$\big(d, D;\ (0, q_0, \ldots, q_{n-1})\big)$ *(the Sun at the origin, fixed-Sun clause of* FixKol*).*

Proof. **Step S8.27: (a).** The tube is defined by a strict bound on $\|z - \mathrm{circ}(\phi)\|$, so it is monotone in the radius.

Step S8.28: (b) Max norm to Euclidean norm. The phase norm is the maximum of the two coordinate max norms (Part 0, conventions), so $\|z - \mathrm{circ}(\phi)\| < \zeta$ gives $|q_{(i,c)} - a_i e(\phi_i)_c| < \zeta$ for $c = 1, 2$, i.e. $|q_i - a_i e(\phi_i)|_\infty < \zeta$; Lemma 8.1(b) gives the Euclidean bound with the factor $\sqrt{2}$. *This factor is not optional*: the planar Euclidean distance of a point of the max-norm tube from the circle may be $\sqrt{2}\zeta$.

Step S8.29: (c). Let $u_i = a_i e(\phi_i)$, so $\ell(u_i) = a_i$ ($\cos^2 + \sin^2 = 1$). Reverse triangle inequality (E5): $|\ell(q_i) - a_i| \leq \ell(q_i - u_i) < \sqrt{2}\zeta$. For $i \neq j$: $\ell(q_j - q_i) \geq \ell(u_j - u_i) - \ell(q_j - u_j) - \ell(q_i - u_i) > \ell(u_j - u_i) - 2\sqrt{2}\zeta$ and $\ell(q_j - q_i) < \ell(u_j - u_i) + 2\sqrt{2}\zeta$; finally $|a_j - a_i| = |\ell(u_j) - \ell(u_i)| \leq \ell(u_j - u_i) \leq \ell(u_j) + \ell(u_i) = a_i + a_j$.

Step S8.30: (d) Numerics. Since $\zeta \le \gamma_a/6$: $\sqrt{2}\zeta \le 0.2358\,\gamma_a < \gamma_a/4$ and $2\sqrt{2}\zeta \le 0.4715\,\gamma_a < \gamma_a/2$ (as $\sqrt{2} < 1.4143$). As $a_i \ge \gamma_a$ and $|a_i - a_j| \ge \gamma_a$: $\ell(q_i) > a_i - \gamma_a/4 \ge \frac{3}{4}\gamma_a \ge d$ and $\ell(q_j - q_i) > \gamma_a - \gamma_a/2 = d$. Upper: $\ell(q_i) < a_{\max} + \gamma_a/4 < D$ and $\ell(q_j - q_i) < 2a_{\max} + \gamma_a/2 < D$. Positivity of d gives NoColl(q) (Definition 1.18; $\ell(q_i - q_j) = \ell(q_j - q_i)$).

Step S8.31: (e) Exact μ-independence. By Lemma 8.10(a), for $Q = \texttt{barycentricPositions}\, m_0\, m\, \mu\, q$: $\ell(Q_i - Q_0) = \ell(q_i)$ and $\ell(Q_j - Q_i) = \ell(q_j - q_i)$ for all planets $i \ne j$ and *all* μ; and PairBounds asks $d \le \ell(Q_l - Q_k) \le D$ for all ordered pairs $k \ne l$ of $\texttt{Fin}(n+1)$, which splits into the three cases (Sun–planet in both orders, planet–planet) covered by (d) and evenness. The fixed-Sun configuration has $Q_0 = 0$, $Q_i = q_i$, the same argument. No estimate $|x_0| = O(\mu)$ is used; the identity $x_i - x_j = q_i - q_j$ makes the bounds independent of the position of the barycentre. □

Remark 8.17 (On collision exclusion from track I). Track I's lemma [I:`bpI:p8:collision-exclusion`] (distances to points that move by $\le c|\mu|$ are bounded below by half the unperturbed distance for $|\mu| \le \delta/2c$) and corollary [I:`bpI:p8:min-distance`] (compact collision-free sets of configurations have a positive minimal separation) are *not* needed for Parts 12–13: after reducing to heliocentric coordinates q the only displaced mass is the Sun, and $Q_i - Q_0 = q_i$ holds exactly. They would be needed for distances to the *barycentre* (not a frozen clause), for which Lemma 8.10(d) gives $\ell(x_0) \le \mu \sum_i m_i \ell(q_i)/m_0 = O(\mu)$. The explicit constants of Definition 8.15 replace the compactness argument of [I:`bpI:p8:min-distance`] for the same purpose on the tube.

Corollary 8.18 (Newton solutions staying in the tube, on any set of times)**.** *Let $z : \mathbb{R} \to \mathrm{Ph}(2n)$, $J \subset \mathbb{R}$, $\mu \ge 0$, $0 < \zeta \le \zeta_*$ with $z(t) \in U_\zeta$ and z having derivative $X_{H_{\mathrm{mov}}}(z(t))$ at t, for all $t \in J$. Then (R1)–(R2) of Prop. 8.11 hold, so* $\texttt{IsNewtonSolutionOn}\,(G \cdot \mathrm{mass}_\mu)\, J\, Q\, V$ *holds, and moreover* $\mathrm{PairBounds}(d, D; Q(t))$ *for every $t \in J$ with the constants of Def. 8.15. The analogous statement holds for H_{fix} with Prop. 8.13.*

Proof. (R1) is Lemma 8.16(d) at each $t \in J$, (R2) is a hypothesis; apply Prop. 8.11 and Lemma 8.16(e). □

8.7. **Positive finite measure.**

Lemma 8.19 ($0 < \mathrm{meas}\, S < \infty$)**.** *Let $G, m_0 > 0$, $m_i > 0$, $a_i > 0$, $\eta > 0$.*

*(a) (*finiteness*) U_η is bounded: every $z \in U_\eta$ has $|z_1|_\infty < a_{\max} + \eta$ and $|z_2|_\infty < P_{\max} + \eta$, $P_{\max} = \max_i m_i\sqrt{Gm_0/a_i}$. Hence every $S \subset U_\eta$ has $\mathrm{meas}\, S \le \big(2(a_{\max}+\eta)\big)^{2n}\big(2(P_{\max}+\eta)\big)^{2n} < \infty$.*

*(b) (*positivity from the chart*) Let Ψ be a c-chart on an open $O \subset \mathrm{Ph}(2n)$ with the lattice compatibility (Q1)–(Q2) of Theorem 2.9 for $\sigma_k = 0$ ($d = 2n$), and let $A \subset O$ be Lebesgue measurable with $A + \iota_k = A$ for all $k \in \mathbb{Z}^{2n}$, and $\mathrm{meas}\, A > 0$. Then $S = \Psi(A)$ is measurable and $0 < \mathrm{meas}\, S = c^{2n}\,\mathrm{meas}(A \cap Q)$ (Q the unit cell, Lemma 2.19); if A is Borel then S is Borel (the frozen* `MeasurableSet`*). If moreover $S \subset U_\eta$ then $0 < \mathrm{meas}\, S < \infty$.*

Proof. **Step S8.32: (a).** For $z \in U_\eta$ pick ϕ with $\|z - \mathrm{circ}(\phi)\| < \eta$. The coordinates of $\mathrm{circ}(\phi)$ are bounded by a_i (position, $|a_i \cos 2\pi\phi_i| \le a_i$) and $m_i\sqrt{Gm_0/a_i}$ (momentum); the triangle inequality gives the bounds. So U_η lies in a box and (E6) gives the measure bound.

Step S8.33: (b). Lemma 2.19 (with its Borel version) gives measurability, the formula and $0 < \mathrm{meas}\,\Psi(A) \iff 0 < \mathrm{meas}\, A$; the last sentence combines with (a). □

Remark 8.20 (Use). In P-PLANAR/P-FIXED, S is the image under the symplectic chart Φ of the chart-level union of whole tori of positive measure (Part 11, [P:`bpP:p1:measure-use`]; the Borel version of the union is Lemma 2.20(B1)); the containment $S \subset U_\eta$ is the tube containment of Part 12 and the finiteness is then free.

8.8. **Assembly of the physical clauses.**

Proposition 8.21 (Physical clauses of MovKol)**.** *Let $G, m_0 > 0$, $m_i > 0$, $a_i > 0$ pairwise distinct, $\eta > 0$, and ζ_*, d, D as in Definition 8.15. Let $\mu \geq 0$ and $S \subset \mathrm{Ph}(2n)$ satisfy:*

(S1) S is measurable and $S \subset U_{\zeta_*}$;

(S2) every $z \in S$ has $w \in \mathbb{R}^{2n}$, K, θ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$ and $\mathrm{IsPhysTorus}(H_{\mathrm{mov}}, w, K)$.

Then $\mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S)$ *(Definition 1.25). The constants d, D depend on a only, so in P-PLANAR they may be fixed* before *μ_0 and μ; the dependence on the coupling is confined to (S1)–(S2).*

Proof. **Step S8.34: Tube and set.** $S \subset U_{\zeta_*} \subset U_\eta$ (Lemma 8.16(a)); S is measurable by (S1).

Step S8.35: Collision. For $z \in S$ and $\phi \in \mathbb{R}^{2n}$: $K(\phi) \in K(\mathbb{R}^{2n}) \subset S \subset U_{\zeta_*}$, so $\mathrm{NoColl}(K(\phi)_1)$ by Lemma 8.16(d).

Step S8.36: Newton. Corollary 8.14(a).

Step S8.37: Pair bounds. Lemma 8.16(e) at $K(\phi)$; as ϕ is arbitrary this covers every point $K(\phi + tw)$ of every orbit. □

Proposition 8.22 (Physical clauses of FixKol)**.** *Under the same hypotheses with H_{fix} and $\varepsilon \in \mathbb{R}$ in place of H_{mov} and μ in (S2),* $\mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S)$ *holds. The proof is that of Prop. 8.21 with Cor. 8.14(b) and the fixed-Sun case of Lemma 8.16(e).*

8.9. **Totality, sources, deviations, forward references.**

Totality. Every division by $\ell(\cdot)^3$ or $\ell(\cdot)$ is guarded by `NoCollision` or by $\ell > 0$ from the Newton predicates (Remark 1.27); the derivative of H is used only on W where H is C^∞ (Lemma 8.2); $M_* = m_0 + \mu \sum m_i > 0$ and $m_i > 0$ guard the barycentric divisions; $1/\ell(q_j - q_i)$ appears in the tube bounds only after $d > 0$.

Sources read. Fej04r pp. 46–47 (Newton equations, Hamiltonian, heliocentric symplectic coordinates, reduction, formula (28) with μ_j, M_j), Fej07e p. 3 (`Formule (28). Remplacer` $m_i m_j$ `par` $m_j m_k$); the revised printing's (28) already has $m_j m_k$.

Deviations. (1) Fej04r works with $G = 1$; G is restored throughout, with the frozen H_{mov} (in which the Kepler part uses planet mass m_i, *not* the reduced mass; the term $\mu|p_i|_2^2/2m_0$ is part of the perturbation, Remark 8.4). (2) The tube sketch of `PLAN.md` ($|q^{(i)}| \in [a_i - \eta, a_i + \eta]$, "barycentric Sun displacement $\leq \mu C$") is replaced by the exact statement with the factor $\sqrt{2}$ of the max-to-Euclidean conversion (route study, `p11-route-study.md`, "Uniform tube and pair bounds"), and the μ-dependence of the barycentre is removed altogether by Lemma 8.10(a). (3) The statements are for arbitrary $J \subset \mathbb{R}$, not only $\mathbb{R}$.

Suspected source errata. None new. Fej04r (28) was printed with $m_i m_j$ in the original and corrected in Fej07e p. 3 (and in the revised printing); in the original-formula direction the reduction (Prop. 8.8(b)) shows the pair potential is $m_j m_k/\ell(X_j - X_k)$. One bookkeeping remark: Fej04r p. 46 writes $\epsilon\omega$ for the reduced symplectic form and F for the Hamiltonian "relative to ω"; this is the scaling $P_i = \mu p_i$, $F_\mu = \mu H_{\mathrm{mov}}$ of Prop. 8.8(b).

Doubts about frozen statements. None. The frozen `IsNewtonSolution` is formulated with the actual masses and G, and `MovingKolmogorovSet` (c) requires `PairBounds` for the barycentric positions of every orbit point; both are proved above. Note the frozen `IsMovingSolutionOn` fixes $G = m_0 = 1$ (D25) while P-PLANAR keeps G, m_0 general; Prop. 8.11 covers both.

Forward references. Part 12 ([P:`bpP:p12:physical`], [P:`bpP:p12:quantifiers`]), Part 13 ([P:`bpP:p13:assembly`]) cite Props. 8.21, 8.22 and Lemma 8.19; the D22/D25 additions (Part 14) cite Props. 8.11, 8.13 and Prop. 8.8(c). Track I cites Lemma 8.5 and Prop. 8.6 if it needs the planar n-body gradient.

Label index. `bpP:p6:plane`, `domain`, `split`, `kepler-part`, `action-reaction`, `hamilton-eqs`, `nbody`, `reduction`, `bary-algebra`, `reconstruction`, `units`, `reconstruction-fixed`, `orbit-transfer`, `tube-constants`, `tube-geometry`, `vs-I`, `newton-in-tube`, `measure-positive`, `movkol-assembly`, `fixkol-assembly`. The plan's names `bpP:p6:tube-geometry`, `measure-positive` are kept; `bpP:p6:reduction` and `bpP:p6:reconstruction` are the plan's.

9. Kepler problem and Poincaré coordinates

The unperturbed part of both physical models is a sum of n decoupled planar Kepler problems (Lemma 8.3). This part gives the coordinates in which that sum is integrable *and* regular at the circular orbits: the Poincaré chart $(\lambda, \Lambda, \xi, \eta)$. The results are:

- `kepler` (Prop. 9.3): the planar Kepler problem of a planet of mass m in the field of a centre of strength $\kappa = Gm_0$, through the mass/strength dictionary of [I:`bpI:p7:scaling`]: the Delaunay map with mass is symplectic (factor one) in the variables (ℓ, g, Λ, C) with $\Lambda = m\sqrt{\kappa a}$, $H_{\mathrm{Kep}} = -\kappa^2 m^3/(2\Lambda^2)$, and the mean values $\langle 1/|q|\rangle$, $\langle |p|^2\rangle$, $\langle |q|\rangle$;
- `regular-kepler` (Prop. 9.4): Kepler's equation in the eccentric longitude $F - v_0 \sin F + v_1 \cos F = \lambda$, with the eccentricity *vector* v as parameter, solved analytically (real, and holomorphically near $v = 0$), with no angle of perihelion;
- `poincare` (Thm. 9.7): the Poincaré chart, real analytic through $\xi = \eta = 0$ (circular orbits), a 1-chart in the sense of Definition 2.3, bijective modulo 2π in λ onto the prograde bound region, with the inverse formulas; Prop. 9.8 is its holomorphic extension near the circular torus;
- `product-chart` (Thm. 9.12): n independent charts; the *same* chart serves H_{mov} and H_{fix}, with $H^{\Psi} = h(\Lambda) + \mu f_{\mathrm{mov}} \circ \Psi$, $h(\Lambda) + \varepsilon f_{\mathrm{fix}} \circ \Psi$; the tube lemma, and the lattice hypotheses (Q1), (Q2) of Theorem 2.9 (Lemmas 9.14, 9.15);
- `eccentric-longitude` (Prop. 9.16): the D25 secular definitions (`ellipsePosition`, `ellipseMomentum`, `keplerMeanLongitude`, `longitudeJacobian`, `secularEccVector`, ...) are exactly the chart in the eccentric longitude F; and the change of variable $\lambda \leftrightarrow F$ in angular averages (Cor. 9.17).

Ownership. The Kepler fundamentals are owned by track I Part 7 and are cited, not repeated: Kepler's equation on a complex tube around the real eccentricity interval, including $e = 0$ ([I:`bpI:p7:kepler-eq`]), the Delaunay orbit maps ([I:`bpI:p7:orbit-maps`], [I:`bpI:p7:orbit-maps-e`]), their identities ([I:`bpI:p7:orbit-identities`]), the circular endpoint ([I:`bpI:p7:circular`]), the mean values ([I:`bpI:p7:means`]), the dictionary ([I:`bpI:p7:scaling`]), and the Delaunay chart ([I:`bpI:p7:symplectic`], [I:`bpI:p7:chart`]). Part 7 owns the Poincaré chart and the product chart.

9.1. **Conventions, external facts, notation.** Throughout $m > 0$ is the planet mass and $\kappa = Gm_0 > 0$ the strength of the centre; $|\cdot|_2$ is the planar length (Definition 1.4); angles λ, ℓ, g, F are in *radians* (period 2π), as in the D25 secular definitions; the passage to the frozen period-one torus parameter is Lemma 2.11 and is made in Lemma 9.15. The frozen sign convention is that of Definition 1.2: a phase point is $z = (z_1, z_2)$ with positions first, and $\Omega = \sum dz_{2,i} \wedge dz_{1,i}$. In the Poincaré coordinates of one planet $(\lambda, \xi; \Lambda, \eta)$ the positions are (λ, ξ) and the momenta (Λ, η) (this is the order of `secularGamma`: $\Gamma = (\xi^2 + \eta^2)/2 = R(\eta, \xi)$, Definition 1.7), so

$$\Omega_{\mathrm{chart}} = d\Lambda \wedge d\lambda + d\eta \wedge d\xi.$$

Write $J_0 v = (-v_1, v_0)$ (`quarterTurn`), $\mathsf{R}(g)$ for the rotation by g, $e_g = \mathsf{R}(g)(1,0) = (\cos g, \sin g)$, and $\mathrm{cr}(q,p) = q_0 p_1 - q_1 p_0$ (`planeCross`). The Delaunay map of track I (strength one, unit mass) is written $\mathsf{Q}^{\mathrm{I}}(\ell, g, L, G)$ ([I:`bpI:p7:orbit-maps`]); it satisfies $(\mathsf{Q}^{\mathrm{I}})^*\big(\sum dq_i \wedge dp_i\big) = d\ell \wedge dL + dg \wedge dG$ ([I:`bpI:p7:symplectic`]).

External facts.

(E1) *Real-analytic inverse function theorem* (as (X2), with analytic regularity): a real-analytic map with invertible differential at z_0 is a real-analytic diffeomorphism from a neighbourhood of z_0 onto an open set; in particular an implicit equation $\Phi(F,x)=0$ with $\partial_F\Phi\neq 0$ has a locally unique real-analytic solution $F(x)$. A bijective real-analytic map between open sets with everywhere invertible differential is a real-analytic diffeomorphism.

(E2) *Holomorphic calculus.* Sums, products, compositions and quotients (nonvanishing denominator) of holomorphic functions of several variables are holomorphic; $\sqrt{\cdot}$ is holomorphic on $\{\mathrm{Re}>0\}$; a locally uniform limit of holomorphic functions is holomorphic ([I:`bpI:p7:x-hol`], [I:`bpI:p7:x-weier`]).

(E3) Banach's fixed point theorem, the mean value inequality on a convex set ([K:`bp:p1:ext-banach`], [K:`bp:p1:ext-mvt`]).

(E4) *Exterior calculus on open sets of* $\mathbb{R}^N$ ([I:`bpI:p7:x-forms`]): d commutes with pull-back; the identity $\Omega(Au,Av)=c\,\Omega(u,v)$ for the differential $A=D\Psi$ holds iff the pulled-back 2-forms satisfy $\Psi^*\Omega=c\,\Omega$ (Lemma 2.1(c)).

(E5) A continuous identity between continuous functions which holds on a dense subset of an open set holds on the whole set.

(E6) *Change of variables in* $\mathbb{R}$ ([I:`bpI:p7:x-change`]).

(E7) Compact sets and open sets: a compact K contained in an open $V\subset\mathbb{R}^N$ has a positive max-norm distance to $\mathbb{R}^N\setminus V$.

9.2. The planar Kepler problem with mass m and strength κ.

Definition 9.1 (Kepler data)**.** For $(q,p)\in\mathbb{R}^2\times\mathbb{R}^2$, $q\neq 0$, put

$$E_{m,\kappa}(q,p)=\frac{|p|_2^2}{2m}-\frac{\kappa m}{|q|_2},\qquad C(q,p)=\mathrm{cr}(q,p),\qquad a=-\frac{\kappa m}{2E_{m,\kappa}},\qquad \Lambda=m\sqrt{\kappa a}\ \ (E_{m,\kappa}<0).$$

The *prograde bound region* is $\mathcal{K}_{m,\kappa}=\{(q,p): q\neq 0,\ E_{m,\kappa}<0,\ C>0\}$ (open: strict inequalities between continuous functions on $\{q\neq 0\}$). On it the eccentricity is $e=\sqrt{1-C^2/\Lambda^2}$ (Lemma 9.2: $0<C\le\Lambda$). The *Delaunay map with mass* is defined on $\mathcal{D}^m=\{(\ell,g,\Lambda,C)\in\mathbb{R}^4:\Lambda>C>0\}$ by

$$\mathsf{Q}_{m,\kappa}(\ell,g,\Lambda,C)=\mathsf{S}_{m,\kappa}\circ\mathsf{Q}^{\mathrm{I}}\Big(\ell,g,\frac{\Lambda}{m},\frac{C}{m}\Big),\qquad \mathsf{S}_{m,\kappa}(Q,P)=\Big(\frac{Q}{\kappa},\ m\kappa P\Big).$$

For $\kappa=m=1$ this is Q^{I}. For $E_{m,\kappa}<0$ and $m=1$ the quantity $\Lambda=\sqrt{\kappa a}$ and the orbit is the one of E_κ in [I:`bpI:p7:scaling`].

Frozen Lean declaration: `KAM.Planetary.keplerEnergy, keplerAxis, keplerAction, planeCross (the case` $\kappa=1$`, n planets componentwise)`

Source: Fej04r (31)–(32), p. 47–48 ($\Lambda_j=\mu_j\sqrt{M_ja_j}$, $F_{\mathrm{Kep}}=-\mu_j^3M_j^2/(2\Lambda_j^2)$; here the mass m_j and strength κ replace the reduced mass and total mass, see the deviations); [I:`bpI:p7:scaling`] for $\mathsf{S}_{m,\kappa}$ (this is its mass dictionary).

Lemma 9.2 (Range of the elements)**.** *For $(q,p)\in\mathcal{K}_{m,\kappa}$: $0<C\le\Lambda$, with equality iff the orbit is circular ($|q|_2$ constant). Equivalently $\Gamma:=\Lambda-C\in[0,\Lambda)$.*

Proof. **Step S9.1: Scaling to strength one.** With $(Q,P)=\mathsf{S}_{m,\kappa}^{-1}(q,p)=(\kappa q,p/(m\kappa))$: $E_1(Q,P)=\frac{1}{m\kappa^2}E_{m,\kappa}(q,p)$ ([I:`bpI:p7:scaling`]: $E_\kappa\circ\mathsf{S}_\kappa=\kappa^2E_1$ and $E_{m,\kappa}=mE_\kappa(q,p/m)$), $\mathrm{cr}(Q,P)=C/m$, and $L:=\Lambda/m=\sqrt{-1/(2E_1(Q,P))}$.

Step S9.2: Lagrange identity. With $r=|Q|_2$, $|Q|_2^2|P|_2^2=(Q\cdot P)^2+\mathrm{cr}(Q,P)^2$ and $\frac12|P|_2^2=E_1+1/r=-\frac{1}{2L^2}+\frac1r$. Hence $\mathrm{cr}(Q,P)^2=2r-r^2/L^2-(Q\cdot P)^2\le 2r-r^2/L^2=L^2-(r-L^2)^2/L^2\le L^2$, i.e. $C\le\Lambda$; equality forces $Q\cdot P=0$ and $r=L^2$, i.e. constant radius, and conversely. □

Proposition 9.3 (Planar Kepler problem, mass m, strength κ)**.** *Let $\mathcal{D}^m$ and $\mathsf{Q}_{m,\kappa}$ be as in Definition 9.1.*

(a) (Delaunay chart) $\mathsf{Q}_{m,\kappa}$ *is real analytic,* 2π*-periodic in* ℓ *and in* g*, induces a real-analytic diffeomorphism* $(\mathbb{R}/2\pi\mathbb{Z})^2 \times \{\Lambda > C > 0\} \to \mathcal{K}^*_{m,\kappa} := \{(q,p) \in \mathcal{K}_{m,\kappa} : C < \Lambda\}$ *(the non-circular part), and* $\mathsf{Q}^*_{m,\kappa}\Big(\sum dp_i \wedge dq_i\Big) = d\Lambda \wedge d\ell + dC \wedge dg$.

(b) (Identities) *For* $(q,p) = \mathsf{Q}_{m,\kappa}(\ell, g, \Lambda, C)$*,* $a = \Lambda^2/(m^2\kappa)$*,* $e = \sqrt{1 - C^2/\Lambda^2}$*,* $u = u_{\mathbb{R}}(\ell, e)$ ([I:`bpI:p7:kepler-eq`](c))*:* $q = a\big((\cos u - e)e_g + \sqrt{1-e^2}\,\sin u\, J_0 e_g\big)$*,* $|q|_2 = a(1 - e\cos u)$*,* $\mathrm{cr}(q,p) = C$*,* $E_{m,\kappa}(q,p) = -\dfrac{\kappa^2 m^3}{2\Lambda^2}$*, and the elements of Definition 9.1 are* Λ, C, a, e.

(c) (Mean values; includes the circular case $e = 0$) *For* $\Lambda > 0$*,* $0 < C \le \Lambda$ *and* $g \in \mathbb{R}$*, with* $(q,p) = \mathsf{Q}_{m,\kappa}(\ell, g, \Lambda, C)$ (extended to $C = \Lambda$ by the same formulas, [I:`bpI:p7:circular`])*:*

$$\frac{1}{2\pi}\int_0^{2\pi} \frac{d\ell}{|q|_2} = \frac{1}{a}, \qquad \frac{1}{2\pi}\int_0^{2\pi} |p|_2^2 d\ell = \frac{m^2\kappa}{a}, \qquad \frac{1}{2\pi}\int_0^{2\pi} |q|_2\, d\ell = a\Big(1 + \frac{e^2}{2}\Big).$$

(d) (Hamiltonian) $H_{\mathrm{Kep},1} := E_{m,\kappa}$ *satisfies* $H_{\mathrm{Kep},1} \circ \mathsf{Q}_{m,\kappa} = h_m(\Lambda) := -\kappa^2 m^3/(2\Lambda^2)$*, and* $h_m'(\Lambda) = \kappa^2 m^3/\Lambda^3 = \sqrt{\kappa}\, a^{-3/2}$ (Kepler's third law)*; the Hamilton equations of* $h_m(\Lambda)$ *in the variables* $(\ell, g; \Lambda, C)$ *are* $\dot\ell = h_m'(\Lambda)$*,* $\dot g = \dot\Lambda = \dot C = 0$.

Source: Fej04r Thm. 62 and (32)–(33), p. 48 (energy $-\mu^3 M^2/(2\Lambda^2)$*, mean motion* $\nu = M/a^{3/2}$*-type formula); Chi05 §2 for the Delaunay chart (via track I); the dictionary is* [I:`bpI:p7:scaling`]*; the mean values are standard* (track I states them for $\kappa = m = 1$).

Proof. Throughout, $L = \Lambda/m$, $G = C/m$, $(Q,P) = \mathsf{Q}^{\mathrm{I}}(\ell, g, L, G)$, $(q,p) = \mathsf{S}_{m,\kappa}(Q,P)$; note $\Lambda > C > 0$ iff $L > G > 0$, i.e. $(L,G) \in \mathcal{C}_0$ of [I:`bpI:p7:orbit-maps`].

Step S9.3: (a) Chart. $\mathsf{S}_{m,\kappa}$ is linear and invertible, and $(\Lambda, C) \mapsto (L,G) = (\Lambda, C)/m$ is a linear isomorphism of $\{\Lambda > C > 0\}$ onto $\mathcal{C}_0$. By [I:`bpI:p7:chart`], Q^{I} induces a real-analytic diffeomorphism of $(\mathbb{R}/2\pi\mathbb{Z})^2 \times \mathcal{C}_0$ onto $\mathcal{C} = \{E_1 < 0,\ \mathrm{cr} > 0,\ \mathrm{cr} < L\}$ ([I:`bpI:p7:elements`]). Under $(q,p) = \mathsf{S}_{m,\kappa}(Q,P)$ one has $E_{m,\kappa}(q,p) = m\kappa^2 E_1(Q,P)$, $\mathrm{cr}(q,p) = m\,\mathrm{cr}(Q,P)$ and (Step S9.1) $\Lambda = mL(Q,P)$; so $\mathsf{S}_{m,\kappa}$ maps $\mathcal{C}$ bijectively onto $\{E_{m,\kappa} < 0,\ C > 0,\ C < \Lambda\} = \mathcal{K}^*_{m,\kappa}$.

Step S9.4: (a) The form. $\sum dp_i \wedge dq_i = \sum (m\kappa\, dP_i) \wedge (dQ_i/\kappa) = -m \sum dQ_i \wedge dP_i$. By [I:`bpI:p7:symplectic`], pulling back by Q^{I} gives $-m(d\ell \wedge dL + dg \wedge dG) = m\, dL \wedge d\ell + m\, dG \wedge dg = d\Lambda \wedge d\ell + dC \wedge dg$.

Step S9.5: (b). [I:`bpI:p7:orbit-identities`] for (Q,P): $|Q|_2 = L^2(1 - e\cos u)$, $E_1(Q,P) = -1/(2L^2)$, $\mathrm{cr}(Q,P) = G$, with $e = \sqrt{1 - G^2/L^2} = \sqrt{1 - C^2/\Lambda^2}$. With $q = Q/\kappa$: $|q|_2 = a(1 - e\cos u)$ for $a = L^2/\kappa = \Lambda^2/(m^2\kappa)$ (this is the a of Definition 9.1: $a = -\kappa m/(2E_{m,\kappa})$ with $E_{m,\kappa} = m\kappa^2 E_1 = -m\kappa^2/(2L^2)$ gives $a = L^2/\kappa$). The formula for q is Q/κ with $Q = \mathsf{R}(g)(X,Y)$, $X = L^2(\cos u - e)$, $Y = L^2\sqrt{1-e^2}\,\sin u$ and $\mathsf{R}(g)(x,y) = xe_g + yJ_0e_g$. $\mathrm{cr}(q,p) = m\,\mathrm{cr}(Q,P) = mG = C$; $E_{m,\kappa}(q,p) = -m\kappa^2/(2L^2) = -\kappa^2 m^3/(2\Lambda^2)$.

Step S9.6: (c). For $C < \Lambda$ and $C = \Lambda$ (circular), [I:`bpI:p7:means`] and [I:`bpI:p7:circular`] give $\langle 1/|Q|_2\rangle = 1/L^2$, $\langle |P|_2^2\rangle = 1/L^2$, $\langle |Q|_2\rangle = L^2(1 + e^2/2)$. Then $1/|q|_2 = \kappa/|Q|_2$ has mean $\kappa/L^2 = 1/a$; $|p|_2^2 = m^2\kappa^2|P|_2^2$ has mean $m^2\kappa^2/L^2 = m^2\kappa/a$; $|q|_2 = |Q|_2/\kappa$ has mean $a(1 + e^2/2)$.

Step S9.7: (d). The value of $E_{m,\kappa} \circ \mathsf{Q}_{m,\kappa}$ is (b). $h_m' = \kappa^2 m^3/\Lambda^3$, and $\Lambda^3 = m^3\kappa^{3/2}a^{3/2}$ gives $\sqrt{\kappa}\, a^{-3/2}$. By (a) the chart $\mathsf{Q}_{m,\kappa}$ is symplectic with factor one in the variables $(\ell, g; \Lambda, C)$ (positions (ℓ, g), momenta (Λ, C), $\Omega = d\Lambda \wedge d\ell + dC \wedge dg$), so the Hamilton equations of $h_m(\Lambda)$ in these variables are $\dot\ell = \partial_\Lambda h_m$, $\dot g = \partial_C h_m = 0$, $\dot\Lambda = -\partial_\ell h_m = 0$, $\dot C = -\partial_g h_m = 0$, and Lemma 2.4 transports them to the Kepler flow. □

9.3. **Kepler's equation in the eccentric longitude.** The Delaunay map is singular at $e = 0$ (g is undefined). The D25 formulas and the Poincaré chart use instead the *eccentric longitude* $F = u + g$

and the *eccentricity vector* $v = e\,e_g$. Substituting $\ell = \lambda - g$, $u = F - g$ in $u - e\sin u = \ell$ gives

$$F - e\sin(F-g) = \lambda, \qquad e\sin(F-g) = v_0\sin F - v_1\cos F \quad (v = e(\cos g, \sin g)),$$

an equation *regular in* $v \in \mathbb{R}^2$: $F - v_0\sin F + v_1\cos F = \lambda$ (`keplerMeanLongitude`).

Proposition 9.4 (Regular Kepler equation)**.** *For $v \in \mathbb{R}^2$, $F \in \mathbb{R}$ put $\Phi(F,v) = F - v_0\sin F + v_1\cos F$ (so Φ is `keplerMeanLongitude` with $v =$ `secularEccVector`) and $J(F,v) = \partial_F\Phi = 1 - v_0\cos F - v_1\sin F = 1 - v\cdot(\cos F, \sin F)$ (`longitudeJacobian`).*

(a) (real) *For $|v|_2 < 1$: $J \geq 1 - |v|_2 > 0$; $F \mapsto \Phi(F,v)$ is a real-analytic increasing bijection of $\mathbb{R}$ with $\Phi(F + 2\pi, v) = \Phi(F,v) + 2\pi$. So for each $(\lambda, v) \in \mathbb{R}\times\{|v|_2 < 1\}$ there is exactly one real $F = F(\lambda, v)$ with $\Phi(F,v) = \lambda$; F is real analytic on $\mathbb{R}\times\{|v|_2 < 1\}$, $F(\lambda + 2\pi, v) = F(\lambda, v) + 2\pi$, $F(\lambda, 0) = \lambda$, and*

$$\partial_\lambda F = \frac{1}{J}, \qquad \partial_{v_0} F = \frac{\sin F}{J}, \qquad \partial_{v_1} F = -\frac{\cos F}{J} \qquad (J = J(F(\lambda,v),v)).$$

(b) (complex) *Let $\mathcal{N}_0 = \{(\lambda,v) \in \mathbb{C}\times\mathbb{C}^2 : |\operatorname{Im}\lambda| < \frac14,\ |v_0| + |v_1| < \frac14\}$ and $\mathcal{S} = \{F \in \mathbb{C} : |\operatorname{Im} F| \leq 1\}$. There is a unique holomorphic $F : \mathcal{N}_0 \to \mathcal{S}$ with $\Phi(F(\lambda,v),v) = \lambda$ (with Φ, J extended to complex arguments by the same formulas); $|J| \geq 1 - \frac14\cosh 1 > \frac35$ on $\mathcal{N}_0$, and the formulas of (a) hold. At real points F is real and equals the F of (a).*

(c) (relation to track I) *For $v = e\,e_g$ with $e \in (0,1)$, $g \in \mathbb{R}$: $F(\lambda,v) = g + u_{\mathbb{R}}(\lambda - g, e)$, where $u_{\mathbb{R}}$ is the solution of Kepler's equation of* [I:`bpI:p7:kepler-eq`]*(c). (The composition of* [I:`bpI:p7:kepler-eq`] *with $v \mapsto (g,e)$ is not defined at $v = 0$, which is why (b) is proved directly: it is the holomorphic continuation of this branch through $v = 0$.)*

Source: [I:`bpI:p7:kepler-eq`] *(Kepler's equation near $e = 0$); the regular form in v is the D25 definition* `keplerMeanLongitude` *(derived from LP17 (10)–(11)); the contraction proof is ours.*

Proof. **Step S9.8: (a) Monotonicity.** $|v\cdot(\cos F, \sin F)| \leq |v|_2$ (Cauchy–Schwarz), so $J \geq 1 - |v|_2 > 0$. Hence $\Phi(\cdot, v)$ is strictly increasing with $\Phi(F,v) - F$ bounded ($\leq |v|_2$), so $\Phi(F,v) \to \pm\infty$ as $F \to \pm\infty$: a bijection of $\mathbb{R}$. $\Phi(F + 2\pi, v) = \Phi(F,v) + 2\pi$ by periodicity of $\sin, \cos$; $\Phi(F, 0) = F$.

Step S9.9: (a) Analyticity. Φ is real analytic on $\mathbb{R}\times\mathbb{R}^2$ with $\partial_F\Phi = J \neq 0$ on $\{|v|_2 < 1\}$; (E1) gives a local real-analytic solution near any (λ_0, v_0), which coincides with the unique global solution; so F is real analytic. Differentiating $\Phi(F(\lambda,v),v) = \lambda$: $J\,\partial_\lambda F = 1$; $J\,\partial_{v_0}F - \sin F = 0$; $J\,\partial_{v_1}F + \cos F = 0$.

Step S9.10: (b) Contraction. Let $\beta_{\lambda,v}(F) = \lambda + v_0\sin F - v_1\cos F$, so $\Phi(F,v) = \lambda \iff \beta_{\lambda,v}(F) = F$. For $F = x + iy \in \mathcal{S}$: $|\sin F|^2 = \sin^2 x + \sinh^2 y$, $|\cos F|^2 = \cos^2 x + \sinh^2 y$, both $\leq \cosh^2 y \leq \cosh^2 1$ ([I:`stepI:p7:ke-trig`]). For $(\lambda,v) \in \mathcal{N}_0$ then $|\operatorname{Im}\beta| \leq |\operatorname{Im}\lambda| + (|v_0| + |v_1|)\cosh 1 < \frac14 + \frac14\cdot 1.55 < 1$, so $\beta_{\lambda,v}(\mathcal{S}) \subset \mathcal{S}$, and $|\beta'| = |v_0\cos F - v_1\sin F| \leq \frac14\cosh 1 < 0.39$, so $\beta_{\lambda,v}$ is a 0.39-contraction of the closed convex set $\mathcal{S}$ ((E3)). It has a unique fixed point $F(\lambda,v) \in \mathcal{S}$.

Step S9.11: (b) Holomorphy, real points. $F_0 = \lambda$ (which lies in $\mathcal{S}$) and $F_{j+1} = \beta_{\lambda,v}(F_j)$ are holomorphic on $\mathcal{N}_0$ with values in $\mathcal{S}$ and converge uniformly (geometric rate 0.39, Banach), so F is holomorphic ((E2)). For real $(\lambda,v) \in \mathcal{N}_0$ the real map $\beta_{\lambda,v}$ preserves $\mathbb{R} \subset \mathcal{S}$ and is a contraction there, so its fixed point is real, equals the real solution of (a) (uniqueness there), and F is real. $|J| \geq 1 - (|v_0| + |v_1|)\cosh 1 > 1 - 0.39 > \frac35$. The derivative formulas are the same implicit differentiation as in (a), valid for holomorphic maps.

Step S9.12: (c) Relation. For $v = e\,e_g$: $v_0\sin F - v_1\cos F = e(\cos g\sin F - \sin g\cos F) = e\sin(F - g)$, so $F' = g + u_{\mathbb{R}}(\lambda - g, e)$ solves $\Phi(F', v) = F' - e\sin(F' - g) = g + u_{\mathbb{R}} - e\sin u_{\mathbb{R}} = g + (\lambda - g) = \lambda$; uniqueness in (a) gives $F' = F$. □

9.4. **The Poincaré chart.**

Definition 9.5 (Poincaré chart of one planet)**.** Let $m > 0$ and $\kappa > 0$. The *Poincaré domain* is $\mathcal{O} = \{(\lambda, \xi, \Lambda, \eta) \in \mathbb{R}^4 : \ \Lambda > 0,\ \Gamma < \Lambda\}$ with $\Gamma = \Gamma(\xi, \eta) = (\xi^2 + \eta^2)/2$ (open, invariant under $\lambda \mapsto \lambda + 2\pi$). On $\mathcal{O}$ put

$$k = 1 - \frac{\Gamma}{\Lambda} \ \in (0, 1], \qquad v = \sqrt{\frac{1 - \Gamma/(2\Lambda)}{\Lambda}}\,(\xi, \eta), \qquad a = \frac{\Lambda^2}{m^2\kappa}, \qquad F = F(\lambda, v) \ \ \text{(Prop. 9.4(a))},$$
$$c = (\cos F, \sin F), \qquad J = 1 - v \cdot c,$$
$$q = a\Big(k\, c - v + \frac{v \cdot c}{1 + k}\, v\Big), \qquad p = \frac{m^2\kappa}{\Lambda J}\Big(J_0 c - \frac{v \cdot c}{1 + k}\, J_0 v\Big),$$

and $\Psi_{m,\kappa}(\lambda, \xi, \Lambda, \eta) = (q, p) \in \mathbb{R}^2 \times \mathbb{R}^2$. For $\kappa = 1$ these are `secularEccVector`, `secularRoot` $(= k = \sqrt{1 - e^2})$, `secularAxis`, `longitudeJacobian`, `ellipsePosition`, `ellipseMomentum`, and $F =$ the solution of `keplerMeanLongitude` $= \lambda$ (Proposition 9.16).

Frozen Lean declaration: `KAM.Planetary.secularEccVector, secularRoot, secularGamma, secularAxis, ellipsePosition, ellipseMomentum, keplerMeanLongitude, longitudeJacobian (for` $\kappa = 1$`)`

Source: Fej04r (31), Thm. 62, pp. 47–48, with the normalisation of Fej07e p. 3: $r_j = \xi_j + i\eta_j = \sqrt{2\Lambda_j}\sqrt{1 - \sqrt{1 - e_j^2}}\, e^{i(g_j + \theta_j)}$ (planar: $\theta_j = 0$); explicit regular form: LP17 (10)–(11) via the D25 definitions.

Lemma 9.6 (Elementary facts on $\mathcal{O}$)**.** *On $\mathcal{O}$: $\Gamma \geq 0$, $0 < k \leq 1$, $|v|_2^2 = 1 - k^2$ (so $|v|_2 < 1$ and F is defined, real analytic, Proposition 9.4(a)); $J \geq 1 - |v|_2 > 0$; the radicand of v is positive; $v \cdot c = e\cos(F - g)$ and $|v|_2 = e$ with $e = \sqrt{1 - k^2}$. For $(\xi, \eta) \neq 0$ write $(\xi, \eta) = \rho\, e_g$ with $\rho = \sqrt{2\Gamma}$: then $v = e\, e_g$ and $\Gamma = \Lambda - C$ with $C = \Lambda k$.*

Proof. $\Gamma < \Lambda$ and $\Lambda > 0$ give $k = 1 - \Gamma/\Lambda > 0$ and $1 - \Gamma/(2\Lambda) > \frac{1}{2}$. $|v|_2^2 = \frac{1 - \Gamma/(2\Lambda)}{\Lambda} 2\Gamma = \frac{2\Gamma}{\Lambda}\big(1 - \frac{\Gamma}{2\Lambda}\big) = 1 - (1 - \frac{\Gamma}{\Lambda})^2 = 1 - k^2$. The rest is Proposition 9.4(a), and for $(\xi, \eta) \neq 0$: $|v|_2/\rho = \sqrt{(1 - \Gamma/2\Lambda)/\Lambda}$ with $|v|_2 = \sqrt{1 - k^2}$ the eccentricity, and the direction of v is that of (ξ, η). □

Theorem 9.7 (Poincaré chart, analytic through $e = 0$)**.** *Let $\Psi = \Psi_{m,\kappa} : \mathcal{O} \to \mathbb{R}^2 \times \mathbb{R}^2$ be as in Definition 9.5.*

- *(a)* (analytic) *Ψ is real analytic on $\mathcal{O}$, including $\xi = \eta = 0$; $\Psi(\lambda + 2\pi, \xi, \Lambda, \eta) = \Psi(\lambda, \xi, \Lambda, \eta)$.*
- *(b)* (Delaunay on $\Gamma > 0$) *Let $(\xi, \eta) = \rho\, e_g$, $\rho > 0$. Then $\Psi(\lambda, \xi, \Lambda, \eta) = \mathsf{Q}_{m,\kappa}(\lambda - g,\ g,\ \Lambda,\ \Lambda - \Gamma)$, with the eccentric longitude $F = g + u$, $u = u_{\mathbb{R}}(\lambda - g, e)$.*
- *(c)* (symplectic) *$\Omega(D\Psi\, u, D\Psi\, w) = \Omega(u, w)$ on all of $\mathcal{O}$, i.e. $\Psi^*\Omega_{\mathrm{cart}} = d\Lambda \wedge d\lambda + d\eta \wedge d\xi$ with $\Omega_{\mathrm{cart}} = \sum dp_i \wedge dq_i$. Thus Ψ is a 1-chart on $\mathcal{O}$ in the sense of Definition 2.3, and $D\Psi$ is invertible.*
- *(d)* (elements) *$\Psi(\mathcal{O}) \subset \mathcal{K}_{m,\kappa}$ and for $(q, p) = \Psi(\lambda, \xi, \Lambda, \eta)$: $E_{m,\kappa}(q, p) = -\kappa^2 m^3/(2\Lambda^2)$, $a(q, p) = \Lambda^2/(m^2\kappa)$, the action of Definition 9.1 is Λ, $\mathrm{cr}(q, p) = \Lambda - \Gamma = \Lambda k$ and, the Laplace–Runge–Lenz vector satisfies $\kappa^{-1}\big[\big(\frac{|p|_2^2}{m^2} - \frac{\kappa}{|q|_2}\big)q - \frac{q \cdot p}{m^2}p\big] = v$ (for $\kappa = 1$ this is `keplerEccVector` $= v$).*
- *(e)* (bijectivity modulo 2π) *$\Psi : \mathcal{O} \to \mathcal{K}_{m,\kappa}$ is surjective, and $\Psi(\lambda', \xi', \Lambda', \eta') = \Psi(\lambda, \xi, \Lambda, \eta)$ iff $(\xi', \Lambda', \eta') = (\xi, \Lambda, \eta)$ and $\lambda' - \lambda \in 2\pi\mathbb{Z}$. The induced map $\mathcal{O}/(2\pi\mathbb{Z} \times \{0\}) \to \mathcal{K}_{m,\kappa}$ (quotient by the λ-translations) is a real-analytic diffeomorphism. Its inverse is $\Lambda = m\sqrt{\kappa a}$, $\Gamma = \Lambda - \mathrm{cr}(q, p)$, $v =$ the rescaled LRL vector of (d), $(\xi, \eta) = v/\sqrt{(1 - \Gamma/(2\Lambda))/\Lambda}$, $c = \big(k\,\mathrm{Id} + \frac{1}{1+k} v v^{\top}\big)^{-1}(q/a + v)$, $F = \arg c$, $\lambda = \Phi(F, v)$.*
- *(f)* (circular torus) *At $\xi = \eta = 0$: $v = 0$, $k = 1$, $F = \lambda$, $q = a(\cos\lambda, \sin\lambda)$, $p = m\sqrt{\kappa/a}\,(-\sin\lambda, \cos\lambda)$, $a = \Lambda^2/(m^2\kappa)$: for $\Lambda = m\sqrt{\kappa a}$ this is `circularPoint` (with $\phi = \lambda/2\pi$, one planet).*

Proof. **Step S9.13: (a).** k, v (positive radicand, Lemma 9.6) and J are real analytic, F is real analytic (Proposition 9.4(a) composed with the analytic $v(\xi,\eta,\Lambda)$, $|v|_2<1$), $1+k\geq 1$, $\Lambda J>0$; q,p are rational expressions in these and in $\cos F,\sin F$. Periodicity: $F(\lambda+2\pi,v)=F+2\pi$ and c is 2π-periodic in F.

Step S9.14: (b) Delaunay on $\Gamma>0$. Put $\ell=\lambda-g$, $C=\Lambda-\Gamma=\Lambda k$, $L=\Lambda/m$, $G=C/m=Lk$. Then $\Lambda>C>0$, $e=\sqrt{1-C^2/\Lambda^2}=\sqrt{1-k^2}$ and $v=e\,e_g$ (Lemma 9.6), so by Proposition 9.4(c) $F=g+u$, $u=u_{\mathbb{R}}(\ell,e)$. Write $\hat e=e_g$, $\hat e^{\perp}=J_0e_g$ and $c=\cos u\,\hat e+\sin u\,\hat e^{\perp}$ ($F=u+g$). Then $v\cdot c=e\cos u$, $J=1-e\cos u$ and, using $e^2/(1+k)=(1-k^2)/(1+k)=1-k$,

$$kc-v+\frac{v\cdot c}{1+k}v=(k\cos u-e+(1-k)\cos u)\hat e+k\sin u\,\hat e^{\perp}=(\cos u-e)\hat e+k\sin u\,\hat e^{\perp},$$

so $q=a((\cos u-e)\hat e+k\sin u\,\hat e^{\perp})$, which is the position of Proposition 9.3(b). Likewise $J_0c=\cos u\,\hat e^{\perp}-\sin u\,\hat e$, $J_0v=e\hat e^{\perp}$ and

$$J_0c-\frac{v\cdot c}{1+k}J_0v=-\sin u\,\hat e+\big(1-\tfrac{e^2}{1+k}\big)\cos u\,\hat e^{\perp}=-\sin u\,\hat e+k\cos u\,\hat e^{\perp}=\mathsf{R}(g)\begin{pmatrix}-\sin u\\ k\cos u\end{pmatrix}.$$

By [I:bpI:p7:orbit-maps], $P=\mathsf{R}(g)\binom{-\sin u}{k\cos u}/(LJ)$ and $p=m\kappa P=\frac{m\kappa}{LJ}(\cdots)=\frac{m^2\kappa}{\Lambda J}(\cdots)$, which is the p of the definition. Hence $\Psi=\mathsf{S}_{m,\kappa}\circ\mathsf{Q}^{\mathrm{I}}(\ell,g,L,G)=\mathsf{Q}_{m,\kappa}(\ell,g,\Lambda,C)$.

Step S9.15: (c) Symplecticity on $\{\Gamma>0\}$. On the open set $\mathcal{A}=\{(\xi,\eta)\neq 0\}\subset\mathcal{O}$ choose locally a real-analytic branch $g(\xi,\eta)=\arg(\xi+i\eta)$ and put $\rho=\sqrt{2\Gamma}$. By (b), $\Psi=\mathsf{Q}_{m,\kappa}\circ T$ with $T(\lambda,\xi,\Lambda,\eta)=(\lambda-g,g,\Lambda,\Lambda-\Gamma)$ (real analytic, values in $\mathcal{D}^m$). By Proposition 9.3(a), $\Psi^*\Omega_{\mathrm{cart}}=T^*(d\Lambda\wedge d\ell+dC\wedge dg)=d\Lambda\wedge(d\lambda-dg)+(d\Lambda-d\Gamma)\wedge dg=d\Lambda\wedge d\lambda-d\Gamma\wedge dg$. Since $\xi=\rho\cos g$, $\eta=\rho\sin g$: $d\xi\wedge d\eta=\rho\,d\rho\wedge dg=d\Gamma\wedge dg$, i.e. $d\eta\wedge d\xi=-d\Gamma\wedge dg$. So $\Psi^*\Omega_{\mathrm{cart}}=d\Lambda\wedge d\lambda+d\eta\wedge d\xi$ on $\mathcal{A}$.

Step S9.16: (c) All of $\mathcal{O}$. The matrix identity $D\Psi^{\top}M_0D\Psi=M_0$ (equivalent to the form identity, (E4)) is an identity between continuous functions on $\mathcal{O}$ (Ψ is C^1 there by (a)), valid on the dense subset $\mathcal{A}$ (the complement $\{\xi=\eta=0\}$ has empty interior); (E5). This is the chart property of Definition 2.3 with $c=1$ (the factor is 1 because the source form is the frozen Ω in the order $(\lambda,\xi;\Lambda,\eta)$). Invertibility: Lemma 2.1(b).

Step S9.17: (d) Elements. On $\mathcal{A}$ by (b) and Proposition 9.3(b): $E_{m,\kappa}=-\kappa^2m^3/(2\Lambda^2)<0$, $\mathrm{cr}=\Lambda k>0$, $q\neq 0$ (as $|q|_2=a(1-e\cos u)>0$), hence $\Psi(\mathcal{A})\subset\mathcal{K}_{m,\kappa}$, and $a(q,p)=-\kappa m/(2E)=\Lambda^2/(m^2\kappa)$, action $m\sqrt{\kappa a}=\Lambda$. At $\xi=\eta=0$ the same identities are [I:bpI:p7:circular] (with $e=0$, $k=1$, $u=F=\lambda$) through the scaling $\mathsf{S}_{m,\kappa}$ of Step S9.5; they can also be checked directly from (f). The LRL identity: for $(Q,P)=\mathsf{S}_{m,\kappa}^{-1}(q,p)$ one computes $\kappa^{-1}\big[(\frac{|p|_2^2}{m^2}-\frac{\kappa}{|q|_2})q-\frac{q\cdot p}{m^2}p\big]=(|P|_2^2-\frac{1}{|Q|_2})Q-(Q\cdot P)P=:A$. On $\mathcal{A}$, in the orbit frame $(\hat e,\hat e^{\perp})$ of Step S9.14 with $c_u=\cos u$, $s_u=\sin u$, $J=1-ec_u$, $L=\Lambda/m$ ([I:bpI:p7:orbit-identities]): $|P|_2^2=\frac{1+ec_u}{L^2J}$, $\frac{1}{|Q|_2}=\frac{1}{L^2J}$, $Q=L^2(c_u-e,\,ks_u)$, $Q\cdot P=Les_u$, $P=\frac{1}{LJ}(-s_u,kc_u)$, hence $A=\frac{ec_u}{J}(c_u-e,\,ks_u)-\frac{es_u}{J}(-s_u,\,kc_u)=\big(\frac{e}{J}(c_u^2-ec_u+s_u^2),\,0\big)=(e,0)$ in the frame, i.e. $A=e\,\hat e=v$. At $\xi=\eta=0$ both sides vanish ($|Q|_2=L^2$, $|P|_2^2=1/L^2$, $Q\cdot P=0$ by (f)).

Step S9.18: (e) Bijectivity. *Injectivity modulo 2π in λ.* Let $\Psi(w')=\Psi(w)$, $w,w'\in\mathcal{O}$. By (d) $\Lambda'=\Lambda$ ($\Lambda=m\sqrt{\kappa a}$ is a function of (q,p)) and $\Gamma'=\Lambda-\mathrm{cr}=\Gamma$. If $\Gamma=0$: $\xi=\eta=\xi'=\eta'=0$ and by (f) $\Psi=(a(\cos\lambda,\sin\lambda),\cdots)$ so $\lambda'-\lambda\in2\pi\mathbb{Z}$. If $\Gamma>0$: $w,w'\in\mathcal{A}$, (ℓ,g), (ℓ',g') as in (b), and by [I:bpI:p7:chart] (injectivity modulo 2π of Q^{I}, transported by Proposition 9.3(a)) $g'-g\in2\pi\mathbb{Z}$, $\ell'-\ell\in2\pi\mathbb{Z}$. Then $(\xi',\eta')=\rho\,e_{g'}=\rho\,e_g=(\xi,\eta)$ and $\lambda'=\ell'+g'\in\lambda+2\pi\mathbb{Z}$. The converse is (a). *Surjectivity.* Let $(q,p)\in\mathcal{K}_{m,\kappa}$, Λ, C its elements, $\Gamma=\Lambda-C\in[0,\Lambda)$ (Lemma 9.2). If $C<\Lambda$: $(q,p)\in\mathcal{K}^*_{m,\kappa}$ is $\mathsf{Q}_{m,\kappa}(\ell,g,\Lambda,C)$ by Proposition 9.3(a), and with $(\xi,\eta)=\sqrt{2\Gamma}\,e_g$, $\lambda=\ell+g$, (b) gives $\Psi(\lambda,\xi,\Lambda,\eta)=(q,p)$. If $C=\Lambda$ (so $\Gamma=0$): in the scaling of Step S9.1, (Q,P) has $\mathrm{cr}=L$ and the

equality case of Step S9.2 gives $|Q|_2 = L^2$, $Q \cdot P = 0$, $|P|_2 = \mathrm{cr}\,/|Q|_2 = 1/L$; so $Q = L^2(\cos\lambda, \sin\lambda)$ for some λ and $P = \frac{1}{L}(-\sin\lambda, \cos\lambda)$ (orthogonal to Q, of length $1/L$, positive cross product $\mathrm{cr}(Q,P) = L > 0$). That is $(q,p) = \Psi(\lambda, 0, \Lambda, 0)$ by (f). *Diffeomorphism.* $D\Psi$ is invertible (c); Ψ is a bijective, real-analytic local diffeomorphism from the quotient by the lattice $2\pi\mathbb{Z}$ in λ onto the open set $\mathcal{K}_{m,\kappa}$ ((E1)). Inverse formulas: Λ, Γ as in (d); v by the LRL identity (d); $(\xi,\eta) = v/s$ with $s = \sqrt{(1 - \Gamma/(2\Lambda))/\Lambda} > 0$ (defined also at $v = 0$); $\frac{q}{a} = kc - v + \frac{v\cdot c}{1+k}v = (k\,\mathrm{Id} + \frac{1}{1+k}vv^\top)c - v$ and $k\,\mathrm{Id} + \frac{1}{1+k}vv^\top$ has eigenvalues k and $k + |v|_2^2/(1+k) = 1$, hence is invertible; c is then the unit vector $\cos F, \sin F$, and $\lambda = \Phi(F, v)$ modulo 2π.

Step S9.19: (f) Circular points. $\Gamma = 0$ gives $v = 0$, $k = 1$, $F = \lambda$ (Proposition 9.4(a)), $J = 1$, $q = a(c - 0 + 0) = a(\cos\lambda, \sin\lambda)$, $p = \frac{m^2\kappa}{\Lambda} J_0 c = m\sqrt{\kappa/a}\, J_0 c$ since $\Lambda = m\sqrt{\kappa a}$. Compare Definition 1.17. □

Proposition 9.8 (Holomorphic extension near the circular orbits)**.** *Let* $\Lambda^\circ > 0$ *and*

$$\mathcal{N}(\Lambda^\circ) = \Big\{(\lambda,\xi,\Lambda,\eta) \in \mathbb{C}^4 :\ |\mathrm{Im}\,\lambda| < \tfrac{1}{4},\ |\Lambda - \Lambda^\circ| < \tfrac{1}{2}\Lambda^\circ,\ |\xi|, |\eta| < \tfrac{1}{16}\sqrt{\Lambda^\circ}\Big\}.$$

The formulas of Definition 9.5 (with $\Gamma = (\xi^2+\eta^2)/2$, $F = F(\lambda, v)$ *of Proposition 9.4(b)) define a holomorphic map* $\Psi : \mathcal{N}(\Lambda^\circ) \to \mathbb{C}^4$, 2π*-periodic in* λ*, real at real points, with* $\mathcal{N}(\Lambda^\circ) \cap \mathbb{R}^4 \subset \mathcal{O}$ *and* $D\Psi^\top M_0 D\Psi = M_0$ *at real points. Hence it satisfies the hypotheses of Lemma 2.7 with* $c = 1$.

Proof. **Step S9.20: Sizes.** On $\mathcal{N}(\Lambda^\circ)$: $|\Lambda| > \frac{1}{2}\Lambda^\circ$ and $|\arg\Lambda| \le \arcsin\frac{1}{2} = \frac{\pi}{6}$; $|\Gamma| < \frac{1}{256}\Lambda^\circ$, so $|\Gamma/\Lambda| < \frac{1}{128}$, hence $|\arg(1 - \Gamma/2\Lambda)| < 0.01$, $|1 - \Gamma/2\Lambda| < 1.01$, $\mathrm{Re}\,k > 0.99$ ($k = 1 - \Gamma/\Lambda$), $|1+k| > 1.9$. The radicand $(1-\Gamma/2\Lambda)/\Lambda$ has argument of modulus $< \frac{\pi}{6} + 0.01 < \frac{\pi}{2}$, so its principal square root s is holomorphic ((E2)), and $|s| \le \sqrt{1.01 \cdot 2/\Lambda^\circ} < 1.5/\sqrt{\Lambda^\circ}$. Therefore $|v_0| + |v_1| \le |s|(|\xi| + |\eta|) < 1.5 \cdot \frac{2}{16} < \frac{1}{4}$, $(\lambda, v) \in \mathcal{N}_0$, and F is holomorphic with $|J| > \frac{3}{5}$ (Proposition 9.4(b)).

Step S9.21: Holomorphy, reality, periodicity. q, p are rational in $(\Lambda, k, v, \cos F, \sin F, J^{-1}, (1+k)^{-1})$, all holomorphic with nonvanishing denominators; periodicity as in Step S9.13; at real points everything is real and $\Gamma < \Lambda$, so the point lies in $\mathcal{O}$ and the real maps agree. The matrix identity at real points is Theorem 9.7(c) (the complex derivative restricts to the real one, [K:`bp:p1:real-restriction`]). □

Remark 9.9. The neighbourhood is small in v by design (a contraction in $|v_0| + |v_1| < \frac{1}{4}$); this is all that the circular-orbit analysis (Parts 2, 5, 8) needs: $|\xi|, |\eta|$ of size $\ll \sqrt{\Lambda}$ is eccentricity $e \lesssim \frac{1}{4}$ (cf. $|v|_2 = e$, Lemma 9.6). For larger eccentricities (not needed in this track) the complex tube of [I:`bpI:p7:kepler-eq`] covers $e \le e_1 < 1$ in the Delaunay form away from $e = 0$.

Remark 9.10 (Normalisation, sign conventions, what the source does not prove). (1) *Normalisation.* Fej04r (31) as printed has $r_j = \xi_j + i\eta_j = \sqrt{\Lambda_j}\sqrt{1 - \sqrt{1-\varepsilon_j^2}}\, e^{i(g_j+\theta_j)}$ with only $\sqrt{\Lambda_j}$, for which the chart is merely conformally symplectic (factor 2 in $d\xi \wedge d\eta$). Fej07e p. 3 ("sans le facteur $\sqrt{2}$, elles ne sont que conformément symplectiques") corrects it to $\sqrt{2\Lambda_j}$, i.e. $\frac{1}{2}(\xi^2+\eta^2) = \Lambda - C = \Gamma$ (Lemma 9.6), which is exactly `secularGamma`: $\Gamma = R(\eta, \xi)$ with Definition 1.7. With this normalisation the oscillator frequencies of the secular quadratic form are the diagonal values of $\partial^2\langle H\rangle/\partial\Gamma^2$ with no extra factor, which is what later fixes the frozen $\Omega = 2\sigma$. (2) *Sign.* Fej04r Thm. 62 writes $\omega = \sum d\lambda_j \wedge d\Lambda_j + d\xi_j \wedge d\eta_j$ (position∧momentum), and Fej07e p. 3 warns that signs vary ("conventions de signe variables quant à la forme symplectique") and gives the a-posteriori test (the perihelion of the larger planet precesses forward). The frozen order $(\lambda, \xi; \Lambda, \eta)$ makes ξ the position and η the momentum, $\Omega = \sum dz_2 \wedge dz_1$; with that order the identity of Theorem 9.7(c) forces $(\xi,\eta) = \rho(\cos g, \sin g)$ (the argument of $\xi + i\eta$ is $+g$, as in Fej07e). If one writes the same chart with the roles of ξ, η exchanged (η position), the formula becomes $(\xi,\eta) = \rho(\cos g, -\sin g)$; this is only a relabelling and is not used. (3) *Source gap.* Fej04r Thm. 62 gives no proof ("Voir

[Poin05, Chap. III], ou [Che88, Appendice]", neither pinned); in particular the analyticity *through the circular orbits* is the whole content of the theorem. The proof above is ours and uses only explicit formulas: the regular Kepler equation (Proposition 9.4), the Delaunay chart of track I on $\Gamma > 0$, and a density argument for the symplectic identity at $\Gamma = 0$.

Source gap: Fej04r Thm. 62 (p. 48) cites Poincaré and Chenciner, not pinned; analyticity at $e = 0$ and bijectivity at the circular orbits are proved here from explicit formulas (Theorem 9.7(a),(c),(e)).

9.5. **The product chart for the two physical models.** Fix $n \geq 1$, $G, m_0 > 0$ ($\kappa = Gm_0$) and $m = (m_i) \in (0,\infty)^n$. Coordinates $w = (w_1, w_2) \in \mathrm{Ph}(2n)$ with $w_1 = (\lambda, \xi)$ and $w_2 = (\Lambda, \eta)$, $\lambda, \xi, \Lambda, \eta \in \mathbb{R}^n$ (fast block first: the order of `assemble`, Definition 1.6, with $a = b = n$). Physical points are $(q, p) \in \mathrm{Ph}(2n)$ with $q^{(i)} = \mathrm{body}(q, i)$.

Definition 9.11 (Product chart). $\mathcal{O}_n = \{w : (\lambda_i, \xi_i, \Lambda_i, \eta_i) \in \mathcal{O}\ \forall i\} = \{\Lambda_i > 0,\ \frac{1}{2}(\xi_i^2 + \eta_i^2) < \Lambda_i\ \forall i\}$ and $\Psi^{(n)}(w) = (q, p)$ with $(q^{(i)}, p^{(i)}) = \Psi_{m_i,\kappa}(\lambda_i, \xi_i, \Lambda_i, \eta_i)$. $h(\Lambda) = -\sum_i \kappa^2 m_i^3/(2\Lambda_i^2)$ and $\mathcal{K}^{(n)} = \{(q, p) : (q^{(i)}, p^{(i)}) \in \mathcal{K}_{m_i,\kappa}\ \forall i\}$; for $\kappa = 1$, $\mathcal{K}^{(n)} = \{\mathrm{ProgradeElliptic}\ m\ z\}$ (`ProgradeElliptic`: $0 < m_i$, $|q^{(i)}|_2 > 0$, `keplerEnergy` < 0, `planeCross` > 0).

Theorem 9.12 (Product Poincaré chart). *(a) $\Psi^{(n)}$ is a real-analytic 1-chart on the open set $\mathcal{O}_n$ (Definition 2.3, $c = 1$), $\mathcal{O}_n + ((2\pi k, 0), 0) = \mathcal{O}_n$, $\Psi^{(n)}(w + ((2\pi k, 0), 0)) = \Psi^{(n)}(w)$ for $k \in \mathbb{Z}^n$, and $\Psi^{(n)}(w') = \Psi^{(n)}(w)$ iff $w' = w + ((2\pi k, 0), 0)$ for some $k \in \mathbb{Z}^n$. It maps $\mathcal{O}_n$ onto $\mathcal{K}^{(n)}$, and induces a real-analytic diffeomorphism of $\mathcal{O}_n/\{(2\pi k, 0, 0, 0)\}$ onto $\mathcal{K}^{(n)}$.*

(b) (Decoupling) *$H_{\mathrm{Kep}} \circ \Psi^{(n)} = h(\Lambda)$ on $\mathcal{O}_n$, with H_{Kep} of Lemma 8.3; h does not depend on λ, ξ, η, nor on any coupling parameter. $\partial h/\partial \Lambda_i = \nu_i(\Lambda_i) := \kappa^2 m_i^3/\Lambda_i^3$ and $\partial^2 h/\partial \Lambda_i^2 = -3\kappa^2 m_i^3/\Lambda_i^4 \neq 0$; h is real analytic on $\{\Lambda_i > 0\}$.*

(c) (The two models share the chart) *Let $\Psi^{(-1)}(W) = \{w \in \mathcal{O}_n : \Psi^{(n)}(w) \in W\}$ (open, W of Lemma 8.2) and*

$$\hat{f}_{\mathrm{mov}} = f_{\mathrm{mov}} \circ \Psi^{(n)}, \qquad \hat{f}_{\mathrm{fix}} = f_{\mathrm{fix}} \circ \Psi^{(n)} \qquad (f \text{ of Lemma 8.3}).$$

These are real analytic on $\Psi^{(-1)}(W)$ and do not depend on μ or ε. For all μ, ε and $w \in \mathcal{O}_n$: $H^{\Psi}_{\mathrm{mov}}(w) = h(\Lambda) + \mu \hat{f}_{\mathrm{mov}}(w)$ and $H^{\Psi}_{\mathrm{fix}}(w) = h(\Lambda) + \varepsilon \hat{f}_{\mathrm{fix}}(w)$ (the Lean total functions; $H^{\Psi} = H \circ \Psi^{(n)}$ because $c = 1$). On $\Psi^{(-1)}(W)$ both are differentiable and Lemma 2.4 gives $D\Psi^{(n)} X_{H^{\Psi}} = X_H \circ \Psi^{(n)}$; orbits transfer by Corollary 2.5.

(d) (Hamilton equations in the chart) *On $\Psi^{(-1)}(W)$ the equations $\dot{w} = X_{H^{\Psi}}(w)$ read $\dot{\lambda}_i = \nu_i(\Lambda_i) + \partial_{\Lambda_i}(\mu \hat{f})$, $\dot{\Lambda}_i = -\mu \partial_{\lambda_i} \hat{f}$, $\dot{\xi}_i = \mu \partial_{\eta_i} \hat{f}$, $\dot{\eta}_i = -\mu \partial_{\xi_i} \hat{f}$ (with $\mu \hat{f}$ replaced by $\varepsilon \hat{f}_{\mathrm{fix}}$ for H_{fix}).*

Proof. **Step S9.22: Product structure.** The i-th pair $(q^{(i)}, p^{(i)})$ depends only on $(\lambda_i, \xi_i, \Lambda_i, \eta_i)$, and $\Omega(v, w) = \sum_i \Omega_i$ with $\Omega_i(v, w) = v_{2,i} \cdot w_{1,i} - v_{1,i} \cdot w_{2,i}$ over the coordinates $(\lambda_i, \xi_i; \Lambda_i, \eta_i)$ (positions λ_i, ξ_i) resp. $(q^{(i)}; p^{(i)})$; the index bookkeeping ($k + 2i$, `finProdFinEquiv`) is a permutation applied to positions and to momenta alike, which does not change $\Omega = \sum dz_2 \wedge dz_1$. Hence $D\Psi^{(n)}$ is block diagonal with blocks $D\Psi_{m_i,\kappa}$ each satisfying the identity of Theorem 9.7(c); the sum gives $\Omega(D\Psi u, D\Psi w) = \Omega(u, w)$. Analyticity, lattice invariance, bijectivity, the diffeomorphism and $\mathcal{O}_n$ open are blockwise Theorem 9.7(a),(c),(e) (the lattice $2\pi\mathbb{Z}^n$ in λ acts coordinatewise).

Step S9.23: Decoupling. $H_{\mathrm{Kep}}(q, p) = \sum_i \left[|p_i|_2^2/(2m_i) - \kappa m_i/|q_i|_2\right] = \sum_i E_{m_i,\kappa}(q_i, p_i)$ (Remark 8.4), and $E_{m_i,\kappa} \circ \Psi_{m_i,\kappa} = -\kappa^2 m_i^3/(2\Lambda_i^2)$ by Theorem 9.7(d). Derivatives are elementary.

Step S9.24: Shared chart. By Lemma 8.3, $H_{\mathrm{mov}} = H_{\mathrm{Kep}} + \mu f_{\mathrm{mov}}$, $H_{\mathrm{fix}} = H_{\mathrm{Kep}} + \varepsilon f_{\mathrm{fix}}$ on all of $\mathrm{Ph}(2n)$ (total functions), with H_{Kep}, f_{mov}, f_{fix} independent of μ, ε; composing with $\Psi^{(n)}$ gives the formulas. $f_{\mathrm{mov}}, f_{\mathrm{fix}}$ are real analytic on W (finite sums of polynomials and $1/|\cdot|_2$ with positive radicand, Lemma 8.2, whose proof gives analyticity as well as C^∞); $\Psi^{(n)}$ is analytic; the composition is analytic by (X1). The domain $\Psi^{(-1)}(W)$ is open since W is open and $\Psi^{(n)}$ continuous. The field statements are Lemma 2.4, Corollary 2.5.

Step S9.25: (d). Definition 1.2: $X_G = (\partial_{z_2} G, -\partial_{z_1} G)$ with $z_1 = (\lambda, \xi)$, $z_2 = (\Lambda, \eta)$. □

Remark 9.13 (Why the same chart serves both models). The Kepler part of H_{mov} is H_{Kep} with the *planet* mass m_i and centre mass m_0, not Fej04r's reduced mass $\mu_i = m_0 m_i/(m_0 + \mu m_i)$ and $M_i = m_0 + \mu m_i$ (Remark 8.4): the term $\mu|p_i|_2^2/(2m_0)$ is part of μf_{mov}. The chart is therefore independent of μ and equal to the chart of the fixed-Sun problem; this is what makes $H_{\mathrm{mov}} = h + \mu f$ and $H_{\mathrm{fix}} = h + \varepsilon f'$ with coupling-independent f, f'. The difference $f_{\mathrm{mov}} - f_{\mathrm{fix}} = |P|_2^2/(2m_0)$, $P = \sum_i p_i$, is analytic on all of $\mathrm{Ph}(2n)$ (a polynomial) and its λ-average is computed in Part 8 from Proposition 9.3(c) and Fej04r Lemma 64.

Lemma 9.14 (The tube in chart coordinates). *Let $a_i > 0$ and $\Lambda_i^* = m_i\sqrt{\kappa a_i}$. For every $\delta > 0$ there is $\eta_\delta > 0$ such that for all $\eta \in (0, \eta_\delta]$: the circular tube U_η (Definition 1.17, with G, m_0, a as in the definition) is contained in $\mathcal{K}^{(n)}$, and every $w \in \mathcal{O}_n$ with $\Psi^{(n)}(w) \in U_\eta$ satisfies $|\Lambda_i - \Lambda_i^*| < \delta$ and $\frac{1}{2}(\xi_i^2 + \eta_i^2) < \delta$ for all i. Moreover* $\mathrm{circ}(\phi) = \Psi^{(n)}\big((2\pi\phi, 0), (\Lambda^*, 0)\big)$.

Proof. **Step S9.26: Circular points.** Theorem 9.7(f) blockwise: with $\Lambda_i = \Lambda_i^*$, $a = a_i$, $\xi = \eta = 0$ and $\lambda_i = 2\pi\phi_i$ the image is $\mathrm{circ}(\phi)$.

Step S9.27: Continuity of the elements. On the open set $\mathcal{K}^{(n)}$ the functions $\Lambda_i(z) = m_i\sqrt{-\kappa^2 m_i/(2E_{m_i,\kappa})}$ and $\Gamma_i(z) = \Lambda_i(z) - \mathrm{cr}(q^{(i)}, p^{(i)})$ are continuous, and by Theorem 9.7(d) they equal Λ_i, Γ_i of any preimage. The set $V_\delta = \{z \in \mathcal{K}^{(n)} : |\Lambda_i(z) - \Lambda_i^*| < \delta,\ \Gamma_i(z) < \delta\ \forall i\}$ is open and contains the compact torus $T = \{\mathrm{circ}(\phi)\}$ (Step S9.26: there $\Lambda_i = \Lambda_i^*$, $\Gamma_i = 0$; T is the continuous image of $[0,1]^n$). By (E7) there is $\eta_\delta > 0$ with $\{z : \mathrm{dist}_\infty(z, T) < \eta_\delta\} \subset V_\delta$; this set is U_{η_δ} (Definition 1.17). For $\eta \le \eta_\delta$, $U_\eta \subset V_\delta \subset \mathcal{K}^{(n)}$, and a preimage w of $z \in U_\eta$ has $\Lambda_i(w) = \Lambda_i(z)$, $\frac{1}{2}(\xi_i^2 + \eta_i^2) = \Gamma_i(z) < \delta$. □

Lemma 9.15 (Lattice compatibility for Part 1). *Let $\Theta_n : \{(\varphi, \psi; I, R) :\ I_i > 0,\ 0 < R_i < I_i\} \to \mathcal{O}_n$ be the oscillator polar chart*

$$\Theta_n((\varphi,\psi),(I,R)) = \big((\varphi,\ \sqrt{2R}\sin\psi),\ (I,\ \sqrt{2R}\cos\psi)\big)$$

(componentwise square roots and sines; the formula of Remark 2.10(1): $\xi = \sqrt{2R}\sin\psi$ is the position, $\eta = \sqrt{2R}\cos\psi$ the momentum; $\Gamma_i = R_i$, $\Lambda_i = I_i$), and $\Phi_n = \Psi^{(n)} \circ \Theta_n$ on $O_0 = \{I_i > 0, 0 < R_i < I_i\}$. Then:

- *(a) Θ_n is a 1-chart on O_0; Φ_n is a 1-chart on O_0; $O_0 + \iota'_k = O_0$ for the full lattice shift $\iota'_k = ((2\pi k, 0), 0)$ of all $2n$ positions, $k \in \mathbb{Z}^{2n}$ (ι'_k adds $2\pi k$ to (φ, ψ)).*
- *(b) $\Phi_n(z + \iota'_k) = \Phi_n(z)$ for $z \in O_0$, $k \in \mathbb{Z}^{2n}$; and if $z, z' \in O_0$, $\Phi_n(z') = \Phi_n(z)$ then $z' = z + \iota'_k$ for some $k \in \mathbb{Z}^{2n}$. This is (Q1) (with $\sigma_k = 0$) and (Q2) of Theorem 2.9 in the radian convention; the period-one version is (c).*
- *(c) In the period-one torus parameter of the frozen predicates: $\Phi_n^{(1)} := \Phi_n \circ S_{2\pi}$ (with $S_{2\pi}$ of Lemma 2.11) is a 2π-chart on $S_{2\pi}^{-1}(O_0)$ satisfying (Q1) with $\sigma_k = 0$ for the lattice $\iota_k = (k, 0)$, $k \in \mathbb{Z}^{2n}$, and (Q2); in particular Corollary 2.13 applies to Φ_n and Theorem 2.9(b) to $\Phi_n^{(1)}$.*
- *(d) Under Θ_n the pulled-back Hamiltonians are $H^{\Phi_n} = H \circ \Psi^{(n)} \circ \Theta_n = h(I) + \mu\hat{f} \circ \Theta_n$ ($I = \Lambda$), i.e. in the variables $(\varphi, \psi; I, R)$ the unperturbed part is $h(I)$, independent of ψ, R.*

Proof. **Step S9.28: (a) Polar chart.** Θ_n is blockwise $(\psi, R) \mapsto (\xi, \eta) = \sqrt{2R}(\sin\psi, \cos\psi)$, with $d\eta \wedge d\xi = dR \wedge d\psi$ (computed: $d\eta \wedge d\xi = \rho\, d\rho \wedge d\psi = dR \wedge d\psi$ for $\rho = \sqrt{2R}$), real analytic for $R > 0$, mapping into $\mathcal{O}_n$ since $\Gamma_i = R_i < I_i = \Lambda_i$, and the identity on the φ, I blocks; so Θ_n is a 1-chart. The composition of 1-charts is a 1-chart (Remark 2.10(3)); $\Theta_n(O_0) \subset \mathcal{O}_n$ (indeed $= \{\Gamma > 0\} \cap \mathcal{O}_n$).

Step S9.29: (b) Invariance. Θ_n is 2π-periodic in ψ and equals Θ_n in φ shifted; precisely $\Theta_n(z + \iota'_k) = \Theta_n(z) + ((2\pi k_\varphi, 0), 0)$ ($k = (k_\varphi, k_\psi)$), and $\Psi^{(n)}$ is invariant under $((2\pi k_\varphi, 0), 0)$ (Theorem 9.12(a)). If $\Phi_n(z') = \Phi_n(z)$ then $\Theta_n(z') = \Theta_n(z) + ((2\pi k_\varphi, 0), 0)$ by the same theorem,

so $\varphi' = \varphi + 2\pi k_\varphi$, $I' = I$, and $\sqrt{2R'}(\sin\psi', \cos\psi') = \sqrt{2R}(\sin\psi, \cos\psi)$ with $R, R' > 0$: then $R' = R$ and $\psi' - \psi \in 2\pi\mathbb{Z}^n$.

Step S9.30: (c) Period one. Lemma 2.11(a): $S_{2\pi}$ is a 2π-chart with $S_{2\pi}(z + \iota_k) = S_{2\pi}(z) + (2\pi k, 0)$; compose with (a), (b) (Remark 2.10(3): $\Phi_n \circ S_{2\pi}$ is a 2π-chart), and note $S_{2\pi}^{-1}(O_0) + \iota_k = S_{2\pi}^{-1}(O_0 + \iota'_k) = S_{2\pi}^{-1}(O_0)$; the injectivity statement (Q2) for $\Phi_n \circ S_{2\pi}$ is that of (b) read through $S_{2\pi}$.

Step S9.31: (d). Theorem 9.12(b),(c) and $c = 1$. □

9.6. **The D25 secular definitions in terms of the chart.** The frozen D25 definitions (`EllipsePosition.lean`, `EllipseMomentum.lean`, `KeplerMeanLongitude.lean`, `LongitudeJacobian.lean`, `SecularEccVector.lean`, `SecularRoot.lean`, `SecularAxis.lean`, `SecularGamma.lean`, `EllipsePhase.lean`, `KeplerAction.lean`, `KeplerAxis.lean`, `KeplerEnergy.lean`, `KeplerEccVector.lean`) are stated for $n = 2$ and $G = m_0 = 1$ ($\kappa = 1$) in the phase of one chart point $w \in \mathtt{Phase}(2+2)$ with $w_1 = (\lambda_1, \lambda_2, \xi_1, \xi_2)$, $w_2 = (\Lambda_1, \Lambda_2, \eta_1, \eta_2)$ (so $\mathtt{fast}\, w_1 = \lambda$, $\mathtt{slow}\, w_1 = \xi$, $\mathtt{fast}\, w_2 = \Lambda$, $\mathtt{slow}\, w_2 = \eta$) and a vector of eccentric longitudes $F \in \mathbb{R}^2$. (The same statement holds for every n componentwise.)

Proposition 9.16 (D25 definitions = the chart in the eccentric longitude)**.** *Let $n = 2$, $\kappa = 1$, $m \in (0,\infty)^2$, $w \in \mathcal{O}_2$, $i < 2$, and write $\Lambda_i = \mathtt{fast}\, w_2\, i$, $\Gamma_i = \mathtt{secularGamma}\, w\, i$. Then (all formulas for the Lean functions, which are total, are guarded by $\Lambda_i > 0$, $\Gamma_i < \Lambda_i$):*

- *(a)* (vector, root, axis) $\Gamma_i = \frac{1}{2}(\xi_i^2 + \eta_i^2) \geq 0$; $\mathtt{secularRoot} = k_i = 1 - \Gamma_i/\Lambda_i \in (0,1]$; $\mathtt{secularEccVector} = v_i$ *of Definition 9.5 with* $|v_i|_2^2 = 1 - k_i^2$ *(eccentricity* $e_i = |v_i|_2$*);* $\mathtt{secularAxis} = (\Lambda_i/m_i)^2 = a_i$.
- *(b)* (Kepler's equation, Jacobian) $F \mapsto \mathtt{keplerMeanLongitude}\, w\, i\, F = \Phi(F, v_i)$ *is a real-analytic increasing bijection of* $\mathbb{R}$ *commuting with* $F \mapsto F + 2\pi$*, with derivative* $\mathtt{longitudeJacobian} = J(F, v_i) \geq 1 - e_i > 0$. *Let $F_i(w)$ be the unique solution of $\Phi(F, v_i) = \lambda_i$ (Proposition 9.4(a);* $\lambda_i = \mathtt{fast}\, w_1\, i$*); F_i is real analytic on $\mathcal{O}_2$.*
- *(c)* (positions and momenta) *For all $F \in \mathbb{R}$ and $i < 2$, with* $\tilde{q}_i(F) = \mathtt{ellipsePosition}\, m\, w\, i\, F$*,* $\tilde{p}_i(F) = \mathtt{ellipseMomentum}\, m\, w\, i\, F$*:* $(\tilde{q}_i(F_i(w)), \tilde{p}_i(F_i(w))) = \Psi_{m_i,1}(\lambda_i, \xi_i, \Lambda_i, \eta_i)$*; i.e.* $\mathtt{ellipsePhase}\, m\, w\, (F_1(w), F_2(w)) = \Psi^{(2)}(w)$. *More generally $\tilde{q}_i(F)$, $\tilde{p}_i(F)$ for* arbitrary *F are the point of the same osculating Kepler ellipse (same E,* cr*, v) at the eccentric longitude F, i.e. $\Psi(\Phi(F, v_i), \xi_i, \Lambda_i, \eta_i)$.*
- *(d)* (Kepler functions of the physical point) *For* $z = \Psi^{(2)}(w)$*:* $\mathtt{ProgradeElliptic}\, m\, z$*;* $\mathtt{keplerEnergy}\, m\, z\, i = -m_i^3/(2\Lambda_i^2)$*;* $\mathtt{keplerAxis}\, m\, z\, i = a_i = \mathtt{secularAxis}\, m\, w\, i$*;* $\mathtt{keplerAction}\, m\, z\, i = \Lambda_i$*;* $\mathtt{planeCross}(q^{(i)}, p^{(i)}) = \Lambda_i - \Gamma_i$*;* $\mathtt{keplerEccVector}\, m\, z\, i = v_i = \mathtt{secularEccVector}\, w\, i$. *Conversely every z with* $\mathtt{ProgradeElliptic}\, m\, z$ *is $\Psi^{(2)}(w)$ for a $w \in \mathcal{O}_2$, unique modulo $\lambda \mapsto \lambda + 2\pi k$.*
- *(e)* (elements) *For $\Gamma_i > 0$: $(\xi_i, \eta_i) = \rho_i e_{g_i}$, g_i the perihelion argument, $F_i = g_i + u_i$, u_i the eccentric anomaly and $\lambda_i = \ell_i + g_i$ (mean longitude = mean anomaly + perihelion argument).*

Proof. **Step S9.32: (a),(b).** The Lean definitions are exactly $\Gamma = R(\eta, \xi)$, $k = 1 - \Gamma/\Lambda$, $v = \sqrt{(1 - \Gamma/(2\Lambda))/\Lambda}\,(\xi, \eta)$, $\Phi(F, v) = F - v_0 \sin F + v_1 \cos F$, $J = 1 - v \cdot (\cos F, \sin F)$, $a = (\Lambda/m)^2$ (`planeDot`, `orbitalUnit`); Lemma 9.6 and Proposition 9.4(a).

Step S9.33: (c). `ellipsePosition` $= a(kc - v + \frac{v \cdot c}{1+k} v)$ and `ellipseMomentum` $= \frac{m^2}{\Lambda J}(J_0 c - \frac{v \cdot c}{1+k} J_0 v)$, $c =$ `orbitalUnit` F (note $\mathtt{fast}\, w_2\, i \cdot$ `longitudeJacobian` $= \Lambda J$): these are the displayed q, p of Definition 9.5 with $\kappa = 1$ and F arbitrary. At $F = F_i(w)$ they are Ψ. For arbitrary F put $\lambda' = \Phi(F, v_i)$: then $F = F(\lambda', v_i)$ by uniqueness, so the same formulas give $\Psi(\lambda', \xi_i, \Lambda_i, \eta_i)$.

Step S9.34: (d). Theorem 9.7(d): $\Psi(w) \in \mathcal{K}_{m_i,1}$ blockwise, i.e. $m_i > 0$, $|q^{(i)}|_2 > 0$ (`planeLength`> 0), $E < 0$ (`keplerEnergy` is $E_{m_i,1}$), cr > 0: this is `ProgradeElliptic`. The values: $E = -m_i^3/(2\Lambda_i^2)$;

`keplerAxis` $= -m_i/(2E) = \Lambda_i^2/m_i^2$; `keplerAction` $= m_i\sqrt{a} = \Lambda_i$; the LRL identity of Theorem 9.7(d) is `keplerEccVector` $= v_i$. Converse and uniqueness: Theorem 9.12(a).

Step S9.35: (e). Step S9.14 and Proposition 9.4(c). □

Corollary 9.17 (Angular averages via the eccentric longitude)**.** *Let $w \in \mathcal{O}_2$ and $\varphi : \mathbb{R}^4 \to \mathbb{R}$ be continuous. Then for each fixed (ξ, Λ, η), as functions of λ_i with the other λ_j fixed,*

$$\frac{1}{2\pi}\int_0^{2\pi} \varphi\big(\Psi^{(2)}(\lambda,\xi,\Lambda,\eta)\big)\,d\lambda_i = \frac{1}{2\pi}\int_0^{2\pi} \varphi\big(\texttt{ellipsePhase}\, m\, w\, F\big)\Big|_{F_i=s} J(s,v_i)\,ds$$

for the unit speed $F_i = s$ and the other $F_j = F_j(w)$ fixed (the integrand is the point of the i-th osculating ellipse at eccentric longitude s), with $J =$ `longitudeJacobian`. In particular, with a_i, e_i and (Proposition 9.3(c)): $\langle 1/|q^{(i)}|_2\rangle_{\lambda_i} = 1/a_i$, $\langle |p^{(i)}|_2^2\rangle_{\lambda_i} = m_i^2/a_i$, $\langle |q^{(i)}|_2\rangle_{\lambda_i} = a_i(1 + e_i^2/2)$, with $a_i = (\Lambda_i/m_i)^2$ and $e_i^2 = |v_i|_2^2 = 1 - k_i^2$.

Proof. **Step S9.36: Change of variable.** For fixed $v = v_i$, $s \mapsto \Phi(s, v)$ is C^1, increasing, with $\Phi(s + 2\pi) = \Phi(s) + 2\pi$ and $\Phi' = J > 0$ (Proposition 9.4(a)); its inverse is $\lambda \mapsto F(\lambda, v)$. Over one period, $\int_{\lambda_*}^{\lambda_*+2\pi} \Xi(F(\lambda, v))\,d\lambda = \int_{F_*}^{F_*+2\pi} \Xi(s)J(s, v)\,ds$ for continuous 2π-periodic Ξ ((E6)), with $\Xi(s) = \varphi(\Psi$ at eccentric longitude $s)$ by Proposition 9.16(c); periodic integrals do not depend on the starting point.

Step S9.37: Means. The mean values: the average over λ_i over a period equals that over $\ell_i = \lambda_i - g_i$ for $\Gamma_i > 0$ (shift by a constant of a periodic function), and Proposition 9.3(c) applies ($\kappa = 1$, $m = m_i$, $\Lambda = \Lambda_i$, $C = \Lambda_i k_i$, $a = a_i$); for $\Gamma_i = 0$ the circular case of the same item ([I:`bpI:p7:circular`]). □

9.7. **Sources, deviations, errata, forward references.**
Sources read. Fej04r pp. 46–48 (Hamiltonian (28), the heliocentric coordinates, (31) and Thm. 62, (32)–(33)), Fej07e p. 3 (`Formule (31)`: factor $\sqrt{2}$; `Formules (35-37)`: sign conventions and the sign test). Chi05 §2 via track I's Part 7 (not re-read here). The D25 frozen Lean files listed in §9.6. Totality. The chart uses $1/\Lambda$, $\sqrt{(1 - \Gamma/2\Lambda)/\Lambda}$, $1/(1 + k)$, $1/J$: guarded on $\mathcal{O}$ by $\Lambda > 0$, $\Gamma < \Lambda$ (so radicand$> \frac{1}{2\Lambda} > 0$, $k \in (0, 1]$, $J \geq 1 - e > 0$; Lemma 9.6). The frozen Lean functions `secularEccVector`, `ellipsePosition`, `ellipseMomentum` use total `Real.sqrt` and division; on $\mathcal{O}_2$ nothing is truncated. The inverse of the matrix $k\mathrm{Id} + vv^\top/(1 + k)$ is used only in the formula of Theorem 9.7(e), where its eigenvalues are $k, 1 > 0$.
Deviations from the source. (1) Fej04r (31) and (32) use the reduced mass μ_j and the corresponding total mass M_j in the Kepler part; the frozen H_{mov} (and Part 6) put $\mu|p_i|_2^2/(2m_0)$ into the perturbation, so the unperturbed Kepler problem has mass m_i and strength Gm_0 and Fej04r's Λ_j is replaced by $m_i\sqrt{Gm_0 a_i}$ (Definition 9.1, Remark 9.13). (2) The planar chart has no nodal angle: $\theta_j = 0$ and the spatial coordinates (p_j, q_j) of (31) are absent. (3) The proof of Thm. 62 is ours (not Poincaré/Chenciner). (4) The D25 secular definitions are stated for $G = m_0 = 1$; the chart is proved for general $\kappa = Gm_0$ (Theorem 9.7), and Proposition 9.16 is the case $\kappa = 1$.
Suspected errata. (1) Fej04r (31), p. 47 (original printing as used in Fej07e): "$r_j = \xi_j + i\eta_j = \sqrt{\Lambda_j}\sqrt{1 - \sqrt{1 - \varepsilon_j^2}}\ldots$" without the factor $\sqrt{2}$ makes the chart only conformally symplectic: already recorded in Fej07e p. 3, reproduced here as Remark 9.10(1). (2) The orchestrator's brief wrote the regularising change as $(\xi, \eta) = \sqrt{2(\Lambda - G)}(\cos, -\sin)(g)$; with the frozen order $(\lambda, \xi; \Lambda, \eta)$ (position ξ) the correct sign is $(\cos g, +\sin g)$ (derived in Step S9.15 and checked numerically in a scratch script); the minus sign belongs to the exchanged-roles convention (Remark 9.10(2)). Not an error in a pinned source.

Doubts about frozen statements. None. The frozen D25 definitions agree with the chart (checked above: formulas, and by a numerical scratch check at one sample point that $\Psi^*\Omega = \Omega$, $E = -m^3/2\Lambda^2$, $\mathrm{cr} = \Lambda - \Gamma$, $\mathrm{LRL} = v$ at a sample point). `secularAxis`$= (\Lambda/m)^2$ requires $Gm_0 = 1$ (D25 normalisation), consistent with Remark 8.12.
External facts used. (E1)–(E7) of §9.1, plus [K:`bp:p1:real-restriction`] and the results of Parts 1 and 6 and track I quoted by label.
Forward references and consumers. Part 8: Corollary 9.17, Proposition 9.3(c), Theorem 9.12(b),(c), Remark 9.13; Parts 2, 5: Proposition 9.8; Parts 12, 13: Lemma 9.15, Lemma 9.14, Theorem 9.12; Part 14: Proposition 9.16, Corollary 9.17. Needed from other parts and not yet written (forward): none beyond the labels of Parts 1, 6 and of track I Part 7 already in the repository.
Label index. `bpP:p7:kepler-data`, `range`, `kepler`, `regular-kepler`, `poincare-def`, `o-facts`, `poincare`, `holomorphic`, `normalisation`, `product-def`, `product-chart`, `same-chart`, `tube`, `lattice`, `d25` (= `eccentric-longitude`), `average`.

10. Averaged perturbation and the secular quadratic form

In the Poincaré chart of Part 7 both physical models are $h(\Lambda) + \mu f$ with a coupling-independent perturbation f (Theorem 9.12(c)). Averaging f over the mean longitudes $\lambda \in \mathbb{T}^n$ gives the *secular function* $S(\xi, \Lambda, \eta)$, whose Taylor expansion in the slow variables (ξ, η) at the circular torus feeds the Birkhoff form (A_2'') of Part 0. This part proves what that expansion needs, and the exact relation between the two models.

- `perturbation-analytic` (Prop. 10.2): on an explicit complex polydisc around the circular torus the pair interaction in the chart is holomorphic for every real λ, and S is holomorphic (real analytic) there;
- `evenness` (Prop. 10.3 and Cor. 10.4): three exact symmetries of the chart; S is even in η and in ξ and invariant under simultaneous rotation, so every odd Taylor coefficient vanishes and the quadratic and quartic parts have a prescribed invariant form;
- `lemma64` (Prop. 10.5): $\langle p^{(j)} \cdot p^{(k)} \rangle_\lambda = 0$ for $j \neq k$ *exactly*, and $\langle |p^{(i)}|^2 \rangle = m_i^2 \kappa / a_i$;
- `laplace` (Def. 10.6, Lemma 10.7, Lemma 10.9): Laplace coefficients, locally holomorphic on $|\alpha| < 1$ from the series, and two series identities;
- `quadratic-form` (Thm. 10.11): C_0 and Q_h with the P5 corrections *and* a further overall factor $\frac{1}{2}$ forced by the erratum's normalisation (suspected erratum E-P-Fej04r-3);
- `identification` (Thm. 10.13): $\langle f_{\mathrm{mov}} \rangle - \langle f_{\mathrm{fix}} \rangle = \delta(\Lambda)$ is a function of Λ alone, *exactly*, so all (ξ, η)-Taylor coefficients of order ≥ 1 agree (Cor. 10.14); this is what lets $n = 2$ P-FIXED follow route (c);
- `quartic-jet` (Prop. 10.16): the interface for the planar torsion (Part 11b, CP11 §8.1).

10.1. **Conventions, external facts, the secular polydisc.** Fix $n \geq 1$, $G, m_0 > 0$, $\kappa = Gm_0$, $m \in (0, \infty)^n$, and the chart $\Psi = \Psi^{(n)}$ with $\hat{f}_{\mathrm{fix}} = f_{\mathrm{fix}} \circ \Psi$, $\hat{f}_{\mathrm{mov}} = f_{\mathrm{mov}} \circ \Psi$ (Theorem 9.12). Coordinates $w = (\lambda, \xi, \Lambda, \eta)$; $u_i = (\xi_i, \eta_i) \in \mathbb{R}^2$; $\Gamma_i = |u_i|^2/2$; $a_i = \Lambda_i^2/(m_i^2 \kappa)$. Angles have period 2π as in Part 7. For continuous φ on $\{(\lambda, w')\}$, $w' = (\xi, \Lambda, \eta)$,

$$\langle \varphi \rangle(w') = (2\pi)^{-n} \int_{[0,2\pi]^n} \varphi(\lambda, w')\, d\lambda, \qquad S_{\mathrm{fix}} = \langle \hat{f}_{\mathrm{fix}} \rangle, \quad S_{\mathrm{mov}} = \langle \hat{f}_{\mathrm{mov}} \rangle.$$

Dictionary to the frozen average. Since $\mathrm{assemble}(I, \varphi, p, q) = ((\varphi, q), (I, p))$ and $w = ((\lambda, \xi), (\Lambda, \eta))$ (Part 7), `secularAverage` $P\, \mu\, I\, p\, q = \langle P \circ \cdot \rangle$ at $(\xi, \Lambda, \eta) = (q, I, p)$: the frozen momentum p is η, the frozen position q is ξ, $I = \Lambda$, and $R_j = \Gamma_j$.

For the axes $0 < a_1^* < \cdots < a_n^*$ (the *increasing* order of the Lean statements; Fej04r uses the decreasing one, see Remark 10.12) put $\Lambda_i^* = m_i \sqrt{\kappa a_i^*}$ and, for $\rho > 0$,

$$\mathcal{N}(\rho) = \{(\xi, \Lambda, \eta) \in \mathbb{C}^{3n} : \ |\xi_i|, |\eta_i| < \rho, \ |\Lambda_i - \Lambda_i^*| < \rho \ \ \forall i\}, \qquad \mathcal{N}_{\mathbb{R}}(\rho) = \mathcal{N}(\rho) \cap \mathbb{R}^{3n}.$$

External facts.

(E1) *Holomorphic parameter integrals.* If $\varphi(\lambda, w')$ is continuous on $[0,2\pi]^n \times U$ and holomorphic in $w' \in U \subset \mathbb{C}^N$ open for each λ, then $w' \mapsto \int \varphi \, d\lambda$ is holomorphic on U (Morera/Fubini, differentiation under the integral); likewise real-analytic parameter dependence follows by restriction ([I:`bpI:p7:x-param`], (E2)).

(E2) *Differentiation under the integral:* if φ is C^k in w' with derivatives jointly continuous on $[0,2\pi]^n \times U$, then $\partial^\alpha \langle \varphi \rangle = \langle \partial^\alpha \varphi \rangle$, $|\alpha| \le k$. Fubini on products of intervals; translation invariance of $\int_0^{2\pi}$ of a 2π-periodic continuous function.

(E3) Taylor expansion of a real-analytic function: the Taylor polynomial of order N at a point has remainder $O(|x|^{N+1})$ locally uniformly; Taylor coefficients are the iterated derivatives; a composition of an analytic function with a real-analytic map is real analytic.

(E4) *First fundamental theorem for $O(2)$* (Weyl): a polynomial in vectors $u_1, \dots, u_r \in \mathbb{R}^2$ invariant under the diagonal action of $O(2)$ is a polynomial in the dot products $u_i \cdot u_j$.

(E5) Pochhammer symbols $(s)_p = s(s+1)\cdots(s+p-1)$, $(s)_{p+1} = (s)_p(s+p)$; binomial series $(1-z)^{-s} = \sum_p \frac{(s)_p}{p!} z^p$ for $|z| < 1$, and $\frac{(s)_p}{p!} = O(p^{s-1})$.

Definition 10.1 (Secular polydisc constants)**.** $\rho_0 := \min_i \min\{\Lambda_i^*/2, \sqrt{\Lambda_i^*}/16\}$, so that $\mathcal{N}(\rho)$, $\rho < \rho_0$, is inside the domain of Proposition 9.8 for each planet. $\rho_* = \rho_*(m, \kappa, a^*) \in (0, \rho_0)$ is the constant of Proposition 10.2.

10.2. **Analyticity of the interaction and of its average.**

Proposition 10.2 (Analytic perturbation near the circular torus)**.** *There is $\rho_* \in (0, \rho_0)$ depending only on (m, κ, a^*) such that, with $s_{ij}(\lambda, w') = \sum_{c=0,1} (q_c^{(i)} - q_c^{(j)})^2$ (complex bilinear square, $q = \Psi$ of Prop. 9.8), for all real λ and $w' \in \mathcal{N}(\rho_*)$ and all $i < j$:*

(a) $\operatorname{Re} s_{ij} \ge \frac{1}{2}(a_j^* - a_i^*)^2$. *Hence for real points $|q^{(i)} - q^{(j)}| \ge (a_j^* - a_i^*)/\sqrt{2}$ for* all *$\lambda \in \mathbb{R}^n$: $\Psi(\lambda, w') \in W$ (Lemma 8.2) and $\mathcal{N}_{\mathbb{R}}(\rho_*) \times \mathbb{R}^n \subset \Psi^{-1}(W)$.*

(b) $\hat{f}_{\mathrm{fix}}(\lambda, w') = -G \sum_{i<j} m_i m_j \, s_{ij}^{-1/2}$ (principal root) is holomorphic in $w' \in \mathcal{N}(\rho_*)$ for each real λ, jointly continuous, 2π-periodic in λ, real at real points; the same holds for $\hat{f}_{\mathrm{mov}} = \hat{f}_{\mathrm{fix}} + |\sum_i p^{(i)}|^2/(2m_0)$.

(c) $S_{\mathrm{fix}}, S_{\mathrm{mov}}$ are holomorphic on $\mathcal{N}(\rho_*)$, real at real points, hence real analytic on $\mathcal{N}_{\mathbb{R}}(\rho_*)$, and $\partial^\alpha_{(\xi,\eta,\Lambda)} S = \langle \partial^\alpha \hat{f} \rangle$ for all α.

Proof. **Step S10.1: Value at the circular torus.** At $w'^* = (0, \Lambda^*, 0)$, Theorem 9.7(f) gives $q^{(i)} = a_i^*(\cos\lambda_i, \sin\lambda_i)$, so $s_{ij} = a_i^{*2} + a_j^{*2} - 2a_i^* a_j^* \cos(\lambda_i - \lambda_j) \ge (a_j^* - a_i^*)^2$.

Step S10.2: Uniformity. s_{ij} is holomorphic in w' on $\mathcal{N}(\rho_0)$ (Prop. 9.8 and $\bar{\Psi}$ real at real points, polynomial in q) and continuous and 2π-periodic in $\lambda \in \mathbb{R}^n$; hence uniformly continuous on $[0,2\pi]^n \times \overline{\mathcal{N}(\rho_0/2)}$ (compact). By Step S10.1 there is $\rho_* < \rho_0/2$ with $\operatorname{Re} s_{ij} \ge \frac{1}{2}(a_j^* - a_i^*)^2$ on $[0,2\pi]^n \times \mathcal{N}(\rho_*)$; it depends only on the data (m, κ, a^*) (through Λ^* and Ψ). This is (a); at real points $|q^{(i)} - q^{(j)}|^2 = s_{ij}$, and NoColl also needs $|q^{(i)}| > 0$, which holds on $\mathcal{O}_n$ (Lemma 9.6, Theorem 9.7(d)).

Step S10.3: Holomorphy of $\hat{f}$. On $\{\operatorname{Re} s > 0\}$ the principal $s^{-1/2}$ is holomorphic ((E2)); composing with the holomorphic s_{ij} gives (b); at real points $s_{ij} > 0$ and $s_{ij}^{-1/2} = 1/|q^{(i)} - q^{(j)}|$ so the formula is $f_{\mathrm{fix}} \circ \Psi$ (Lemma 8.3). The extra term of $\hat{f}_{\mathrm{mov}}$ is a polynomial in the components of $p^{(i)} = \Psi_{m_i,\kappa}$-momenta, holomorphic by Prop. 9.8.

Step S10.4: The average. By (E1) S is holomorphic on $\mathcal{N}(\rho_*)$; real at real points; the derivative formula is (E2) (the w'-derivatives of a holomorphic function are jointly continuous in (λ, w') by Cauchy's formula on a slightly smaller polydisc, uniformly in the compact λ-torus). □

10.3. **Symmetries and evenness.** Write R_θ for the rotation of $\mathbb{R}^2$ by θ, $\sigma = \mathrm{diag}(1,-1)$, $\mathbb{1} = (1,\dots,1)$, and for $\xi,\eta \in \mathbb{R}^n$ put $R_\theta(\xi,\eta) = (\xi\cos\theta - \eta\sin\theta,\ \xi\sin\theta + \eta\cos\theta)$ (planetwise; this rotates each u_i by θ).

Proposition 10.3 (Symmetries of the chart)**.** *For $w \in \mathcal{O}_n$ write $\Psi(w) = (q,p)$ (blocks $q^{(i)}, p^{(i)}$). Then:*

(S1) $\Psi(\lambda + \pi\mathbb{1}, -\xi, \Lambda, -\eta) = (-q, -p)$;
(S2) $\Psi(\lambda + \theta\mathbb{1}, R_\theta(\xi,\eta), \Lambda) = (R_\theta q^{(i)}, R_\theta p^{(i)})_i$ *for every* $\theta \in \mathbb{R}$;
(S3) $\Psi(-\lambda, \xi, \Lambda, -\eta) = (\sigma q^{(i)}, -\sigma p^{(i)})_i$.

(The domain $\mathcal{O}_n$ is invariant under all three maps.) The Kepler functions $f_{\mathrm{fix}}, f_{\mathrm{mov}}$ of Lemma 8.3 satisfy $f(-q,-p) = f(q,p) = f(Rq, \pm Rp)$ for $R \in O(2)$ acting blockwise.

Proof. It suffices to treat one planet. Put $\Phi(F,v) = F - v_0\sin F + v_1\cos F$, $c = (\cos F, \sin F)$.

Step S10.5: (S1). $\Phi(F+\pi, -v) = \Phi(F,v) + \pi$, so $F(\lambda+\pi, -v) = F(\lambda, v) + \pi$ (uniqueness, Prop. 9.4(a)), $c \mapsto -c$, J and k are unchanged, $v \mapsto -v$. In $q = a(kc - v + \frac{v\cdot c}{1+k}v)$ and $p = \frac{m^2\kappa}{\Lambda J}(J_0 c - \frac{v\cdot c}{1+k}J_0 v)$ every term is odd under $(c,v) \mapsto (-c,-v)$ ($v\cdot c$ is even, the remaining factor is linear in c or v, resp. quadratic factor $v\cdot c\, v$ odd).

Step S10.6: (S2). $v \mapsto R_\theta v$ for $(\xi,\eta) \mapsto R_\theta(\xi,\eta)$ (v is a positive multiple of (ξ,η) with Γ, Λ unchanged); expanding, $\Phi(F+\theta, R_\theta v) = \Phi(F,v) + \theta$ (the $\cos\theta, \sin\theta$ terms combine to $-v_0\sin F + v_1\cos F$), so $F(\lambda+\theta, R_\theta v) = F + \theta$, $c \mapsto R_\theta c$, J unchanged, and $R_\theta J_0 = J_0 R_\theta$, $(R_\theta v)\cdot(R_\theta c) = v\cdot c$: $q, p \mapsto R_\theta q, R_\theta p$.

Step S10.7: (S3). With $\bar v = \sigma v$ ($\eta \mapsto -\eta$): $\Phi(-F, \bar v) = -\Phi(F,v)$, so $F(-\lambda, \bar v) = -F(\lambda, v)$, $c \mapsto \sigma c$, J unchanged, $\bar v \cdot \sigma c = v\cdot c$, $q \mapsto \sigma q$; $J_0\sigma = -\sigma J_0$ gives $p \mapsto -\sigma p$.

Step S10.8: The Kepler functions. f_{fix} depends on $|q_i - q_j|$ and $f_{\mathrm{mov}} - f_{\mathrm{fix}} = |\sum p_i|^2/(2m_0)$; both are invariant under $(q,p) \mapsto (\pm Rq, \pm' Rp)$ for any isometry R. □

Corollary 10.4 (Evenness and invariance of the secular function)**.** *Let $S \in \{S_{\mathrm{fix}}, S_{\mathrm{mov}}\}$ on $\mathcal{N}_{\mathbb{R}}(\rho_*)$. Then $S(-\xi,\Lambda,-\eta) = S(\xi,\Lambda,\eta) = S(\xi,\Lambda,-\eta) = S(-\xi,\Lambda,\eta)$ and $S(R_\theta(\xi,\eta),\Lambda) = S(\xi,\Lambda,\eta)$. Consequently the Taylor expansion of S at $(\xi,\eta) = 0$ has only terms of even total degree, $(\xi,\eta) = 0$ is a critical point (the origin is the circular torus, Fej04r p. 50), and the degree-$2d$ homogeneous part is an $O(2)$-invariant polynomial in $(u_1,\dots,u_n)$ with coefficients real-analytic in Λ; the degree-two part is*

$$\xi^\top A(\Lambda)\xi + \eta^\top A(\Lambda)\eta, \qquad A = A^\top. \tag{1}$$

Proof. **Step S10.9: Invariances.** Each of (S1)–(S3) shifts or reflects the λ variables and maps $\hat f$ to itself (Prop. 10.3); the λ-average of a 2π-periodic continuous function is invariant under translation $\lambda \mapsto \lambda + c\mathbb{1}$ and under $\lambda \mapsto -\lambda$ ((E2)). This gives the invariances of S; the fourth equality combines the first and second.

Step S10.10: Taylor structure. S is real analytic in (ξ,η) (Prop. 10.2(c)); evenness under $(\xi,\eta) \mapsto -(\xi,\eta)$ kills odd-degree parts. The degree-$2d$ part of an $O(2)$-invariant analytic function is invariant, because $O(2)$ acts linearly and preserves homogeneity.

Step S10.11: Quadratic part. Write the degree-two part as $\xi^\top A\xi + 2\xi^\top B\eta + \eta^\top C\eta$. Evenness in η alone gives $B = 0$ (it kills terms odd in η); the rotation $\theta = \pi/2$, $(\xi,\eta) \mapsto (-\eta,\xi)$, gives $\eta^\top A\eta + \xi^\top C\xi = \xi^\top A\xi + \eta^\top C\eta$, so $C = A$. □

10.4. **Exact orthogonality of momenta (Fej04r Lemma 64).**

Proposition 10.5 (Lemma 64, exactly)**.** *For every $w \in \mathcal{O}_n$ (no neighbourhood, no collision hypothesis), $\nu_i = \kappa^2 m_i^3/\Lambda_i^3$:*

(a) $m_i\nu_i\,\partial_{\lambda_i} q^{(i)}(w) = p^{(i)}(w)$ *and* $\partial_{\lambda_k} q^{(i)} = 0$ *for* $k \neq i$*;*
(b) $\frac{1}{2\pi}\int_0^{2\pi} p^{(i)}\,d\lambda_i = 0 \in \mathbb{R}^2$*;*
(c) for $j \neq k$*:* $(2\pi)^{-n}\int_{[0,2\pi]^n} p^{(j)}\cdot p^{(k)}\,d\lambda = 0$*;*
(d) $(2\pi)^{-n}\int |p^{(i)}|^2 d\lambda = \frac{1}{2\pi}\int_0^{2\pi} |p^{(i)}|^2 d\lambda_i = m_i^2\kappa/a_i = m_i^4\kappa^2/\Lambda_i^2$*;*
(e) hence $(2\pi)^{-n}\int \big|\sum_i p^{(i)}\big|^2 d\lambda = \sum_i m_i^4\kappa^2/\Lambda_i^2$*.*

Source: Fej04r Lemme 64, pp. 49–50 (cross terms; proof: $Y_j = \mu_j\,dX_j/dt \propto \partial X_j/\partial\lambda_j$ and Fubini). The virial value (d) is ours (the source drops the diagonal term, which is not needed for $\langle F_{\rm per}\rangle = \langle F_{\rm princ}\rangle$; here it is).

Proof. **Step S10.12: (a).** $H_{\rm Kep}$ is differentiable at $\Psi(w)$ (no pair terms; $q^{(i)} \neq 0$) and $H_{\rm Kep}^{\Psi} = h(\Lambda)$ (Thm. 9.12(b)). By Lemma 2.4, $D\Psi\, X_h = X_{H_{\rm Kep}} \circ \Psi$ with X_h having entries ν_i in the λ_i slots and 0 elsewhere (Thm. 9.12(d), $\mu = 0$). The position part of $X_{H_{\rm Kep}}$ is $(p^{(i)}/m_i)_i$ (Prop. 8.6 with $\mu = 0$), the position part of $D\Psi X_h$ is $(\sum_k \nu_k \partial_{\lambda_k} q^{(i)})_i = \nu_i \partial_{\lambda_i} q^{(i)}$, since $q^{(i)}$ depends only on $(\lambda_i, \xi_i, \Lambda_i, \eta_i)$.

Step S10.13: (b). $q^{(i)}$ is real analytic and 2π-periodic in λ_i, so $\int_0^{2\pi} \partial_{\lambda_i} q^{(i)} d\lambda_i = 0$ (fundamental theorem of calculus); use (a) and $m_i\nu_i > 0$.

Step S10.14: (c). $p^{(j)}\cdot p^{(k)}$ is continuous, $p^{(j)}$ depends on λ_j only, $p^{(k)}$ on λ_k only; by Fubini the integral is $\big(\frac{1}{2\pi}\int p^{(j)} d\lambda_j\big)\cdot\big(\frac{1}{2\pi}\int p^{(k)} d\lambda_k\big) = 0$.

Step S10.15: (d). The first equality is Fubini. For $\Gamma_i > 0$ write $p^{(i)} = \mathsf{Q}_{m_i,\kappa}(\ell, g, \Lambda, C)$ with $\ell = \lambda - g$ (Step S9.14); the period integral over λ_i equals that over ℓ (shift of a periodic function), and Prop. 9.3(c) gives $m^2\kappa/a$. For $\Gamma_i = 0$ the circular case of the same item. And $a_i = \Lambda_i^2/(m_i^2\kappa)$.

Step S10.16: (e). Expand $|\sum p^{(i)}|^2 = \sum |p^{(i)}|^2 + 2\sum_{j<k} p^{(j)}\cdot p^{(k)}$ and use (c),(d). □

10.5. **Laplace coefficients.**

Definition 10.6 (Laplace coefficients)**.** For $s > 0$, $k \in \mathbb{Z}$ and $\alpha \in [0,1)$, $b_s^{(k)}(\alpha) = \frac{1}{2\pi}\int_0^{2\pi} \cos(kt)\,(1+\alpha^2 - 2\alpha\cos t)^{-s} dt$, so $(1+\alpha^2-2\alpha\cos t)^{-s} = \sum_{k\in\mathbb{Z}} b_s^{(k)}(\alpha) e^{ikt}$ (Poincaré normalisation, as in Fej04r Def. 66).

Source: Fej04r Def. 66, p. 50; CP11 App. B footnote 72 ($b_{h,k}$, $\beta_k^{(r)} = b_{r/2,k}$).

Lemma 10.7 (Series; local holomorphy)**.** *For $k \geq 0$ and $\alpha \in [0,1)$,*

$$b_s^{(k)}(\alpha) = \sum_{q\geq 0} \frac{(s)_{q+k}\,(s)_q}{(q+k)!\,q!}\;\alpha^{2q+k}, \qquad b_s^{(-k)} = b_s^{(k)}.$$

The series has radius of convergence ≥ 1; it defines a holomorphic function on $\{|\alpha| < 1\}$, real and positive on $[0,1)$, and the Laplace coefficients are understood as this function wherever a complex argument is used. For $0 < \alpha < 1$ they are real analytic in α, and $\alpha \mapsto b_s^{(k)}(a_i/a_j)$ is real analytic on $\{0 < a_i < a_j\}$.

Proof. **Step S10.17: Expansion.** $1+\alpha^2-2\alpha\cos t = (1-\alpha e^{it})(1-\alpha e^{-it})$, so by (E5) $(\cdot)^{-s} = \sum_{p,q\geq 0} \frac{(s)_p(s)_q}{p!q!}\alpha^{p+q} e^{i(p-q)t}$, absolutely and uniformly in t for $\alpha < 1$ (coefficients are $O((pq)^{s-1})$). Integrating against $e^{-ikt}/(2\pi)$ termwise keeps $p - q = k$, i.e. $p = q + k$. Evenness in k is evenness of cos.

Step S10.18: Convergence. The coefficient of α^{2q+k} is $O(q^{2s-2}(q+k)^{s-1})$, polynomial growth, so the radius is ≥ 1; positivity: all coefficients are positive for $s > 0$. □

Remark 10.8 (Fej04r Lemma 76 is not used). Lemma 76 (p. 56) asserts meromorphy on the Riemann sphere with poles $0, \pm 1, \infty$. This is false as printed: $b^{(0)}_{1/2}$ has a logarithmic singularity at $\alpha = 1$ (suspected erratum E-P-Fej04r-2, recorded in Part 0). Only the local holomorphy of Lemma 10.7 on the unit disc is used here and in Part 9–10.

Lemma 10.9 (Two identities)**.** *For $\alpha \in [0,1)$ with $\beta = b^{(0)}_{1/2}$ and $b = b^{(1)}_{1/2}$:*

(I1) $\alpha\beta'' + 2\beta' = b^{(1)}_{3/2}$;

(I2) $\alpha^2 b'' + 2\alpha b' - 2b = \alpha\, b^{(2)}_{3/2}$.

Proof. Compare coefficients of the series of Lemma 10.7, using $(\frac{3}{2})_p = (\frac{1}{2})_{p+1}/\frac{1}{2} = (2p+1)(\frac{1}{2})_p$ and $(\frac{3}{2})_{p-1} = 2(\frac{1}{2})_p$.

Step S10.19: (I1). $\beta = \sum_q \big(\frac{(1/2)_q}{q!}\big)^2 \alpha^{2q}$ and $(\alpha\partial^2 + 2\partial)\alpha^{2q} = 2q(2q+1)\alpha^{2q-1}$. The coefficient of α^{2q-1} $(q \ge 1)$ on the left is $2q(2q+1)\big(\frac{(1/2)_q}{q!}\big)^2$; on the right $(p = q-1)$ it is $\frac{(3/2)_q (3/2)_{q-1}}{q!(q-1)!} = \frac{(2q+1)(1/2)_q \cdot 2(1/2)_q}{q!(q-1)!}$, equal.

Step S10.20: (I2). $b = \sum_q c_q \alpha^{2q+1}$, $c_q = \frac{(1/2)_{q+1}(1/2)_q}{(q+1)!q!}$, and $(\alpha^2\partial^2 + 2\alpha\partial - 2)\alpha^m = (m+2)(m-1)\alpha^m$, which for $m = 2q+1$ is $2q(2q+3)$. The coefficient of α^{2q+1} in $\alpha b^{(2)}_{3/2}$ is $\frac{(3/2)_{q+1}(3/2)_{q-1}}{(q+1)!(q-1)!} = \frac{(2q+3)(1/2)_{q+1}\cdot 2(1/2)_q}{(q+1)!(q-1)!} = 2q(2q+3)c_q$. □

10.6. **The quadratic part: C_0 and Q_h.**

Definition 10.10 (Secular coefficients)**.** Let $a_1 < \cdots < a_n$, $\alpha_{ij} = a_i/a_j < 1$ for $i < j$ (planet i inner, j outer), and

$$\mathfrak{a}_{ij} = \frac{G m_i m_j\, a_i}{4a_j^2}\, b^{(1)}_{3/2}(\alpha_{ij}) > 0, \qquad \mathfrak{b}_{ij} = \frac{G m_i m_j\, a_i}{4a_j^2}\, b^{(2)}_{3/2}(\alpha_{ij}) > 0 \qquad (i<j),$$

$\mathfrak{a}_{ji} = \mathfrak{a}_{ij}$, $\mathfrak{b}_{ji} = \mathfrak{b}_{ij}$, and

$$C_0(a) = -\sum_{i<j} \frac{G m_i m_j}{a_j}\, b^{(0)}_{1/2}(\alpha_{ij}), \quad \hat q_{jj} = -\sum_{k\neq j} \mathfrak{a}_{jk}, \quad \hat q_{jk} = \mathfrak{b}_{jk}\ (j \neq k), \quad Q_h = \Delta^{-1}\hat q\,\Delta^{-1},\ \Delta = \operatorname{diag}(\sqrt{\Lambda_i}).$$

All are real analytic functions of Λ on $\{a_1 < \cdots < a_n\}$ ($a = a(\Lambda)$, Lemma 10.7).

Theorem 10.11 (Quadratic part of the secular function)**.** *In the situation of Prop. 10.2, with a^* increasing, for $(\xi,\Lambda,\eta) \in \mathcal{N}_{\mathbb{R}}(\rho_*)$ (and also $a(\Lambda)$ increasing there, shrinking ρ_* if needed):*

$$S_{\mathrm{fix}}(\xi,\Lambda,\eta) = C_0(a) + \xi^{\top} Q_h \xi + \eta^{\top} Q_h \eta + O\big(|(\xi,\eta)|^4\big),$$

the error being real analytic and $O(|(\xi,\eta)|^4)$ locally uniformly in Λ. Equivalently the matrix A of (1) is Q_h. If $\varrho \in SO(n)$ diagonalises $Q_h = \varrho\,\mathrm{diag}(\sigma)\varrho^{\top}$ then in the symplectic variables $(\tilde\xi,\tilde\eta) = \varrho^{\top}(\xi,\eta)$ the quadratic part is $\sum_j \sigma_j(\tilde\xi_j^2 + \tilde\eta_j^2) = \sum_j 2\sigma_j \tilde\Gamma_j$, i.e. $\Omega_j = 2\sigma_j$ in the frozen sense.

Source: Fej04r Formule 67 and (35)–(37), pp. 50–51; Prop. 73 proof, p. 55 (the matrix $\hat q = \Delta^{\top} Q_h \Delta$); CP11 (B.1)–(B.2), p. 62; corrections P5 (`statement.md`) and the factor $\frac{1}{2}$, see Remark 10.12. The derivation is ours (the source cites Laplace–Lagrange and [LR95], not pinned).

Source gap: Fej04r p. 50: "Le calcul complet est un peu long", detailed only in [LR95], not pinned. The proof below is a complete replacement.

Proof. **Step S10.21: Pair decomposition.** $\hat f_{\mathrm{fix}} = \sum_{i<j} V_{ij}$, $V_{ij} = -Gm_im_j/|q^{(i)} - q^{(j)}|$ depends on (λ_i,λ_j) and on the data $(u_i,\Lambda_i,u_j,\Lambda_j)$ only. So $S_{\mathrm{fix}} = \sum_{i<j} S_{ij}$, $S_{ij} = \langle V_{ij}\rangle$ with the average over (λ_i,λ_j) (Fubini). Corollary 10.4 applies to each S_{ij} (as the $n=2$ function of (u_i,u_j)): its

degree-two part is $A_{ij}|u_i|^2 + A'_{ij}|u_j|^2 + B_{ij}\, u_i \cdot u_j$ (invariants of $O(2)$, (E4); the quadratic invariants of two vectors are spanned by the three dot products).

Step S10.22: Scaling. Let $\Phi(r_1,r_2) = \frac{1}{2\pi}\int_0^{2\pi}(r_1^2 + r_2^2 - 2r_1r_2\cos t)^{-1/2}dt = \frac{1}{r_2}b^{(0)}_{1/2}(r_1/r_2)$ for $r_1 < r_2$ and $\Phi(r_1,r_2) = \Phi(r_2,r_1)$: real analytic on $\{r_1 \neq r_2\}$ and homogeneous of degree -1.

Step S10.23: C_0**.** At $u = 0$ both planets are circular (Thm. 9.7(f)): $S_{ij}(0) = -Gm_im_j\Phi(a_i,a_j) = -Gm_im_ja_j^{-1}b^{(0)}_{1/2}(\alpha_{ij})$.

Step S10.24: Diagonal coefficients. Let $u_j = 0$ (j circular, any $i \neq j$, $u_i = (\xi_i,\eta_i)$). Averaging over λ_j first, which by rotation invariance of the circle gives $\Phi(|q^{(i)}|,a_j)$, independent of the direction of $q^{(i)}$. By Prop. 9.3(b), $|q^{(i)}| = a_i(1 - e\cos u)$, $d\lambda_i = (1 - e\cos u)\,du$ over a period (u the eccentric anomaly, or $F-g$; [I:`bpI:p7:orbit-identities`]); with $e^2 = |v_i|^2 = 2\Gamma_i/\Lambda_i - \Gamma_i^2/\Lambda_i^2 =: E$ (Lemma 9.6),

$$S_{ij}\big|_{u_j=0} = -Gm_im_j\,\frac{1}{2\pi}\int_0^{2\pi} h(e\cos u)\,du, \qquad h(\varepsilon) = \Phi\big(a_i(1-\varepsilon),a_j\big)(1-\varepsilon).$$

Only even powers of $e\cos u$ survive and $\langle\cos^2\rangle = \frac12$, so the right side is $-Gm_im_j\big[h(0) + \frac14 h''(0)E + O(E^2)\big]$ with $h''(0) = a_i^2\Phi_{11} + 2a_i\Phi_1 = (\,r_1^2\partial_{r_1}^2 + 2r_1\partial_{r_1})\Phi\,|_{(a_i,a_j)}$. For $i<j$ (inner eccentric): with $\beta = b^{(0)}_{1/2}$, $h''(0) = a_j^{-1}(\alpha^2\beta'' + 2\alpha\beta') = \alpha\, b^{(1)}_{3/2}(\alpha)/a_j$ by (I1). For $i>j$ (outer eccentric, inner circular of radius $c = a_j$): $g(r) = r^{-1}\beta(c/r)$, $r^2g'' + 2rg' = 2c\beta'/r^2 + c^2\beta''/r^3$, at $r = a_i$ equal to $\alpha\, b^{(1)}_{3/2}(\alpha)/a_i$ with $\alpha = c/a_i$ again by (I1). Since $E = |u|^2/\Lambda_i + O(|u|^4)$, the coefficient of $|u_i|^2$ is $-\mathfrak{a}_{ij}/\Lambda_i$ in both cases ($\mathfrak{a}_{ij} = Gm_im_j\frac{\alpha}{4a_{\max}}b^{(1)}_{3/2}$ with $a_{\max}$ the larger axis, α the ratio).

Step S10.25: Cross coefficient. Let $i<j$. By (E2), B_{ij} is determined by $\partial_{\xi_i}\partial_{\xi_j}\langle V_{ij}\rangle = \langle\partial_{\xi_i}\partial_{\xi_j}V_{ij}\rangle = B_{ij}$ at 0. Use $v = (1-\Gamma/2\Lambda)^{1/2}\Lambda^{-1/2}u$, so to first order $v_i = u_i/\sqrt{\Lambda_i}$, and write $v_i = (s,0)$, $v_j = (t,0)$. To first order (Prop. 9.4(a): $\partial_vF = (\sin F, -\cos F)$ at $v=0$, $F=\lambda$, $J=1$) the position of planet i is $q = a[(1 - v\cdot c)c - 2(v\cdot J_0c)J_0c]$, $c = (\cos\lambda,\sin\lambda)$: in polar coordinates $r_i = a_i(1 - s\cos\lambda_i)$, $\theta_i = \lambda_i + 2s\sin\lambda_i$, and likewise for j with t. Write $V_{ij} = U(r_i,r_j,\theta_i-\theta_j)$ with $U = -Gm_im_j(r_i^2 + r_j^2 - 2r_ir_j\cos\vartheta)^{-1/2} = -Gm_im_j\big[B_0 + 2\sum_{k\ge1}B_k\cos k\vartheta\big]$, $B_k(r_i,r_j) = r_j^{-1}b^{(k)}_{1/2}(r_i/r_j)$ (Def. 10.6), homogeneous of degree -1. The mixed derivative at $s=t=0$ is $U_{r_ir_j}r_i'r_j' - U_{r_i\vartheta}r_i'\theta_j' + U_{\vartheta r_j}\theta_i'r_j' - U_{\vartheta\vartheta}\theta_i'\theta_j'$ with $r_i' = -a_i\cos\lambda_i$, $\theta_i' = 2\sin\lambda_i$, etc. Averaging over (λ_i,λ_j) with $\vartheta = \lambda_i - \lambda_j$ and $\langle\cos\lambda_i\cos\lambda_j\cos k\vartheta\rangle = \langle\sin\lambda_i\sin\lambda_j\cos k\vartheta\rangle = \frac14\delta_{k1}$, $\langle\sin\lambda_i\cos\lambda_j\sin k\vartheta\rangle = -\langle\cos\lambda_i\sin\lambda_j\sin k\vartheta\rangle = \frac14\delta_{k1}$, only B_1 survives:

$$\partial_s\partial_t\langle V_{ij}\rangle = Gm_im_j\Big[-\tfrac12 a_ia_j\partial_i\partial_jB_1 - a_i\partial_iB_1 - a_j\partial_jB_1 - 2B_1\Big] = Gm_im_j\Big[-\tfrac12 a_ia_j\partial_i\partial_jB_1 - B_1\Big]$$

by Euler's relation $r_i\partial_iB_1 + r_j\partial_jB_1 = -B_1$. With $b = b^{(1)}_{1/2}$, $B_1 = r_j^{-1}b(r_i/r_j)$, $\partial_i\partial_jB_1 = -r_j^{-3}(2b' + \alpha b'')$; the bracket is $a_j^{-1}\big[\alpha b' + \frac12\alpha^2b'' - b\big] = \frac{\alpha}{2a_j}b^{(2)}_{3/2}(\alpha)$ by (I2). Thus $\partial_s\partial_t\langle V_{ij}\rangle = 2\mathfrak{b}_{ij}$, and $B_{ij} = \partial_{\xi_i}\partial_{\xi_j}S = 2\mathfrak{b}_{ij}/\sqrt{\Lambda_i\Lambda_j}$.

Step S10.26: Assembly. $S_{\mathrm{fix}} - C_0$ has degree-two part $\sum_{i<j}\big[-\mathfrak{a}_{ij}(|u_i|^2/\Lambda_i + |u_j|^2/\Lambda_j) + 2\mathfrak{b}_{ij}\,u_i\cdot u_j/\sqrt{\Lambda_i\Lambda_j}\big] = \sum_{j,k}(Q_h)_{jk}(\xi_j\xi_k + \eta_j\eta_k)$ with $Q_h = \Delta^{-1}\hat q\Delta^{-1}$. The remainder is analytic with vanishing jet up to order three (Cor. 10.4), whence $O(|(\xi,\eta)|^4)$ by (E3). For the shrinking of ρ_*: $a(\Lambda)$ is continuous.

Step S10.27: Diagonalisation. $(\xi,\eta)\mapsto(\varrho^\top\xi,\varrho^\top\eta)$ is symplectic for $d\eta\wedge d\xi$ and leaves Λ and the form Γ-sum invariant in the sense $\sum_j(\tilde\xi_j^2 + \tilde\eta_j^2) = \sum_j(\xi_j^2+\eta_j^2)$; $\tilde\Gamma_j = (\tilde\xi_j^2+\tilde\eta_j^2)/2$. □

Remark 10.12 (Factor $\frac{1}{2}$ against the source, dictionary, numerical evidence). (1) *Index order.* Fej04r (34) orders $a_1 > \cdots > a_n$ and writes $C_1(a_j, a_k) = -\frac{a_k}{2a_j^2} b_{3/2}^{(1)}(\frac{a_k}{a_j})$, $C_2 = \frac{a_k}{2a_j^2} b_{3/2}^{(2)}$, $j < k$ (outer j, inner k). The dictionary to Definition 10.10 is $j \leftrightarrow n+1-j$; each of C_1, C_2 is $-2\mathfrak{a}/(Gm_im_j)$, $+2\mathfrak{b}/(Gm_im_j)$ for the pair. The corrected entries of `statement.md` P5 (decreasing order, $\hat{q} = \Delta^\top Q_h \Delta$) are $2\hat{q}$ in the present notation. (2) *Factor $\frac{1}{2}$ (suspected erratum E-P-Fej04r-3).* (35)–(37) as printed give $\langle F_{\mathrm{per}}\rangle = C_0 + Q_h^{\mathrm{F}} \cdot (\xi^2 + \eta^2)$ with $Q_h^{\mathrm{F}} = 2Q_h$. This is the right constant for the *printed* (31), $r = \sqrt{\Lambda}\sqrt{1-\sqrt{1-\varepsilon^2}}\, e^{i\cdot}$, $|r|^2 = \Gamma$; with the erratum's $\sqrt{2\Lambda_j}$ (Fej07e p. 3; the frozen chart, $|u|^2 = 2\Gamma$) every coefficient is halved, which Theorem 10.11 proves. In frozen terms $\langle f\rangle = C_0 + \sum_j \Omega_j \tilde{\Gamma}_j$ with $\Omega_j = 2\sigma_j =$ the eigenvalues of $2Q_h$, i.e. of the matrix printed in Fej04r (36)–(37) in the ξ-basis; `statement.md` P7 "$\Omega_j = 2\sigma_j$" is correct only if σ_j are the eigenvalues of the *present* Q_h (half of the printed). Positivity/definiteness (Part 9) is unaffected. (3) CP11 (B.2), p. 62 writes $C_2 = \frac{\alpha}{b}\beta_2^{(3)}$ (no $\frac{1}{2}$) while Fej04r (37) has $\frac{1}{2}$ in C_2; with (B.1) read literally the off-diagonal term is twice too large (suspected erratum E-P-CP11-1; the diagonal part of (B.1) agrees with Theorem 10.11). (4) *Numerical check* (scratch script in `.drafts/P/bp-scratch-bpP-p8/`, three planets $m = (0.7, 1.9, 0.4)$, $a = (4.1, 2.3, 1.0)$, $G = 1.3$, $m_0 = 0.9$, torus averages by trapezoid rule with 48^2 nodes of the chart of Def. 9.5): $(S_{\mathrm{fix}} - C_0)/(\xi^\top Q_h \xi + \eta^\top Q_h \eta) = 1.0000128,\ 1.0000021$ at amplitudes $10^{-3}, 5\cdot 10^{-4}$ (error $O(|u|^2)$); C_0 agrees to $4\cdot 10^{-14}$; $\langle p^{(1)} \cdot p^{(2)}\rangle = 10^{-18}$; (S1)–(S3) hold to $5\cdot 10^{-16}$. This is evidence, not a proof; the proof is Theorem 10.11.

10.7. **The exact identification of the two models.**

Theorem 10.13 (Identification of $\langle H_{\mathrm{mov}}\rangle$ and $\langle H_{\mathrm{fix}}\rangle$). *Let $w' = (\xi, \Lambda, \eta)$ with $(\lambda, \xi, \Lambda, \eta) \in \mathcal{O}_n$, and put*

$$\delta(\Lambda) = \frac{1}{2m_0} \sum_{i=1}^{n} \frac{m_i^4 \kappa^2}{\Lambda_i^2} = \frac{G}{2} \sum_i \frac{m_i^2}{a_i} \ \Big(= \tfrac{\kappa G}{2} \sum_i \tfrac{m_i^4}{\Lambda_i^2} \Big).$$

Then $\lambda \mapsto \hat{f}_{\mathrm{mov}}(\lambda, w') - \hat{f}_{\mathrm{fix}}(\lambda, w') = |\sum_i p^{(i)}|^2/(2m_0)$ is continuous, and

$$(2\pi)^{-n} \int_{[0,2\pi]^n} \big(\hat{f}_{\mathrm{mov}} - \hat{f}_{\mathrm{fix}}\big)\, d\lambda = \delta(\Lambda).$$

Consequently $\hat{f}_{\mathrm{fix}}(\cdot, w')$ is integrable iff $\hat{f}_{\mathrm{mov}}(\cdot, w')$ is, and then $S_{\mathrm{mov}} = S_{\mathrm{fix}} + \delta(\Lambda)$; in particular on $\mathcal{N}_{\mathbb{R}}(\rho_*)$ (where both are analytic, Prop. 10.2). With the same average for the full Hamiltonians: $\langle H_{\mathrm{mov}}^{\Psi}\rangle = h(\Lambda) + \mu S_{\mathrm{mov}} = h + \mu S_{\mathrm{fix}} + \mu\delta$ and $\langle H_{\mathrm{fix}}^{\Psi}\rangle = h + \varepsilon S_{\mathrm{fix}}$.

Source: Ours (`statement.md` P7 item 4 gives the plan; Fej04r Lemma 64 gives the cross terms); Lemma 8.3 for the difference.

Proof. **Step S10.28: Difference.** $f_{\mathrm{mov}} - f_{\mathrm{fix}} = |P|^2/(2m_0)$ identically (Lemma 8.3), a polynomial in the continuous functions $p^{(i)}(\lambda_i)$.

Step S10.29: Average. Divide Prop. 10.5(e) by $2m_0$ and use $\kappa/m_0 = G$ for the second form.

Step S10.30: Integrability. A bounded continuous function is integrable on the compact torus, so integrability of one of $\hat{f}_{\mathrm{fix}}, \hat{f}_{\mathrm{mov}}$ is that of the other, and the averages subtract. (Lean: where neither is integrable both Bochner averages are 0 and the difference is 0 rather than δ; this never occurs on $\mathcal{N}_{\mathbb{R}}(\rho_*)$, where $\hat{f}$ is continuous on the torus, Prop. 10.2(a).) □

Corollary 10.14 (Same jets, same quadratic form, same Hamilton equations). *On $\mathcal{N}_{\mathbb{R}}(\rho_*)$, for every multi-index α in (ξ, η) with $\alpha \neq 0$ and every Λ-derivative, $\partial_{(\xi,\eta)}^{\alpha} S_{\mathrm{mov}} = \partial_{(\xi,\eta)}^{\alpha} S_{\mathrm{fix}}$ (all orders, not only quadratic). Therefore:*

- *(a) $S_{\mathrm{mov}} = C_0 + \delta + \xi^\top Q_h \xi + \eta^\top Q_h \eta + O(|(\xi,\eta)|^4)$: same Q_h, same σ_j, same $\Omega_j = 2\sigma_j$; only the constant term $P_0 = C_0 + \delta$ (resp. C_0) differs, by an action-only function; and the quartic jets (Prop. 10.16) coincide.*

(b) *For $\mu=\varepsilon$ the secular equations $\dot\xi_i=\mu\partial_{\eta_i}S$, $\dot\eta_i=-\mu\partial_{\xi_i}S$ (Thm. 9.12(d), averaged) of the two models are* identical*; Λ is constant in both, and the frequencies $\dot\lambda_i$ differ by $\mu\partial_{\Lambda_i}\delta$.*

(c) *(*Transfer to the frozen Birkhoff form.*) Take $P_{\mathrm{mov}}=\hat f_{\mathrm{mov}}$, $P_{\mathrm{fix}}=\hat f_{\mathrm{fix}}$ (no μ-dependence), assembled as in the dictionary of §10.1. If* HasBirkhoff$(P_{\mathrm{mov}},V,r_0,\mu_0,P_0,\Omega,\beta)$ *(Def. 1.14) holds, then* HasBirkhoff$(P_{\mathrm{fix}},V,r_0,\mu_0,P_0-\delta,\Omega,\beta)$ *holds, and conversely, provided the points $(I,p,q)=(\Lambda,\eta,\xi)$ with $I\in V$, $p_j^2+q_j^2<r_0^2$ lie in $\mathcal{N}_{\mathbb{R}}(\rho_*)$; the hypotheses (H5), (H6) on β are literally the same. If a slow-variable symplectic change $\chi(\Lambda;\cdot)$ (rotation ϱ, Birkhoff map of Part 11b), depending on Λ only as a parameter, is applied before averaging, $S_{\mathrm{mov}}\circ\chi-S_{\mathrm{fix}}\circ\chi=\delta(\Lambda)$ still.*

Proof. δ does not depend on (ξ,η) by Theorem 10.13, so all derivatives of order ≥ 1 in (ξ,η) of the difference vanish. (a),(b) follow from Theorem 10.11 and Theorem 9.12(d). (c): $P_{\mathrm{av}}^{\mathrm{mov}}-P_{\mathrm{av}}^{\mathrm{fix}}=\delta(I)$ (frozen average of Def. 1.12, $I=\Lambda$) on the integrability set, so the integrability clause (i) transfers and the clause (ii) expansion with $P_0\to P_0-\delta$; Ω,β unchanged, χ acts on (ξ,η) only. □

Remark 10.15 (Use in Part 11). For $n=2$ the hypotheses of P-G on $(\Omega,\beta,\det\beta)$ for $H_{\mathrm{fix}}=h+\varepsilon\hat f_{\mathrm{fix}}$ coincide with those for $H_{\mathrm{mov}}=h+\mu\hat f_{\mathrm{mov}}$; hence $n=2$ P-FIXED follows P-PLANAR route (c) with f_{mov} replaced by f_{fix} (the Kepler part h and the fast frequency map are the same, Lemma 8.3). The additional constant is the only change (P_0, not used by the KAM argument beyond the energy normalisation).

10.8. **Quartic data for the planar torsion (interface for Part 11b).** Part 11b evaluates the second-order Birkhoff invariants τ of CP11 §8.1, pp. 32–34, (8.1)–(8.4) (the frozen β of Def. 1.14). The input is the jet of S to order four in $u=(u_1,u_2)$ ($n=2$, planar: no $(\bar p,\bar q)$) as an analytic function of the axes. CP11 (6.9)–(6.10) and (B.5)–(B.6), pp. 21, 62–63, write $\langle f\rangle=C_0+Q_h\cdot\frac{\eta^2+\xi^2}{2}+\frac12F_h\cdot\eta^2\xi^2$-type quartic with coefficients r_1,r_2,r_3,r_4 (functions of (a_i,a_j) via $\beta_k^{(9)}=b_{9/2}^{(k)}$, $\beta_k^{(r)}:=b_{r/2,k}$); their ordering is $(\eta,\xi)=(p,q)$ and $a_1<a_2$, the Lean order. The statements below fix the interface; the quartic computation is left to Part 11b.

Proposition 10.16 (Quartic jet)**.** *Let $n\geq 2$ and $\Lambda\in\mathcal{N}_{\mathbb{R}}(\rho_*)$-range. The degree-four part of S_{fix} (and of S_{mov}, Cor. 10.14) in (ξ,η) is*

$$\mathcal{Q}_4=\sum_{i<j}\Big[c_1^{ij}|u_i|^4+c_2^{ij}|u_j|^4+c_3^{ij}|u_i|^2|u_j|^2+c_4^{ij}(u_i\cdot u_j)^2+c_5^{ij}|u_i|^2(u_i\cdot u_j)+c_6^{ij}|u_j|^2(u_i\cdot u_j)\Big],$$

with $c_\nu^{ij}=c_\nu^{ij}(\Lambda_i,\Lambda_j)$ real analytic. Moreover:

(a) *(*scaling*) In the eccentricity vectors v of Def. 9.5 the pair average is $S_{ij}=-Gm_im_ja_j^{-1}\mathcal{W}(\alpha_{ij};v_i,v_j)$ with $\mathcal{W}$ real analytic on $(0,1)\times\{|v|<\frac14\}$ and independent of m,G,κ. The passage to u is exact: $v_i=(1-|u_i|^2/(4\Lambda_i))^{1/2}u_i/\sqrt{\Lambda_i}$, so if $\mathcal{W}_2=A\,|v_i|^2+A'|v_j|^2+B\,v_i\cdot v_j$ is the quadratic part and $\mathcal{W}_4$ the quartic part in v, then $\mathcal{Q}_4^{(ij)}=\mathcal{W}_4(u/\sqrt{\Lambda})-\big(\frac{A}{4}\frac{|u_i|^4}{\Lambda_i^2}+\frac{A'}{4}\frac{|u_j|^4}{\Lambda_j^2}+\frac{B}{8}\frac{u_i\cdot u_j}{\sqrt{\Lambda_i\Lambda_j}}\big(\frac{|u_i|^2}{\Lambda_i}+\frac{|u_j|^2}{\Lambda_j}\big)\big)$ times the appropriate prefactor $-Gm_im_ja_j^{-1}$.*

(b) *(*diagonal terms, exact*) c_1^{ij} and c_2^{ij} are*

$$c=-Gm_im_j\Big(\frac{h^{(4)}(0)}{64}-\frac{h''(0)}{16}\Big)\frac{1}{\Lambda^2},\quad h(\varepsilon)=\Phi\big(a(1-\varepsilon),a'\big)(1-\varepsilon),$$

with $(a,\Lambda)=(a_i,\Lambda_i)$, $a'=a_j$ for c_1 and (a_j,Λ_j), $a'=a_i$ for c_2.

(c) *(*provenance for the remaining ones*) $c_3,\dots,c_6$ follow by the same method as Theorem 10.11 (Fourier expansion of $U(r_i,r_j,\vartheta)$ in the orbit variables, now to second order in v using $q=a(kc-v+\frac{v\cdot c}{1+k}v)$) and are linear combinations of $b_s^{(k)}(\alpha)$, $s\leq\frac92$, with polynomial*

coefficients; the correspondence with CP11 is $c_1^{ij} = \frac{m_i m_j}{2\Lambda_i^2} r_1(a_i, a_j)$, $c_2^{ij} = \frac{m_i m_j}{2\Lambda_j^2} r_1(a_j, a_i)$ *($G = m_0 = 1$, units of CP11), and r_2, r_3, r_4 for the mixed monomials of (B.5), after the substitution $(\eta, \xi) \mapsto u$ and the $O(2)$-relations $\eta_i^2\xi_j^2 + \xi_i^2\eta_j^2 = |u_i|^2|u_j|^2 - (\dots)$ (CP11 use a non-symmetric tensor F_h).*

Source: CP11 §8.1, pp. 32–34 ((8.1) $\tau_{11}, \tau_{12}, \tau_{22}$ for the planar block, asymptotics (8.2)–(8.4)); App. B (B.5)–(B.6), pp. 62–64 (quartic tensor F_h and $r_1, \dots, r_4$); (6.9)–(6.10), p. 21. Symmetry structure: Cor. 10.4, Weyl.
Source gap: CP11's proof of (B.1)–(B.17) is in App. B only for the structure; the coefficients are printed without derivation of the $\beta^{(9)}$ polynomials.

Proof. **Step S10.31: Structure.** By Cor. 10.4 and Prop. 10.2, the degree-four part is an $O(2)$-invariant polynomial, analytic in Λ (Taylor coefficients of a jointly analytic function). By pair decomposition (Step S10.21) it is a sum over pairs of $O(2)$-invariants of (u_i, u_j), i.e. polynomials in $|u_i|^2, |u_j|^2, u_i \cdot u_j$ ((E4)) of degree two: the six monomials listed.

Step S10.32: (a). $q^{(i)} = a_i \tilde{q}(\lambda_i, v_i)$ with $\tilde{q}$ independent of a, m, κ (Def. 9.5), so $V_{ij} = -Gm_i m_j a_j^{-1} \big|\alpha\tilde{q}(\lambda_i, v_i) - \tilde{q}(\lambda_j, v_j)\big|^{-1}$ and the average over (λ_i, λ_j) is a function of (α, v_i, v_j), real analytic for $\alpha < 1$ by the argument of Prop. 10.2 (a real α-nbhd inside the separation condition). The conversion is the expansion of $(1-t)^{1/2}$ with $t = |u|^2/(4\Lambda)$.

Step S10.33: (b). For $u_j = 0$ the computation of Step S10.24 gives $S_{ij} = -Gm_i m_j \sum_l E^l \frac{h^{(2l)}(0)}{(2l)!} \frac{\binom{2l}{l}}{4^l}$, i.e. the coefficient of E^2 is $h^{(4)}(0)/64$, and $E = |u|^2/\Lambda - |u|^4/(4\Lambda^2)$ gives $|u|^4$ coefficient $-Gm_i m_j\big(h^{(4)}/64 - h''/16\big)/\Lambda^2$.

Step S10.34: (c). The method is described; the printed formulas are compared in Remark 10.17. □

Remark 10.17 (Check of CP11 (B.6) r_1; suspected erratum E-P-CP11-2). c_1, c_2 of Prop. 10.16(b) were evaluated numerically (mpmath, Φ by quadrature, $m = (0.7, 1.9)$, $G = \kappa = 1$) and independently by the series $\sum_k \binom{2k}{k}^2 4^{-2k}\alpha^{2k}\langle r^{2k}\rangle$ and by brute-force torus averages; the three agree. For $a = (1, 3)$: $c_1 = 0.0065937$, $c_2 = -0.0028633$, while $\frac{m_1 m_2}{2\Lambda_i^2} r_1$ with r_1 of (B.6) as printed gives 0.007418 and −0.002975; for $a = (1, 10)$ the agreement is $1 + 4 \cdot 10^{-5}$ (the leading terms $\frac{3}{16}\alpha^2/a_2$, (8.2) of CP11, are right), for $a = (1, 2)$ the printed value has the wrong sign (0.0574 against −0.0310). So the printed r_1 is correct only asymptotically (as $\alpha \to 0$, which is all (8.2)–(8.4) use) and *must not be used for the torsion at finite axis ratio*; either the printed coefficients contain misprints or CP11's (η, ξ) differ from Poincaré variables beyond leading order (footnote 73 says they coincide in the planar case). Part 11b should recompute $c_1, \dots, c_6$ from the method above (exactly, and checkable by quadrature), not from (B.6). The quartic quantities CP11 uses for τ in (8.1) are all of $r_1, \dots, r_4$ (equivalently $c_1, \dots, c_6$), C_1, C_2 (equivalently $\mathfrak{a}, \mathfrak{b}$) and the eigenvector matrix U_h of Q_h (7.19), (7.27).

10.9. **Sources, deviations, errata, forward references.**
Sources read. Fej04r pp. 46–55 (Thm. 62, §6.2, Lemma 64, Def. 66, Formule 67, Prop. 73 proof incl. $\hat{q}$ on p. 55, Lemma 76 p. 56), Fej07e p. 3, CP11 pp. 21–22, 31–34, 62–65 (rendered pages 62–63 read as images), `docs/tracks/P/statement.md` P5–P7.
Deviations. (1) Lean (increasing) axis order throughout; Fej04r's decreasing order by $j \mapsto n+1-j$. (2) Q_h halved against the printed formula (normalisation of the frozen chart). (3) Identification is proved to all orders and *exactly*, not only at quadratic order, and the diagonal virial term is included. (4) δ is expressed in the action Λ and in the axis a.

Suspected errata (for the ledger). E-P-Fej04r-3: (35)–(37) printed hold for $|r|^2 = \Gamma$ (printed (31)); under Fej07e p. 3's $\sqrt{2\Lambda}$ every coefficient is halved (Remark 10.12(2)). E-P-Fej04r-1 (P5) confirmed numerically up to that factor. E-P-Fej04r-2 (Lemma 76) unchanged. E-P-CP11-1: (B.2) C_2 has no $\frac{1}{2}$ (p. 62). E-P-CP11-2: r_1 of (B.6), p. 63, disagrees with the exact quartic coefficient away from $\alpha \to 0$ (Remark 10.17); r_2, r_3, r_4 not checked.
Doubts about frozen statements. None. `statement.md` P7 "$\Omega_j = 2\sigma_j$" needs the qualifier of Remark 10.12(2).
External facts used. (E1)–(E5), (E2); Parts 1, 6, 7 and [I:`bpI:p7:orbit-identities`].
Forward references and consumers. Part 9 (definiteness of Q_h, Lemmas 74–75: uses Def. 10.10 and positivity Lemma 10.7); Part 10 (analytic continuation: Lemma 10.7, no global meromorphy); Parts 11, 11a, 11b (Thm. 10.11, Cor. 10.14, Prop. 10.16, Remark 10.15); Part 12 (Prop. 10.2).
Needed from other parts: none.
Label index. `rho`, `perturbation-analytic`, `symmetries`, `evenness`, `lemma64`, `laplace-def`, `laplace`, `identities`, `coeffs`, `quadratic-form`, `identification`, `same-jets`, `quartic`.

11. The spectrum of the secular matrix

Part 8 reduced the secular function to $S = C_0 + \xi^\top Q_h \xi + \eta^\top Q_h \eta + O(|(\xi,\eta)|^4)$ with the real analytic symmetric $n \times n$ matrix Q_h (Definition 10.10, Theorem 10.11). This part studies the spectrum of Q_h and turns it into the frozen first-order Birkhoff data $\Omega_j = 2\sigma_j$ of Definition 1.14.

- `lemmas-70-72` (Lemma 11.1, Lemma 11.2, Corollary 11.3): eigenvalue perturbation of $\mathrm{Diag}(d) + \varepsilon B$. Fej04r's Lemma 71 is re-proved in a sharper, fully explicit form (eigenvectors included, no sorting, no dependence on $\|D\|$);
- `definite` (Lemma 11.5, Lemma 11.4, Proposition 11.6): Q_h is negative definite for $n \geq 2$ with an explicit bound; Lemma 74 is replaced by a one-line integration by parts that gives the one inequality needed, $b_s^{(1)} > b_s^{(2)}$;
- `diagonalize` (Proposition 11.7, Proposition 11.8, Theorem 11.9): on a patch where the spectrum is simple, the eigenvalues σ_j and the rotation $\varrho \in SO(n)$ are real analytic; the rotation of (ξ,η) has an explicit symplectic lift to the full chart, under which the secular function is simply composed with the rotation; relation to $\Omega_j = 2\sigma_j$ and the frozen R_j;
- `interface` (Definition 11.11, Proposition 11.12, Lemma 11.13, Corollary 11.14): a single analytic function Φ_4 of the axes whose non-vanishing means *simple spectrum and non-resonance up to order four*; its zero set, the consequence of one good point, and the explicit case $n = 2$ (the spectrum is always simple there, and non-resonance is the exclusion of four ratios).

Global continuation of the eigenvalues over the whole domain of axes is *not* used (Part 10; Fej04r Lemma 76 is false as printed, Remark 10.8). Everything here is local, or an identity between analytic functions.

11.1. **Conventions and external facts.** $|x|$ is the Euclidean norm of $x \in \mathbb{R}^n$ or $\mathbb{C}^n$, $\|M\|$ the induced operator norm, $\|M\|_F$ the Frobenius norm ($\|M\| \leq \|M\|_F$), e_j the unit vectors, Sym_n the real symmetric matrices. For a matrix, $|M_{jk}| \leq \|M\|$ and every column has Euclidean norm $\leq \|M\|$. The data are as in Part 8: $n \geq 1$, G, m_0, m, κ, increasing axes $a_1 < \cdots < a_n$, $\Lambda_i = m_i\sqrt{\kappa a_i}$; $\mathfrak{a}_{ij}, \mathfrak{b}_{ij}, \hat{q}, Q_h = \Delta^{-1}\hat{q}\Delta^{-1}$ are those of Definition 10.10. Note that $\hat{q}$ is a function of a (and G, m) and Δ a function of $\Lambda(a)$, so $Q_h = Q_h(a)$ is a real analytic function of a on $\{a_1 < \cdots < a_n\}$ (Definition 10.10). The eigenvalues of Q_h are labelled in *increasing* order $\sigma_1 < \cdots < \sigma_n$ wherever they are simple; this labelling is unrelated to the order of the planets (Fej04r's decreasing planet order corresponds to $j \mapsto n + 1 - j$ in the matrix entries only, Remark 10.12(1)).

External facts.

(E1) *Real spectral theorem.* A real symmetric matrix M has an orthonormal basis of eigenvectors: $M = \varrho\,\mathrm{diag}(\lambda)\varrho^{\top}$ with $\varrho \in O(n)$, and $\det(sI - M) = \prod_j (s - \lambda_j)$. Mathlib: `Matrix.IsHermitian.spectral_theorem`, `Matrix.IsHermitian.eigenvalues`, `eigenvectorBasis`, `eigenvectorUnitary`, `Matrix.IsHermitian.charpoly_eq`, `det_eq_prod_eigenvalues` (for $\mathbb{R}$, `Matrix.isHermitian_iff_isSymm`). Mathlib's `eigenvalues` is indexed by an arbitrary finite type; `eigenvalues`$_0$ is antitone. Neither is analytic in parameters; the blueprint therefore never uses them as functions of a except through Proposition 11.7.

(E2) *Analytic implicit function theorem.* If $p(w, s)$ is real analytic near $(w^0, s^0) \in \mathbb{R}^m \times \mathbb{R}$, $p(w^0, s^0) = 0$, $\partial_s p(w^0, s^0) \neq 0$, then there is a real analytic $s(w)$ near w^0 with $s(w^0) = s^0$ and $p(w, s(w)) = 0$, and it is the only continuous solution near (w^0, s^0). (Mathlib has the one-variable analytic inverse function theorem, `analyticAt_localInverse`; the version with parameters was not found and is quoted.)

(E3) *Fundamental theorem of symmetric polynomials.* Every symmetric polynomial in $\sigma_1, \dots, \sigma_n$ with real coefficients is a polynomial in the elementary symmetric polynomials; the latter are, up to sign, the coefficients of $\det(sI - M) = \prod(s - \sigma_j)$. Mathlib: `MvPolynomial.esymmAlgEquiv`; `Matrix.charpoly`.

(E4) *Identity theorem.* A real analytic function on a connected open set $\mathcal{A} \subset \mathbb{R}^m$ that vanishes on a nonempty open subset vanishes identically; if it is not identically zero its zero set is closed, has empty interior and Lebesgue measure zero.

(E5) Integration by parts and positivity of the integral of a continuous nonnegative function that is positive somewhere.

11.2. **Eigenvalue perturbation: Fej04r Lemmas 70, 71, Corollary 72.**

Lemma 11.1 (Fej04r Lemma 70)**.** *Let $D = \mathrm{diag}(\delta_1, \dots, \delta_n)$ with $\delta_j \in \mathbb{R}$, $A \in M_n(\mathbb{C})$, and $\sigma \in \mathbb{C}$ an eigenvalue of $D + A$. Then $\min_j |\sigma - \delta_j| \le \|A\|$.*

Source: Fej04r Lemme 70, p. 53 (the norm "as in (23)" is not specified; we take the Euclidean operator norm). Proof ours, by eigenvectors.

Proof. **Step S11.1: Eigenvector.** Let $(D+A)x = \sigma x$, $|x| = 1$. If $\sigma = \delta_j$ for some j there is nothing to prove. Otherwise $(D - \sigma)x = -Ax$, so $x = -(D - \sigma)^{-1}Ax$ and $1 \le \max_j |\sigma - \delta_j|^{-1}\|A\|$. □

Lemma 11.2 (Fej04r Lemma 71, sharpened: second-order perturbation with eigenvectors)**.** *Let $n \ge 2$, $d \in \mathbb{R}^n$ with $g := \min_{j \neq k} |d_j - d_k| > 0$ (no ordering assumed), $D = \mathrm{diag}(d)$, $B \in \mathrm{Sym}_n$ with $B_{jj} = 0$ for all j, $\varepsilon \in \mathbb{R}$, and $\tau := |\varepsilon|\|B\|/g$. Assume $8\sqrt{n}\,\tau \le 1$. Then there are $\varrho \in SO(n)$ with columns $v^{(1)}, \dots, v^{(n)}$ and $\lambda \in \mathbb{R}^n$ with $(D + \varepsilon B)\varrho = \varrho\,\mathrm{diag}(\lambda)$ and, for all $j \neq k$,*

(a) $|\lambda_j - d_j| \le 2\tau^2 g = 2\varepsilon^2\|B\|^2/g$;

(b) $|v^{(j)} - e_j| \le 3\tau$; hence $\|\varrho - I\|_F \le 3\sqrt{n}\,\tau \le \frac{3}{8}$;

(c) $|\lambda_j - \lambda_k| \ge \frac{15}{16} g$: the spectrum of $D + \varepsilon B$ is simple, and these are all its eigenvalues.

Source: Fej04r Lemme 71, p. 53 (printed constants $\frac{3}{\sigma}(1 + \|D\|/\sigma)^2$ in the statement against $\frac{2}{\sigma}(\cdots)$ at the end of the proof are inconsistent, and the proof only says "ε small", see §11.6). The statement here and its proof are ours; the idea (eliminate the first-order term) is the source's.

Proof. Put $N = D + \varepsilon B$ (symmetric) and note $\tau \le \frac{1}{8\sqrt{2}} < \frac{1}{8}$.

Step S11.2: Localisation. Let λ be an eigenvalue of N with unit eigenvector v. By Lemma 11.1 (with $A = \varepsilon B$) there is j with $|\lambda - d_j| \le |\varepsilon|\|B\| = \tau g$. For $k \neq j$, $|\lambda - d_k| \ge g - \tau g \ge \frac{7}{8} g$. Write $j = j(\lambda)$.

Step S11.3: Eigenvector equations. Let $b \in \mathbb{R}^{n-1}$ be the j-th column of B without its (zero) j-th entry, B' the submatrix of B with row and column j removed, $\Delta' = \mathrm{diag}(\lambda - d_k)_{k \neq j}$ (invertible by Step S11.2) and $u = (v_k)_{k \neq j}$. Row $k \neq j$ of $Nv = \lambda v$ reads $(\lambda - d_k)v_k = \varepsilon(B_{kj}v_j + \sum_{l \neq j} B_{kl}v_l)$,

i.e. $\Delta' u = \varepsilon(v_j b + B'u)$; row j reads $(\lambda - d_j)v_j = \varepsilon\, b\cdot u$ (as $B_{jj}=0$ and B is symmetric). Since $\|\Delta'^{-1}\| \le \frac{8}{7g}$ and $|b|, \|B'\| \le \|B\|$: $|u| \le \frac{8\tau}{7}(|v_j|+|u|)$, i.e. $|u| \le \frac{8\tau/7}{1-8\tau/7}|v_j| \le \frac{4\tau}{3}|v_j|$. In particular $v_j \neq 0$ (else $u=0$, $v=0$). Put $w = u/v_j$: $|w| \le \frac{4}{3}\tau \le 2\tau$.

Step S11.4: Estimates (a), (b). From row j: $|\lambda - d_j| = |\varepsilon\, b\cdot w| \le |\varepsilon|\|B\||w| \le \tau g \cdot 2\tau$. Normalise $v_j > 0$: $v = (1+|w|^2)^{-1/2}(e_j + w)$, so $|v - e_j| \le (1-v_j) + v_j|w| \le \frac{1}{2}|w|^2 + |w| \le 2\tau + 2\tau^2 \le 3\tau$.

Step S11.5: The map $\lambda \mapsto j(\lambda)$ is a bijection. Take an orthonormal eigenbasis of N ((E1)) and for each member the index j of Step S11.2. If two members v, v' had the same index, then $v\cdot v' = v_j v'_j(1 + w\cdot w')$ with $|w\cdot w'| \le 4\tau^2 < 1$ and $v_j v'_j \neq 0$, so $v\cdot v' \neq 0$, contradicting orthogonality. So the assignment is injective, hence bijective; order the basis so that member j has index j, and fix the signs so that $v_j^{(j)} > 0$.

Step S11.6: (c). $|\lambda_j - \lambda_k| \ge g - 2\cdot 2\tau^2 g \ge g(1 - \frac{4}{64})$.

Step S11.7: $\varrho \in SO(n)$. ϱ is orthogonal; $\|\varrho - I\| \le \|\varrho - I\|_F \le \sqrt{n}\, 3\tau < 2$. If $\det\varrho = -1$, a real orthogonal matrix has an odd number of eigenvalues -1 (non-real ones come in conjugate pairs of product 1), so $\varrho x = -x$ for some unit x and $|(\varrho - I)x| = 2$, a contradiction. Hence $\det\varrho = 1$. □

Corollary 11.3 (Fej04r Corollary 72, quantitative)**.** *Let $n \ge 2$, $\hat D \in \mathrm{Sym}_{n-1}$ with eigenvalues $\delta_1, \dots, \delta_{n-1}$, $\delta_n := 0$, and $\sigma := \min_{j\neq k}|\delta_j - \delta_k| > 0$ (so the δ_j are distinct and nonzero for $j < n$). Let $D = \mathrm{diag}(\hat D, 0)$, $K > 0$, and $0 < \varepsilon$ with $32\sqrt{n}\,\varepsilon K \le \sigma$. Let $A \in \mathrm{Sym}_n$ (it may depend on ε) with $\|A\| \le K$. Then $D + \varepsilon A$ has n simple eigenvalues $\lambda_1, \dots, \lambda_n$, pairwise at distance $\ge \sigma/4$, with*

$$|\lambda_n - \varepsilon A_{nn}| \le \frac{16\,\varepsilon^2 K^2}{\sigma}.$$

In particular, if $A_{nn} = c_1 + c_2\varepsilon^\beta$ $(0 \le \beta < 2)$, then $\lambda_n = \varepsilon(c_1 + c_2\varepsilon^\beta) + O(\varepsilon^2)$, with the constant in O explicit; and if $|c_1 + c_2\varepsilon^\beta| \ge c > 0$ and $16\varepsilon K^2 < c\,\sigma$ this eigenvalue is nonzero. If $\hat D$ is diagonal, $D + \varepsilon A$ is conjugated to a diagonal matrix by some $\varrho \in SO(n)$ with $\|\varrho - I\|_F \le 12\sqrt{n}\,\varepsilon K/\sigma$ (in general $\|\varrho - \varrho_0\|_F$ is bounded by the same quantity, with $\varrho_0 = \mathrm{diag}(\hat\varrho, 1)$, $\hat\varrho \in SO(n-1)$ diagonalising $\hat D$).

Source: Fej04r Corollaire 72, p. 54. The printed statement leaves implicit that A_ε is bounded as $\varepsilon \to 0$ (we assume $\|A\| \le K$) and that β plays no role in the proof; the explicit remainder is ours.

Proof. **Step S11.8: Diagonalise $\hat D$.** Take $\hat\varrho \in SO(n-1)$ ((E1), flipping a sign if needed) with $\hat\varrho^\top \hat D \hat\varrho = \mathrm{diag}(\delta_1, \dots, \delta_{n-1})$ and $\varrho_0 = \mathrm{diag}(\hat\varrho, 1)$. Then $\varrho_0^\top(D + \varepsilon A)\varrho_0 = \mathrm{diag}(\delta) + \varepsilon A'$, $A' = \varrho_0^\top A \varrho_0$, $\|A'\| = \|A\| \le K$, $A'_{nn} = A_{nn}$.

Step S11.9: Split off the diagonal. $A' = \mathrm{Diag}(A') + B$ with $B \in \mathrm{Sym}_n$ of zero diagonal, $\|B\| \le 2K$ (diagonal entries are $\le \|A'\|$). Put $d_j = \delta_j + \varepsilon A'_{jj}$, so $\mathrm{diag}(\delta) + \varepsilon A' = \mathrm{diag}(d) + \varepsilon B$. Its gap is $g \ge \sigma - 2\varepsilon K \ge \sigma/2$.

Step S11.10: Apply Lemma 11.2. $\tau = \varepsilon\|B\|/g \le 4\varepsilon K/\sigma$ and $8\sqrt{n}\tau \le 32\sqrt{n}\varepsilon K/\sigma \le 1$. Part (a) with $j = n$, $d_n = \varepsilon A_{nn}$ gives $|\lambda_n - \varepsilon A_{nn}| \le 2\varepsilon^2\|B\|^2/g \le 2\varepsilon^2 \cdot 4K^2 \cdot \frac{2}{\sigma} = 16\varepsilon^2K^2/\sigma$. Part (c) gives the separation $\ge \frac{15}{16}g \ge \frac{15}{32}\sigma \ge \sigma/4$. Part (b): $\|\varrho_1 - I\|_F \le 3\sqrt{n}\tau \le 12\sqrt{n}\varepsilon K/\sigma$ for the rotation ϱ_1 of the lemma; $\varrho = \varrho_0\varrho_1$ and $\|\varrho - \varrho_0\|_F = \|\varrho_1 - I\|_F$. □

11.3. Negative definiteness of Q_h: Fej04r Proposition 73, Lemmas 74, 75.

Lemma 11.4 (Hadamard; Fej04r Lemma 75)**.** *Let $Q \in \mathrm{Sym}_n$ with $q_{jk} \ge 0$ for $j \neq k$, and $\delta q_j := -q_{jj} - \sum_{l\neq j} q_{jl}$. Then for all $x \in \mathbb{R}^n$*

$$x^\top Q x = -\sum_j \delta q_j\, x_j^2 - \sum_{j<k} q_{jk}(x_j - x_k)^2.$$

Hence if $\delta q_j \ge 0$ for all j then $Q \le 0$, and if $c := \min_j \delta q_j > 0$ then $x^\top Q x \le -c|x|^2$.
Source: Fej04r Lemme 75, p. 55 (hypotheses there: $q_{jj} < 0 < q_{jk}$, $\delta q_j > 0$). Proof: expand.

Proof. **Step S11.11: Expansion.** $-\sum_j \delta q_j x_j^2 = \sum_j q_{jj} x_j^2 + \sum_j \sum_{l \ne j} q_{jl} x_j^2$ and $\sum_{j<k} q_{jk}(x_j - x_k)^2 = \sum_j \sum_{l \ne j} q_{jl} x_j^2 - 2\sum_{j<k} q_{jk} x_j x_k$. The sum of the two with the sign as stated is $\sum_j q_{jj} x_j^2 + 2\sum_{j<k} q_{jk} x_j x_k = x^\top Q x$. □

Lemma 11.5 (Laplace coefficients: $b_s^{(1)} > b_s^{(2)}$; the case of Fej04r Lemma 74 that is needed)**.** *For every $s > 0$ and $\alpha \in (0,1)$, with $A(t) = 1 + \alpha^2 - 2\alpha\cos t$ (Definition 10.6),*

$$b_s^{(1)}(\alpha) - b_s^{(2)}(\alpha) = \frac{2s\alpha}{\pi} \int_0^\pi \sin^2 t\,(1 - \cos t)\, A(t)^{-s-1}\, dt > 0.$$

In particular $b_{3/2}^{(1)}(\alpha) > b_{3/2}^{(2)}(\alpha) > 0$.

Source: Fej04r Lemme 74, p. 54 states $b_s^{(j)} > b_s^{(j+1)}$ for all $s \in \mathbb{R}_+ \setminus \mathbb{N}$, $j \in \mathbb{N}$, by an induction on s using three-term recurrences that are only quoted from [Poin05, LR95]; the proof is not complete in the source as printed. Only $j = 1$, $s = \frac{3}{2}$ is used; the proof below is ours and does not need the recurrences.

Proof. **Step S11.12: Reduction to $[0,\pi]$.** $w(t) = A(t)^{-s}$ is even and 2π-periodic, so $b_s^{(k)} = \frac{1}{\pi}\int_0^\pi \cos(kt)\, w(t)\, dt$.

Step S11.13: Integration by parts. Let $H(t) = \sin t\,(1 - \cos t)$. Then $H'(t) = \cos t - \cos 2t$, $H(0) = H(\pi) = 0$, and $H > 0$ on $(0,\pi)$. So $b_s^{(1)} - b_s^{(2)} = \frac{1}{\pi}\int_0^\pi H' w\, dt = -\frac{1}{\pi}\int_0^\pi H w'\, dt$, and $-w'(t) = 2s\alpha \sin t\, A(t)^{-s-1} \ge 0$ on $[0,\pi]$. This is the stated formula ((E5)); the integrand is continuous, ≥ 0 and positive on $(0,\pi)$. For the last claim use Lemma 10.7 (positivity of $b_s^{(k)}$ from the series). □

Proposition 11.6 (Fej04r Proposition 73)**.** *Let $n \ge 2$, $a_1 < \cdots < a_n$ (so $0 < \alpha_{ij} = a_i/a_j < 1$ for $i < j$), all masses positive. With $\mathfrak{a}_{ij}, \mathfrak{b}_{ij}$ of Definition 10.10 put*

$$c_{\hat q}(a) := \min_j \sum_{k \ne j} (\mathfrak{a}_{jk} - \mathfrak{b}_{jk}) > 0.$$

Then for all $x \in \mathbb{R}^n$: $x^\top \hat q x \le -\sum_j \big(\sum_{k\ne j}(\mathfrak{a}_{jk} - \mathfrak{b}_{jk})\big) x_j^2 \le -c_{\hat q}|x|^2$, and

$$x^\top Q_h x \le -\frac{c_{\hat q}(a)}{\max_i \Lambda_i}|x|^2.$$

So $\hat q$ and Q_h are negative definite, every eigenvalue satisfies $\sigma_j \le -c_{\hat q}/\max_i \Lambda_i < 0$, and $\Omega_j = 2\sigma_j < 0$. The bound $c_{\hat q}/\max \Lambda$ is continuous, hence locally uniformly positive, on $\{a_1 < \cdots < a_n\}$. (For $n = 1$ there is no pair and $Q_h = 0$.)
Source: Fej04r Prop. 73 and proof, pp. 54–55 (proof via Lemmas 74, 75 and a maximum-component argument for definiteness). The source's remark that Herman omitted the sign of the Newtonian potential is consistent with Part 8, where $V_{ij} = -Gm_im_j/|q^{(i)} - q^{(j)}|$. The quantitative proof is ours.

Proof. **Step S11.14: Strong diagonal dominance.** Definition 10.10: $\hat q_{jk} = \mathfrak{b}_{jk} > 0$ $(j \ne k)$, $\hat q_{jj} = -\sum_{k\ne j} \mathfrak{a}_{jk}$, hence $\delta\hat q_j = \sum_{k\ne j}(\mathfrak{a}_{jk} - \mathfrak{b}_{jk})$. For $j < k$: $\mathfrak{a}_{jk} - \mathfrak{b}_{jk} = \frac{Gm_jm_ka_j}{4a_k^2}\big(b_{3/2}^{(1)} - b_{3/2}^{(2)}\big)(\alpha_{jk}) > 0$ by Lemma 11.5, and the same for $k < j$ by symmetry of $\mathfrak{a}, \mathfrak{b}$. As $n \ge 2$ every j has at least one $k \ne j$, so $\delta\hat q_j > 0$.

Step S11.15: Lemma 11.4. gives the first two inequalities (the dropped term $-\sum_{j<k} \hat q_{jk}(x_j - x_k)^2 \le 0$).

Step S11.16: Congruence. $x^\top Q_h x = (\Delta^{-1}x)^\top \hat{q}(\Delta^{-1}x) \le -c_{\hat{q}}|\Delta^{-1}x|^2 \le -c_{\hat{q}}|x|^2/\max_i \Lambda_i$ since $\Delta^{-1} = \mathrm{diag}(\Lambda_i^{-1/2})$. An eigenvalue is $x^\top Q_h x$ for a unit eigenvector x. Continuity: $\mathfrak{a}_{jk}, \mathfrak{b}_{jk}$ are continuous in a (Lemma 10.7). □

11.4. **Local analytic diagonalisation; the symplectic rotation.**

Proposition 11.7 (Local analytic spectral decomposition)**.** *Let $\mathcal{W} \subset \mathbb{R}^m$ be open, $M : \mathcal{W} \to \mathrm{Sym}_n$ real analytic, $w^0 \in \mathcal{W}$ such that $M(w^0)$ has n distinct eigenvalues $\sigma_1^0 < \cdots < \sigma_n^0$ (equivalently* $\mathrm{discr}(M(w^0)) \neq 0$*, `Matrix.discr`). Then there are an open ball $\mathcal{W}' \ni w^0$ and real analytic $\sigma_j : \mathcal{W}' \to \mathbb{R}$, $\varrho : \mathcal{W}' \to SO(n)$ with $\sigma_1 < \cdots < \sigma_n$ on $\mathcal{W}'$, $M(w) = \varrho(w)\,\mathrm{diag}(\sigma(w))\varrho(w)^\top$ and $\sigma_j(w^0) = \sigma_j^0$. The spectrum of $M(w)$ is simple on $\mathcal{W}'$. On a connected open set, σ_j is unique, and ϱ is unique up to multiplying columns by ± 1 (with* $\det = 1$*).*

Proof. **Step S11.17: Eigenvalues.** $p(w,s) = \det(sI - M(w))$ is real analytic in (w,s) and $\partial_s p(w^0, \sigma_j^0) = \prod_{k\neq j}(\sigma_j^0 - \sigma_k^0) \neq 0$. By (E2) there are analytic $\sigma_j(w)$ near w^0 with $p(w, \sigma_j(w)) = 0$, $\sigma_j(w^0) = \sigma_j^0$. Shrinking to a ball the $\sigma_j(w)$ stay in disjoint intervals around the σ_j^0, so they are n distinct real roots of the degree-n polynomial $p(w,\cdot)$: they are all of its roots, they are the eigenvalues of $M(w)$ with $\sigma_1 < \cdots < \sigma_n$, and $M(w)$ has simple spectrum.

Step S11.18: Eigenprojections by the adjugate. Fix w, an orthonormal eigenbasis v_k ($M(w)v_k = \sigma_k v_k$). In this basis $\mathrm{adj}(\sigma_j I - M) = \mathrm{diag}\big(\prod_{l\neq k}(\sigma_j - \sigma_l)\big)_k$ has only its (j,j) entry nonzero, so $\mathrm{adj}(\sigma_j I - M) = p_s(\sigma_j)\, v_j v_j^\top$ with $p_s = \partial_s p(w,\cdot) \neq 0$. Hence $\Pi_j(w) := \mathrm{adj}(\sigma_j(w)I - M(w))/\partial_s p(w, \sigma_j(w)) = v_j v_j^\top$ is the orthogonal projection onto the σ_j-eigenspace, and *real analytic in w* (adjugate: polynomial in the entries).

Step S11.19: Eigenvectors. $\mathrm{tr}\,\Pi_j(w^0) = 1$, so some diagonal entry $\Pi_j(w^0)_{kk} > 0$, with $k = k(j)$; on a smaller ball $\Pi_j(w)_{kk} > 0$ and $v_j(w) := \Pi_j(w)e_k/\sqrt{\Pi_j(w)_{kk}}$ is analytic, of unit norm, an eigenvector for $\sigma_j(w)$ with $\Pi_j v_j = v_j$, hence $v_j = \pm$ the vectors of the eigenbasis. The $v_j(w)$ are orthonormal.

Step S11.20: $SO(n)$ and uniqueness. $\det[v_1|\ldots|v_n]$ is continuous with values ± 1, so constant on the ball; replace v_1 by $-v_1$ if it equals -1. Uniqueness: an eigenvalue branch is a continuous root of p, unique by (E2) and the ordering; eigenvectors for a simple eigenvalue are determined up to sign, locally constant. □

Proposition 11.8 (Symplectic lift of a Λ-dependent rotation)**.** *Let $\mathcal{W} \subset \mathbb{R}^n$ be open (actions Λ) and $\varrho : \mathcal{W} \to SO(n)$ real analytic. On $\mathbb{R}^n \times \mathcal{W} \times \mathbb{R}^n \times \mathbb{R}^n \ni (\lambda, \Lambda, \xi, \eta)$ with the chart form $\Omega_{\mathrm{chart}} = d\Lambda \wedge d\lambda + d\eta \wedge d\xi$ (Theorem 9.12, Part 7) define*

$$\Phi_\varrho(\lambda,\Lambda,\xi,\eta) = (\tilde{\lambda},\Lambda,\tilde{\xi},\tilde{\eta}), \quad \tilde{\xi} = \varrho(\Lambda)^\top\xi,\ \tilde{\eta} = \varrho(\Lambda)^\top\eta,\ \tilde{\lambda}_l = \lambda_l - \eta^\top\big(\partial_{\Lambda_l}\varrho\big)\varrho^\top\xi \quad (l = 1,\ldots,n).$$

Then Φ_ϱ is a real analytic diffeomorphism of $\mathbb{R}^n \times \mathcal{W} \times \mathbb{R}^n \times \mathbb{R}^n$ onto itself with inverse $\xi = \varrho\tilde{\xi}$, $\eta = \varrho\tilde{\eta}$, $\lambda_l = \tilde{\lambda}_l + \tilde{\eta}^\top(\varrho^\top\partial_{\Lambda_l}\varrho)\tilde{\xi}$; it is symplectic, $\Phi_\varrho^*\Omega_{\mathrm{chart}} = \Omega_{\mathrm{chart}}$ (a 1-chart in the sense of Definition 2.3); $\Phi_\varrho(\lambda + c, \Lambda, \xi, \eta) = \Phi_\varrho(\lambda,\Lambda,\xi,\eta) + (c,0,0,0)$ for every $c \in \mathbb{R}^n$ (in particular it commutes with 2π-shifts of the angles); and it preserves Λ and $|\xi|, |\eta|$ (Euclidean norms).
Source: Ours; the lift of a parameter-dependent symplectic map is standard (Maurer–Cartan). Numerically confirmed ($n = 3$, Jacobian by finite differences, $\|J^\top\Omega J - \Omega\|_\infty = 10^{-5}$, scratch script `.drafts/P/bp-scratch-bpP-p9/chk.py`).

Proof. **Step S11.21: Bijectivity.** $\varrho(\Lambda)$ is orthogonal; the displayed inverse is read off, with $H_l(\Lambda,\tilde{\xi},\tilde{\eta}) := \tilde{\eta}^\top\Omega_l\tilde{\xi}$, $\Omega_l := \varrho^\top\partial_{\Lambda_l}\varrho$, since $\eta^\top(\partial_l\varrho)\varrho^\top\xi = \tilde{\eta}^\top\varrho^\top(\partial_l\varrho)\tilde{\xi} = H_l$. All maps are analytic. The commutation with angle translations holds because $\tilde{\lambda} - \lambda$ does not depend on λ.

Step S11.22: Antisymmetry. $\varrho^\top\varrho = I$ gives $\Omega_l^\top = -\Omega_l$; $d\varrho = \varrho\sum_l \Omega_l d\Lambda_l$, $d\varrho^\top = -\sum_l \Omega_l\varrho^\top d\Lambda_l$.

Step S11.23: Pull-back computation. Substitute $\xi = \varrho\tilde\xi$, $\eta = \varrho\tilde\eta$ in $d\eta^\top \wedge d\xi = \sum_i d\eta_i \wedge d\xi_i$:

$$d\eta^\top \wedge d\xi = d\tilde\eta^\top \wedge d\tilde\xi + \sum_l \big(d\tilde\eta^\top \Omega_l \tilde\xi + \tilde\eta^\top \Omega_l d\tilde\xi\big) \wedge d\Lambda_l - \sum_{l,m} \tilde\eta^\top \Omega_l \Omega_m \tilde\xi \; d\Lambda_l \wedge d\Lambda_m,$$

using $\varrho^\top d\varrho = \sum \Omega_l d\Lambda_l$ and $d\varrho^\top \varrho = -\sum \Omega_l d\Lambda_l$ (the mixed terms $d\tilde\eta^\top \varrho^\top \wedge d\varrho\tilde\xi$, $\tilde\eta^\top d\varrho^\top \wedge \varrho\, d\tilde\xi$ and $\tilde\eta^\top d\varrho^\top \wedge d\varrho\,\tilde\xi$). The first bracket is $d_{(\tilde\xi,\tilde\eta)} H_l$, the differential of H_l in the variables $(\tilde\xi,\tilde\eta)$ only. The full differential is $dH_l = d_{(\tilde\xi,\tilde\eta)} H_l + \sum_m \tilde\eta^\top (\partial_m \Omega_l)\tilde\xi \, d\Lambda_m$, so

$$d\eta^\top \wedge d\xi = d\tilde\eta^\top \wedge d\tilde\xi + \sum_l dH_l \wedge d\Lambda_l - \sum_{l,m} \tilde\eta^\top \big(\partial_m \Omega_l + \Omega_m \Omega_l\big)\tilde\xi \; d\Lambda_m \wedge d\Lambda_l.$$

Step S11.24: Maurer–Cartan. $\partial_m \Omega_l = \partial_m \varrho^\top \, \partial_l \varrho + \varrho^\top \partial_m \partial_l \varrho = -\Omega_m \Omega_l + \varrho^\top \partial_m \partial_l \varrho$ (as $\partial_m \varrho^\top = -\Omega_m \varrho^\top$), so $\partial_m \Omega_l + \Omega_m \Omega_l = \varrho^\top \partial_m \partial_l \varrho$ is symmetric in (m,l) and the last sum vanishes against $d\Lambda_m \wedge d\Lambda_l$.

Step S11.25: Conclusion. $d\eta^\top \wedge d\xi = d\tilde\eta^\top \wedge d\tilde\xi - \sum_l d\Lambda_l \wedge dH_l$, hence $\sum_l d\Lambda_l \wedge d\lambda_l + d\eta \wedge d\xi = \sum_l d\Lambda_l \wedge d(\lambda_l - H_l) + d\tilde\eta \wedge d\tilde\xi = \sum_l d\Lambda_l \wedge d\tilde\lambda_l + d\tilde\eta \wedge d\tilde\xi$, i.e. $\Phi_\varrho^* \Omega_{\mathrm{chart}} = \Omega_{\mathrm{chart}}$. Euclidean norms are preserved since ϱ is orthogonal. □

Theorem 11.9 (Diagonalisation and the frozen Birkhoff data)**.** *In the situation of Prop. 10.2 and Theorem 10.11 with $n \geq 2$, let $\mathcal{W} \subset \{\Lambda : a(\Lambda)$ increasing$\}$ be open, and suppose that $Q_h(\Lambda)$ has simple spectrum on $\mathcal{W}$ and σ_j, ϱ are the analytic functions of Proposition 11.7 on $\mathcal{W}$ (this holds on a ball around every Λ^0 where $\operatorname{discr} Q_h \neq 0$; if $\mathcal{W}$ is connected the labelling by increasing order is global on $\mathcal{W}$). Then:*

*(a) (*Quadratic part.*) With $(\tilde\xi,\tilde\eta) = \Phi_\varrho$-image of Prop. 11.8,*

$$\xi^\top Q_h \xi + \eta^\top Q_h \eta = \sum_j \sigma_j(\tilde\xi_j^2 + \tilde\eta_j^2) = \sum_j \Omega_j \tilde R_j, \qquad \tilde R_j := \tfrac12(\tilde\eta_j^2 + \tilde\xi_j^2), \quad \Omega_j := 2\sigma_j < 0.$$

*(b) (*Secular function of the rotated problem.*) For $S \in \{S_{\mathrm{fix}}, S_{\mathrm{mov}}\}$ and $\hat f \in \{\hat f_{\mathrm{fix}}, \hat f_{\mathrm{mov}}\}$ (Prop. 10.2) the angle-average of $\hat f \circ \Phi_\varrho^{-1}$ is*

$$\big\langle \hat f \circ \Phi_\varrho^{-1} \big\rangle(\Lambda,\tilde\xi,\tilde\eta) = S(\Lambda, \varrho\tilde\xi, \varrho\tilde\eta) = C_0 + [\delta] + \sum_j \Omega_j(\Lambda)\tilde R_j + O(|(\tilde\xi,\tilde\eta)|^4),$$

where $[\delta] = 0$ for fix and $= \delta(\Lambda)$ for mov (Theorem 10.13), locally uniformly in $\Lambda \in \mathcal{W}$, on the set where $(\xi,\Lambda,\eta) = (\varrho\tilde\xi, \Lambda, \varrho\tilde\eta) \in \mathcal{N}_{\mathbb{R}}(\rho_*)$; for $\mu > 0$ the Hamiltonians $H_{\mathrm{fix}}^\Psi, H_{\mathrm{mov}}^\Psi$ transformed by Φ_ϱ^{-1} are $h(\Lambda) + \mu\, \hat f \circ \Phi_\varrho^{-1}$ (h depends on Λ only, which Φ_ϱ preserves).

*(c) (*Frozen dictionary.*) With $(I,p,q) = (\Lambda,\tilde\eta,\tilde\xi)$ as in §10.1, $P_0 = C_0(+\delta)$, $\Omega_j(\mu, I) = 2\sigma_j(a(I))$ (independent of μ) and $R_j = \frac12(p_j^2 + q_j^2)$ of Definition 1.14, the zeroth- and first-order terms of* HasBirkhoff *for $P = \hat f \circ \Phi_\varrho^{-1}$ are those of (b).* The matrix β and the integrability clause (i) are not treated here *(Part 11b, Corollary 10.14(c)). The domains: if $\tilde\xi_j^2 + \tilde\eta_j^2 < \delta_0^2$ for all j then $|\xi|, |\eta| < \sqrt{n}\,\delta_0$, so $(\xi,\Lambda,\eta) \in \mathcal{N}_{\mathbb{R}}(\rho_*)$ as soon as $\sqrt{n}\,\delta_0 \leq \rho_*$ and $|\Lambda - \Lambda^*|_\infty < \rho_*$.*

*(d) (*Transport.*) $\Psi \circ \Phi_\varrho^{-1}$ is a symplectic chart of Part 1 (Definition 2.3) on the preimage of $\mathcal{O}_n$, satisfies the lattice compatibility (Q1)–(Q2) of Theorem 2.9 whenever Ψ does (the angle-shift of Φ_ϱ is independent of λ), and the torus-transport theorem applies to it.*

Source: Fej04r Formule (36)–(37), pp. 50–51 and the diagonalisation around Prop. 73, pp. 54–55; CP09 Thm. 4 (properly degenerate form $\sum \Omega_j(u_j^2 + v_j^2)/2$), CP10 ($A_2''$). The factor: Remark 10.12(2) (halving of Q_h, E-P-Fej04r-3). The lift (b), (d) is ours; Fej04r and CP11 pass to "Birkhoff variables" without mentioning the Λ-dependence of the rotation.

Source gap: The sources treat the diagonalising rotation as a plain symplectic change of the slow variables; because ϱ depends on the actions Λ it is symplectic on the full phase space only after the angle correction of Prop. 11.8. The correction is a shift of λ independent of λ and so does not affect any average over λ (part (b)).

Proof. **Step S11.26: (a).** $Q_h = \varrho\,\mathrm{diag}(\sigma)\varrho^\top$ gives $\xi^\top Q_h \xi = \sum_j \sigma_j \tilde\xi_j^2$, similarly for η; and $\sigma_j(\tilde\xi_j^2 + \tilde\eta_j^2) = 2\sigma_j \tilde R_j$. Negativity: Prop. 11.6.

Step S11.27: (b). $\Phi_\varrho^{-1}(\tilde\lambda, \Lambda, \tilde\xi, \tilde\eta) = (\tilde\lambda + H(\Lambda, \tilde\xi, \tilde\eta), \Lambda, \varrho\tilde\xi, \varrho\tilde\eta)$, with H independent of $\tilde\lambda$. Hence $\hat f \circ \Phi_\varrho^{-1}(\tilde\lambda, \cdot) = \hat f(\tilde\lambda + H, \Lambda, \varrho\tilde\xi, \varrho\tilde\eta)$, which is continuous and 2π-periodic in $\tilde\lambda$ (Prop. 10.2(b)). The average over $\tilde\lambda \in [0, 2\pi]^n$ is invariant under the translation by the constant vector H (periodicity, (E2)), so it equals $S(\Lambda, \varrho\tilde\xi, \varrho\tilde\eta)$. Insert Theorem 10.11, Theorem 10.13 and (a): the remainder $O(|(\xi,\eta)|^4)$ becomes $O(|(\tilde\xi,\tilde\eta)|^4)$ since $|\varrho\tilde\xi| = |\tilde\xi|$. The statement on $H^\Psi \circ \Phi_\varrho^{-1}$ is Theorem 9.12(b),(c) and Φ_ϱ^{-1} preserving Λ.

Step S11.28: (c). Read off from (a), (b) and Definition 1.14; the norm comparison is $|\xi|_\infty \le |\xi| = |\tilde\xi| \le \sqrt{n}\max_j|\tilde\xi_j|$ and $\max_j|\tilde\xi_j| < \delta_0$.

Step S11.29: (d). Compositions of symplectic charts are symplectic charts (the pull-back of Ω_{cart} by $\Psi \circ \Phi_\varrho^{-1}$ is $(\Phi_\varrho^{-1})^*\Omega_{\mathrm{chart}} = \Omega_{\mathrm{chart}}$ by Prop. 11.8, Theorem 9.12). (Q1): $\Phi_\varrho^{-1}(z + 2\pi e_k) = \Phi_\varrho^{-1}(z) + 2\pi e_k$ and Part 7's Ψ has the same property; (Q2): if $\Psi\Phi_\varrho^{-1}(z') = \Psi\Phi_\varrho^{-1}(z) + \sigma_k$ then, by the property of Ψ, $\Phi_\varrho^{-1}z' = \Phi_\varrho^{-1}z + 2\pi m$, hence $z' = z + 2\pi m$ as Φ_ϱ is injective and commutes with the shifts. □

Remark 11.10 (Dictionary: Fej04r, CP09/CP10, frozen). (1) Frozen Ω_j multiplies $R_j = \frac12(p_j^2 + q_j^2)$ (Definition 1.14; CP09 Thm. 4 writes $\Omega_j(u_j^2 + v_j^2)/2$). Under the frozen chart (Part 7, $|u|^2 = 2\Gamma$) $S = C_0 + \sum \sigma_j(\tilde\xi_j^2 + \tilde\eta_j^2) = C_0 + \sum 2\sigma_j\tilde R_j$, so $\Omega_j = 2\sigma_j$ with σ_j the eigenvalues of the *halved* Q_h of Definition 10.10; they are the eigenvalues of the matrix printed in Fej04r (36)–(37) in the ξ-basis, i.e. of $Q_h^{\mathrm F} = 2Q_h$ (suspected erratum E-P-Fej04r-3, Remark 10.12(2)). Both sets of eigenvalues are negative (Prop. 11.6). (2) Sign: $\Omega_j < 0$ for all j. CP10 (A_2''), (8) only requires $|\sum_j \Omega_j k_j| \ge \mathrm{const} > 0$ for $0 < |k|_1 \le 4$, which is insensitive to the overall sign. (3) Coordinate permutation: the Lean order of the entries of Q_h is that of increasing axes (Q_h index i = planet i); the eigenvalue labelling is increasing in σ and independent of it; the frozen $\Omega : \mathbb{R}\times\mathbb{R}^a \to \mathbb{R}^b$ is total, so Part 12 sets it to 0 off the patch (nothing is claimed there). (4) $n = 1$: there is no secular problem, $Q_h = 0$ and $\Omega_1 = 0$ (no pair); statements with $n \ge 2$ above do not apply.

11.5. Distinct and non-resonant spectrum: the interface for Parts 10–11.

Definition 11.11 (Non-resonance up to order four; the function Φ_4)**.** Let $K_4 = \{k \in \mathbb{Z}^n : 0 < |k|_1 \le 4\}$. For $\sigma \in \mathbb{R}^n$ put

$$\mathrm{NR}_4(\sigma) :\Longleftrightarrow \ \sigma_j \text{ pairwise distinct and } k\cdot\sigma \neq 0 \ \ \forall k \in K_4,$$

and

$$\mathcal{R}_k(\sigma) = \prod_{\pi\in S_n}\sum_j k_j\sigma_{\pi(j)} \quad (k\in K_4), \qquad \mathcal{D}(\sigma) = \prod_{j<l}(\sigma_j - \sigma_l)^2, \qquad \Phi_4(\sigma) = \mathcal{D}(\sigma)\prod_{k\in K_4}\mathcal{R}_k(\sigma).$$

For $a \in \{a_1 < \cdots < a_n\}$ write $\Phi_4(a) := \Phi_4(\sigma(a))$ for the eigenvalues $\sigma(a)$ of $Q_h(a)$ in any order. (Since $\sigma \mapsto 2\sigma$ rescales, $\mathrm{NR}_4(\sigma) \Leftrightarrow \mathrm{NR}_4(\Omega)$, $\Omega = 2\sigma$; and the quantitative form $|\Omega\cdot k| \ge \alpha > 0$ on K_4 of CP10 (8) holds on every compact subset of $\{\mathrm{NR}_4\}$ by continuity.)

Source: CP10 Remark 1 and (A_2'), pp. 548–549, formula (8) with 6 replaced by 4; Prop. 2 (27), p. 553.

Proposition 11.12 (The good set). *(a) Φ_4 is a symmetric polynomial in σ; hence $a \mapsto \Phi_4(a)$ is a polynomial in the coefficients of* $\det(sI - Q_h(a))$ *and is* real analytic on $\{a_1 < \cdots < a_n\}$, *and independent of the labelling.*

(b) $\Phi_4(a) \neq 0$ if and only if $\mathrm{NR}_4(\sigma(a))$*; then $Q_h(a)$ has simple spectrum, all $\Omega_j = 2\sigma_j \neq 0$, and Propositions 11.7, Theorem 11.9 apply on a ball around a.*

(c) Let $\mathcal{A} \subset \{a_1 < \cdots < a_n\}$ be open and connected. Either $\Phi_4 \equiv 0$ on $\mathcal{A}$ or $\mathcal{G} := \{a \in \mathcal{A} : \Phi_4(a) \neq 0\}$ is open, dense and of full Lebesgue measure in $\mathcal{A}$. So: one *point $a^0 \in \mathcal{A}$ with* $\mathrm{NR}_4(\sigma(a^0))$ *gives an open dense full-measure subset on every point of which the eigenvalues are analytic and distinct and Ω is non-resonant to order four, uniformly on compact subsets of $\mathcal{G}$.*

Proof. **Step S11.30: (a).** $\mathcal{D}$ is symmetric (squares), and $\mathcal{R}_k$ is invariant under $\sigma \mapsto \sigma \circ \pi'$ because the product runs over all of S_n; so Φ_4 is a symmetric polynomial and, by (E3), a polynomial Π in the elementary symmetric polynomials of σ, which up to sign are the coefficients of $\det(sI - Q_h(a))$ ((E1)); these are polynomials in the entries of $Q_h(a)$, real analytic by Definition 10.10.

Step S11.31: (b). $\mathcal{D} \neq 0$ iff the σ_j are distinct; $\mathcal{R}_k \neq 0$ iff $\sum k_j \sigma_{\pi(j)} \neq 0$ for all π, iff $k' \cdot \sigma \neq 0$ for all rearrangements k' of k. K_4 is invariant under rearrangement of coordinates, so the conjunction over $k \in K_4$ is $k \cdot \sigma \neq 0$ for all $k \in K_4$. Simple spectrum: Prop. 11.7 (with discr $= \mathcal{D} \neq 0$).

Step S11.32: (c). Identity theorem (E4) for the analytic function Φ_4 on connected $\mathcal{A}$: if it is not identically zero, it does not vanish on any open set; its zero set is closed, nowhere dense and null. Uniformity on compacts: continuity of $a \mapsto \sigma(a)$ (continuity of roots; or locally Prop. 11.7) and finiteness of K_4. □

Lemma 11.13 (The top eigenvalue is always simple). *For $n \geq 2$ and increasing axes, the largest eigenvalue $\sigma_{\max}$ of Q_h (the one closest to 0) is simple and has an eigenvector with all entries nonzero of one sign.*

Proof. **Step S11.33: Positivity.** Off the diagonal, $(Q_h)_{jk} = \mathfrak{b}_{jk}/\sqrt{\Lambda_j \Lambda_k} > 0$ for *all* $j \neq k$ (Definition 10.10, including non-adjacent planets). Take c with $M = Q_h + cI$ entrywise > 0. Let x be a unit eigenvector for $\sigma_{\max}$; $|x|$ (entrywise) satisfies $|x|^\top M |x| \geq x^\top M x = \max_{|y|=1} y^\top M y$, so $|x|$ is also a maximiser, hence an eigenvector for $\sigma_{\max}$, and $M|x| = (\sigma_{\max} + c)|x|$ has strictly positive entries, hence $|x| > 0$ entrywise. So every eigenvector for $\sigma_{\max}$ has all entries nonzero. If the eigenspace had dimension ≥ 2 it would contain a nonzero vector with a vanishing first entry, a contradiction. Sign: $|x|$ and x are both eigenvectors of a simple eigenvalue, so $x = \pm|x|$. □

Corollary 11.14 ($n = 2$: explicit). *Let $n = 2$, $\alpha = a_1/a_2$, $\mathfrak{a} = \mathfrak{a}_{12}$, $\mathfrak{b} = \mathfrak{b}_{12}$. Then*

$$Q_h = \begin{pmatrix} -\mathfrak{a}/\Lambda_1 & \mathfrak{b}/\sqrt{\Lambda_1\Lambda_2} \\ \mathfrak{b}/\sqrt{\Lambda_1\Lambda_2} & -\mathfrak{a}/\Lambda_2 \end{pmatrix}, \quad \operatorname{tr} Q_h = -\mathfrak{a}\Big(\frac{1}{\Lambda_1} + \frac{1}{\Lambda_2}\Big), \quad \det Q_h = \frac{\mathfrak{a}^2 - \mathfrak{b}^2}{\Lambda_1\Lambda_2} > 0,$$

$$\operatorname{tr}^2 - 4\det = \mathfrak{a}^2\Big(\frac{1}{\Lambda_1} - \frac{1}{\Lambda_2}\Big)^2 + \frac{4\mathfrak{b}^2}{\Lambda_1\Lambda_2} > 0.$$

Hence for every $a_1 < a_2$ the eigenvalues $\sigma_1 < \sigma_2 < 0$ are distinct and real analytic with ϱ analytic on all of $\{a_1 < a_2\}$ (no patch needed for $n = 2$; the discriminant never vanishes)*, and, with $t = \sigma_1/\sigma_2 > 1$,*

$$\mathrm{NR}_4(\sigma) \iff t \notin \{2, 3\} \iff \big(2\operatorname{tr}^2 - 9\det\big)\big(3\operatorname{tr}^2 - 16\det\big) \neq 0.$$

So for $n = 2$, Φ_4 (up to the nonvanishing factors $\mathcal{D}$ and powers of σ_j) is $(2\operatorname{tr}^2 - 9\det)(3\operatorname{tr}^2 - 16\det)$*, an explicit analytic function of a built from $b^{(1)}_{3/2}, b^{(2)}_{3/2}$; Part 10 must show that it does not vanish identically, or at one point.*

Proof. **Step S11.34: Matrix.** Definition 10.10 with $n=2$: $\hat{q}_{11}=\hat{q}_{22}=-\mathfrak{a}$ (as $\mathfrak{a}_{21}=\mathfrak{a}_{12}$), $\hat{q}_{12}=\mathfrak{b}$; conjugate by Δ^{-1}. $\det = (\mathfrak{a}^2-\mathfrak{b}^2)/(\Lambda_1\Lambda_2) > 0$ by Lemma 11.5; the discriminant of a 2×2 symmetric matrix is $(q_{11}-q_{22})^2+4q_{12}^2$, here >0 because $\mathfrak{b}>0$.

Step S11.35: Resonances. $\sigma_j<0$ (Prop. 11.6). For $k\in K_4$ with a zero entry, $k\cdot\sigma\neq 0$. For $k_1k_2>0$, $k\cdot\sigma<0$. For $k_1k_2<0$ ($|k|_1\le 4$): $k\cdot\sigma=0 \iff \sigma_1/\sigma_2=-k_2/k_1\in\{1,2,\frac{1}{2},3,\frac{1}{3}\}$ (from $(1,-1),(2,-2)\mapsto 1$; $(1,-2),(2,-1)\mapsto 2,\frac{1}{2}$; $(1,-3),(3,-1)\mapsto 3,\frac{1}{3}$). Distinctness excludes 1; with $\sigma_1<\sigma_2<0$, $t=\sigma_1/\sigma_2>1$ so $\frac{1}{2},\frac{1}{3}$ are impossible. Finally $\mathrm{tr}^2/\det = t+1/t+2$ equals $\frac{9}{2}$ at $t=2$ and $\frac{16}{3}$ at $t=3$, and $t\mapsto t+1/t$ is injective on $(1,\infty)$. □

11.6. **Sources, deviations, errata, forward references.**
Sources read. Fej04r pp. 52–56 ("7.1 Lemmes sur les valeurs propres", Lemmas 70–75, Prop. 73 with its proof, start of 7.3); Fej07e (`fejoz04_erratum.txt`); CP10 pp. 548–549, 553 (non-resonance of order four, Prop. 2); CP09 Thm. 4 for the form $\sum\Omega_j(u_j^2+v_j^2)/2$ (via Part 8); Part 8 of this blueprint in full; Mathlib names in (E1)–(E3) were checked in `.lake/packages/mathlib`.
Deviations. (1) Lemma 11.2 is a new statement: no ordering of d, no hypothesis on $\|D\|$ (the printed bound $\propto(1+\|D\|/\sigma)^2$ is replaced by an absolute constant), eigenvectors included; the proof is a direct eigenvector elimination, not the conjugation $e^{-\varepsilon U}(D+\varepsilon A)e^{\varepsilon U}$ (whose bound $\|U\|\le\|A\|/\sigma$ is not elementary for the operator norm, while for the Frobenius norm the product estimates change). (2) Corollary 11.3 adds the boundedness hypothesis on A_ε. (3) Lemma 74 is replaced by Lemma 11.5 (one inequality, complete proof). (4) The lift Prop. 11.8 and the explicit Φ_4 are not in the sources. (5) Eigenvalue labelling by increasing value; Q_h halved against the print (Part 8).
Suspected errata (for the ledger). E-P-Fej04r-4 (new): Lemma 71, p. 53, statement versus proof: the statement has $\frac{3}{\sigma}$, the last line of the proof $\frac{2}{\sigma}$, and "ε assez petit" is not quantified. E-P-Fej04r-5 (new, minor): Lemma 74, p. 54 is stated for all $j\in\mathbb{N}$ and $s\in\mathbb{R}_+\setminus\mathbb{N}$ with an induction on s whose recurrences are quoted, not proved, (not completed in the source as printed); not used beyond $j=1$. E-P-Fej04r-6 (minor): Cor. 72 does not say that A_ε is bounded. Already recorded: E-P-Fej04r-3 (factor $\frac{1}{2}$), E-P-Fej04r-2 (Lemma 76).
Doubts about frozen statements. None. The Birkhoff form is stated for the given (p,q) (Definition 1.14): with ϱ depending on the action it is reached from the physical chart only through Φ_ϱ (Prop. 11.8), so the *perturbation* handed to P-G is $\hat{f}\circ\Phi_\varrho^{-1}$, not $\hat{f}$.
External facts used. (E1)–(E5); (E2); Parts 7, 8, 0 and Part 1 (chart, torus transport).
Forward references and consumers. Part 10: Prop. 11.12(c) (must exhibit one point with $\Phi_4\neq 0$ on the connected domain of axes, or a point of each patch), Corollary 11.14 ($n=2$: show $(2\,\mathrm{tr}^2-9\det)(3\,\mathrm{tr}^2-16\det)\not\equiv 0$; asymptotically $\sigma_{1,2}\sim$ explicit, giving ratios $\neq 2,3$), Cor. 11.3 (induction for $n\ge 3$), Lemma 11.13. Parts 11b/12: Prop. 11.7, Theorem 11.9 (the rotation Φ_ϱ and $\Omega=2\sigma$), Definition 11.11. Part 13 (fix): same, by Corollary 10.14. No forward references to unwritten parts in this part.
Label index. `lemma70`, `lemma71`, `cor72`, `l75`, `l74`, `definite`, `local-analytic`, `lift`, `diagonalize`, `dict`, `good`, `good-set`, `top-simple`, `n2`.